Bony framework

Bones that contribute significantly to the boundaries of the temporal and infratemporal fossae include the temporal, zygomatic, and sphenoid bones, and the maxilla and mandible (Figs. 8.132 and 8.133).

Parts of the frontal and parietal bones are also involved.

Temporal bone

The squamous part of the temporal bone forms part of the bony framework of the temporal and infratemporal fossae.

The tympanic part of the temporal bone forms the posteromedial corner of the roof of the infratemporal fossa, and also articulates with the head of the mandible to form the temporomandibular joint.

The lateral surface of the squamous part of the temporal bone is marked by two surface features on the medial wall of the temporal fossa:

- a transversely oriented **supramastoid crest**, which extends posteriorly from the base of the zygomatic process and marks the posteroinferior border of the temporal fossa; and
- a vertically oriented **groove for the middle temporal artery**, a branch of the superficial temporal artery.

Two features that participate in forming the temporomandibular joint on the inferior aspect of the root of the zygomatic process are the articular tubercle and the mandibular fossa. Both are elongate from medial to lateral. Posterior to the mandibular fossa is the external acoustic meatus. The tympanic part of the temporal bone is a flat concave plate of bone that curves inferiorly from the back of the mandibular fossa and forms part of the wall of the external auditory meatus.

When viewed from inferiorly, there is a distinct **tympanosquamous fissure** between the tympanic and

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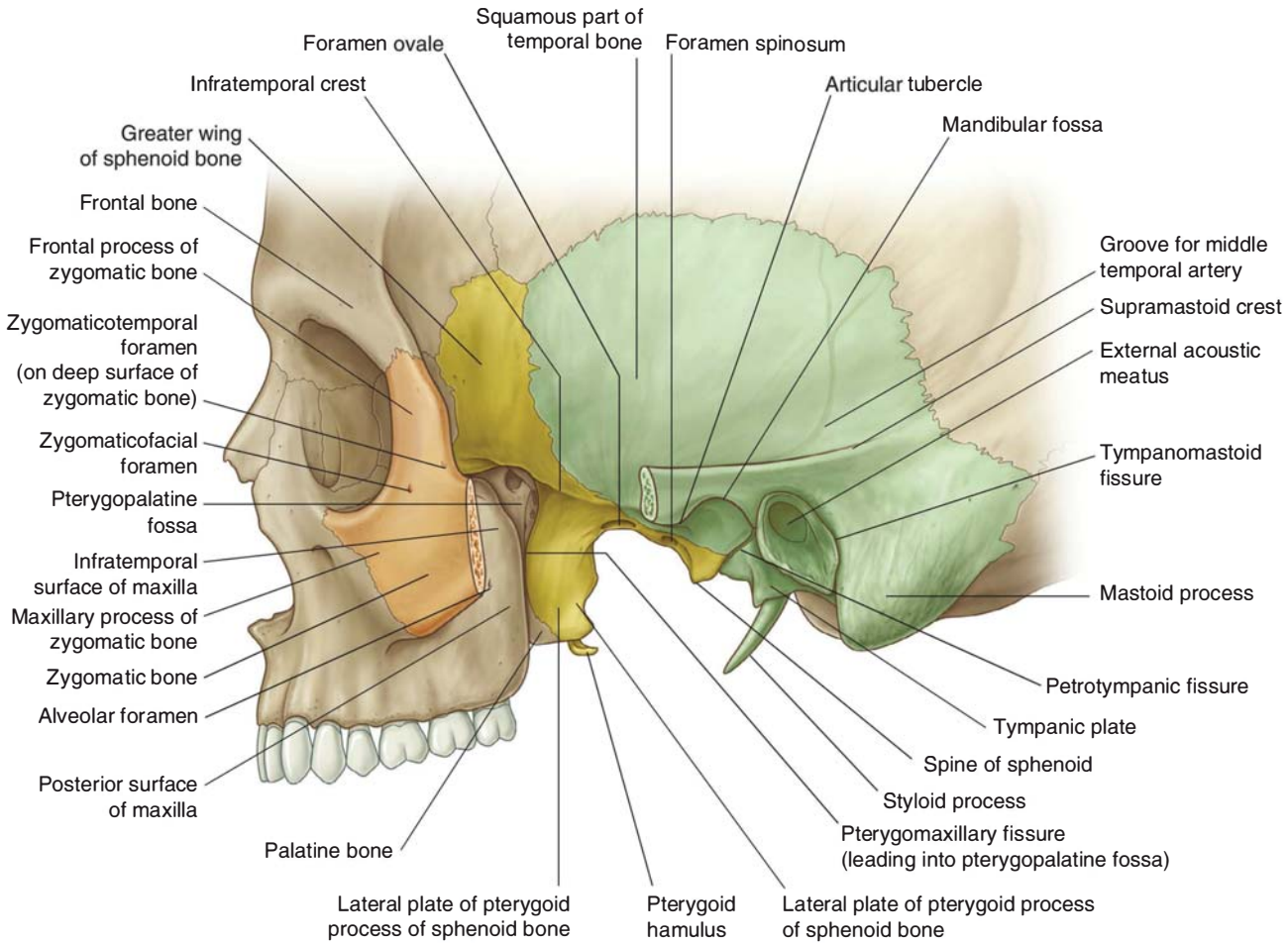


Fig. 8.132 Bony features related to the temporal and infratemporal fossae.

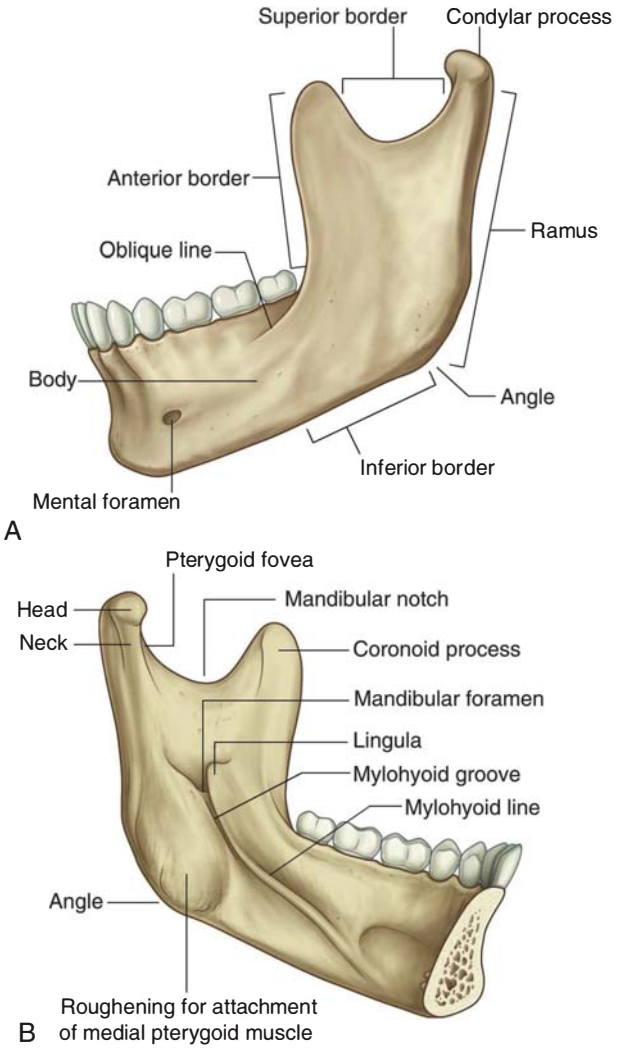


Fig. 8.133 Mandible. **A.** Lateral view of left side. **B.** Medial view of left side.

squamous parts of the temporal bone. Medially, a small slip of bone from the petrous part of the temporal bone insinuates itself into the fissure and forms a **petrotympanic fissure** between it and the tympanic part (Fig. 8.132).

The chorda tympani nerve exits the skull and enters the infratemporal fossa through the medial end of the petrotympanic fissure.

Sphenoid bone

The parts of the sphenoid bone that form part of the bony framework of the infratemporal fossa are the lateral plate of the pterygoid process and the greater wing (Fig. 8.132). The greater wing also forms part of the medial wall of the temporal fossa.

The greater wings extend one on each side from the body of the sphenoid. They project laterally from the body and curve superiorly. The inferior and lateral surfaces form the roof of the infratemporal fossa and the medial wall of the temporal fossa, respectively.

The sharply angled boundary between the lateral and inferior surfaces of the greater wing is the **infratemporal crest** (Fig. 8.132). Two apertures (the foramen ovale and the foramen spinosum) pass through the base of the greater wing and allow the mandibular nerve [V_3] and the middle meningeal artery, respectively, to pass between the middle cranial fossa and infratemporal fossa. In addition, one or more small sphenoidal emissary foramina penetrate the base of the greater wing anteromedial to the foramen ovale and allow emissary veins to pass between the pterygoid plexus of veins in the infratemporal fossa and the cavernous sinus in the middle cranial fossa.

Projecting vertically downward from the greater wing immediately medial to the foramen spinosum is the irregularly shaped **spine of the sphenoid**, which is the attachment site for the cranial end of the sphenomandibular ligament.

The lateral plate of the pterygoid process is a vertically oriented sheet of bone that projects posterolaterally from the pterygoid process (Fig. 8.132). Its lateral and medial surfaces provide attachment for the lateral and medial pterygoid muscles, respectively.

Maxilla

The posterior surface of the maxilla contributes to the anterior wall of the infratemporal fossa (Fig. 8.132). This surface is marked by a foramen for the posterosuperior alveolar nerve and vessels. The superior margin forms the inferior border of the inferior orbital fissure.

Zygomatic bone

The zygomatic bone is a quadrangular-shaped bone that forms the palpable bony prominence of the cheek:

- A **maxillary process** extends anteromedially to articulate with the zygomatic process of the maxilla.
- A **frontal process** extends superiorly to articulate with the zygomatic process of the frontal bone.
- A **temporal process** extends posteriorly to articulate with the zygomatic process of the temporal bone to complete the zygomatic arch.

A small zygomaticofacial foramen on the lateral surface of the zygomatic bone transmits the zygomaticofacial nerve and vessels onto the cheek.

A thin plate of bone extends posteromedially from the frontal process and contributes to the lateral wall of the orbit on one side and the anterior wall of the temporal fossa on the other. A zygomaticotemporal foramen on the

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temporal fossa surface of the plate where it attaches to the frontal process is for the zygomaticotemporal nerve.

Ramus of mandible

The **ramus of the mandible** is quadrangular in shape and has medial and lateral surfaces and condylar and coronoid processes (Fig. 8.133).

The lateral surface of the ramus of the mandible is generally smooth except for the presence of a few obliquely oriented ridges. Most of the lateral surface provides attachment for the masseter muscle.

The posterior and inferior borders of the ramus intersect to form the **angle of the mandible**, while the superior border is notched to form the **mandibular notch**. The anterior border is sharp and is continuous below with the **oblique line** on the body of the mandible.

The **coronoid process** extends superiorly from the junction of the anterior and superior borders of the ramus. It is a flat, triangular process that provides attachment for the temporalis muscle.

The **condylar process** extends superiorly from the posterior and superior borders of the ramus. It consists of:

- the **head of the mandible**, which is expanded medially and participates in forming the temporomandibular joint; and
- the **neck of the mandible**, which bears a shallow depression (the **pterygoid fovea**) on its anterior surface for attachment of the lateral pterygoid muscle.

The medial surface of the ramus of the mandible is the lateral wall of the infratemporal fossa (Fig. 8.133B). Its most distinctive feature is the **mandibular foramen**, which is the superior opening of the mandibular canal. The inferior alveolar nerve and vessels pass through this foramen.

Immediately anterosuperior to the mandibular foramen is a triangular elevation (the **lingula**) for attachment of the mandibular end of the sphenomandibular ligament.

An elongate groove (the **mylohyoid groove**) extends anteroinferiorly from the mandibular foramen. The **nerve to the mylohyoid** is in this groove.

Posteroinferior to the mylohyoid groove and mandibular foramen, the medial surface of the ramus of the mandible is roughened for attachment of the medial pterygoid muscle.

Temporomandibular joints

The temporomandibular joints, one on each side, allow opening and closing of the mouth and complex chewing or side-to-side movements of the lower jaw.

Each joint is synovial and is formed between the head of the mandible and the articular fossa and articular tubercle of the temporal bone (Fig. 8.134A).

Unlike most other synovial joints where the articular surfaces of the bones are covered by a layer of hyaline cartilage, those of the temporomandibular joint are covered

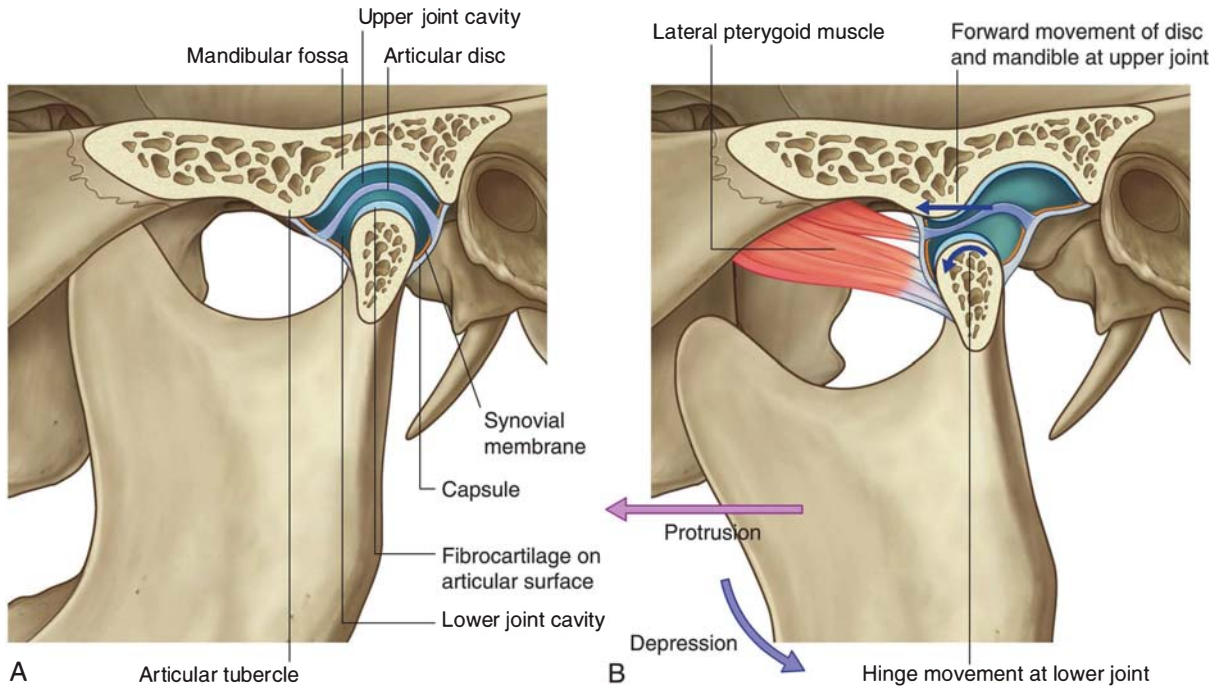


Fig. 8.134 Temporomandibular joint. **A.** Mouth closed. **B.** Mouth open.

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by fibrocartilage. In addition, the joint is completely divided by a fibrous **articular disc** into two parts:

- The lower part of the joint allows mainly the hinge-like depression and elevation of the mandible.
- The upper part of the joint allows the head of the mandible to translocate forward (protrusion) onto the articular tubercle and backward (retraction) into the mandibular fossa.

Opening the mouth involves both depression and protrusion (Fig. 8.134B).

The forward or protrusive movement allows greater depression of the mandible by preventing backward movement of the angle of the mandible into structures in the neck.

Joint capsule

The **synovial membrane** of the joint capsule lines all nonarticular surfaces of the upper and lower compartments of the joint and is attached to the margins of the articular disc.

The **fibrous membrane** of the joint capsule encloses the temporomandibular joint complex and is attached:

- above along the anterior margin of the articular tubercle,
- laterally and medially along the margins of the articular fossa,
- posteriorly to the region of the tympanosquamous suture, and
- below around the upper part of the neck of the mandible.

The articular disc attaches around its periphery to the inner aspect of the fibrous membrane.

Extracapsular ligaments

Three extracapsular ligaments are associated with the temporomandibular joint—the lateral, sphenomandibular, and the stylomandibular ligaments (Fig. 8.135):

- The **lateral ligament** is closest to the joint, just lateral to the capsule, and runs diagonally backward from the margin of the articular tubercle to the neck of the mandible.
- The **sphenomandibular ligament** is medial to the temporomandibular joint, runs from the spine of the sphenoid bone at the base of the skull to the lingula on the medial side of the ramus of the mandible.
- The **stylomandibular ligament** passes from the styloid process of the temporal bone to the posterior margin and angle of the mandible.

Movements of the mandible

A chewing or grinding motion occurs when the movements at the temporomandibular joint on one side are coordinated with a reciprocal set of movements at the joint on the other side. Movements of the mandible include depression, elevation, protrusion, and retraction (Fig. 8.136):

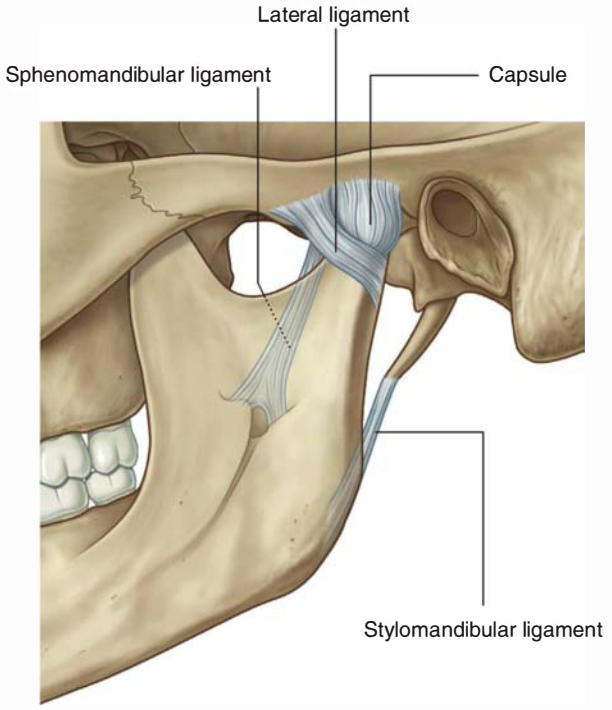


Fig. 8.135 Ligaments associated with the temporomandibular joint.

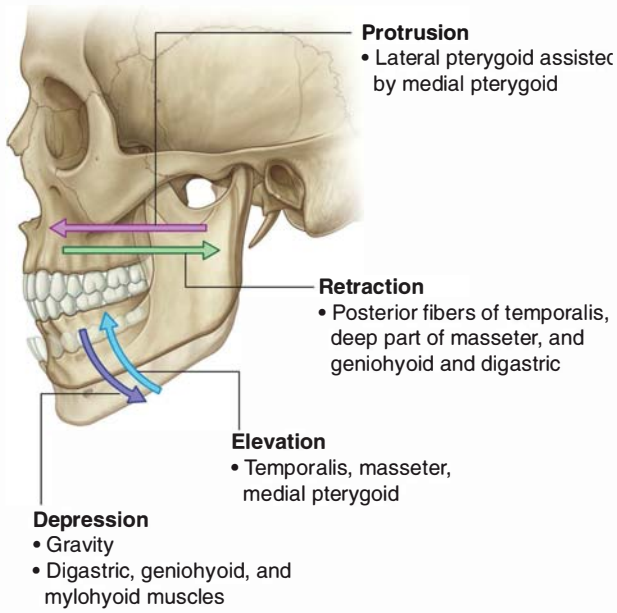


Fig. 8.136 Movements of the temporomandibular joint.

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- Depression is generated by the digastric, geniohyoid, and mylohyoid muscles on both sides, is normally assisted by gravity, and, because it involves forward movement of the head of the mandible onto the articular tubercle, the lateral pterygoid muscles are also involved.
- Elevation is a very powerful movement generated by the temporalis, masseter, and medial pterygoid muscles and also involves movement of the head of the mandible into the mandibular fossa.
- Protraction is mainly achieved by the lateral pterygoid muscle, with some assistance by the medial pterygoid.
- Retraction is carried out by the geniohyoid and digastric muscles, and by the posterior and deep fibers of the temporalis and masseter muscles, respectively.

Except for the geniohyoid muscle, which is innervated by the C1 spinal nerve, all muscles that move the temporomandibular joints are innervated by the mandibular nerve [V_3] by branches that originate in the infratemporal fossa.

Masseter muscle

The **masseter** muscle is a powerful muscle of mastication that elevates the mandible (Fig. 8.137 and Table 8.11). It overlies the lateral surface of the ramus of the mandible.

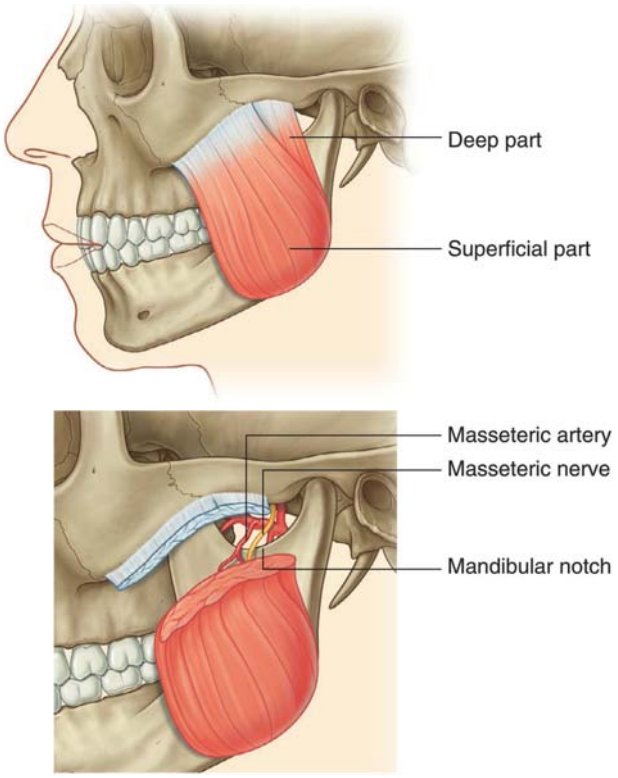


Fig. 8.137 Masseter muscle.

Table 8.11 Muscles of mastication

Muscle	Origin	Insertion	Innervation	Function
Masseter	Zygomatic arch and maxillary process of the zygomatic bone	Lateral surface of ramus of mandible	Masseteric nerve from the anterior trunk of the mandibular nerve [V_3]	Elevation of mandible
Temporalis	Bone of temporal fossa and temporal fascia	Coronoid process of mandible and anterior margin of ramus of mandible almost to last molar tooth	Deep temporal nerves from the anterior trunk of the mandibular nerve [V_3]	Elevation and retraction of mandible
Medial pterygoid	Deep head—medial surface of lateral plate of pterygoid process and pyramidal process of palatine bone; superficial head—tuberosity of the maxilla and pyramidal process of palatine bone	Medial surface of mandible near angle	Nerve to medial pterygoid from the mandibular nerve [V_3]	Elevation and side-to-side movements of the mandible
Lateral pterygoid	Upper head—roof of infratemporal fossa; lower head—lateral surface of lateral plate of the pterygoid process	Capsule of temporomandibular joint in the region of attachment to the articular disc and to the pterygoid fovea on the neck of mandible	Nerve to lateral pterygoid directly from the anterior trunk of the mandibular nerve [V_3] or from the buccal branch	Protrusion and side-to-side movements of the mandible

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The masseter muscle is quadrangular in shape and is anchored above to the zygomatic arch and below to most of the lateral surface of the ramus of the mandible.

The more **superficial part** of the masseter originates from the maxillary process of the zygomatic bone and the anterior two-thirds of the zygomatic process of the maxilla. It inserts into the angle of the mandible and related posterior part of the lateral surface of the ramus of the mandible.

The **deep part** of the masseter originates from the medial aspect of the zygomatic arch and the posterior part of its inferior margin and inserts into the central and upper part of the ramus of the mandible as high as the coronoid process.

The masseter is innervated by the masseteric nerve from the mandibular nerve [V_3] and supplied with blood by the masseteric artery from the maxillary artery.

The masseteric nerve and artery originate in the infratemporal fossa and pass laterally over the margin of the mandibular notch to enter the deep surface of the masseter muscle.

Temporal fossa

The temporal fossa is a narrow fan-shaped space that covers the lateral surface of the skull (Fig. 8.138A):

- Its upper margin is defined by a pair of temporal lines that arch across the skull from the zygomatic process of the frontal bone to the supramastoid crest of the temporal bone.
- It is limited laterally by the **temporal fascia**, which is a tough, fan-shaped aponeurosis overlying the temporalis muscle and attached by its outer margin to the superior temporal line and by its inferior margin to the zygomatic arch.
- Anteriorly, it is limited by the posterior surface of the frontal process of the zygomatic bone and the posterior surface of the zygomatic process of the frontal bone, which separate the temporal fossa behind from the orbit in front.

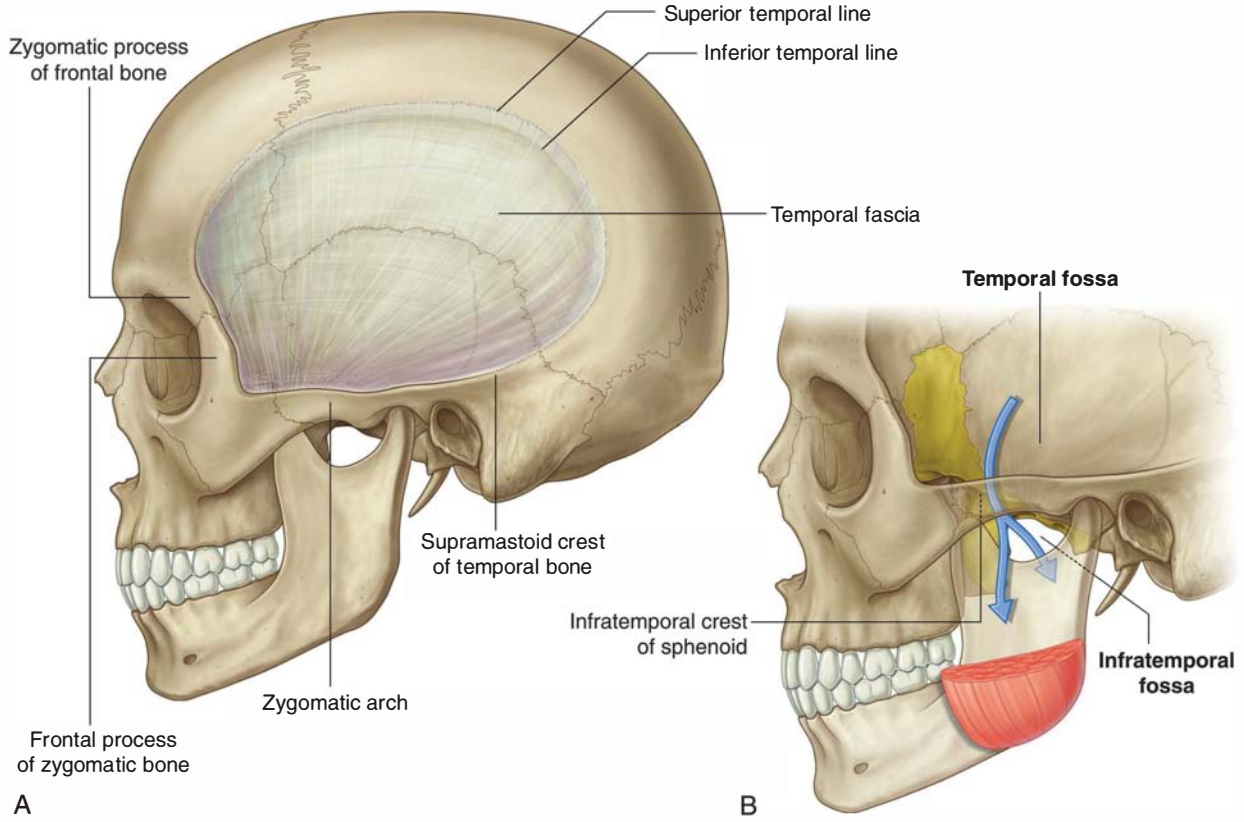


Fig. 8.138 Temporal fossa. **A.** Lateral view. **B.** Lateral view showing the infratemporal fossa.

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- Its inferior margin is marked by the zygomatic arch laterally and by the infratemporal crest of the greater wing of the sphenoid medially (Fig. 8.138B)—between these two features, the floor of the temporal fossa is open medially to the infratemporal fossa and laterally to the region containing the masseter muscle.

Contents

The major structure in the temporal fossa is the temporalis muscle.

Also passing through the fossa is the zygomaticotemporal branch of the maxillary nerve [V_2], which enters the region through the zygomaticotemporal foramen on the temporal fossa surface of the zygomatic bone.

Temporalis muscle

The **temporalis muscle** is a large, fan-shaped muscle that fills much of the temporal fossa (Fig. 8.139). It originates from the bony surfaces of the fossa superiorly to the inferior temporal line and is attached laterally to the surface of the temporal fascia. The more anterior fibers are oriented vertically while the more posterior fibers are oriented horizontally. The fibers converge inferiorly to form a tendon, which passes between the zygomatic arch and the infratemporal crest of the greater wing of the sphenoid to insert on the coronoid process of the mandible.

The temporalis muscle attaches down the anterior surface of the coronoid process and along the related margin of the ramus of the mandible, almost to the last molar tooth.

The temporalis is a powerful elevator of the mandible. Because this movement involves posterior translocation of the head of the mandible from the articular tubercle of the temporal bone and back into the mandibular fossa, the temporalis also retracts the mandible or pulls it posteriorly. In addition, the temporalis participates in side-to-side movements of the mandible.

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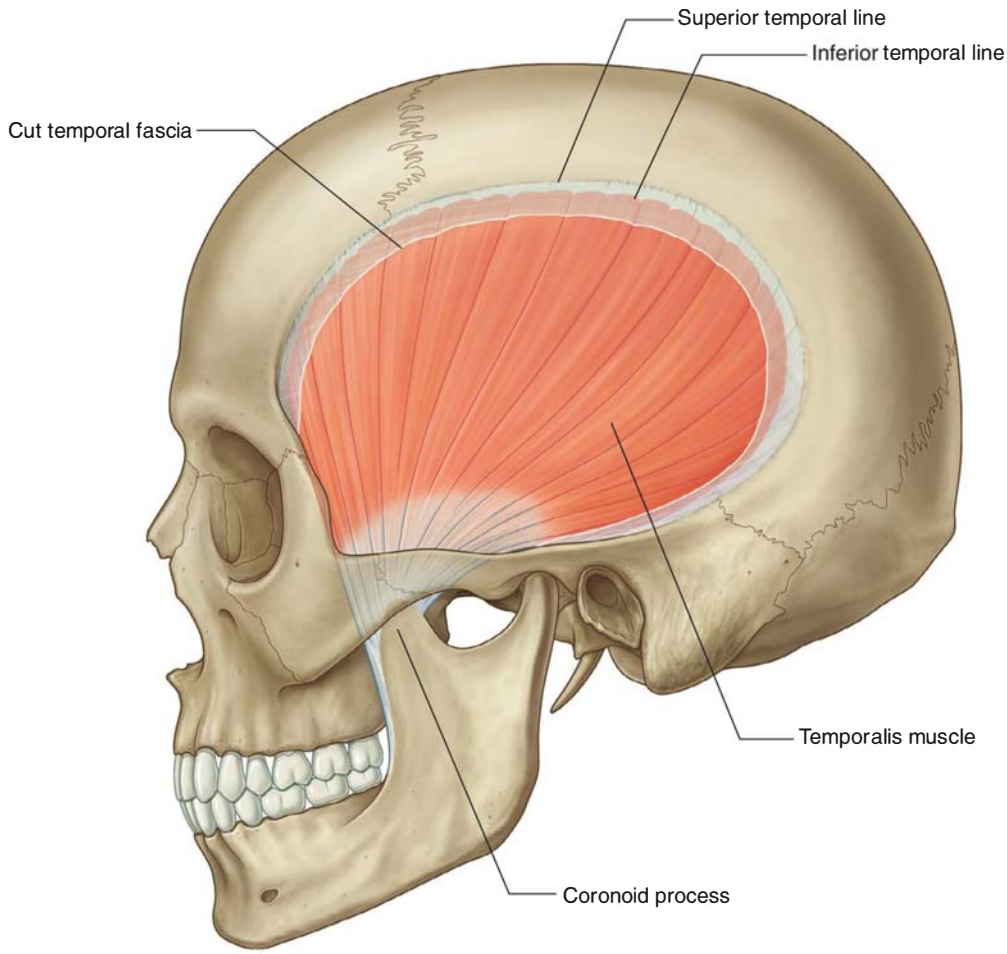


Fig. 8.139 Temporalis muscle. Lateral view.

The temporalis is innervated by deep temporal nerves that originate from the mandibular nerve [V_3] in the infratemporal fossa and then pass into the temporal fossa.

Blood supply of the temporalis is by deep temporal arteries, which travel with the nerves, and the middle temporal artery, which penetrates the temporal fascia at the posterior end of the zygomatic arch.

Deep temporal nerves

The deep temporal nerves, usually two in number, originate from the anterior trunk of the mandibular nerve [V_3] in the infratemporal fossa (Fig. 8.140). They pass superiorly and around the infratemporal crest of the greater wing of the sphenoid to enter the temporal fossa deep to the temporalis muscle, and supply the temporalis muscle.

Zygomaticotemporal nerve

The zygomaticotemporal nerve is a branch of the zygomatic nerve (see Fig. 8.84, p. 933). The zygomatic nerve is a branch of the maxillary nerve [V_2], which originates in the pterygopalatine fossa and passes into the orbit.

The zygomaticotemporal nerve enters the temporal fossa through one or more small foramina on the temporal fossa surface of the zygomatic bone.

Branches of the zygomaticotemporal nerve pass superiorly between the bone and the temporalis muscle to penetrate the temporal fascia and supply the skin of the temple (Fig. 8.140).

Deep temporal arteries

Normally two in number, these vessels originate from the maxillary artery in the infratemporal fossa and travel with the deep temporal nerves around the infratemporal crest of the greater wing of the sphenoid to supply the temporalis muscle (Fig. 8.140). They anastomose with branches of the middle temporal artery.

Middle temporal artery

The middle temporal artery originates from the superficial temporal artery just superior to the root of the zygomatic arch between this structure and the external ear (Fig. 8.140). It penetrates the temporalis fascia, passes under

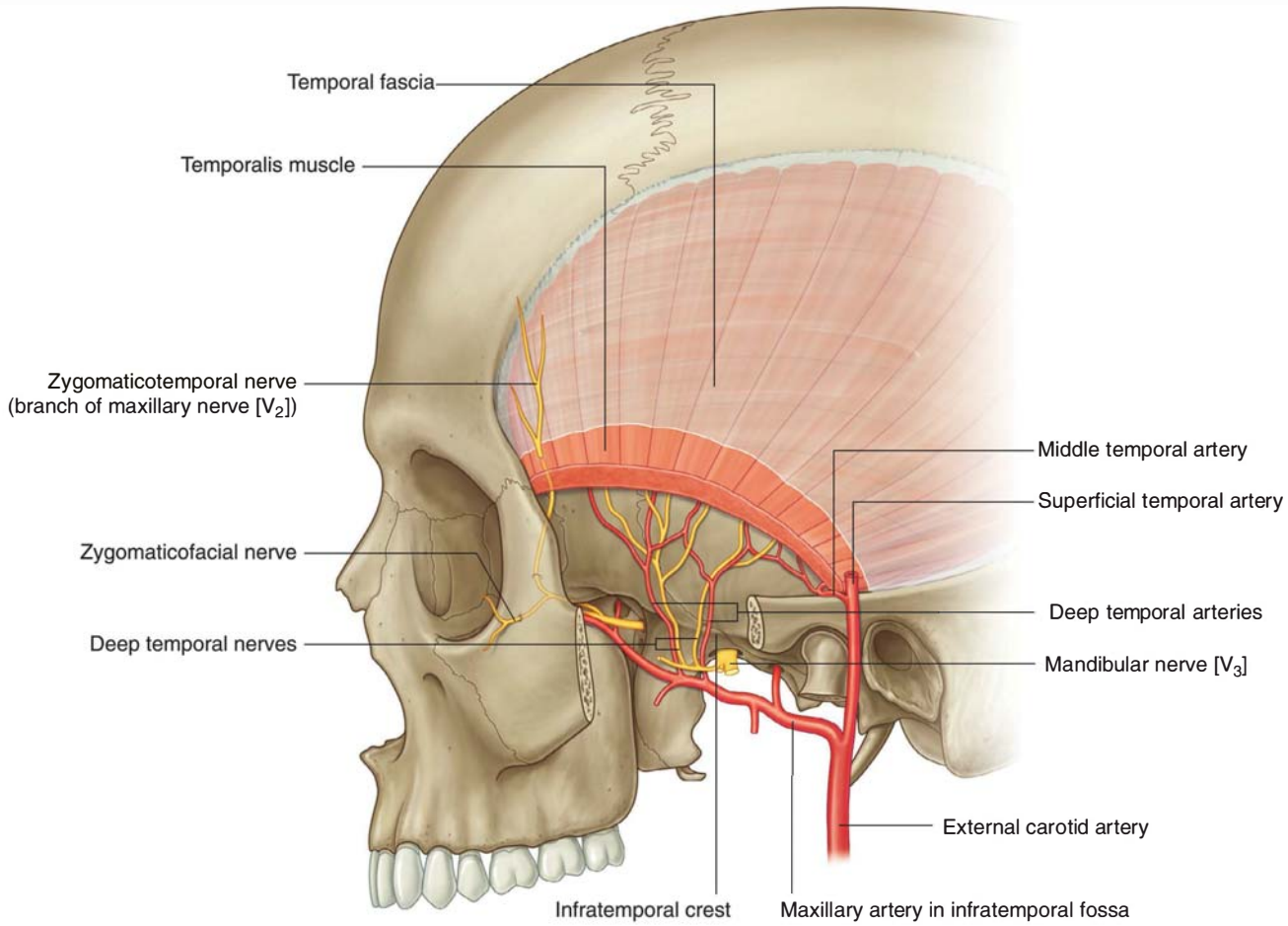


Fig. 8.140 Nerves and arteries of the temporal fossa.

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the margin of the temporalis muscle, and travels superiorly on the deep surface of the temporalis muscle.

The middle temporal artery supplies the temporalis and anastomoses with branches of the deep temporal arteries.

Infratemporal fossa

The wedge-shaped infratemporal fossa is inferior to the temporal fossa and between the ramus of the mandible laterally and the wall of the pharynx medially. It has a roof, a lateral wall, and a medial wall, and is open to the neck posteroinferiorly (Fig. 8.141):

- The **roof** is formed by the inferior surfaces of the greater wing of the sphenoid and the temporal bone, contains the foramen spinosum, foramen ovale, and the petrotympanic fissure, and lateral to the infratemporal crest of the greater wing of the sphenoid, is open superiorly to the temporal fossa.
- The **lateral wall** is the medial surface of the ramus of the mandible, which contains the opening to the mandibular canal.
- The **medial wall** is formed anteriorly by the lateral plate of the pterygoid process and more posteriorly by the pharynx and by two muscles of the soft palate (tensor and levator veli palatini muscles), and contains the pterygomaxillary fissure anteriorly, which allows structures to pass between the infratemporal and pterygopalatine fossae.
- The **anterior wall** is formed by part of the posterior surface of the maxilla and contains the alveolar foramen, and the upper part opens as the inferior orbital fissure into the orbit.

Contents

Major contents of the infratemporal fossa include the sphenomandibular ligament, medial and lateral pterygoid muscles (Table 8.11), the maxillary artery, the mandibular nerve [V_3], branches of the facial nerve [VII] and the glossopharyngeal nerve [IX], and the pterygoid plexus of veins.

Sphenomandibular ligament

The sphenomandibular ligament is an extracapsular ligament of the temporomandibular joint. It is attached

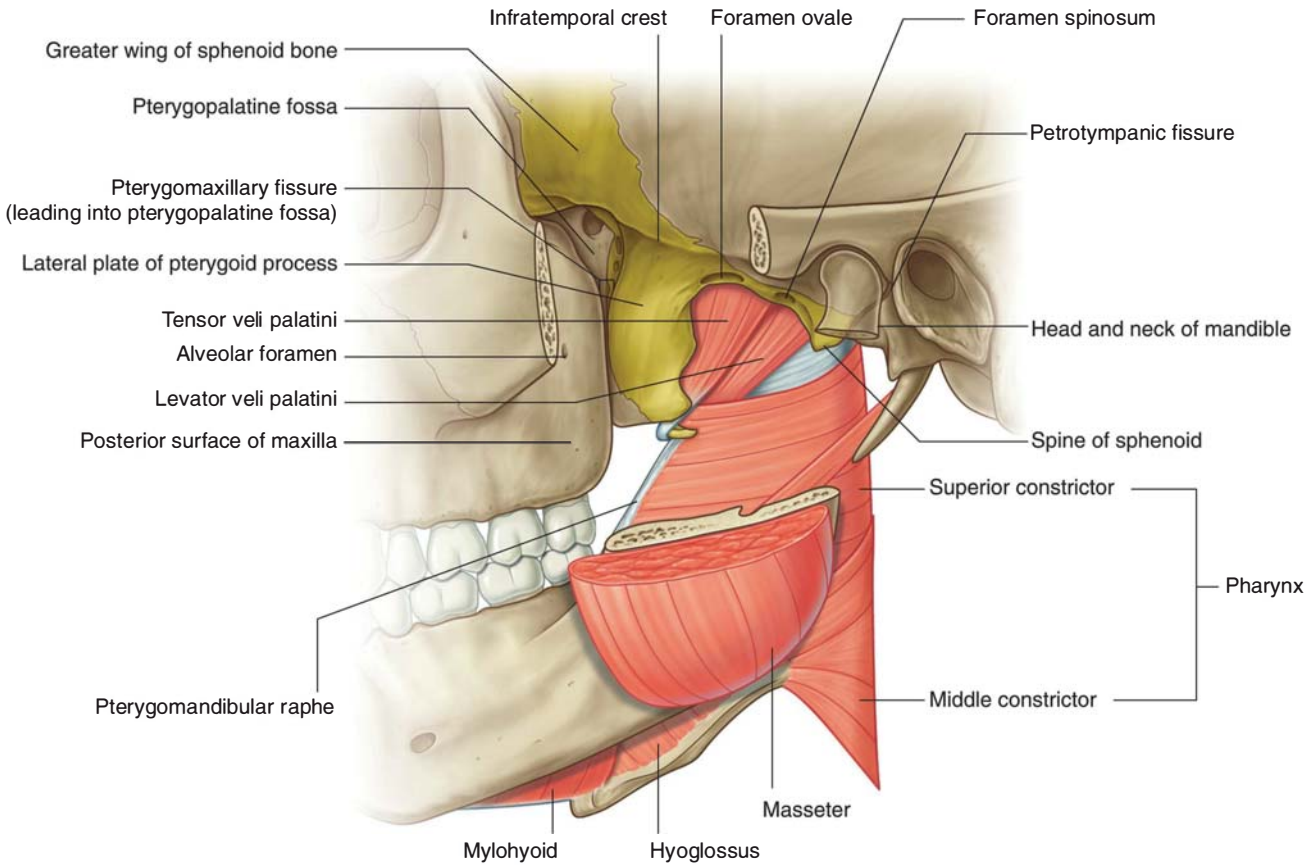


Fig. 8.141 Borders of the infratemporal fossa.

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superiorly to the spine of the sphenoid bone and expands inferiorly to attach to the lingula of the mandible and the posterior margin of the mandibular foramen (Fig. 8.142).

Medial pterygoid

The **medial pterygoid** muscle is quadrangular in shape and has deep and superficial heads (Fig. 8.142):

- The **deep head** is attached above to the medial surface of the lateral plate of the pterygoid process and the associated surface of the pyramidal process of the palatine bone, and descends obliquely downward, medial to the sphenomandibular ligament, to attach to the roughened medial surface of the ramus of the mandible near the angle of the mandible.
- The **superficial head** originates from the tuberosity of the maxilla and adjacent pyramidal process of the palatine bone and joins with the deep head to insert on the mandible.

The medial pterygoid mainly elevates the mandible. Because it passes obliquely backward to insert into the mandible, it also assists the lateral pterygoid muscle in protruding the lower jaw.

The medial pterygoid is innervated by the nerve to the medial pterygoid from the mandibular nerve [V_3].

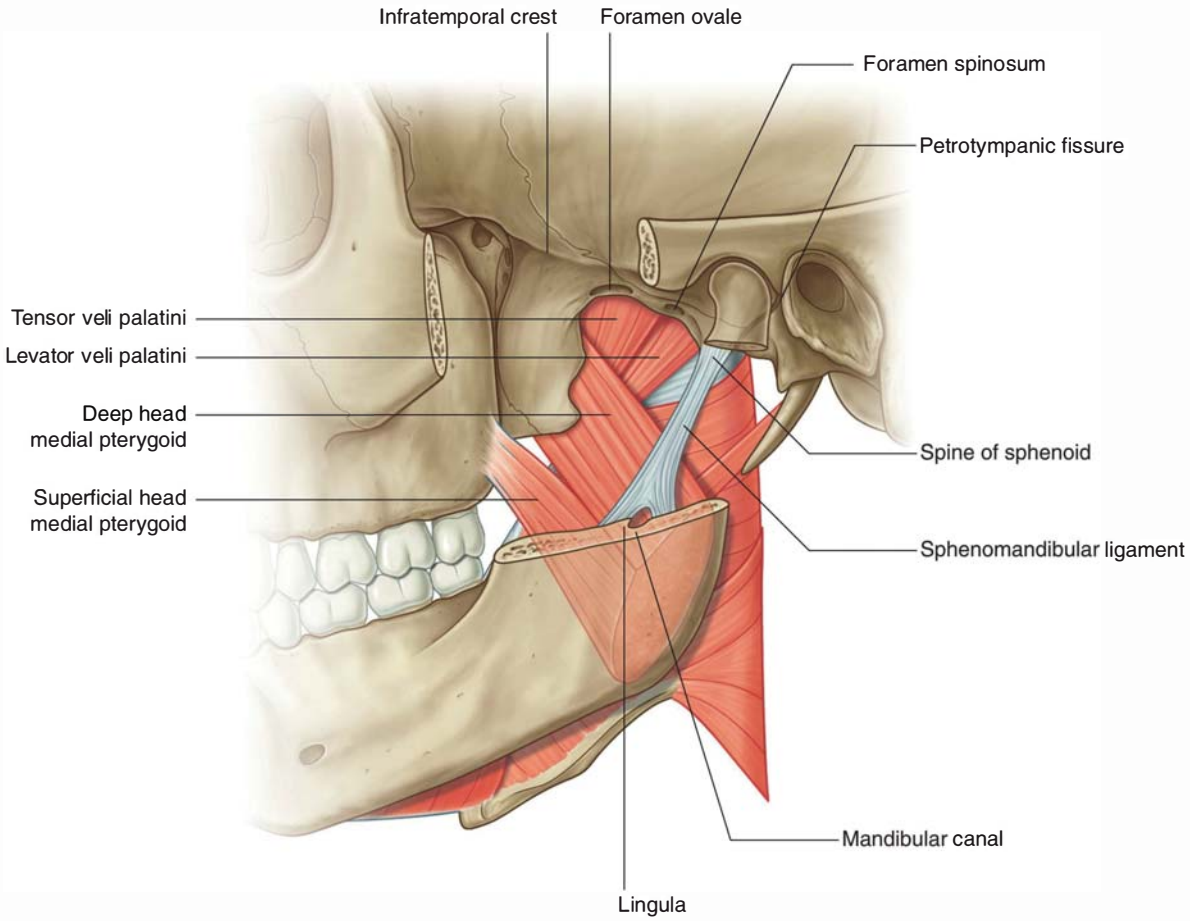


Fig. 8.142 Medial pterygoid muscle.

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Lateral pterygoid

The lateral pterygoid is a thick triangular muscle and like the medial pterygoid muscle has two heads (Fig. 8.143):

- The **upper head** originates from the roof of the infratemporal fossa (inferior surface of the greater wing of the sphenoid and the infratemporal crest) lateral to the foramen ovale and foramen spinosum.
- The **lower head** is larger than the upper head and originates from the lateral surface of the lateral plate of the pterygoid process, and the inferior part insinuates itself between the cranial attachments of the two heads of the medial pterygoid.

The fibers from both heads of the lateral pterygoid muscle converge to insert into the pterygoid fovea of the neck of the mandible and into the capsule of the temporomandibular joint in the region where the capsule is attached internally to the articular disc.

Unlike the medial pterygoid muscle whose fibers tend to be oriented vertically, those of the lateral pterygoid are oriented almost horizontally. As a result, when the lateral pterygoid contracts it pulls the articular disc and head of the mandible forward onto the articular tubercle and is therefore the major protruder of the lower jaw.

The lateral pterygoid is innervated by the nerve to the lateral pterygoid from the mandibular nerve [V_3].

When the lateral and medial pterygoids contract on only one side, the chin moves to the opposite side. When opposite movements at the two temporomandibular joints are coordinated, a chewing movement results.

Mandibular nerve [V_3]

The mandibular nerve [V_3] is the largest of the three divisions of the trigeminal nerve [V].

Unlike the ophthalmic [V_1] and maxillary [V_2] nerves, which are purely sensory, the mandibular nerve [V_3] is both motor and sensory.

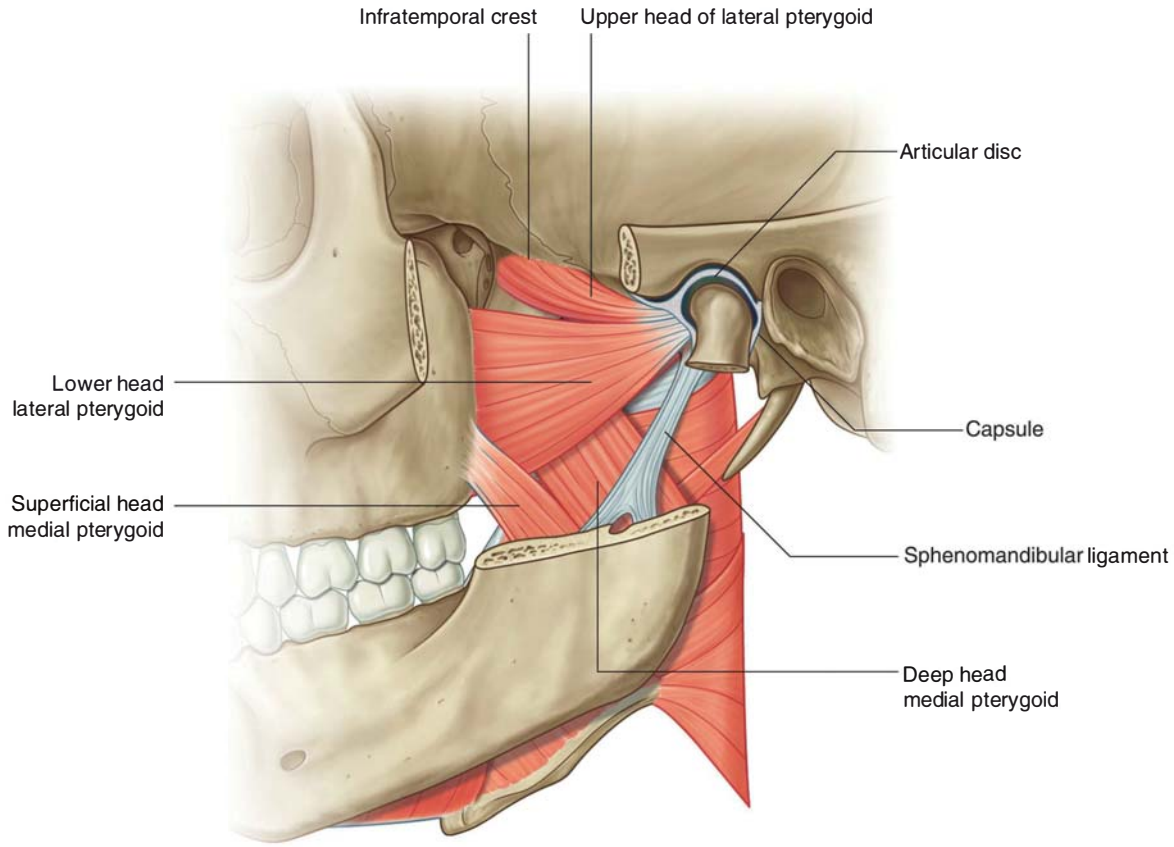


Fig. 8.143 Lateral pterygoid muscle.

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In addition to carrying general sensation from the teeth and gingivae of the mandible, the anterior two-thirds of the tongue, mucosa on the floor of the oral cavity, the lower lip, skin over the temple and lower face, and part of the cranial dura mater, the mandibular nerve [V_3] also carries motor innervation to most of the muscles that move the mandible, one of the muscles (tensor tympani) in the middle ear, and one of the muscles of the soft palate (tensor veli palatini).

All branches of the mandibular nerve [V_3] originate in the infratemporal fossa.

Like the ophthalmic [V_1] and maxillary [V_2] nerves, the sensory part of the mandibular nerve [V_3] originates from the trigeminal ganglion in the middle cranial fossa (Fig. 8.144):

- The sensory part of the mandibular nerve [V_3] drops vertically through the foramen ovale and enters the infratemporal fossa between the tensor veli palatini muscle and the upper head of the lateral pterygoid muscle.
- The small motor root of the trigeminal nerve [V] passes medial to the trigeminal ganglion in the cranial cavity, then passes through the foramen ovale and immediately joins the sensory part of the mandibular nerve [V_3].

Branches

Soon after the sensory and motor roots join, the mandibular nerve [V_3] gives rise to a small meningeal branch and to the nerve to the medial pterygoid, and then divides into anterior and posterior trunks (Fig. 8.144):

- Branches from the anterior trunk are the buccal, masseteric, and deep temporal nerves, and the nerve to the lateral pterygoid, all of which, except the buccal nerve (which is predominantly sensory) are motor nerves.
- Branches from the posterior trunk are the auriculotemporal, lingual, and inferior alveolar nerves, all of which, except a small nerve (nerve to the mylohyoid) that branches from the inferior alveolar nerve, are sensory nerves.

Meningeal branch

The meningeal branch originates from the medial side of the mandibular nerve [V_3] and ascends to leave the infratemporal fossa with the middle meningeal artery and reenter the cranial cavity through the foramen spinosum (Fig. 8.144). It is sensory for the dura mater, mainly of the middle cranial fossa, and also supplies the mastoid cells that communicate with the middle ear.

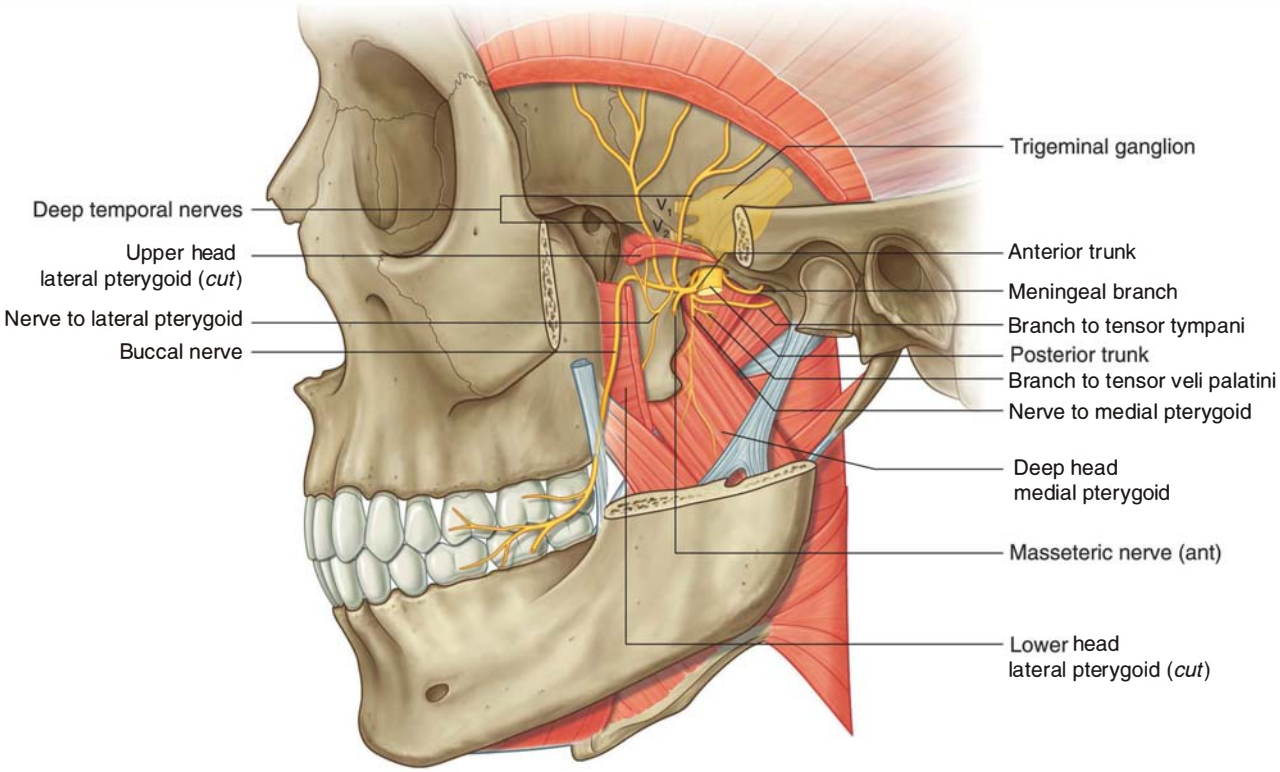


Fig. 8.144 Mandibular nerve [V_3]—anterior trunk. Meningeal branch and nerve to medial pterygoid.

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Nerve to medial pterygoid

The nerve to the medial pterygoid also originates medially from the mandibular nerve [V_3] (Fig. 8.144). It descends to enter and supply the deep surface of the medial pterygoid muscle. Near its origin from the mandibular nerve [V_3], it has two small branches:

- One of these supplies the tensor veli palatini.
- The other ascends to supply the tensor tympani muscle, which occupies a small bony canal above and parallel to the pharyngotympanic tube in the temporal bone.

Buccal nerve

The buccal nerve is a branch of the anterior trunk of the mandibular nerve [V_3] (Fig. 8.144). It is predominantly a sensory nerve, but may also carry the motor innervation to the lateral pterygoid muscle and to part of the temporalis muscle.

The buccal nerve passes laterally between the upper and lower heads of the lateral pterygoid and then descends around the anterior margin of the insertion of the temporalis muscle to the anterior margin of the ramus of the mandible, often slipping through the tendon of the temporalis. It continues into the cheek lateral to the buccinator muscle to supply general sensory nerves to the adjacent skin and oral mucosa and the buccal gingivae of the lower molars.

Masseteric nerve

The masseteric nerve is a branch of the anterior trunk of the mandibular nerve [V_3] (Fig. 8.144; also see Fig. 8.137). It passes laterally over the lateral pterygoid muscle and through the mandibular notch to penetrate and supply the masseter muscle.

Deep temporal nerves

The deep temporal nerves, usually two in number, originate from the anterior trunk of the mandibular nerve [V_3] (Fig. 8.144; also see Fig. 8.140). They pass laterally above the lateral pterygoid muscle and curve around the infratemporal crest to ascend in the temporal fossa and supply the temporalis muscle from its deep surface.

Nerve to lateral pterygoid

The nerve to the lateral pterygoid may originate directly as a branch from the anterior trunk of the mandibular nerve [V_3] or from its buccal branch (Fig. 8.144). From its origin, it passes directly into the deep surface of the lateral pterygoid muscle.

Auriculotemporal nerve

The auriculotemporal nerve is the first branch of the posterior trunk of the mandibular nerve [V_3] and originates as two roots, which pass posteriorly around the middle meningeal artery ascending from the maxillary artery to the foramen spinosum (Fig. 8.145).

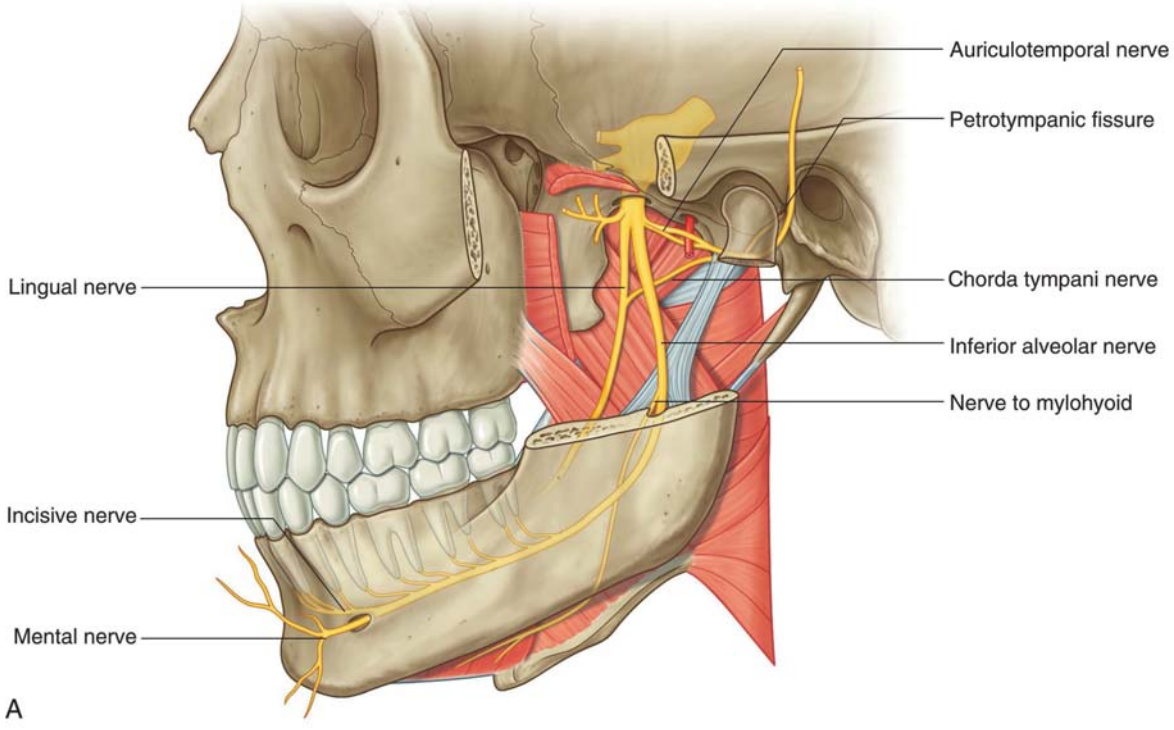


Fig. 8.145 Mandibular nerve [V_3]—posterior trunk. **A.** Lateral view.

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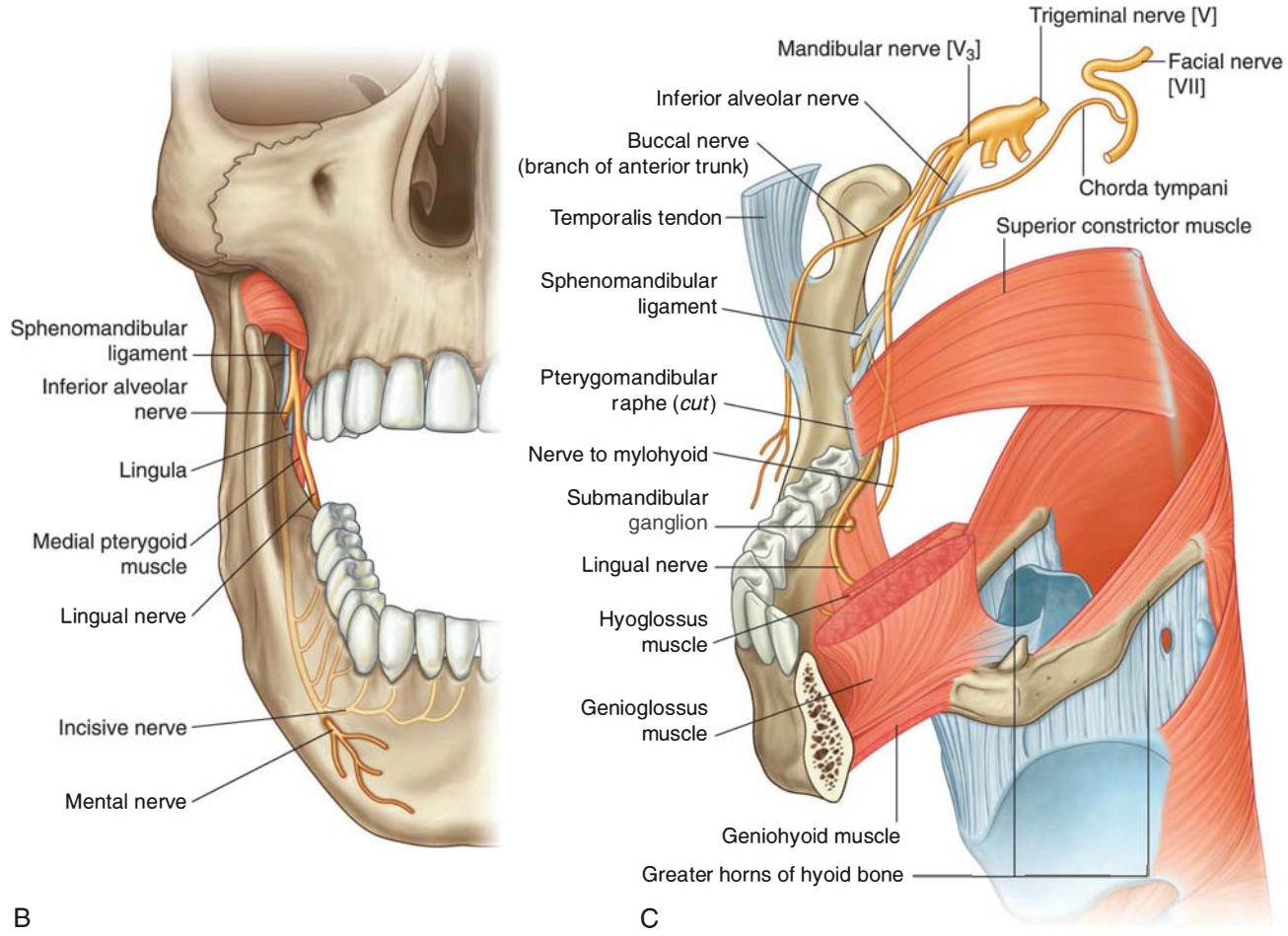


Fig. 8.145, cont'd **B.** Anterior view. **C.** Anteromedial view.

The auriculotemporal nerve passes first between the tensor veli palatini muscle and the upper head of the lateral pterygoid muscle, and then between the sphenomandibular ligament and the neck of the mandible. It curves laterally around the neck of the mandible and then ascends deep to the parotid gland between the temporomandibular joint and ear.

The terminal branches of the auriculotemporal nerve carry general sensation from skin over a large area of the temple. In addition, the auriculotemporal nerve contributes to sensory innervation of the external ear, the external auditory meatus, tympanic membrane, and temporomandibular joint. It also delivers postganglionic parasympathetic nerves from the glossopharyngeal nerve [IX] to the parotid gland.

Lingual nerve

The **lingual nerve** is a major sensory branch of the posterior trunk of the mandibular nerve [V_3] (Fig. 8.145A,B). It carries general sensation from the anterior two-thirds of the tongue, oral mucosa on the floor of the oral cavity, and lingual gingivae associated with the lower teeth.

The lingual nerve is joined high in the infratemporal fossa by the chorda tympani branch of the facial nerve [VII] (Fig. 8.145C), which carries:

- taste from the anterior two-thirds of the tongue, and
- parasympathetic fibers to all salivary glands below the level of the oral fissure.

The lingual nerve first descends between the tensor veli palatini muscle and the lateral pterygoid muscle, where it is joined by the chorda tympani nerve, and then descends across the lateral surface of the medial pterygoid muscle to enter the oral cavity.

The lingual nerve enters the oral cavity between the posterior attachment of the mylohyoid muscle to the mylohyoid line and the attachment of the superior constrictor of the pharynx to the pterygomandibular raphe. As the lingual nerve enters the floor of the oral cavity,

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it is in a shallow groove on the medial surface of the mandible immediately inferior to the last molar tooth. In this position, it is palpable through the oral mucosa and in danger when one is operating on the molar teeth and gingivae (Fig. 8.145C).

The lingual nerve passes into the tongue on the lateral surface of the hyoglossus muscle where it is attached to the **submandibular ganglion**. This ganglion is where the preganglionic parasympathetic fibers carried from the infratemporal fossa into the floor of the oral cavity on the lingual nerve synapse with postganglionic parasympathetic fibers (see Fig. 8.146).

Inferior alveolar nerve

The **inferior alveolar nerve**, like the lingual nerve, is a major sensory branch of the posterior trunk of the mandibular nerve [V_3] (Fig. 8.145A–C). In addition to innervating all lower teeth and much of the associated gingivae, it also supplies the mucosa and skin of the lower lip and skin of the chin. It has one motor branch, which innervates the mylohyoid muscle and the anterior belly of the digastric muscle.

The inferior alveolar nerve originates deep to the lateral pterygoid muscle from the posterior trunk of the mandibular nerve [V_3] in association with the lingual nerve. It descends on the lateral surface of the medial pterygoid muscle, passes between the sphenomandibular ligament and the ramus of the mandible, and then enters the mandibular canal through the mandibular foramen. Just before entering the mandibular foramen, it gives origin to the **nerve to the mylohyoid** (Fig. 8.145C), which lies in the mylohyoid groove inferior to the foramen and continues anteriorly below the floor of the oral cavity to innervate the mylohyoid muscle and the anterior belly of the digastric muscle.

The inferior alveolar nerve passes anteriorly within the mandibular canal of the lower jaw. The mandibular canal and its contents are inferior to the roots of the molar teeth, and the roots can sometimes curve around the canal making extraction of these teeth difficult.

The inferior alveolar nerve supplies branches to the three molar teeth and the second premolar tooth and associated labial gingivae, and then divides into its two terminal branches:

- the **incisive nerve**, which continues in the mandibular canal to supply the first premolar, incisor, and canine teeth, and related gingivae; and
- the **mental nerve**, which exits the mandible through the mental foramen and supplies the lower lip and chin (Fig. 8.145A,B). The mental nerve is palpable and sometimes visible through the oral mucosa adjacent to the roots of the premolar teeth.

Chorda tympani and the lesser petrosal nerve

Branches of two cranial nerves join branches of the mandibular nerve [V_3] in the infratemporal fossa (Fig. 8.146). These are the chorda tympani branch of the facial nerve [VII] and the lesser petrosal nerve, a branch of the tympanic plexus in the middle ear, which had its origin from a branch of the glossopharyngeal nerve [IX] (see Fig. 8.121, p. 964).

Chorda tympani

The chorda tympani (Fig. 8.146) carries taste from the anterior two-thirds of the tongue and parasympathetic innervation to all salivary glands below the level of the oral fissure.

The chorda tympani originates from the facial nerve [VII] within the temporal bone and in association with the mastoid wall of the middle ear, passes anteriorly through a small canal, and enters the lateral aspect of the middle ear. As it continues anterosuperiorly across the middle ear, it is separated from the tympanic membrane by the handle of the malleus. It leaves the middle ear through the medial end of the petrotympanic fissure, enters the infratemporal fossa, descends medial to the spine of the sphenoid and then to the lateral pterygoid muscle, and joins the lingual nerve.

Preganglionic parasympathetic fibers carried in the chorda tympani synapse with postganglionic parasympathetic fibers in the submandibular ganglion, which "hangs off" the lingual nerve in the floor of the oral cavity (Fig. 8.146).

Postganglionic parasympathetic fibers leave the submandibular ganglion and either:

- reenter the lingual nerve to travel with its terminal branches to reach target tissues, or
- pass directly from the submandibular ganglion into glands (Fig. 8.146).

The taste (SA) fibers do not pass through the ganglion and are distributed with terminal branches of the lingual nerve.

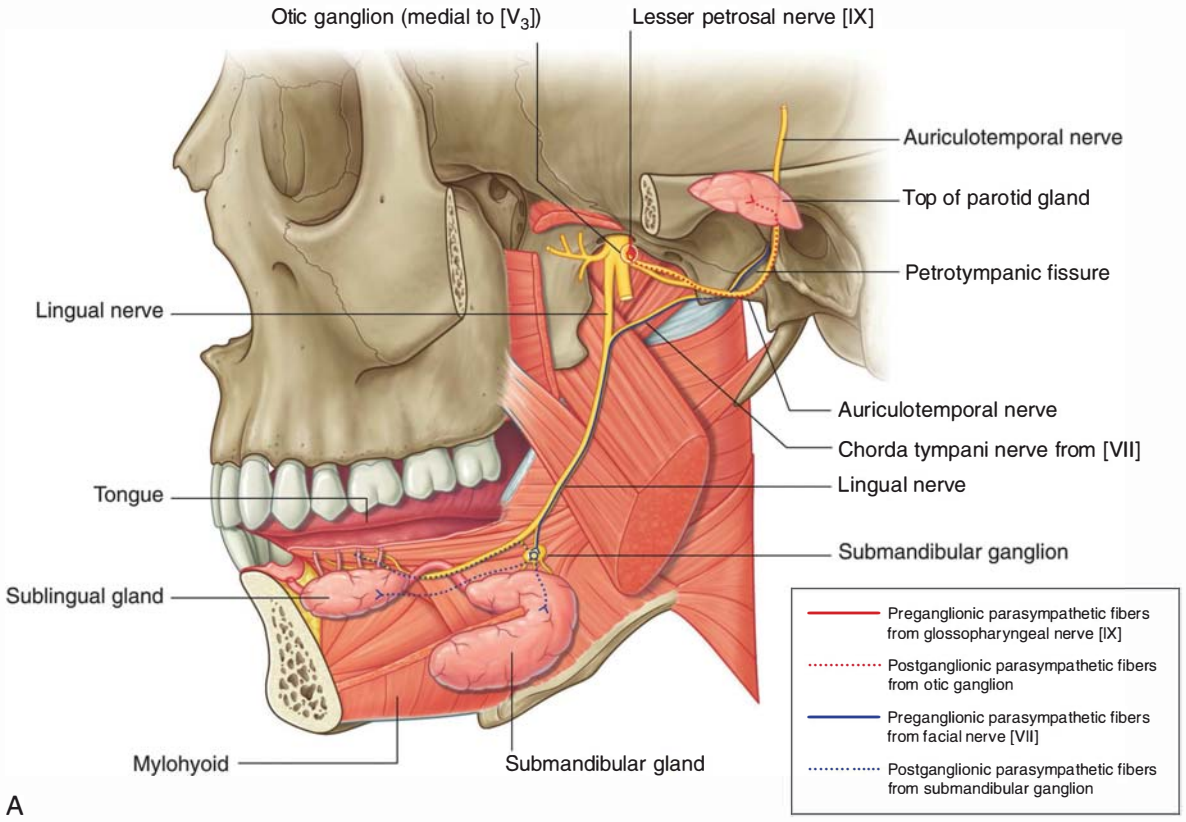
In the clinic

Lingual nerve injury

A lingual nerve injury proximal to where the chorda tympani joins it in the infratemporal fossa will produce loss of general sensation from the anterior two-thirds of the tongue, oral mucosa, gingivae, the lower lip, and the chin.

If a lingual nerve lesion is distal to the site where it is joined by the chorda tympani, secretion from the salivary glands below the oral fissure and taste from the anterior two-thirds of the tongue will also be lost.

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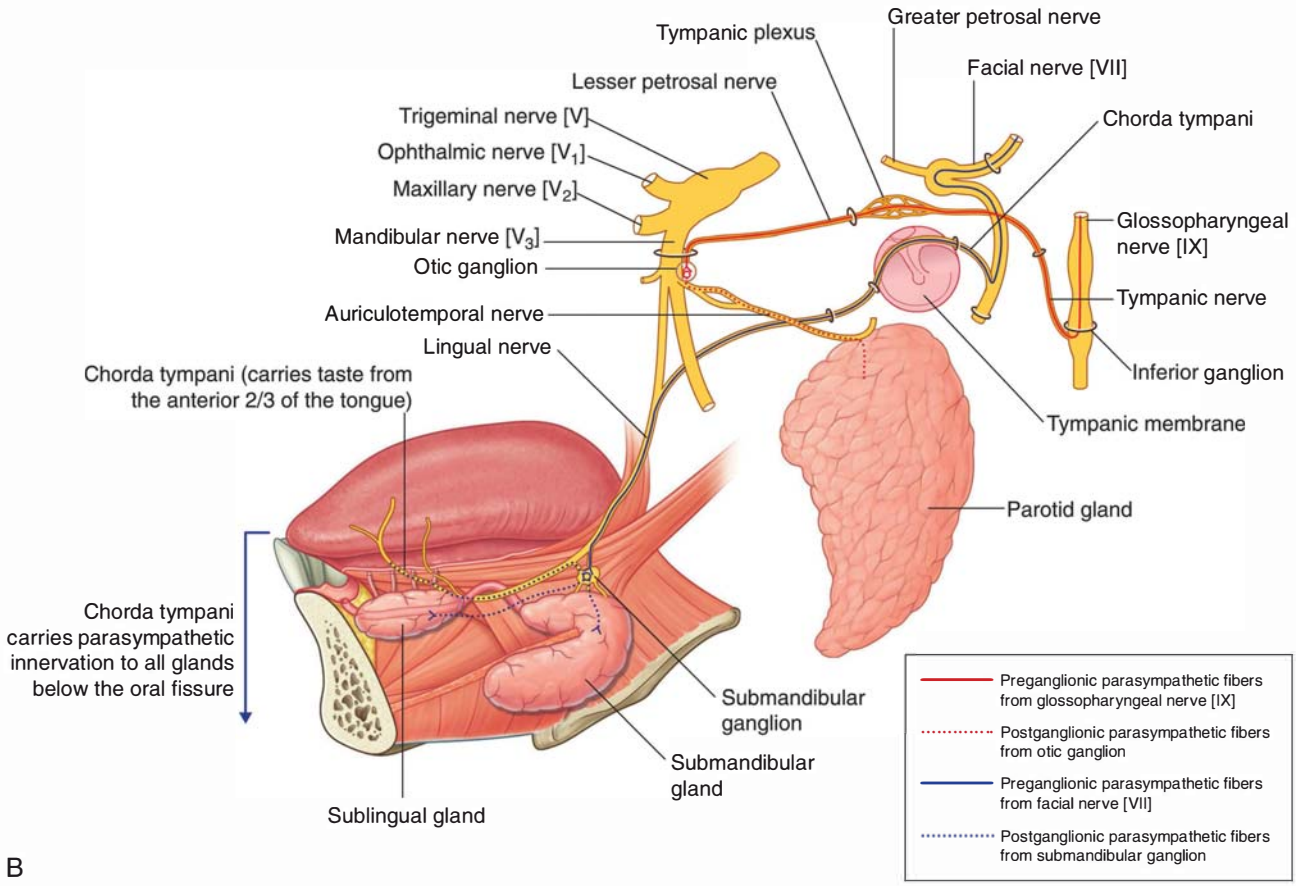


Fig. 8.146 Chorda tympani and lesser petrosal nerves. **A.** Course after emerging from the skull. **B.** Course of parasympathetic fibers.

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Lesser petrosal nerve

The lesser petrosal nerve carries mainly parasympathetic fibers destined for the parotid gland (Fig. 8.146). The preganglionic parasympathetic fibers are located in the glossopharyngeal nerve [IX] as it exits the jugular foramen at the base of the skull. Branching from the glossopharyngeal nerve [IX] either within or immediately outside the jugular foramen is the tympanic nerve (Fig. 8.146B).

The tympanic nerve reenters the temporal bone through a small foramen on the ridge of bone separating the jugular foramen from the carotid canal and ascends through a small bony canal (inferior tympanic canaliculus) to the promontory located on the labyrinthine (medial) wall of the middle ear. Here it participates in the formation of the tympanic plexus. The lesser petrosal nerve is a branch of this plexus (Fig. 8.146B).

The lesser petrosal nerve contains mainly preganglionic parasympathetic fibers. It leaves the middle ear and enters the middle cranial fossa through a small opening on the anterior surface of the petrous part of the temporal bone just lateral and inferior to the opening for the greater petrosal nerve, a branch of the facial nerve [VII]. The lesser petrosal nerve then passes medially and descends through the foramen ovale with the mandibular nerve [V_3].

In the infratemporal fossa, the preganglionic parasympathetic fibers synapse with cell bodies of postganglionic parasympathetic fibers in the otic ganglion located on the medial side of the mandibular nerve [V_3] around the origin of the nerve to the medial pterygoid. Postganglionic parasympathetic fibers leave the otic ganglion and join the auriculotemporal nerve, which carries them to the parotid gland.

In the clinic

Dental anesthesia

Anesthesia of the inferior alveolar nerve is widely practiced by most dentists. The inferior alveolar nerve is one of the largest branches of the mandibular nerve [V_3], carries the sensory branches from the teeth and mandible, and receives sensory information from the skin over most of the mandible.

The inferior alveolar nerve passes into the mandibular canal, courses through the body of the mandible, and eventually emerges through the mental foramen into the chin.

Dental procedures require perineuronal infiltration of the inferior alveolar nerve by local anesthetic. To anesthetize this nerve the needle is placed lateral to the anterior arch of the fauces (palatoglossal arch) in the oral cavity and is advanced along the medial border around the inferior third of the ramus of the mandible so that anesthetic can be deposited in this region.

It is also possible to anesthetize the infra-orbital and buccal nerves, depending on where the anesthesia is needed.

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Maxillary artery

The maxillary artery is the largest branch of the external carotid artery in the neck and is a major source of blood supply for the nasal cavity, the lateral wall and roof of the oral cavity, all teeth, and the dura mater in the cranial cavity. It passes through and supplies the infratemporal fossa and then enters the pterygopalatine fossa, where it gives origin to terminal branches (Fig. 8.147).

The maxillary artery originates within the substance of the parotid gland and then passes forward, between the neck of the mandible and sphenomandibular ligament, into the infratemporal fossa. It ascends obliquely through the infratemporal fossa to enter the pterygopalatine fossa by passing through the pterygomaxillary fissure. This part of the vessel may pass either lateral or medial to the lower head of the lateral pterygoid. If it passes medial to the lower head, the maxillary artery then loops laterally between the upper and lower heads of the lateral pterygoid to access the **pterygomaxillary fissure**.

Branches

Branches of the maxillary artery are as follows (Fig. 8.147):

- The first part of the maxillary artery (the part between the neck of the mandible and the sphenomandibular ligament) gives origin to two major branches (the middle meningeal and inferior alveolar arteries) and a number of smaller branches (deep auricular, anterior tympanic, and accessory meningeal).
- The second part of the maxillary artery (the part related to the lateral pterygoid muscle) gives origin to deep temporal, masseteric, buccal, and pterygoid branches, which course with branches of the mandibular nerve [V_3].
- The third part of the maxillary artery is in the pterygopalatine fossa (see Fig. 8.154)

Middle meningeal artery

The middle meningeal artery ascends vertically from the maxillary artery and passes through the foramen spinosum to enter the cranial cavity (Fig. 8.147). In the infratemporal fossa, it passes superiorly between the sphenomandibular ligament on the medial side and the lateral pterygoid muscle on the lateral side. Just inferior to the foramen spinosum, it passes between the two roots of the auriculotemporal nerve at their origin from the mandibular nerve [V_3] (Fig. 8.147).

The middle meningeal artery is the largest of the meningeal vessels and supplies much of the dura mater, bone, and related bone marrow of the cranial cavity walls.

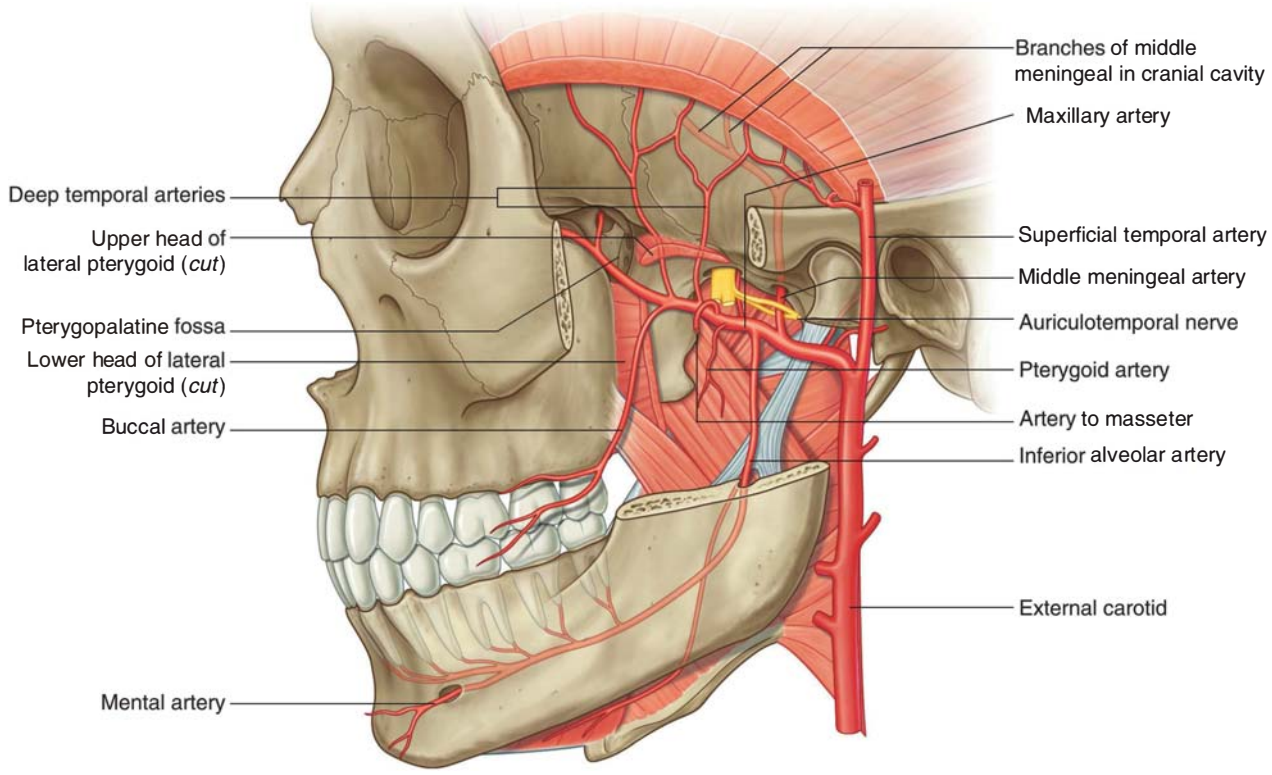


Fig. 8.147 Maxillary artery.

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Within the cranial cavity, the middle meningeal artery and its branches travel in the periosteal (outer) layer of dura mater, which is tightly adherent to the bony walls. As major branches of the middle meningeal artery pass superiorly up the walls of the cranial cavity, they can be damaged by lateral blows to the head. When the vessels are torn, the leaking blood, which is under arterial pressure, slowly separates the dura mater from its attachment to the bone, resulting in an extradural hematoma.

Inferior alveolar artery

The inferior alveolar artery descends from the maxillary artery to enter the mandibular foramen and canal with the inferior alveolar nerve (Fig. 8.147). It is distributed with the inferior alveolar nerve and supplies all lower teeth, and contributes to the supply of the buccal gingivae, chin, and lower lip.

Before entering the mandible, the inferior alveolar artery gives origin to a small mylohyoid branch, which accompanies the nerve to the mylohyoid.

Deep auricular, anterior tympanic, and accessory meningeal arteries

The deep auricular, anterior tympanic, and accessory meningeal arteries are small branches from the first part of the maxillary artery and contribute to the blood supply of the external acoustic meatus, deep surface of the tympanic membrane, and cranial dura mater, respectively.

The accessory meningeal branch also contributes small branches to surrounding muscles in the infratemporal fossa before ascending through the foramen ovale into the cranial cavity to supply the dura mater.

Branches from the second part

Deep temporal arteries, usually two in number, originate from the second part of the maxillary artery and travel with the deep temporal nerves to supply the temporalis muscle in the temporal fossa (Fig. 8.147).

Numerous pterygoid arteries also originate from the second part of the maxillary artery and supply the pterygoid muscles.

The masseteric artery, also from the second part of the maxillary artery, accompanies the masseteric nerve laterally through the mandibular notch to supply the masseter muscle.

The buccal artery is distributed with the buccal nerve and supplies skin, muscle, and oral mucosa of the cheek.

Pterygoid plexus

The **pterygoid plexus** is a network of veins between the medial and lateral pterygoid muscles, and between the lateral pterygoid and temporalis muscles (Fig. 8.148).

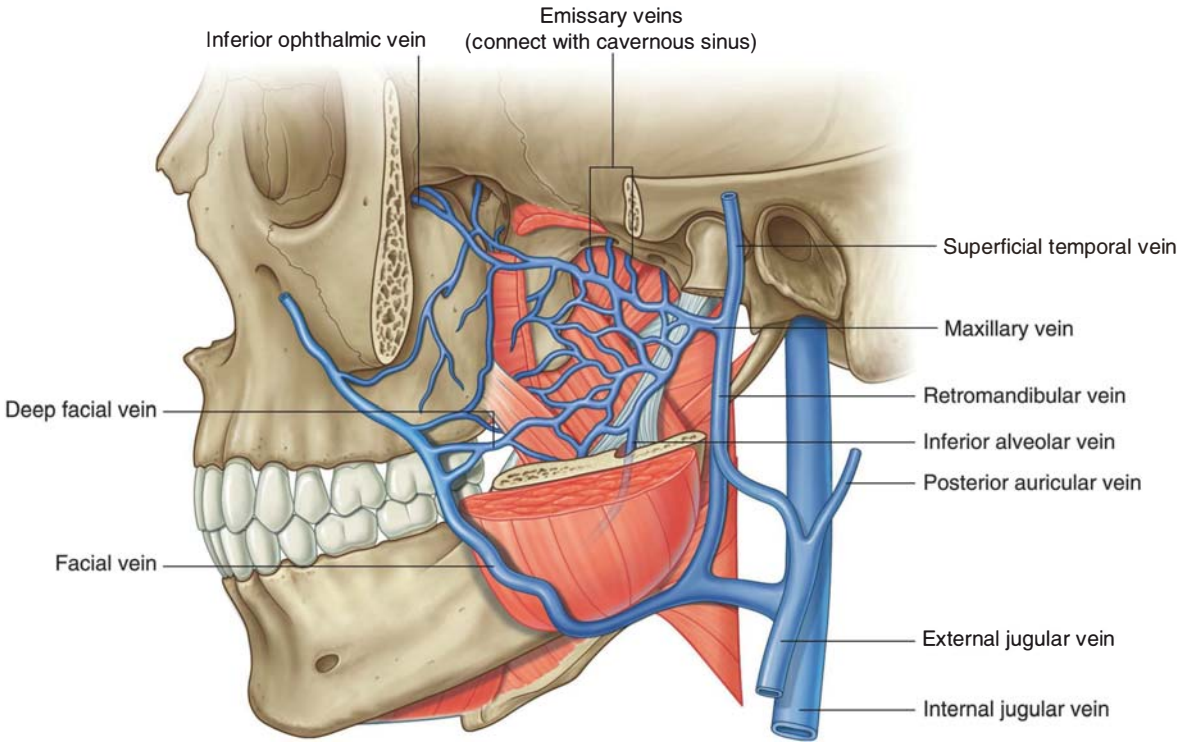


Fig. 8.148 Pterygoid plexus of veins.

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Veins that drain regions supplied by arteries branching from the maxillary artery in the infratemporal fossa and pterygopalatine fossa connect with the pterygoid plexus. These tributary veins include those that drain the nasal cavity, roof and lateral wall of the oral cavity, all teeth, muscles of the infratemporal fossa, paranasal sinuses, and nasopharynx. In addition, the inferior ophthalmic vein from the orbit can drain through the inferior orbital fissure into the pterygoid plexus.

Significantly, small emissary veins often connect the pterygoid plexus in the infratemporal fossa to the cavernous sinus in the cranial cavity. These emissary veins, which pass through the foramen ovale, through the cartilage that fills the foramen lacerum, and through a small sphenoidal foramen on the medial side of the lateral plate of the pterygoid process at the base of the skull, are a route by which infections can spread into the cranial cavity from structures, such as the teeth, that are drained by the pterygoid plexus. Also, because there are no valves in veins of the head and neck, anesthetic inadvertently injected under pressure into veins of the pterygoid plexus can backflow into tissues or into the cranial cavity.

The pterygoid plexus connects:

- posteriorly, via a short maxillary vein, with the retromandibular vein in the neck; and
- anteriorly, via a deep facial vein, with the facial vein on the face.

PTERYGOPALATINE FOSSA

The pterygopalatine fossa is an inverted teardrop-shaped space between bones on the lateral side of the skull immediately posterior to the maxilla (Fig. 8.149).

Although small in size, the pterygopalatine fossa communicates via fissures and foramina in its walls with the:

- middle cranial fossa,
- infratemporal fossa,
- floor of the orbit,
- lateral wall of the nasal cavity,
- oropharynx, and
- roof of the oral cavity.

Because of its strategic location, the pterygopalatine fossa is a major site of distribution for the maxillary nerve [V_2] and for the terminal part of the maxillary artery. It also contains the pterygopalatine ganglion where preganglionic parasympathetic fibers originating in the facial nerve [VII] synapse with postganglionic parasympathetic fibers and these fibers, along with sympathetic fibers originating from the T1 spinal cord level join branches of the maxillary nerve [V_2].

All the upper teeth receive their innervation and blood supply from the maxillary nerve [V_2] and the terminal part of the maxillary artery, respectively, that pass through the pterygopalatine fossa.

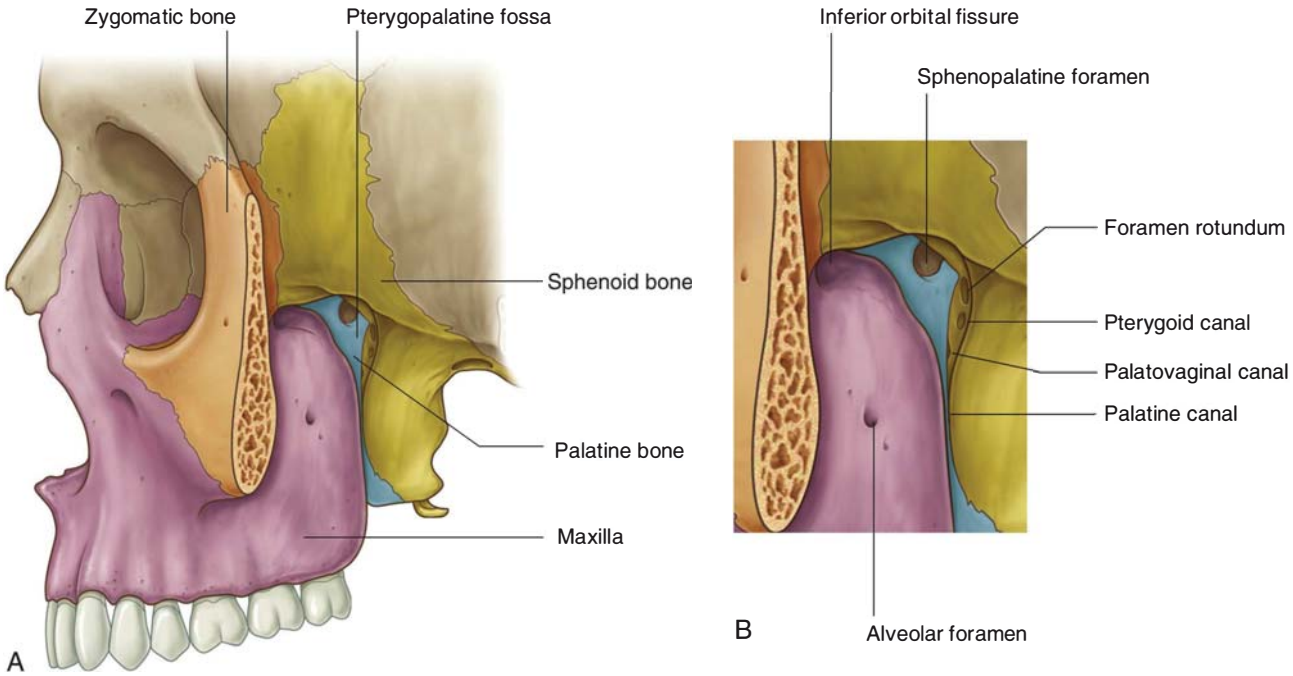


Fig. 8.149 Pterygopalatine fossa. **A.** Anterolateral view. **B.** Lateral view.

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Skeletal framework

The walls of the pterygopalatine fossa are formed by parts of the palatine, maxilla, and sphenoid bones (Fig. 8.149):

- The anterior wall is formed by the posterior surface of the maxilla.
- The medial wall is formed by the lateral surface of the palatine bone.
- The posterior wall and roof are formed by parts of the sphenoid bone.

Sphenoid bone

The part of the sphenoid bone that contributes to the formation of the pterygopalatine fossa is the anterosuperior surface of the pterygoid process (Fig. 8.150). Opening onto this surface are two large foramina:

- The maxillary nerve [V_2] passes through the most lateral and superior of these—**the foramen rotundum**—which communicates posteriorly with the middle cranial fossa (Fig. 8.150B).
- The greater petrosal nerve from the facial nerve [VII] and sympathetic fibers from the internal carotid plexus join to form the nerve of the pterygoid canal that passes forward into the pterygopalatine fossa through the more medial and inferior foramen—**the anterior opening of the pterygoid canal**.

Pterygoid canal

The **pterygoid canal** (Fig. 8.150A) is a bony canal running horizontally through the root of the pterygoid process of the sphenoid bone. It opens anteriorly into the pterygopalatine fossa. Posteriorly it continues through the cartilage filling the foramen lacerum and opens into

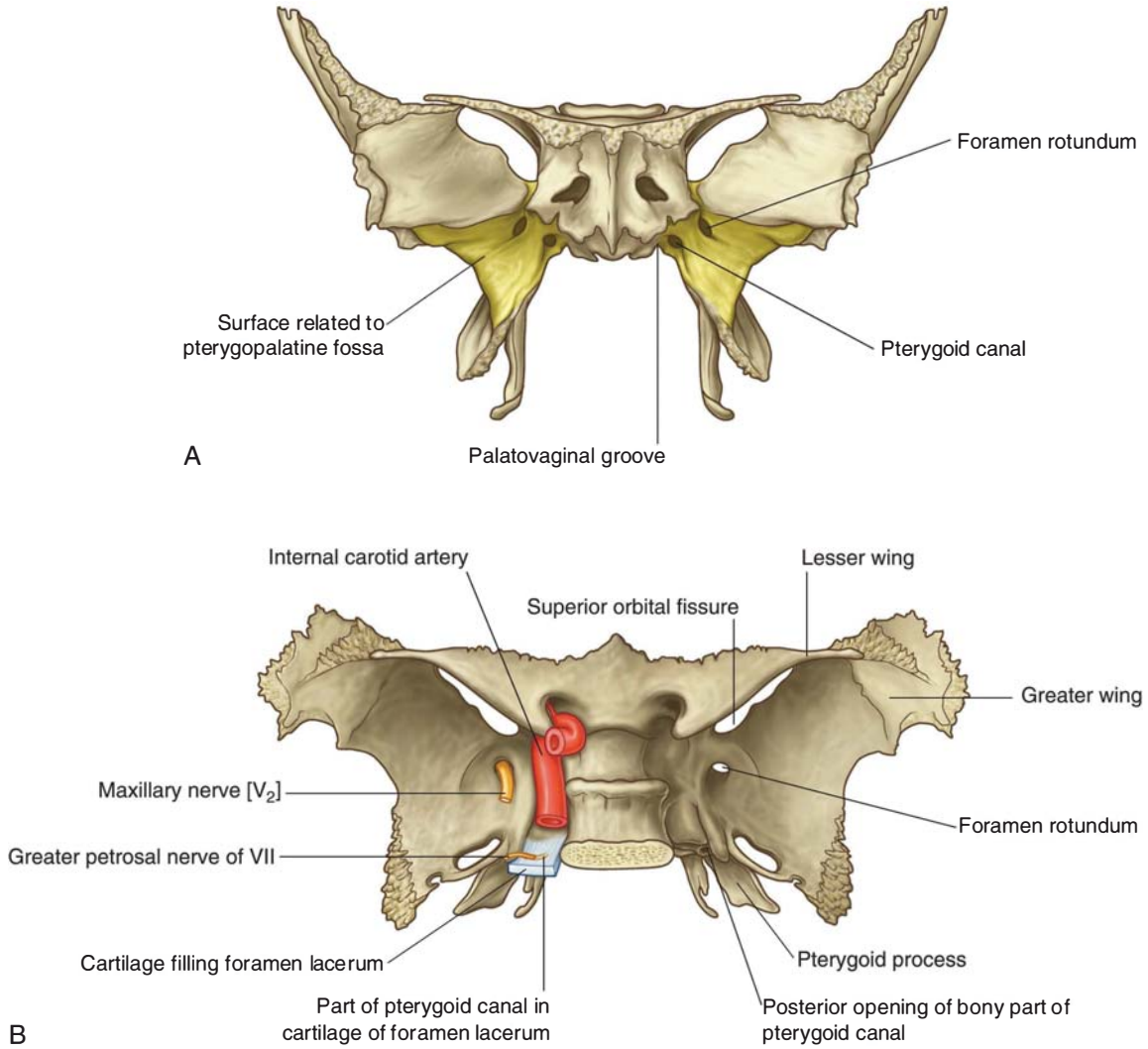


Fig. 8.150 Sphenoid bone. **A.** Anterior view. **B.** Posterosuperior view.

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the middle cranial fossa just anteroinferior to the internal carotid artery as the vessel enters the cranial cavity through the carotid canal (Fig. 8.150B).

Gateways

Seven foramina and fissures provide apertures through which structures enter and leave the pterygopalatine fossa (Fig. 8.151):

- The foramen rotundum and pterygoid canal communicate with the middle cranial fossa and open onto the posterior wall.
- A small **palatovaginal canal** opens onto the posterior wall and leads to the nasopharynx.
- The palatine canal leads to the roof of the oral cavity (hard palate) and opens inferiorly.
- The sphenopalatine foramen opens onto the lateral wall of the nasal cavity and is in the medial wall.
- The lateral aspect of the pterygopalatine fossa is continuous with the infratemporal fossa via a large gap (the **pterygomaxillary fissure**) between the posterior surface of the maxilla and pterygoid process of the sphenoid bone.
- The superior aspect of the anterior wall of the fossa opens into the floor of the orbit via the inferior orbital fissure.

Contents

The maxillary nerve [V_2] and terminal part of the maxillary artery enter and branch within the pterygopalatine fossa. In addition, the nerve of the pterygoid canal enters the fossa carrying:

- preganglionic parasympathetic fibers from the greater petrosal branch of the facial nerve [VII], and
- postganglionic sympathetic fibers from the deep petrosal branch of the carotid plexus.

The preganglionic parasympathetic fibers synapse in the pterygopalatine ganglion and both the sympathetic and postganglionic parasympathetic fibers pass with branches of the maxillary nerve [V_2] out of the fossa and into adjacent regions.

In addition to nerves and arteries, veins and lymphatics also pass through the pterygopalatine fossa.

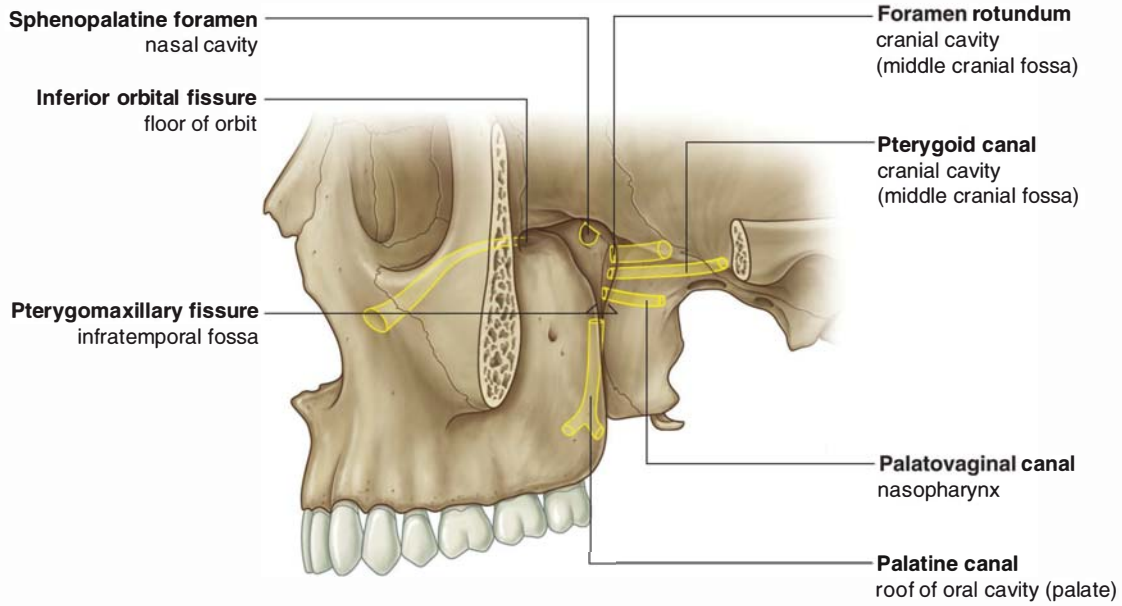


Fig. 8.151 Gateways of the pterygopalatine fossa.

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Maxillary nerve [V_2]

The maxillary nerve [V_2] is purely sensory. It originates from the trigeminal ganglion in the cranial cavity, exits the middle cranial fossa, and enters the pterygopalatine fossa through the foramen rotundum (Fig. 8.152). It passes anteriorly through the fossa and exits as the infra-orbital nerve through the inferior orbital fissure.

While passing through the pterygopalatine fossa, the maxillary nerve [V_2] gives rise to the zygomatic nerve, the posterior superior alveolar nerve, and two ganglionic branches (Fig. 8.152). The two ganglionic branches originate from its inferior surface and pass through the pterygopalatine ganglion.

Postganglionic parasympathetic fibers, arising in the pterygopalatine ganglion, join the general sensory branches of the maxillary nerve [V_2] in the pterygopalatine ganglion, as do postganglionic sympathetic fibers from the carotid plexus. The three types of fibers leave the ganglion as orbital, palatine, nasal, and pharyngeal branches.

Branches

Orbital branches. The **orbital branches** are small and pass through the inferior orbital fissure to contribute to the supply of the orbital wall and of the sphenoidal and ethmoidal sinuses.

Greater and lesser palatine nerves. The **greater** and **lesser palatine nerves** (Fig. 8.152) pass inferiorly from the pterygopalatine ganglion, enter and pass through the palatine canal, and enter the oral surface of the palate through the greater and lesser palatine foramina.

The greater palatine nerve passes forward on the roof of the oral cavity to innervate mucosa and glands of the hard palate and the adjacent gingiva, almost as far forward as the incisor teeth.

In the palatine canal, the greater palatine nerve gives origin to **posterior inferior nasal nerves**, which pass medially through small foramina in the perpendicular plate of the palatine bone and contribute to the innervation of the lateral nasal wall.

After passing through the lesser palatine foramen, the lesser palatine nerve passes posteriorly to supply the soft palate.

Nasal nerves. The nasal nerves (Fig. 8.152), approximately seven in number, pass medially through the sphenopalatine foramen to enter the nasal cavity. Most pass anteriorly to supply the lateral wall of the nasal cavity, while others pass across the roof to supply the medial wall.

One of the nerves passing across the roof to supply the medial wall of the nasal cavity (the **nasopalatine nerve**) is the largest of the nasal nerves and passes anteriorly

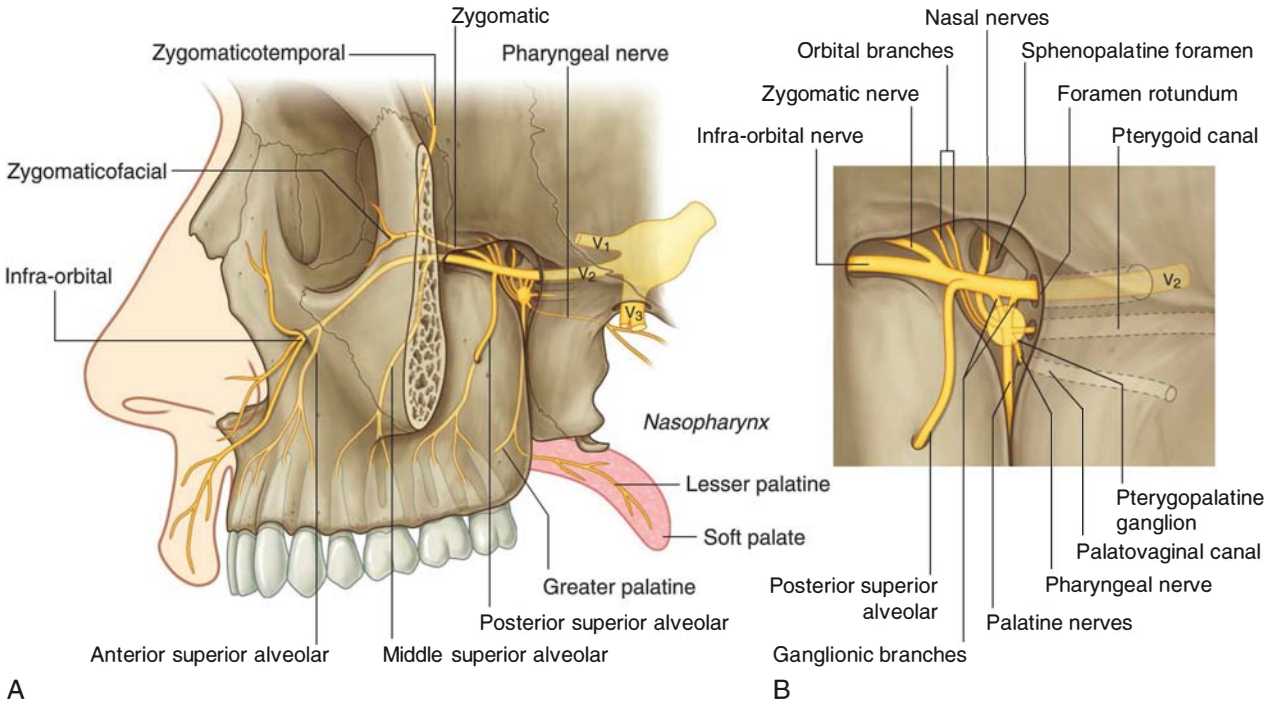


Fig. 8.152 Maxillary nerve [V_2]. **A.** Terminal branches. **B.** In relationship to the pterygopalatine ganglion.

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down the nasal septum, through the incisive canal and fossa in the hard palate to enter the roof of the oral cavity and supply mucosa, gingiva, and glands adjacent to the incisor teeth.

Pharyngeal nerve. The **pharyngeal nerve** (Fig. 8.152) passes posteriorly from the pterygopalatine ganglion, and leaves the fossa through the palatovaginal canal, which it then exits to supply the mucosa and glands of the nasopharynx.

Zygomatic nerve. The **zygomatic nerve** (Fig. 8.152) originates directly from the maxillary nerve [V_2] in the pterygopalatine fossa, which it leaves to enter the orbit through the inferior orbital fissure. It passes forward on the lateral orbital wall and divides into zygomaticotemporal and zygomaticofacial branches:

- The **zygomaticotemporal branch** continues forward at the base of the lateral orbital wall, passes through a small bony canal in the zygomatic bone to enter the temporal fossa through a small foramen in the lateral orbital margin on the posterior surface of the frontal process of the zygomatic bone, and passes superficially to supply skin over the temple.
- The **zygomaticofacial branch** also passes forward at the base of the lateral orbital wall and leaves through a small bony canal, in the orbital margin, which opens via multiple small foramina on the anterolateral surface of the zygomatic bone, and its branches supply the adjacent skin.

Posterior superior alveolar nerve. The **posterior superior alveolar nerve** (Fig. 8.152) originates from the maxillary nerve [V_2] in the pterygopalatine fossa and passes laterally out of the fossa through the pterygomaxillary fissure to enter the infratemporal fossa. It continues laterally and inferiorly to enter the posterior surface of the maxilla through a small alveolar foramen approximately midway between the last molar tooth and the inferior orbital fissure. It then passes inferiorly just deep to the mucosa of the maxillary sinus to join the **superior dental plexus**.

The posterior superior alveolar nerve supplies the molar teeth and adjacent buccal gingivae, and contributes to the supply of the maxillary sinus.

Infra-orbital nerve. The infra-orbital nerve (Fig. 8.152) is the anterior continuation of the maxillary nerve [V_2] that leaves the pterygopalatine fossa through the inferior orbital fissure. It lies first in the infra-orbital groove in the floor of the orbit and then continues forward in the infra-orbital canal.

While in the infra-orbital groove and canal, the infra-orbital nerve gives origin to **middle** and **anterior superior alveolar nerves**, respectively, which ultimately join the **superior alveolar plexus** to supply the upper teeth:

- The middle superior alveolar nerve also supplies the maxillary sinus.
- The anterior superior alveolar nerve also gives origin to a small nasal branch, which passes medially through the lateral wall of the nasal cavity to supply parts of the areas of the nasal floor and walls.

The infra-orbital nerve exits the infra-orbital canal through the infra-orbital foramen inferior to the orbital margin and divides into nasal, palpebral, and superior labial branches:

- Nasal branches supply skin over the lateral aspect of the external nose and part of the nasal septum.
- Palpebral branches supply skin of the lower eyelid.
- Superior labial branches supply skin over the cheek and upper lip, and the related oral mucosa.

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Nerve of the pterygoid canal and the pterygopalatine ganglion

The nerve of the pterygoid canal (Fig. 8.153) is formed in the middle cranial fossa by the union of:

- the greater petrosal nerve (a branch of the facial nerve [VII]), and
- the deep petrosal nerve (a branch of the internal carotid plexus).

The nerve of the pterygoid canal passes into the pterygopalatine fossa and joins the pterygopalatine ganglion. It carries mainly preganglionic parasympathetic and postganglionic sympathetic fibers.

Greater petrosal nerve

The greater petrosal nerve, which originates from the geniculate ganglion of the facial nerve [VII] in the temporal bone, exits the temporal bone through a small canal that opens via a fissure onto the anterior surface of the petrous part of the temporal bone. It passes anteromedially along the posterior margin of the middle cranial fossa and then under the internal carotid artery to reach the superior surface of the cartilage filling the foramen lacerum.

As the greater petrosal nerve passes under the internal carotid artery, it is joined by the deep petrosal nerve to form the nerve of the pterygoid canal.

The greater petrosal nerve carries parasympathetic innervation to all glands above the oral fissure, including:

- mucous glands in the nasal cavity,
- salivary glands in the upper half of the oral cavity, and
- the lacrimal gland in the orbit.

The greater petrosal nerve also carries some taste (SA) fibers from the soft palate in the lesser palatine nerve.

Deep petrosal nerve

The **deep petrosal nerve** is formed by postganglionic sympathetic fibers that originate in the **superior cervical sympathetic ganglion** in the neck and leave the ganglion as the **internal carotid nerve**.

Preganglionic fibers that synapse in the ganglion are from the T1 spinal nerve.

The internal carotid nerve forms the internal carotid plexus around the internal carotid artery as the internal carotid artery passes through the skull and into the cranial cavity. Some of the fibers from the internal carotid plexus converge to form the deep petrosal nerve, which leaves the internal carotid plexus in the middle cranial fossa and joins the greater petrosal branch of the facial nerve [VII].

The deep petrosal nerve carries postganglionic sympathetic fibers destined mainly for blood vessels.

Pterygopalatine ganglion

The nerve of the pterygoid canal enters the superior surface of the cartilage that fills the foramen lacerum and passes anteriorly through the cartilage to enter the pterygoid canal in the root of the pterygoid process of the sphenoid bone. It passes through the canal and into the

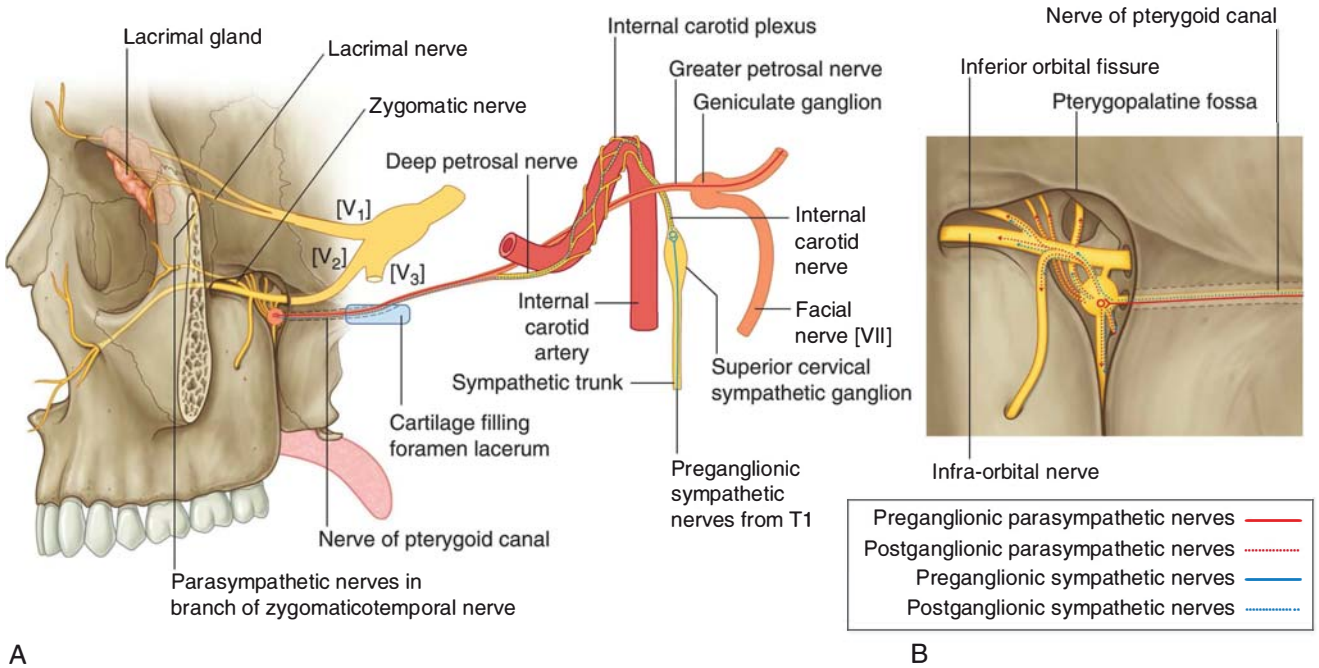


Fig. 8.153 Nerve of the pterygoid canal. **A.** Overview. **B.** In relationship to the pterygopalatine ganglion.

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pterygopalatine fossa where it joins the pterygopalatine ganglion formed around the branches of the maxillary nerve [V_2] (Fig. 8.153).

The **pterygopalatine ganglion** is the largest of the four parasympathetic ganglia in the head and is formed by the cell bodies of the postganglionic neurons associated with preganglionic parasympathetic fibers of the facial nerve [VII] carried by the greater petrosal nerve and the nerve of the pterygoid canal.

The postganglionic parasympathetic fibers that originate in the pterygopalatine ganglion, together with postganglionic sympathetic fibers passing through the ganglion, join fibers from the ganglionic branches of the maxillary nerve [V_2] to form orbital, palatine, nasal, and pharyngeal branches, which leave the ganglion.

Other postganglionic parasympathetic and sympathetic fibers pass superiorly through the ganglionic branches of the maxillary nerve [V_2] to enter the main trunk of the maxillary nerve and be distributed with the zygomatic, posterior superior alveolar, and infra-orbital nerves. Of these, the postganglionic parasympathetic and sympathetic fibers that pass into the orbit with the zygomatic nerve are particularly important because they ultimately innervate the lacrimal gland.

Innervation of the lacrimal gland

Approximately midway along the orbital wall, the postganglionic parasympathetic and sympathetic fibers leave the zygomaticotemporal branch of the zygomatic nerve and form a special autonomic nerve, which travels up the lateral orbital wall to join the lacrimal nerve (Fig. 8.153; also see Fig. 8.84).

The lacrimal nerve is a major general sensory branch of the ophthalmic nerve [V_1], which passes forward in the orbit at the margin between the lateral wall and roof.

The postganglionic parasympathetic and sympathetic fibers pass with the lacrimal nerve to the lacrimal gland.

A lesion anywhere along the course of parasympathetic fibers that leave the brain as part of the facial nerve [VII] and are ultimately carried to the lacrimal gland along branches of the ophthalmic nerve [V_1] results in "dry eye" and can eventually lead to loss of vision in the affected eye.

Maxillary artery

The maxillary artery is a major branch of the external carotid artery in the neck. It originates adjacent to the neck of the mandible, passes forward through the infratemporal fossa, and then enters the pterygopalatine fossa through the pterygomaxillary fissure (Fig. 8.154).

The part of the maxillary artery in the pterygopalatine fossa (the third part) is anterior to the pterygopalatine ganglion and gives origin to branches that accompany branches of the maxillary nerve [V_2] and the pterygopalatine ganglion.

Branches of the maxillary artery include the posterior superior alveolar, infra-orbital, greater palatine, pharyngeal, and sphenopalatine arteries, and the artery of the pterygoid canal (Fig. 8.154). Collectively, these branches supply much of the nasal cavity, the roof of the oral cavity,

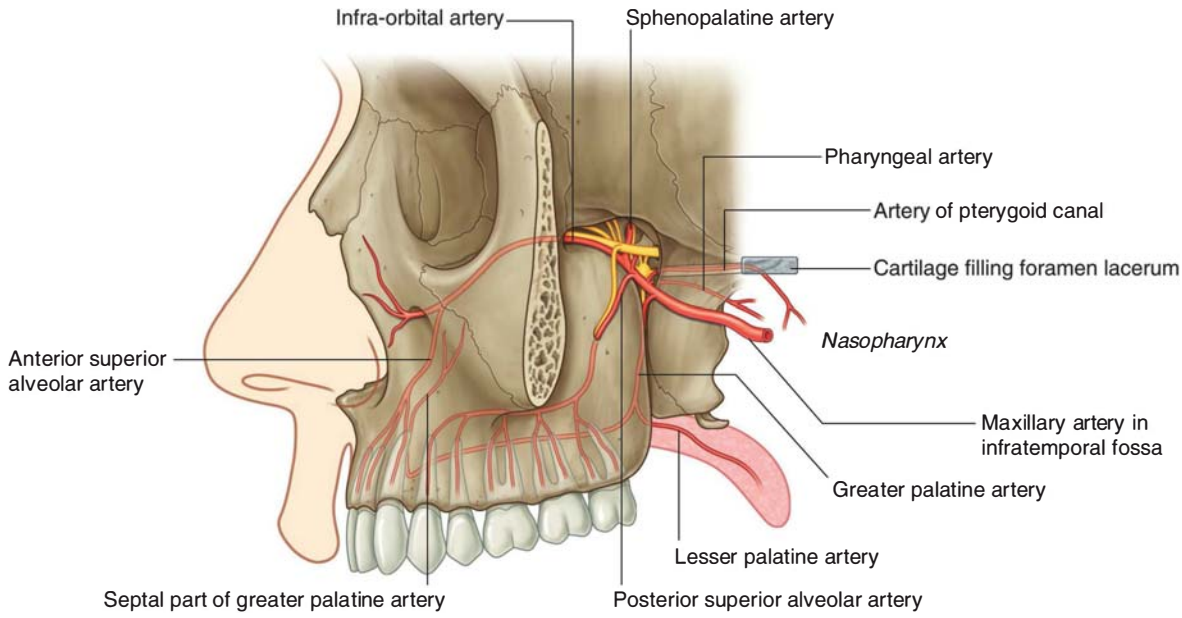


Fig. 8.154 Maxillary artery in the pterygopalatine fossa.

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and all upper teeth. In addition, they contribute to the blood supply of the sinuses, oropharynx, and floor of the orbit.

Branches

Posterior superior alveolar artery. The **posterior superior alveolar artery** (Fig. 8.154) originates from the maxillary artery as it passes through the pterygomaxillary fissure. It meets the posterior superior alveolar nerve, accompanies it through the alveolar foramen on the infratemporal surface of the maxilla, and supplies the molar and premolar teeth, adjacent gingiva, and the maxillary sinus.

Infra-orbital artery. The infra-orbital artery (Fig. 8.154) passes forward with the infra-orbital nerve and leaves the pterygopalatine fossa through the inferior orbital fissure. With the infra-orbital nerve, it lies in the infra-orbital groove and infra-orbital canal, and emerges through the infra-orbital foramen to supply parts of the face.

Within the infra-orbital canal, the infra-orbital artery gives origin to:

- branches that contribute to the blood supply of structures near the floor of the orbit—the inferior rectus and inferior oblique muscles, and the lacrimal sac; and
- **anterior superior alveolar arteries** (Fig. 8.154), which supply the incisor and canine teeth and the maxillary sinus.

Greater palatine artery. The **greater palatine artery** (Fig. 8.154) passes inferiorly with the palatine nerves into the palatine canal. It gives origin to a **lesser palatine branch** (Fig. 8.154), which passes through the lesser palatine foramen to supply the soft palate, and then continues through the greater palatine foramen to supply the hard palate. The latter vessel passes forward on the inferior surface of the palate to enter the incisive fossa and pass superiorly through the incisive canal to supply the anterior aspect of the septal wall of the nasal cavity.

Pharyngeal branch. The **pharyngeal branch** (Fig. 8.154) of the maxillary artery travels posteriorly and leaves the pterygopalatine fossa through the palatovaginal canal with the pharyngeal nerve. It supplies the posterior aspect of the roof of the nasal cavity, the sphenoidal sinus, and the pharyngotympanic tube.

Sphenopalatine artery. The **sphenopalatine artery** (Fig. 8.154) is the terminal branch of the maxillary artery. It leaves the pterygopalatine fossa medially through the sphenopalatine foramen and accompanies the nasal nerves, giving off:

- posterior lateral nasal arteries, which supply the lateral wall of the nasal cavity and contribute to the supply of the paranasal sinuses; and
- posterior septal branches, which travel medially across the roof to supply the nasal septum—the largest of these branches passes anteriorly down the septum to anastomose with the end of the greater palatine artery.

Artery of pterygoid canal. The **artery of the pterygoid canal** passes posteriorly into the pterygoid canal. It supplies surrounding tissues and terminates, after passing inferiorly through cartilage filling the foramen lacerum, in the mucosa of the nasopharynx.

Veins

Veins that drain areas supplied by branches of the terminal part of the maxillary artery generally travel with these branches back into the pterygopalatine fossa.

The veins coalesce in the pterygopalatine fossa and then pass laterally through the pterygomaxillary fissure to join the pterygoid plexus of veins in the infratemporal fossa (Fig. 8.155).

The infra-orbital vein, which drains the inferior aspect of the orbit, may pass directly into the infratemporal fossa through the lateral aspect of the inferior orbital fissure, so bypassing the pterygopalatine fossa.

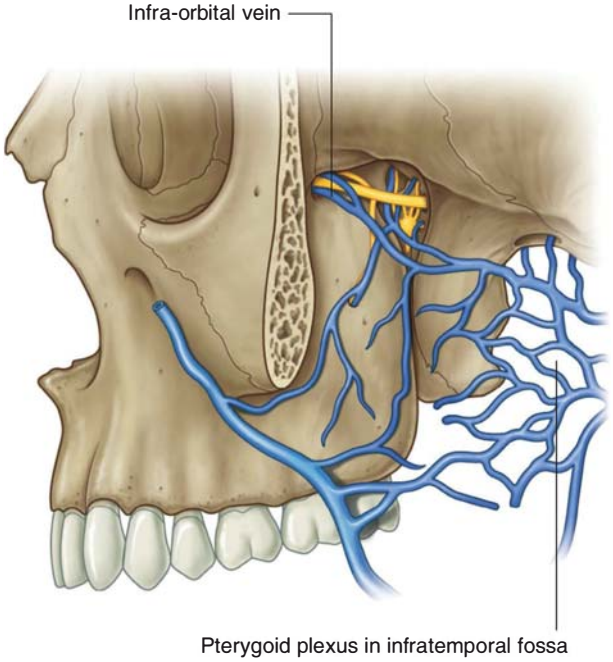


Fig. 8.155 Veins of the pterygopalatine fossa.

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NECK

The neck is a tube providing continuity from the head to the trunk. It extends anteriorly from the lower border of the mandible to the upper surface of the manubrium of the sternum, and posteriorly from the superior nuchal line on the occipital bone of the skull to the intervertebral disc between the CVII and TI vertebrae. Within the tube, four compartments provide longitudinal organization (Fig. 8.156):

- The visceral compartment is anterior and contains parts of the digestive and respiratory systems, and several endocrine glands.
- The vertebral compartment is posterior and contains the cervical vertebrae, spinal cord, cervical nerves, and muscles associated with the vertebral column.
- The two vascular compartments, one on each side, are lateral and contain the major blood vessels and the vagus nerve [X].

All these compartments are contained within unique layers of cervical fascia.

For descriptive purposes the neck is divided into anterior and posterior triangles (Fig. 8.157):

- The boundaries of the **anterior triangle** are the anterior border of the sternocleidomastoid muscle, the inferior border of the mandible, and the midline of the neck.
- The boundaries of the **posterior triangle** are the posterior border of the sternocleidomastoid muscle, the anterior border of the trapezius muscle, and the middle one-third of the clavicle.

Fascia

The fascia of the neck has a number of unique features.

The **superficial fascia** in the neck contains a thin sheet of muscle (the **platysma**), which begins in the superficial fascia of the thorax, runs upward to attach to the mandible and blend with the muscles on the face, is innervated by the cervical branch of the facial nerve [VII], and is only found in this location.

Deep to the superficial fascia, the deep cervical fascia is organized into several distinct layers (Fig. 8.156). These include:

- an investing layer, which surrounds all structures in the neck;
- the prevertebral layer, which surrounds the vertebral column and the deep muscles associated with the back;
- the pretracheal layer, which encloses the viscera of the neck; and
- the carotid sheaths, which receive a contribution from the other three fascial layers and surround the two major neurovascular bundles on either side of the neck.

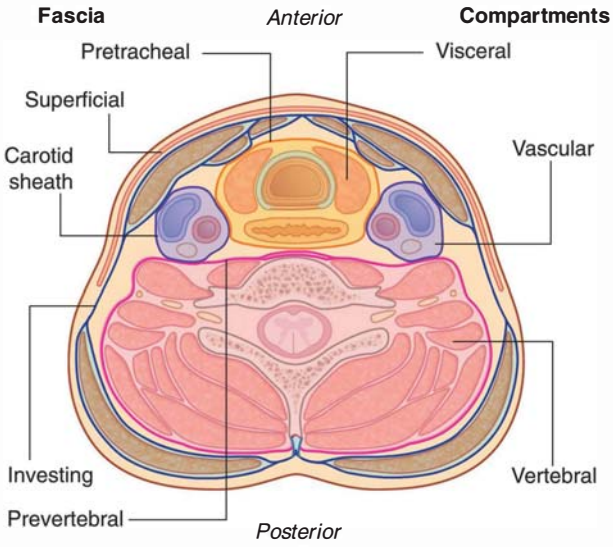


Fig. 8.156 Compartments of the neck.

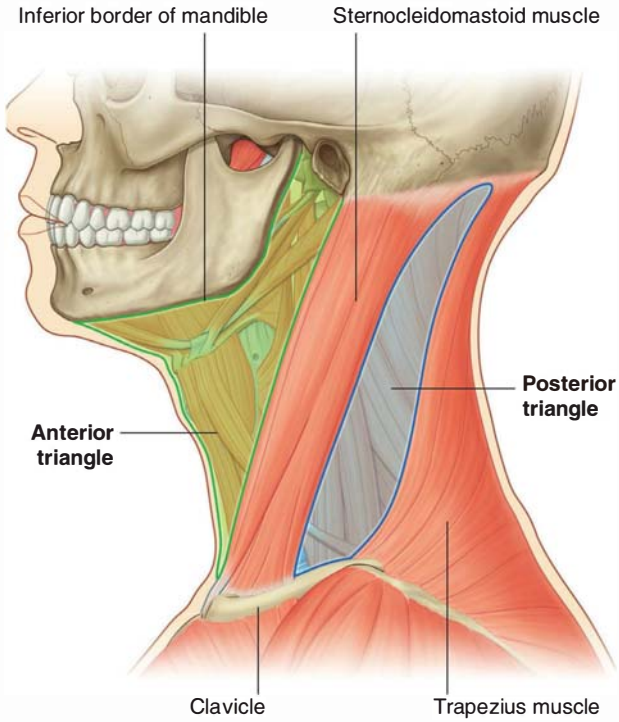


Fig. 8.157 Anterior and posterior triangles of the neck.

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Investing layer

The **investing layer** completely surrounds the neck (Fig. 8.158).

Attaching posteriorly to the ligamentum nuchae and the spinous process of the CVII vertebra, this fascial layer splits as it passes forward to enclose the trapezius muscle, reunites into a single layer as it forms the roof of the posterior triangle, splits again to surround the sternocleidomastoid muscle, and reunites again to join its twin from the other side.

Anteriorly, the investing fascia surrounds the infrahyoid muscles.

The investing fascia is attached:

- superiorly to the external occipital protuberance and the superior nuchal line,
- laterally to the mastoid process and zygomatic arch, and
- inferiorly to the spine of the scapula, the acromion, the clavicle, and the manubrium of the sternum.

The external and anterior jugular veins, and the lesser occipital, great auricular, transverse cervical, and supraclavicular nerves, all branches of the cervical plexus, pierce the investing fascia.

Prevertebral layer

The prevertebral layer is a cylindrical layer of fascia that surrounds the vertebral column and the muscles associated with it (Fig. 8.158). Muscles in this group include the prevertebral muscles, the anterior, middle, and posterior scalene muscles, and the deep muscles of the back.

The prevertebral fascia is attached posteriorly along the length of the ligamentum nuchae, and superiorly forms a continuous circular line attaching to the base of the skull. This circle begins:

- anteriorly as the fascia attaches to the basilar part of the occipital bone, the area of the jugular foramen, and the carotid canal;
- continues laterally, attaching to the mastoid process; and
- continues posteriorly along the superior nuchal line ending at the external occipital protuberance, where it associates with its partner from the opposite side.

Anteriorly, the prevertebral fascia is attached to the anterior surfaces of the transverse processes and bodies of vertebrae CI to CVII.

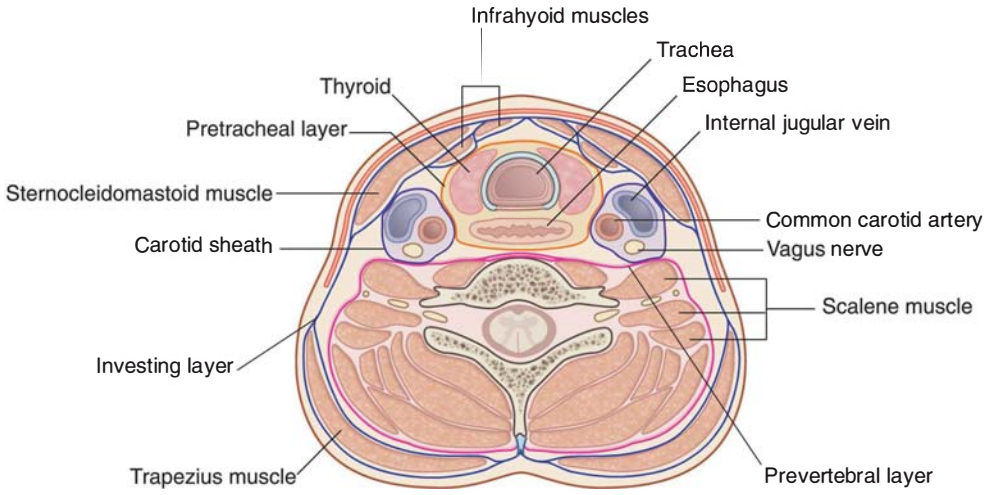


Fig. 8.158 Fascia of neck, transverse view.

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The prevertebral fascia passing between the attachment points on the transverse processes is unique. In this location, it splits into two layers, creating a longitudinal fascial space containing loose connective tissue that extends from the base of the skull through the thorax (Figs. 8.158 and 8.159).

There is one additional specialization of the prevertebral fascia in the lower region of the neck. The prevertebral fascia in an anterolateral position extends from the anterior and middle scalene muscles to surround the brachial plexus and subclavian artery as these structures pass into the axilla. This fascial extension is the **axillary sheath**.

Pretracheal layer

The **pretracheal layer** consists of a collection of fascias that surround the trachea, esophagus, and thyroid gland (Fig. 8.158). Anteriorly, it consists of a pretracheal fascia that crosses the neck, just posterior to the infrahyoid muscles, and covers the trachea and the thyroid gland. The pretracheal fascia begins superiorly at the hyoid bone and ends inferiorly in the upper thoracic cavity. Laterally, this fascia encloses the thyroid gland and more posteriorly is continuous with fascia that surrounds the esophagus.

Posterior to the pharynx, the pretracheal layer is referred to as the buccopharyngeal fascia and separates the pharynx from the prevertebral layer (Fig. 8.159).

The buccopharyngeal fascia begins superiorly at the base of the skull and merges with fascia covering the esophagus that then continues inferiorly into the thoracic cavity.

Carotid sheath

Each **carotid sheath** is a column of fascia that surrounds the common carotid artery, the internal carotid artery, the internal jugular vein, and the vagus nerve as these structures pass through the neck (Fig. 8.158).

It receives contributions from the investing, prevertebral, and pretracheal layers, though the extent of each component's contribution varies.

Fascial compartments

The arrangement of the various layers of cervical fascia organizes the neck into four longitudinal compartments (Fig. 8.156):

- The first compartment is the largest, includes the other three, and consists of the area surrounded by the investing layer.
- The second compartment consists of the vertebral column and the deep muscles associated with this structure, and is the area contained within the prevertebral layer.
- The third compartment (the visceral compartment) contains the pharynx, the trachea, the esophagus, and the thyroid gland, which are surrounded by the pretracheal layer.
- Finally, there is a compartment (the carotid sheath) consisting of the neurovascular structures that pass from the base of the skull to the thoracic cavity, and the sheath enclosing these structures receives contributions from the other cervical fascias.

Fascial spaces

Between the fascial layers in the neck are spaces that may provide a conduit for the spread of infections from the neck to the mediastinum.

Three spaces could be involved in this process (Fig. 8.159):

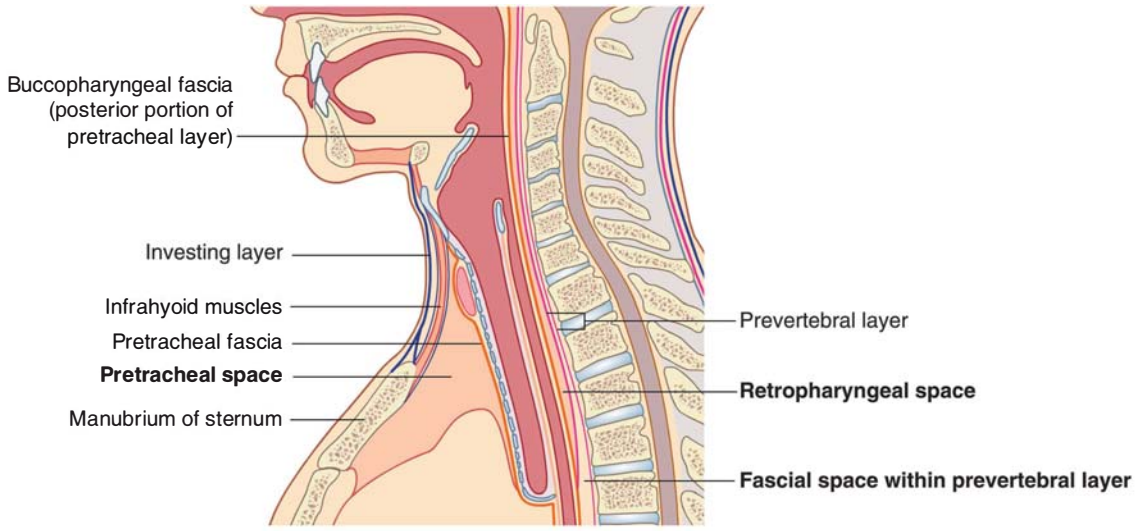


Fig. 8.159 Fascia of the neck, sagittal view.

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- The first is the **pretracheal space** between the investing layer of cervical fascia (covering the posterior surface of the infrahyoid muscles) and the pretracheal fascia (covering the anterior surface of the trachea and the thyroid gland), which passes between the neck and the anterior part of the superior mediastinum.
- The second is the **retropharyngeal space** between the buccopharyngeal fascia (on the posterior surface of the pharynx and esophagus) and the prevertebral fascia (on the anterior surface of the transverse processes and bodies of the cervical vertebrae), which extends from the base of the skull to the upper part of the posterior mediastinum.
- The **third space** is within the prevertebral layer covering the anterior surface of the transverse processes and bodies of the cervical vertebrae. This layer splits into two laminae to create a fascial space that begins at the base of the skull and extends through the posterior mediastinum to the diaphragm.

Superficial venous drainage

The external jugular and anterior jugular veins are the primary venous channels for superficial venous drainage of the neck (Fig. 8.160).

External jugular veins

The external jugular vein is formed posterior to the angle of the mandible as the **posterior auricular vein** and the **retromandibular vein** join:

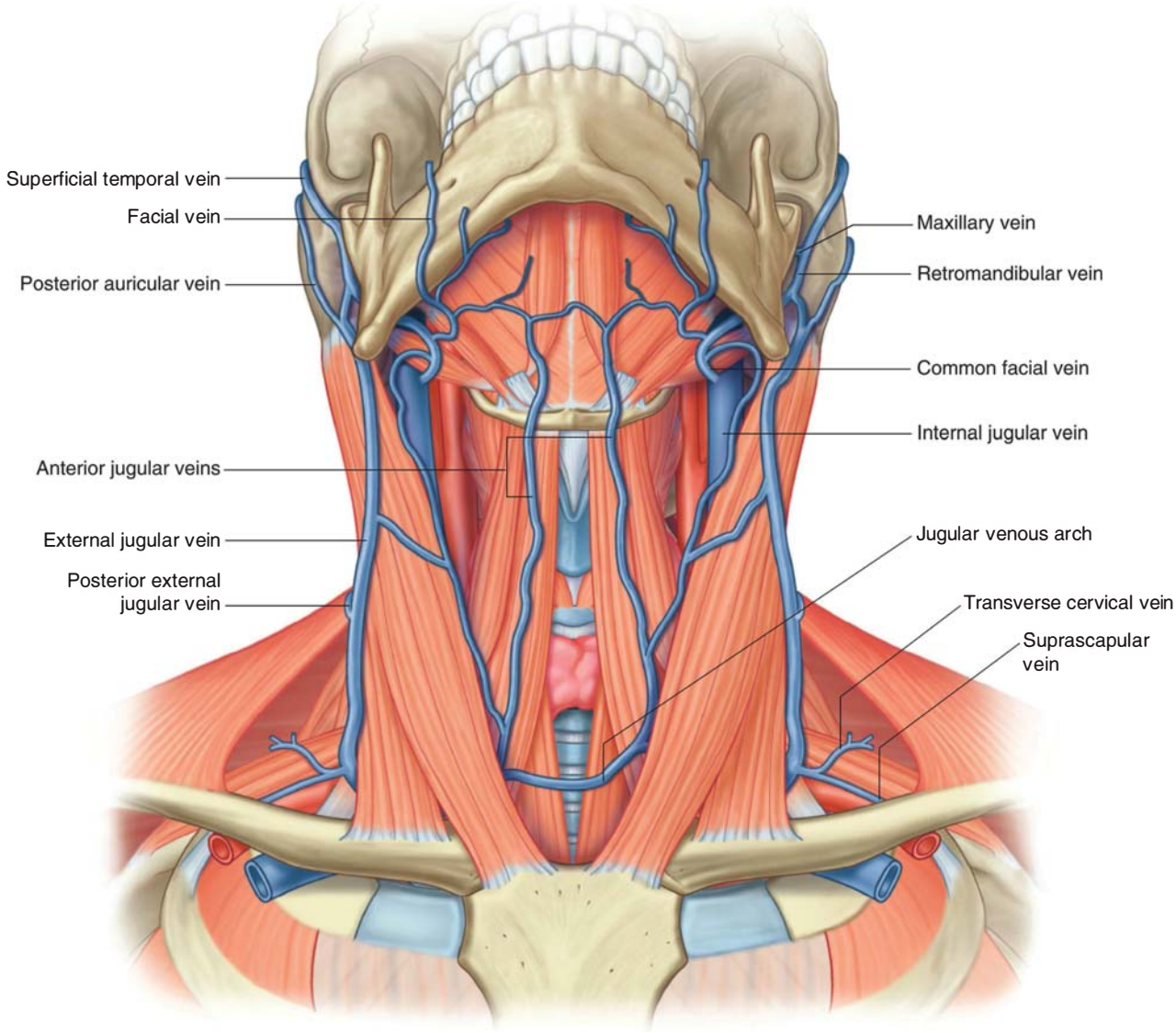


Fig. 8.160 Superficial veins of neck.

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- The posterior auricular vein drains the scalp behind and above the ear.
- The retromandibular vein is formed when the **superficial temporal** and **maxillary veins** join in the substance of the parotid gland and it descends to the angle of mandible, where it divides into an anterior and a posterior division (Fig. 8.160)—the posterior division joins the posterior auricular vein to form the external jugular vein, and the anterior division joins the **facial vein** to form the common facial vein, which passes deep and becomes a tributary to the internal jugular vein.

Once formed, the external jugular vein passes straight down the neck in the superficial fascia and is superficial to the sternocleidomastoid muscle throughout its course, crossing it diagonally as it descends.

Reaching the lower part of the neck, just superior to the clavicle and immediately posterior to the sternocleidomastoid muscle, the external jugular vein pierces the investing layer of cervical fascia, passes deep to the clavicle, and enters the **subclavian vein**.

Tributaries received by the external jugular vein along its course include the **posterior external jugular vein** (draining superficial areas of the back of the neck) and the **transverse cervical** and **suprascapular veins** (draining the posterior scapular region).

Anterior jugular veins

The **anterior jugular veins**, although variable and inconsistent, are usually described as draining the anterior aspect of the neck (Fig. 8.160). These paired venous channels, which begin as small veins, come together at or just superior to the hyoid bone. Once formed, each anterior jugular vein descends on either side of the midline of the neck.

Inferiorly, near the medial attachment of the sternocleidomastoid muscle, each anterior jugular vein pierces the investing layer of cervical fascia to enter the subclavian vein. Occasionally, the anterior jugular vein may enter the external jugular vein immediately before the external jugular vein enters the subclavian vein.

Often, the right and left anterior jugular veins communicate with each other, being connected by a **jugular venous arch** in the area of the suprasternal notch.

In the clinic

Fascial planes of the head and neck

The neck contains a series of compartments, which are bound by tight fascia. All these compartments are within the overall investing layer of cervical fascia. From a clinical perspective the importance of these compartments is that infection tends to spread within compartments or within the spaces between the various fascial layers. For example, if infection arises in the pretracheal space it may spread inferiorly into the superior mediastinum and lie anterior to the pericardium.

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In the clinic

Central venous access

In most instances, access to peripheral veins of the arm and the leg will suffice for administering intravenous drugs and fluids and for obtaining blood for analysis. In certain circumstances it is necessary to place larger-bore catheters in the central veins, for example, for dialysis, parenteral nutrition, or the administration of drugs that have a tendency to produce phlebitis.

"Blind puncture" of the subclavian and jugular veins to obtain central venous access used to be standard practice. However, subclavian vein puncture is not without complications. As the subclavian vein passes inferiorly, posterior to the clavicle, it passes over the apex of the lung. Any misplacement of a needle into or through this structure may puncture the apical pleura, producing a pneumothorax. Inadvertent arterial puncture and vein laceration may also produce a hemopneumothorax.

A puncture of the internal jugular vein (Fig. 8.161) carries fewer risks, but local hematoma and damage to the carotid artery are again important complications.

Current practice is to identify major vessels using ultrasound and to obtain central venous access under direct vision to avoid any significant complication.

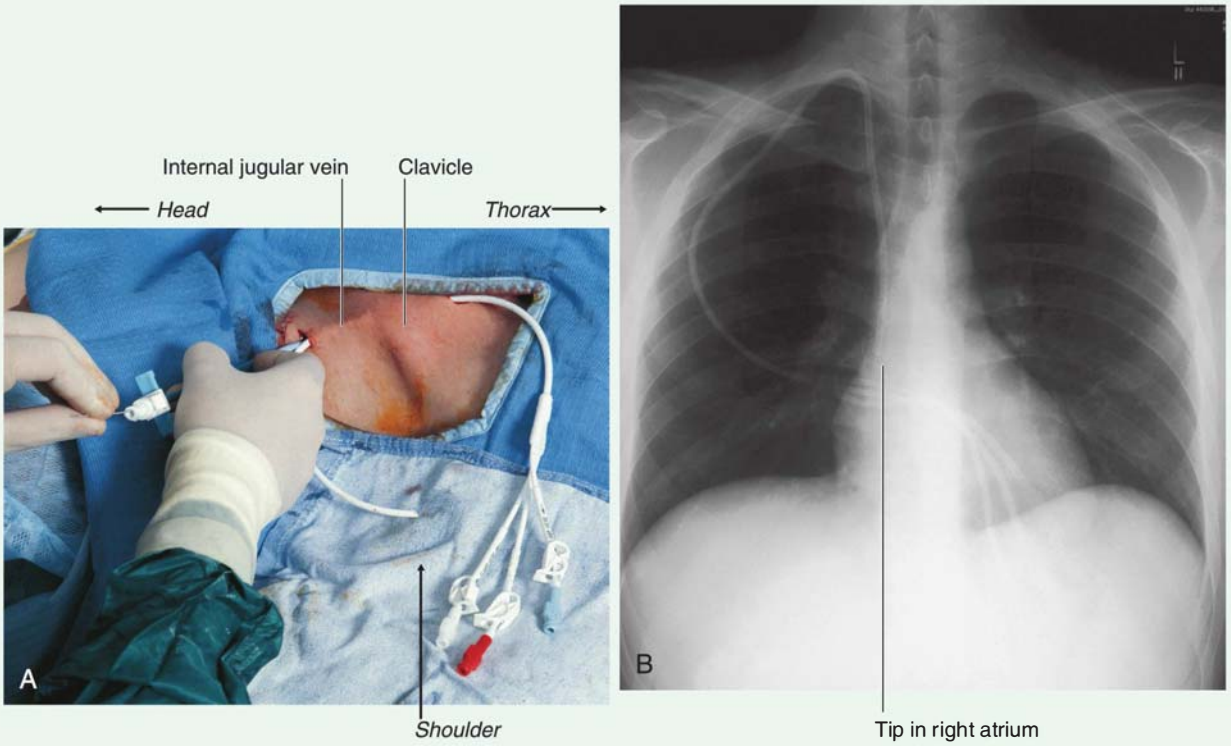


Fig. 8.161 Placing a central venous catheter in the neck. **A.** Clinical procedure. **B.** Chest radiograph showing that the tip of the catheter is in the origin of the right atrium.

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Anterior triangle of the neck

The anterior triangle of the neck is outlined by the anterior border of the sternocleidomastoid muscle laterally, the inferior border of the mandible superiorly, and the midline of the neck medially (Fig. 8.162). It is further subdivided into several smaller triangles as follows:

- The **submandibular triangle** is outlined by the inferior border of the mandible superiorly and the anterior and posterior bellies of the digastric muscle inferiorly.
- The **submental triangle** is outlined by the hyoid bone inferiorly, the anterior belly of the digastric muscle laterally, and the midline.
- The **muscular triangle** is outlined by the hyoid bone superiorly, the superior belly of the omohyoid muscle, and the anterior border of the sternocleidomastoid muscle laterally, and the midline.
- The **carotid triangle** is outlined by the superior belly of the omohyoid muscle anteroinferiorly, the stylohyoid muscle and posterior belly of the digastric superiorly, and the anterior border of the sternocleidomastoid muscle posteriorly.

Each of these triangles contains numerous structures that can be identified as being within a specific triangle, passing into a specific triangle from outside the area, originating in one triangle and passing to another triangle, or passing through several triangles while passing through the region.

A discussion of the anterior triangle of the neck must therefore combine a systemic approach, describing the muscles, vessels, and nerves in the area, with a regional approach, describing the contents of each triangle.

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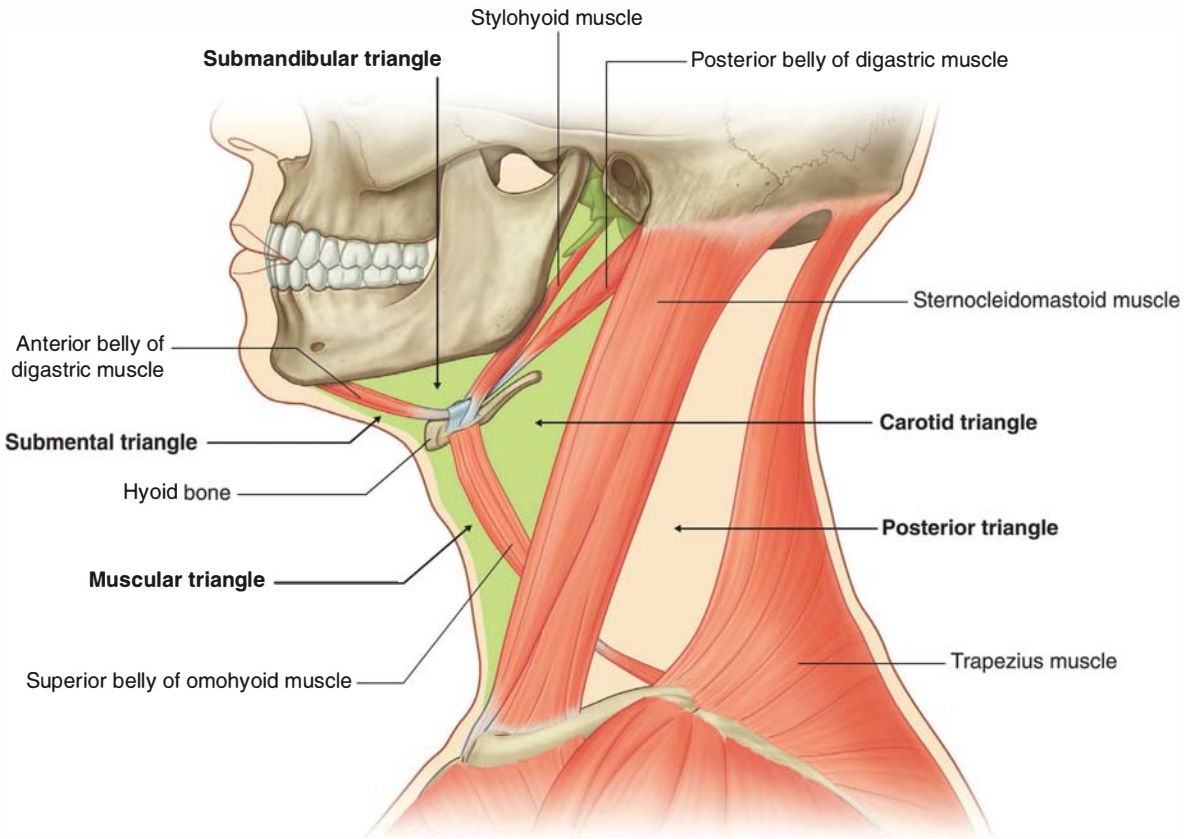


Fig. 8.162 Borders and subdivisions of the anterior triangle of the neck.

Muscles

The muscles in the anterior triangle of the neck (Table 8.12) can be grouped according to their location relative to the hyoid bone:

- Muscles superior to the hyoid are classified as **suprahyoid muscles** and include the stylohyoid, digastric, mylohyoid, and geniohyoid.
- Muscles inferior to the hyoid are **infrahyoid muscles** and include the omohyoid, sternohyoid, thyrohyoid, and sternothyroid.

Suprahyoid muscles

The four pairs of suprahyoid muscles are related to the submental and submandibular triangles (Fig. 8.162). They pass in a superior direction from the hyoid bone to the skull or mandible and raise the hyoid, as occurs during swallowing.

Table 8.12 Anterior triangle of neck (suprahyoid and infrahyoid muscles)

Muscle	Origin	Insertion	Innervation	Function
Stylohyoid	Base of styloid process	Lateral area of body of hyoid bone	Facial nerve [VII]	Pulls hyoid bone upward in a posterosuperior direction
Digastric —Anterior belly	Digastric fossa on lower inside of mandible	Attachment of tendon between two bellies to body of hyoid bone	Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V_3]	Opens mouth by lowering mandible; raises hyoid bone
—Posterior belly	Mastoid notch on medial side of mastoid process of temporal bone	Same as anterior belly	Facial nerve [VII]	Pulls hyoid bone upward and back
Mylohyoid	Mylohyoid line on mandible	Body of hyoid bone and fibers from muscle on opposite side	Mylohyoid nerve from inferior alveolar branch of mandibular nerve [V_3]	Support and elevation of floor of mouth; elevation of hyoid
Geniohyoid	Inferior mental spine on inner surface of mandible	Anterior surface of body of hyoid bone	Branch from anterior ramus of C1 (carried along the hypoglossal nerve [XII])	Fixed mandible elevates and pulls hyoid bone forward; fixed hyoid bone pulls mandible downward and inward
Sternohyoid	Posterior aspect of sternoclavicular joint and adjacent manubrium of sternum	Body of hyoid bone medial to attachment of omohyoid muscle	Anterior rami of C1 to C3 through the ansa cervicalis	Depresses hyoid bone after swallowing
Omohyoid	Superior border of scapula medial to suprascapular notch	Lower border of body of hyoid bone just lateral to attachment of sternohyoid	Anterior rami of C1 to C3 through the ansa cervicalis	Depresses and fixes hyoid bone
Thyrohyoid	Oblique line on lamina of thyroid cartilage	Greater horn and adjacent aspect of body of hyoid bone	Fibers from anterior ramus of C1 carried along hypoglossal nerve [XII]	Depresses hyoid bone, but when hyoid bone is fixed raises larynx
Sternothyroid	Posterior surface of manubrium of sternum	Oblique line on lamina of thyroid cartilage	Anterior rami of C1 to C3 through the ansa cervicalis	Draws larynx (thyroid cartilage) downward

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Stylohyoid

The **stylohyoid muscle** arises from the base of the styloid process and passes anteroinferiorly to attach to the lateral area of the body of the hyoid bone (Fig. 8.163). During swallowing it pulls the hyoid bone posterosuperiorly and it is innervated by the facial nerve [VII].

Digastric

The **digastric muscle** has two bellies connected by a tendon, which attaches to the body of the hyoid bone (Fig. 8.163):

- The **posterior belly** arises from the mastoid notch on the medial side of the mastoid process of the temporal bone.
- The **anterior belly** arises from the digastric fossa on the lower inside of the mandible.

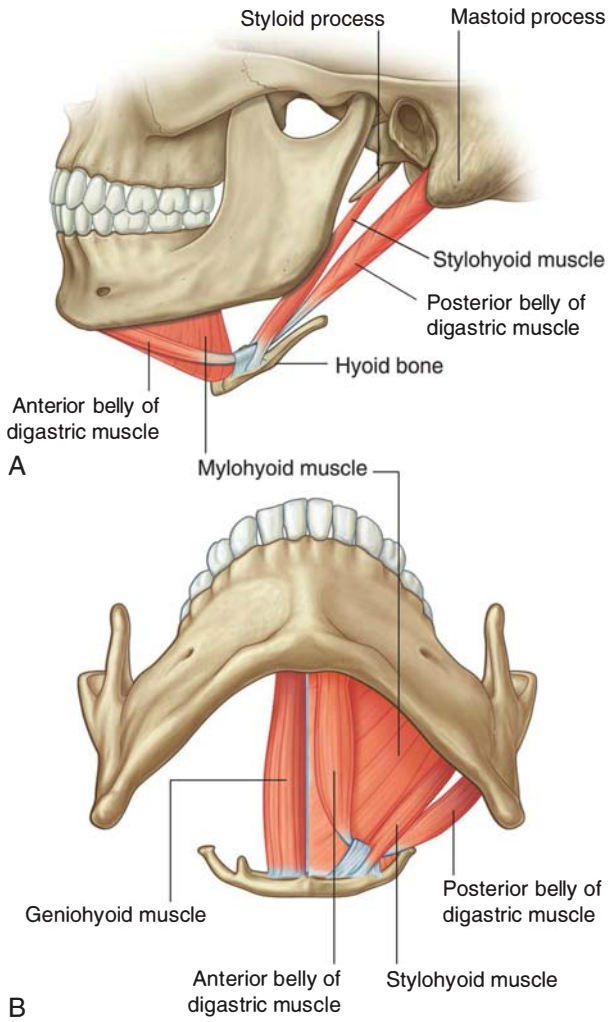


Fig. 8.163 Suprahyoid muscles. **A.** Lateral view. **B.** Inferior view.

The tendon between the two bellies, which is attached to the body of the hyoid bone, is the point of insertion of both bellies. Because of this arrangement, the muscle has multiple actions depending on which bone is fixed:

- When the mandible is fixed, the digastric muscle raises the hyoid bone.
- When the hyoid bone is fixed, the digastric muscle opens the mouth by lowering the mandible.

Innervation of the digastric muscle is from two different cranial nerves.

The innervation of the posterior belly of the digastric muscle is by the facial nerve [VII], whereas the anterior belly of the muscle is innervated by the mandibular division [V_3] of the trigeminal nerve [V].

Mylohyoid

The **mylohyoid muscle** is superior to the anterior belly of the digastric and, with its partner from the opposite side, forms the floor of the mouth (Fig. 8.163). It originates from the mylohyoid line on the medial surface of the body of the mandible and inserts into the hyoid bone and also blends with the mylohyoid muscle from the opposite side.

This mylohyoid muscle supports and elevates the floor of the mouth and elevates the hyoid bone. It is innervated by the mandibular division [V_3] of the trigeminal nerve [V].

Geniohyoid

The **geniohyoid muscle** is superior to the floor of the oral cavity and is not generally considered a muscle of the anterior triangle of the neck; however, it can be regarded as a suprahyoid muscle. It is the final muscle in the suprahyoid group (Fig. 8.163). A narrow muscle, it is superior to the medial part of each mylohyoid muscle. The muscles from each side are next to each other in the midline.

The geniohyoid arises from the inferior mental spine of the mandible and passes backward and downward to insert on the body of the hyoid bone.

It has two functions depending on which bone is fixed:

- Fixation of the mandible elevates and pulls the hyoid bone forward.
- Fixation of the hyoid bone pulls the mandible downward and inward.

The geniohyoid is innervated by a branch from the anterior ramus of C1 carried along the hypoglossal nerve [XII].

Infrahyoid muscles

The four infrahyoid muscles are related to the muscular triangle (Fig. 8.162). They attach the hyoid bone to inferior

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structures and depress the hyoid bone. They also provide a stable point of attachment for the suprahyoid muscles. Because of their appearance, they are sometimes referred to as the **"strap muscles."**

Sternohyoid

The sternohyoid muscle is a long, thin muscle originating from the posterior aspect of the sternoclavicular joint and adjacent manubrium of the sternum (Fig. 8.164). It ascends to insert onto the body of the hyoid bone. It depresses the hyoid bone and is innervated by the anterior rami of C1 to C3 through the ansa cervicalis.

Omohyoid

Lateral to the sternohyoid muscle is the omohyoid muscle (Fig. 8.164). This muscle consists of two bellies with an intermediate tendon in both the posterior and anterior triangles of the neck:

- The **inferior belly** begins on the superior border of the scapula, medial to the suprascapular notch, and passes forward and upward across the posterior triangle ending at the intermediate tendon.
- The **superior belly** begins at the intermediate tendon and ascends to attach to the body of the hyoid bone just lateral to the attachment of the sternohyoid.
- The intermediate tendon is attached to the clavicle, near its medial end, by a fascial sling.

The omohyoid depresses and fixes the hyoid bone. It is innervated by the anterior rami of C1 to C3 through the ansa cervicalis.

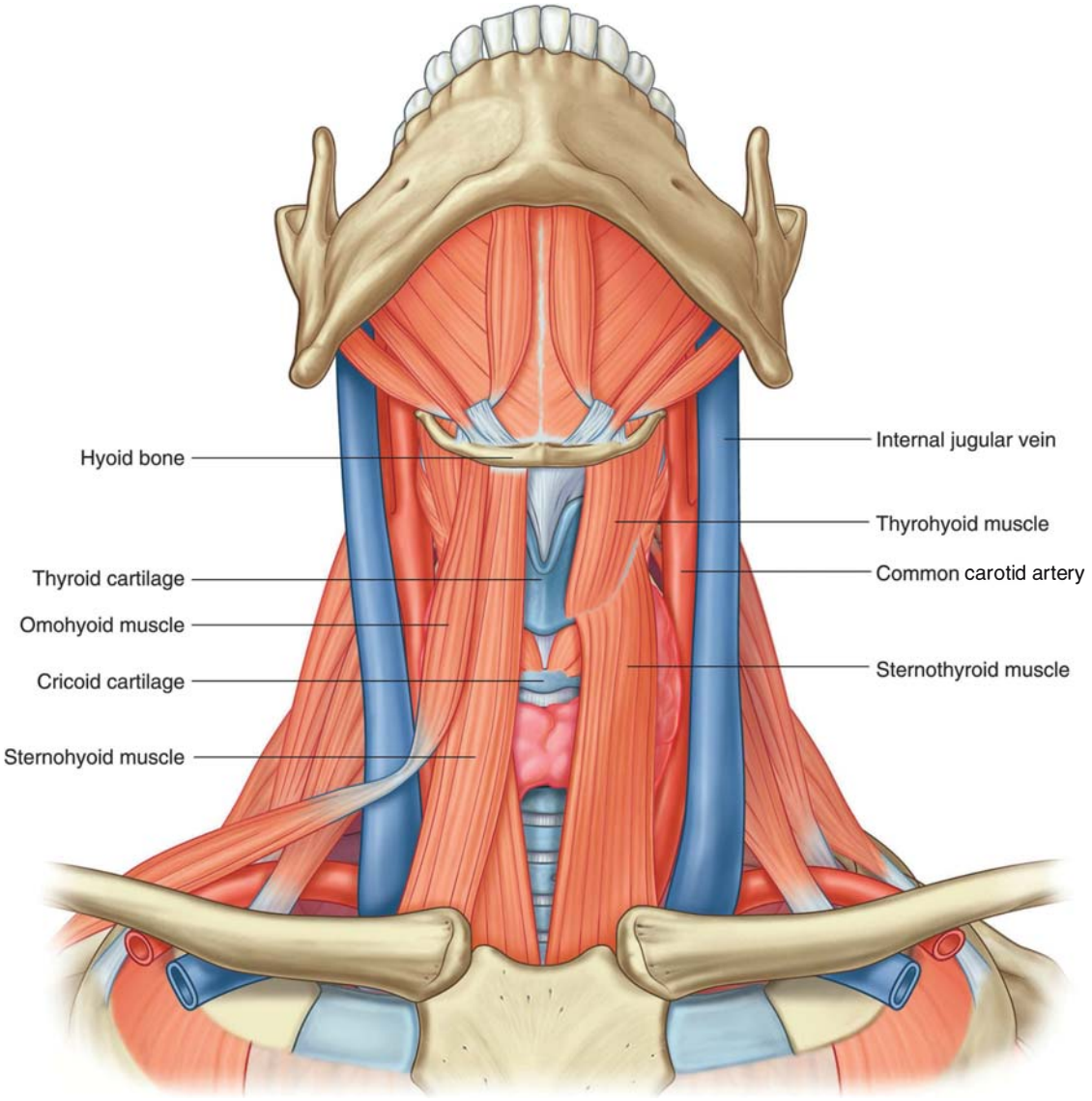


Fig. 8.164 Infrahyoid muscles.

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Thyrohyoid

The thyrohyoid muscle is deep to the superior parts of the omohyoid and sternohyoid (Fig. 8.164). Originating at the oblique line on the lamina of the thyroid cartilage it passes upward to insert into the greater horn and adjacent aspect of the body of the hyoid bone.

The thyrohyoid muscle has variable functions depending on which bone is fixed. Generally, it depresses the hyoid, but when the hyoid is fixed it raises the larynx (e.g., when high notes are sung). It is innervated by fibers from the anterior ramus of C1 that travel with the hypoglossal nerve [XII].

Sternothyroid

Lying beneath the sternohyoid and in continuity with the thyrohyoid, the sternothyroid is the last muscle in the infrahyoid group (Fig. 8.164). It arises from the posterior surface of the manubrium of the sternum and passes upward to attach to the oblique line on the lamina of the thyroid cartilage.

The sternothyroid muscle draws the larynx (thyroid cartilage) downward and is innervated by the anterior rami of C1 to C3 through the ansa cervicalis.

Vessels

Passing through the anterior triangle of the neck are the common carotid arteries and their branches, the external and internal carotid arteries. These vessels supply all structures of the head and neck.

Associated with this arterial system are the internal jugular vein and its tributaries. These vessels receive blood from all structures of the head and neck.

Carotid system

Common carotid arteries

The **common carotid arteries** are the beginning of the carotid system (Fig. 8.165):

- The **right common carotid artery** originates from the brachiocephalic trunk immediately posterior to the right sternoclavicular joint and is entirely in the neck throughout its course.
- The **left common carotid artery** begins in the thorax as a direct branch of the arch of the aorta and passes superiorly to enter the neck near the left sternoclavicular joint.

Both right and left common carotid arteries ascend through the neck, just lateral to the trachea and esophagus, within a fascial compartment (the carotid sheath). They give off no branches as they pass through the neck.

Near the superior edge of the thyroid cartilage each common carotid artery divides into its two terminal branches—the **external** and **internal carotid arteries** (Fig. 8.166).

The superior part of each common carotid artery and its division into external and internal carotid arteries

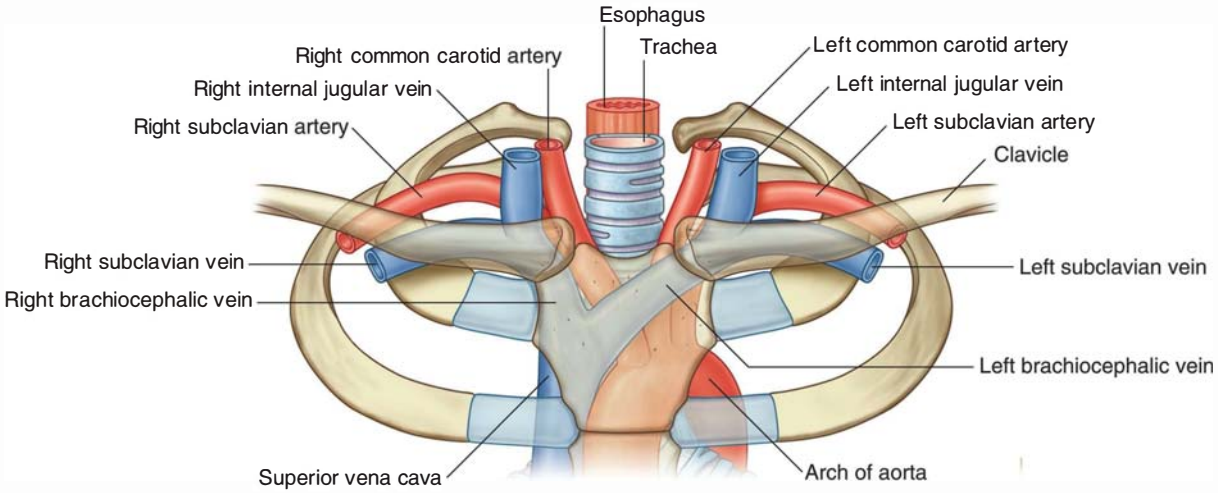


Fig. 8.165 Origin of common carotid arteries.

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occurs in the carotid triangle (Fig. 8.166), which is a subdivision of the anterior triangle of the neck (see Fig. 8.162).

At the bifurcation, the common carotid artery and the beginning of the internal carotid artery are dilated. This dilation is the **carotid sinus** (Fig. 8.167) and contains receptors that monitor changes in blood pressure and are innervated by a branch of the glossopharyngeal nerve [IX].

Another accumulation of receptors in the area of the bifurcation is responsible for detecting changes in blood chemistry, primarily oxygen content. This is the **carotid body** and is innervated by branches from both the glossopharyngeal [IX] and vagus [X] nerves.

Internal carotid arteries

After its origin, the internal carotid artery ascends toward the base of the skull (Fig. 8.167). It gives off no branches in the neck and enters the cranial cavity through the carotid canal in the petrous part of the temporal bone.

The internal carotid arteries supply the cerebral hemispheres, the eyes and the contents of the orbits, and the forehead.

External carotid arteries

The external carotid arteries begin giving off branches immediately after the bifurcation of the common carotid arteries (Fig. 8.167 and Table 8.13) as follows:

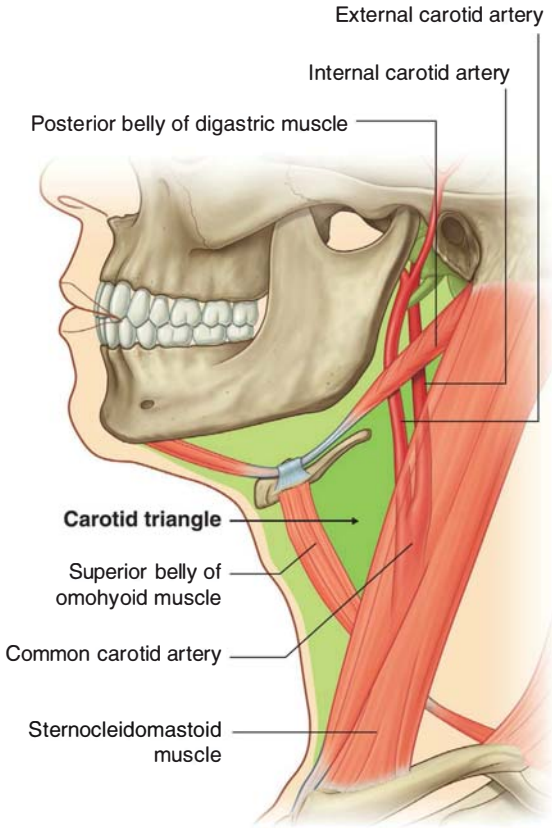


Fig. 8.166 Carotid triangle.

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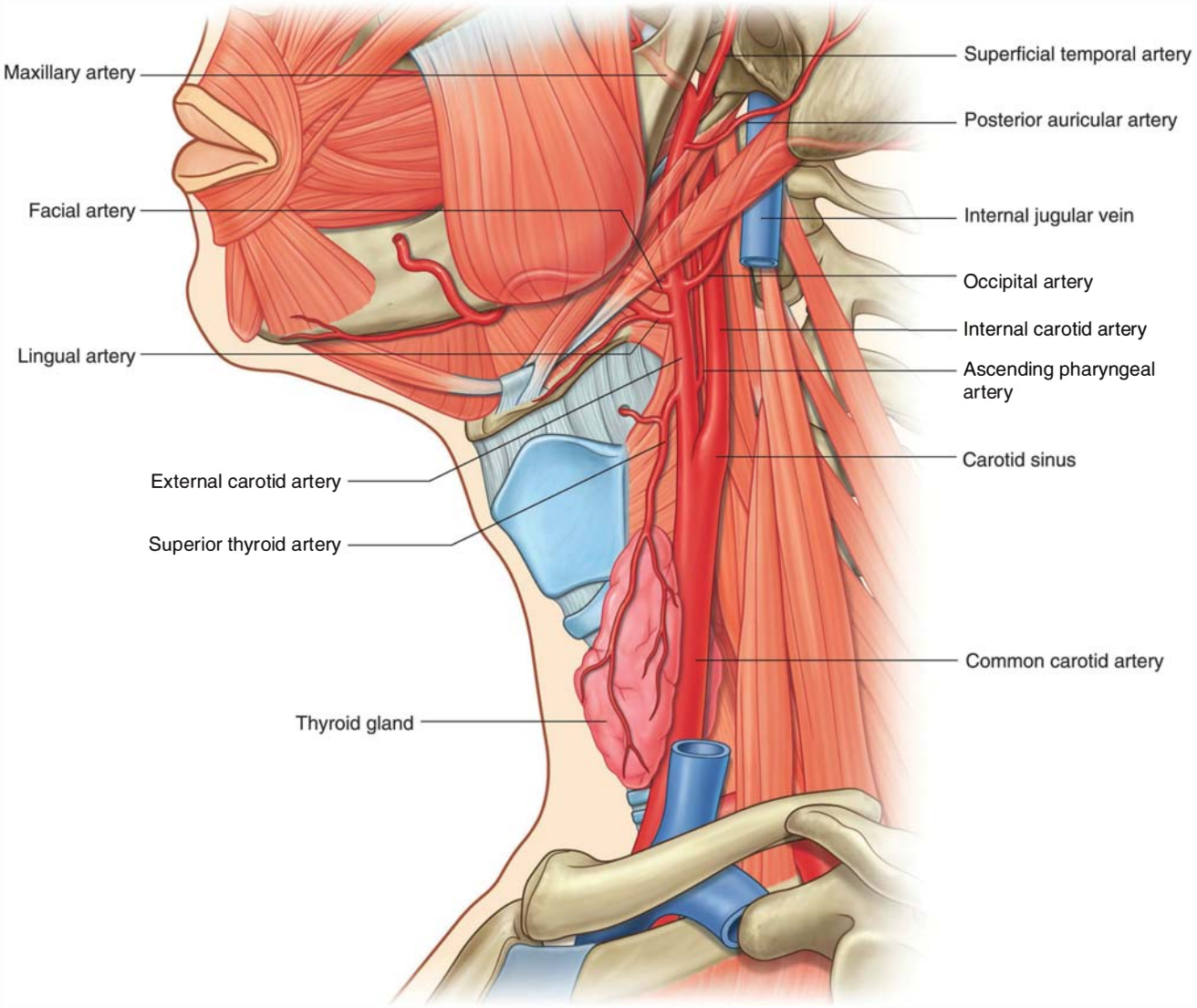


Fig. 8.167 Carotid system.

Table 8.13 Branches of the external carotid artery

Branch	Supplies
Superior thyroid artery	Thyrohyoid muscle, internal structures of the larynx, sternocleidomastoid and cricothyroid muscles, thyroid gland
Ascending pharyngeal artery	Pharyngeal constrictors and stylopharyngeus muscle, palate, tonsil, pharyngotympanic tube, meninges in posterior cranial fossa
Lingual artery	Muscles of the tongue, palatine tonsil, soft palate, epiglottis, floor of mouth, sublingual gland
Facial artery	All structures in the face from the inferior border of the mandible anterior to the masseter muscle to the medial corner of the eye, the soft palate, palatine tonsil, pharyngotympanic tube, submandibular gland
Occipital artery	Sternocleidomastoid muscle, meninges in posterior cranial fossa, mastoid cells, deep muscles of the back, posterior scalp
Posterior auricular artery	Parotid gland and nearby muscles, external ear and scalp posterior to ear, middle and inner ear structures
Superficial temporal artery	Parotid gland and duct, masseter muscle, lateral face, anterior part of external ear, temporalis muscle, parietal and temporal fossae
Maxillary artery	External acoustic meatus, lateral and medial surface of tympanic membrane, temporomandibular joint, dura mater on lateral wall of skull and inner table of cranial bones, trigeminal ganglion and dura in vicinity, mylohyoid muscle, mandibular teeth, skin on chin, temporalis muscle, outer table of bones of skull in temporal fossa, structures in infratemporal fossa, maxillary sinus, upper teeth and gingivae, infra-orbital skin, palate, roof of pharynx, nasal cavity

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- The **superior thyroid artery** is the first branch—it arises from the anterior surface near or at the bifurcation and passes in a downward and forward direction to reach the superior pole of the thyroid gland.
- The **ascending pharyngeal artery** is the second and smallest branch—it arises from the posterior aspect of the external carotid artery and ascends between the internal carotid artery and the pharynx.
- The **lingual artery** arises from the anterior surface of the external carotid artery just above the superior thyroid artery at the level of the hyoid bone, passes deep to the hypoglossal nerve [XII], and passes between the middle constrictor and hyoglossus muscles.
- The **facial artery** is the third anterior branch of the external carotid artery—it arises just above the lingual artery, passes deep to the stylohyoid and posterior belly of the digastric muscles, continues deep between the submandibular gland and mandible, and emerges over the edge of the mandible just anterior to the masseter muscle, to enter the face.
- The **occipital artery** arises from the posterior surface of the external carotid artery, near the level of origin of the facial artery, passes upward and posteriorly deep to the posterior belly of the digastric muscle, and emerges on the posterior aspect of the scalp.
- The **posterior auricular artery** is a small branch arising from the posterior surface of the external carotid artery and passes upward and posteriorly.
- The **superficial temporal artery** is one of the terminal branches and appears as an upward continuation of the external carotid artery—beginning posterior to the neck of the mandible, it passes anterior to the ear, crosses the zygomatic process of the temporal bone, and above this point divides into anterior and posterior branches.
- The **maxillary artery** is the larger of the two terminal branches of the external carotid artery—arising posterior to the neck of the mandible, it passes through the parotid gland, continues medial to the neck of the mandible and into the infratemporal fossa, and continues through this area into the pterygopalatine fossa.

Veins

Collecting blood from the skull, brain, superficial face, and parts of the neck, the **internal jugular vein** begins as a dilated continuation of the **sigmoid sinus**, which is a dural venous sinus. This initial dilated part is referred to as the **superior bulb of jugular vein** and receives another dural venous sinus (the **inferior petrosal sinus**) soon after it is formed. It exits the skull through the jugular foramen associated with the glossopharyngeal [IX], vagus [X], and accessory [XI] nerves, and enters the carotid sheath.

The internal jugular vein traverses the neck within the carotid sheath, initially posterior to the internal carotid artery, but passes to a more lateral position farther down. It remains lateral to the common carotid artery through the rest of the neck with the vagus nerve [X] posterior and partially between the two vessels.

The paired internal jugular veins join with the subclavian veins posterior to the sternal end of the clavicle to form the right and left **brachiocephalic veins** (Fig. 8.165).

Tributaries to each internal jugular vein include the inferior petrosal sinus, and the **facial**, **lingual**, **pharyngeal**, **occipital**, **superior thyroid**, and **middle thyroid veins**.

In the clinic

Jugular venous pulse

The jugular venous pulse is an important clinical sign that enables the physician to assess the venous pressure and waveform and is a reflection of the functioning of the right side of the heart.

Nerves

Numerous cranial and peripheral nerves:

- pass through the anterior triangle of the neck as they continue to their final destination,
- send branches to structures in or forming boundaries of the anterior triangle of the neck, and
- while in the anterior triangle of the neck, send branches to nearby structures.

The cranial nerves in these categories include the facial [VII], glossopharyngeal [IX], vagus [X], accessory [XI], and hypoglossal [XII].

Branches of spinal nerves in these categories include the transverse cervical nerve from the cervical plexus and the upper and lower roots of the ansa cervicalis.

Facial nerve [VII]

After emerging from the stylomastoid foramen, the facial nerve [VII] gives off branches that innervate two muscles associated with the anterior triangle of the neck:

- the posterior belly of the digastric, and
- the stylohyoid.

The facial nerve [VII] also innervates the platysma muscle that overlies the anterior triangle and part of the posterior triangle of the neck.

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Glossopharyngeal nerve [IX]

The glossopharyngeal nerve [IX] leaves the cranial cavity through the jugular foramen. It begins its descent between the internal carotid artery and the internal jugular vein, lying deep to the styloid process and the muscles associated with the styloid process. As the glossopharyngeal nerve [IX] completes its descent, it passes forward between the internal and external carotid arteries, and curves around the lateral border of the stylopharyngeus muscle (Fig. 8.168). At this point, it continues in an anterior direction, deep to the hyoglossus muscle, to reach the base of the tongue and the area of the palatine tonsil.

As the glossopharyngeal nerve [IX] passes through the area of the anterior triangle of the neck it innervates the stylopharyngeus muscle, sends a branch to the carotid sinus, and supplies sensory branches to the pharynx.

Vagus nerve [X]

The vagus nerve [X] exits the cranial cavity through the jugular foramen between the glossopharyngeal [IX] and accessory [XI] nerves.

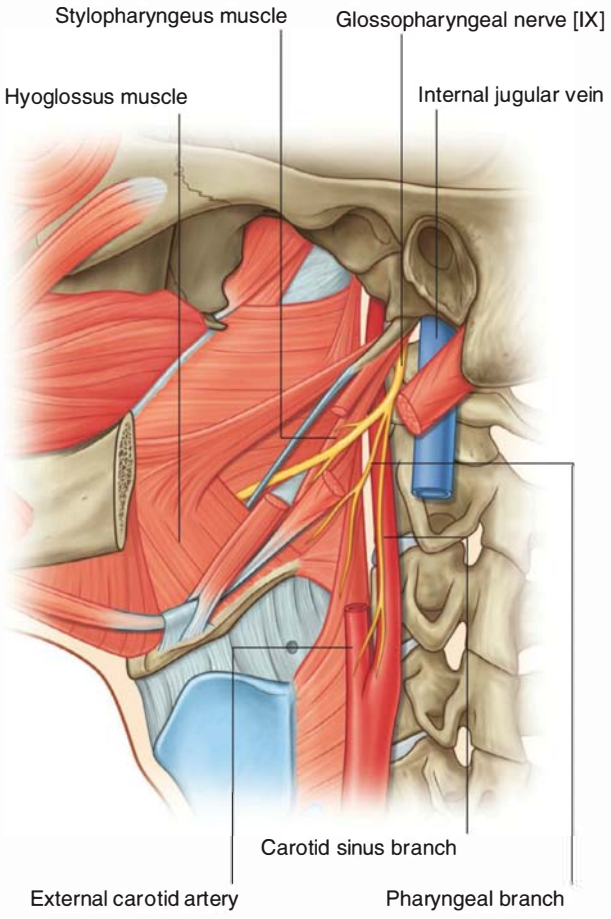


Fig. 8.168 Glossopharyngeal nerve [IX] in the anterior triangle of the neck.

Outside the skull the vagus nerve [X] enters the carotid sheath and descends through the neck enclosed in this structure medial to the internal jugular vein and posterior to the internal carotid and common carotid arteries (Fig. 8.169).

Branches of the vagus nerve [X] as it passes through the anterior triangle of the neck include a motor branch to the pharynx, a branch to the carotid body, the superior laryngeal nerve (which divides into external and internal laryngeal branches), and possibly a cardiac branch.

Accessory nerve [XI]

The accessory nerve [XI] is the most posterior of the three cranial nerves exiting the cranial cavity through the jugular foramen. It begins its descent medial to the internal jugular vein, emerging from between the internal jugular vein and internal carotid artery to cross the lateral surface

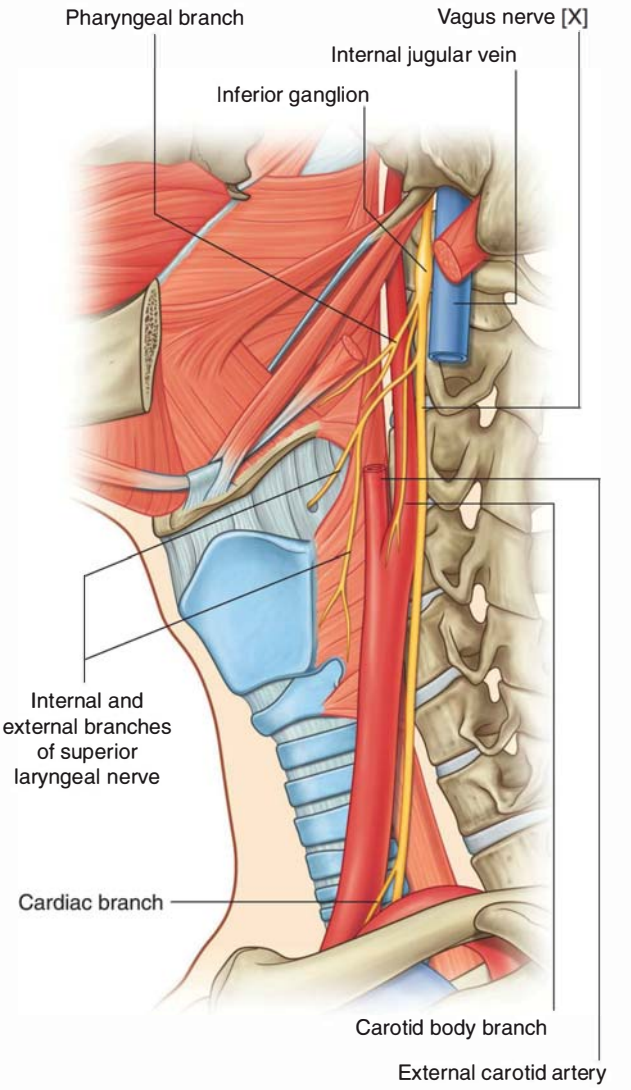


Fig. 8.169 Vagus nerve [X] in the anterior triangle of the neck.

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of the internal jugular vein as it passes downward and backward to disappear either into or beneath the anterior border of the sternocleidomastoid muscle (Fig. 8.170).

The accessory nerve gives off no branches as it passes through the anterior triangle of the neck.

Hypoglossal nerve [XII]

The hypoglossal nerve [XII] leaves the cranial cavity through the hypoglossal canal and is medial to the internal jugular vein and internal carotid artery immediately outside the skull. As it descends, it passes outward between the internal jugular vein and internal carotid artery (Fig. 8.171). At this point it passes forward, hooking around the occipital artery, across the lateral surfaces of the internal and external carotid arteries and the lingual artery, and then continues deep to the posterior belly of the digastric and stylohyoid muscles. It passes over the surface of the hyoglossus muscle and disappears deep to the mylohyoid muscle.

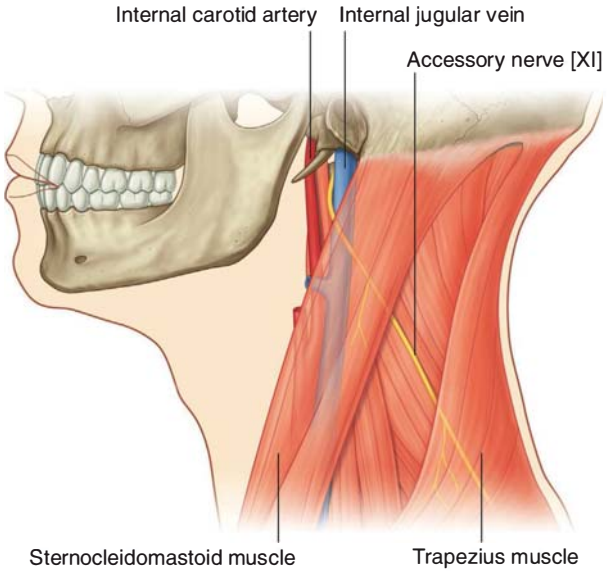


Fig. 8.170 Accessory nerve [XI] in the posterior triangle of the neck.

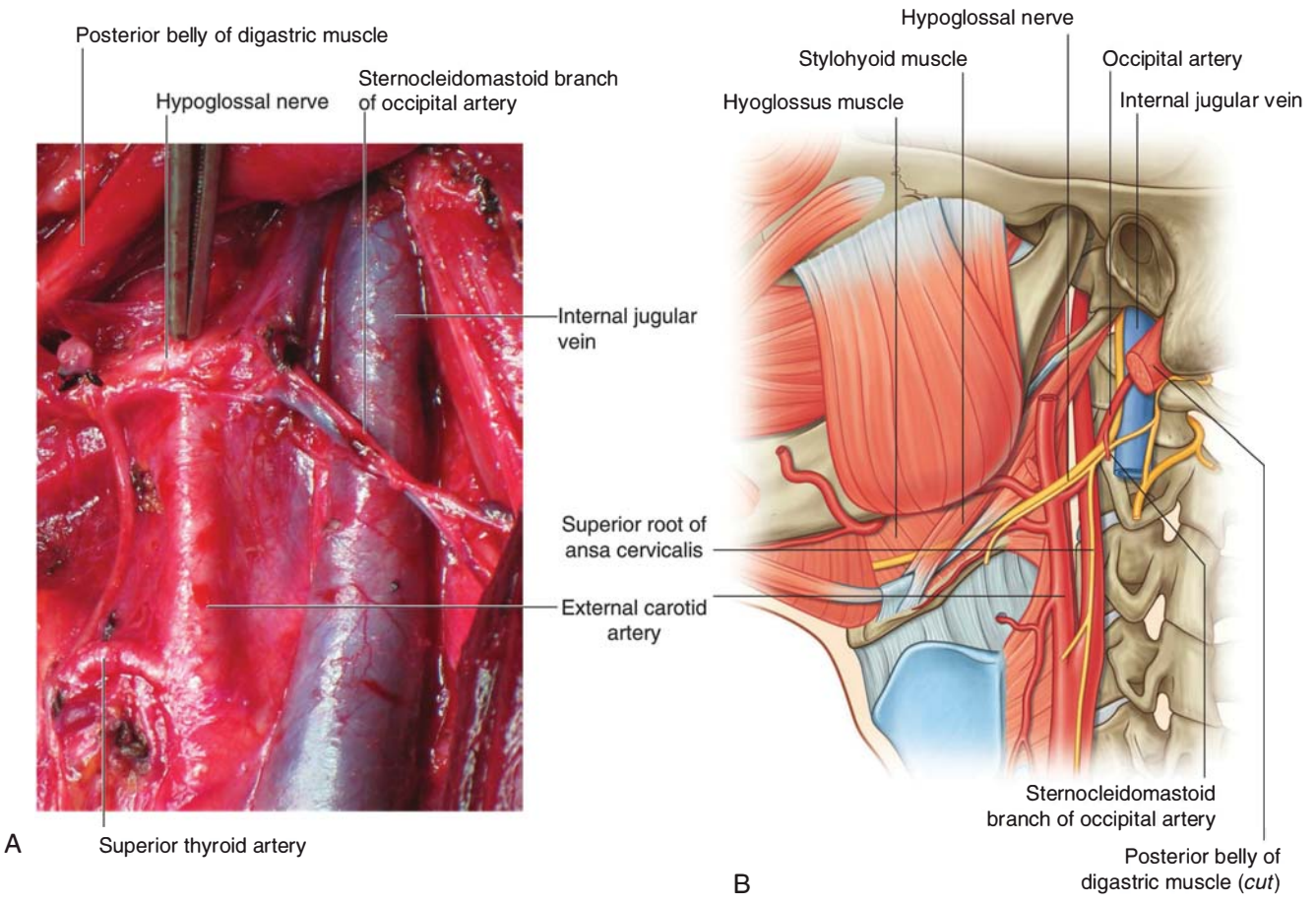


Fig. 8.171 Hypoglossal nerve [XII]. **A.** Surgical view of hypoglossal nerve in anterior triangle of the neck. **B.** Diagram.

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The hypoglossal nerve [XII], which supplies the tongue, does not give off any branches as it passes through the anterior triangle of the neck.

Transverse cervical nerve

The transverse cervical nerve is a branch of the cervical plexus arising from the anterior rami of cervical nerves C2 and C3. It emerges from beneath the posterior border of the sternocleidomastoid muscle, near the middle of the muscle, and loops around the sternocleidomastoid to cross its anterior surface in a transverse direction (Fig. 8.172). It continues across the neck and provides cutaneous innervation to this area.

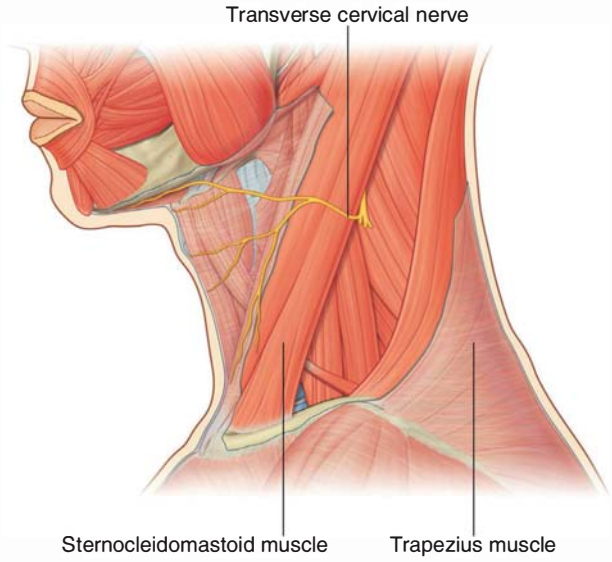


Fig. 8.172 Transverse cervical nerve in the anterior triangle of the neck.

Ansa cervicalis

The ansa cervicalis is a loop of nerve fibers from cervical nerves C1 to C3 that innervate the "strap muscles" in the anterior triangle of the neck (Fig. 8.173). It begins as

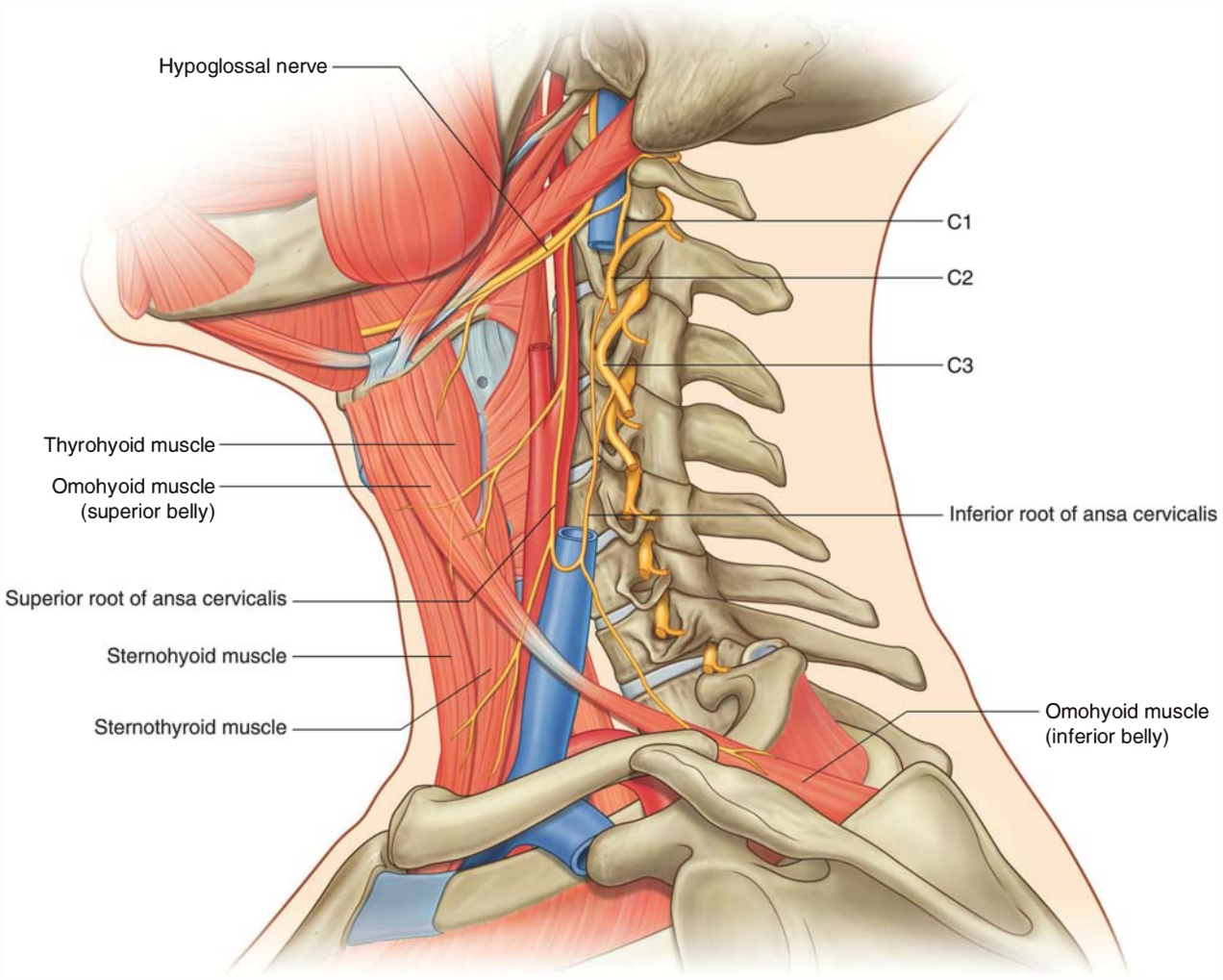


Fig. 8.173 Ansa cervicalis.

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branches from the cervical nerve C1 join the hypoglossal nerve [XII] soon after it leaves the skull.

As the hypoglossal nerve [XII] completes its descent and begins to pass forward across the internal and external carotid arteries, some of the cervical nerve fibers leave it and descend between the internal jugular vein and the internal, and then common, carotid arteries. These nerve fibers are the **superior root** of the ansa cervicalis and innervate the superior belly of the omohyoid muscle, and the upper parts of the sternohyoid and sternothyroid muscles.

Completing the loop is a direct branch from the cervical plexus containing nerve fibers from the second and third cervical nerves C2 and C3 (Fig. 8.173). This is the **inferior root** of the ansa cervicalis. It descends either medial or lateral to the internal jugular vein before turning medially to join the superior root. At this location, the ansa cervicalis gives off branches that innervate the inferior belly of the omohyoid, and the lower parts of the sternohyoid and sternothyroid muscles.

Elements of the gastrointestinal and respiratory systems

The esophagus, trachea, pharynx, and larynx lie in the neck and are related to the anterior triangles.

Esophagus

The esophagus is part of the gastrointestinal system and has only a short course in the lower neck. It begins at vertebral level CVI, where it is continuous with the pharynx above and courses inferiorly to pass through the thoracic inlet. It lies directly anterior to the vertebral column (Fig. 8.174B).

Trachea

The trachea is part of the lower airway and, like the esophagus, begins at vertebral level CVI, where it is continuous with the larynx above (Fig. 8.174B). The trachea lies directly anterior to the esophagus and passes inferiorly in the midline to enter the thorax.

Pharynx and larynx

The pharynx is a common pathway for air and food, and it connects respiratory and digestive compartments in the head with similar compartments in the lower neck (see pp. 1040–1052).

The larynx is the upper end of the lower airway. It is continuous with the trachea below and the pharynx posterosuperiorly (see pp. 1052–1069).

Thyroid and parathyroid glands

The thyroid and parathyroid glands are endocrine glands positioned anteriorly in the neck.

Both glands begin as pharyngeal outgrowths that migrate caudally to their final positions as development continues.

The thyroid gland is a large, unpaired gland, while the parathyroid glands, usually four in number, are small and are on the posterior surface of the thyroid gland.

Thyroid gland

The thyroid gland is anterior in the neck below and lateral to the thyroid cartilage (Fig. 8.174). It consists of two lateral **lobes** (which cover the anterolateral surfaces of the trachea, the cricoid cartilage, and the lower part of the thyroid cartilage) with an **isthmus** that connects the lateral lobes and crosses the anterior surfaces of the second and third tracheal cartilages.

Lying deep to the sternohyoid, sternothyroid, and omohyoid muscles, the thyroid gland is in the visceral compartment of the neck. This compartment also includes the pharynx, trachea, and esophagus and is surrounded by the pretracheal layers of fascia.

The thyroid gland arises as a median outgrowth from the floor of the pharynx near the base of the tongue. The foramen cecum of the tongue indicates the site of origin and the thyroglossal duct marks the path of migration of the thyroid gland to its final adult location. The thyroglossal duct usually disappears early in development, but remnants may persist as a cyst or as a connection to the foramen cecum (i.e., a fistula).

There may also be functional thyroid gland:

- associated with the tongue (a lingual thyroid),
- anywhere along the path of migration of the thyroid gland, or
- extending upward from the gland along the path of the thyroglossal duct (a pyramidal lobe).

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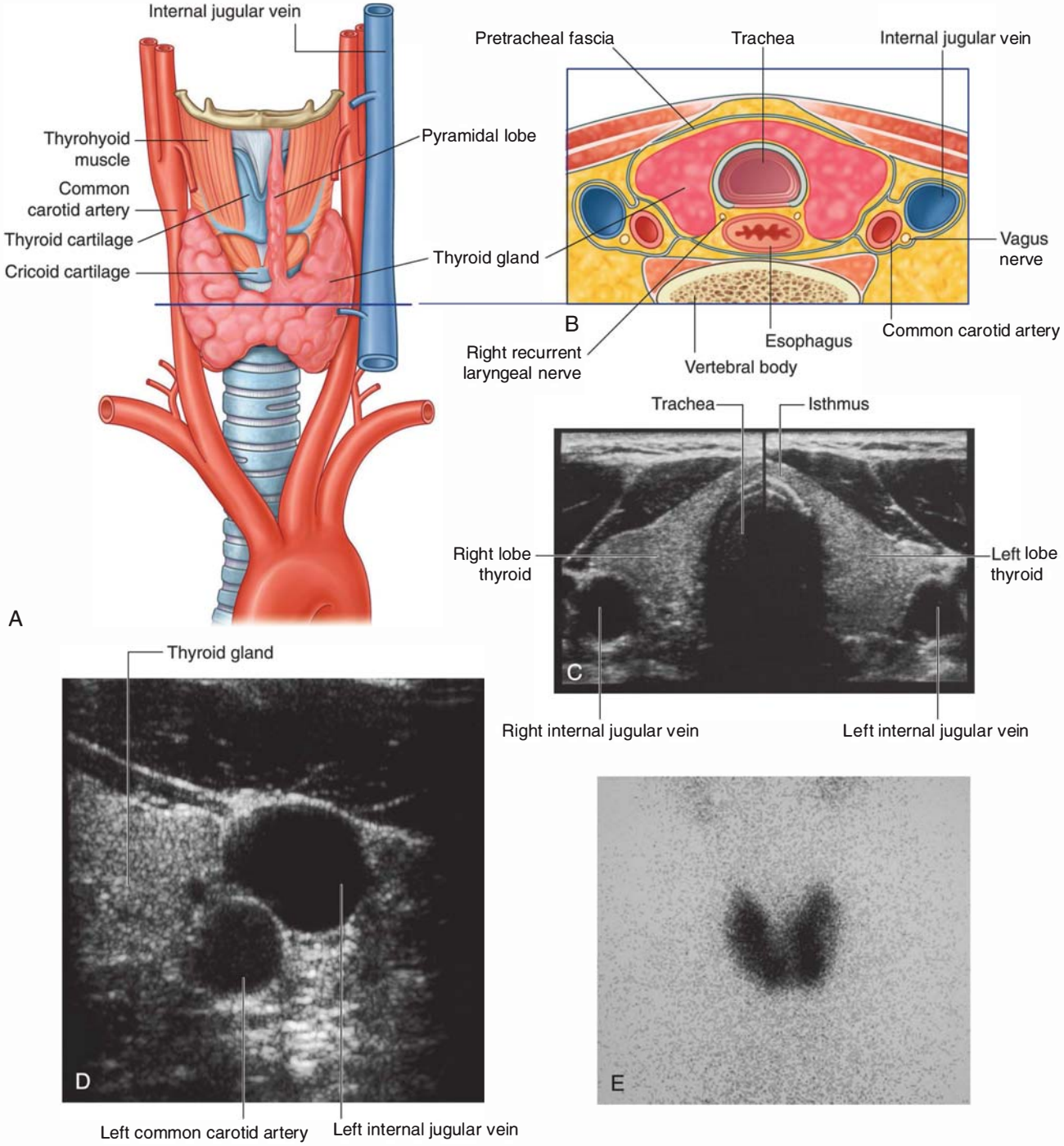


Fig. 8.174 Thyroid gland in the anterior triangle of neck. **A.** Anterior view. **B.** Transverse view. **C.** Ultrasound scan—compound axial view of the neck. **D.** Ultrasound scan—axial view of the neck. **E.** Nuclear medicine scan—normal thyroid uptake of pertechnetate in the neck.

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Arterial supply

Two major arteries supply the thyroid gland.

Superior thyroid artery. The superior thyroid artery is the first branch of the external carotid artery (Fig. 8.175). It descends, passing along the lateral margin of the thyrohyoid muscle, to reach the superior pole of the lateral lobe of the gland where it divides into anterior and posterior glandular branches:

- The **anterior glandular branch** passes along the superior border of the thyroid gland and anastomoses with its twin from the opposite side across the isthmus (Fig. 8.175).
- The **posterior glandular branch** passes to the posterior side of the gland and may anastomose with the inferior thyroid artery (Fig. 8.176).

Inferior thyroid artery. The **inferior thyroid artery** is a branch of the **thyrocervical trunk**, which arises from the first part of the subclavian artery (Figs. 8.175 and 8.176). It ascends along the medial edge of the anterior scalene muscle, passes posteriorly to the carotid sheath, and reaches the inferior pole of the lateral lobe of the thyroid gland.

At the thyroid gland the inferior thyroid artery divides into an:

- inferior branch, which supplies the lower part of the thyroid gland and anastomoses with the posterior branch of the superior thyroid artery, and
- an ascending branch, which supplies the parathyroid glands.

Occasionally, a small **thyroid ima artery** arises from the brachiocephalic trunk or the arch of the aorta and ascends on the anterior surface of the trachea to supply the thyroid gland.

Venous and lymphatic drainage

Three veins drain the thyroid gland (Fig. 8.175):

- The **superior thyroid vein** primarily drains the area supplied by the superior thyroid artery.
- The **middle** and **inferior thyroid veins** drain the rest of the thyroid gland.

The superior and middle thyroid veins drain into the internal jugular vein, and the inferior thyroid veins

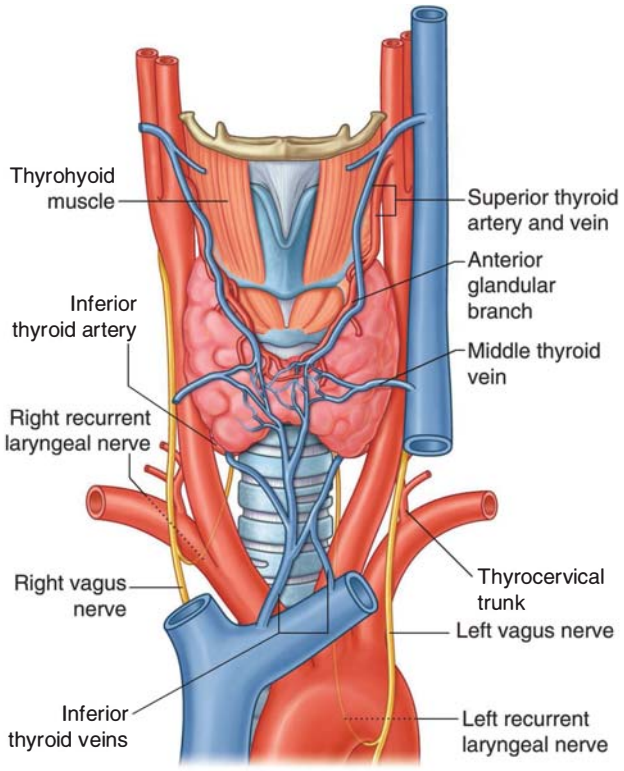


Fig. 8.175 Vasculature of the thyroid: anterior view.

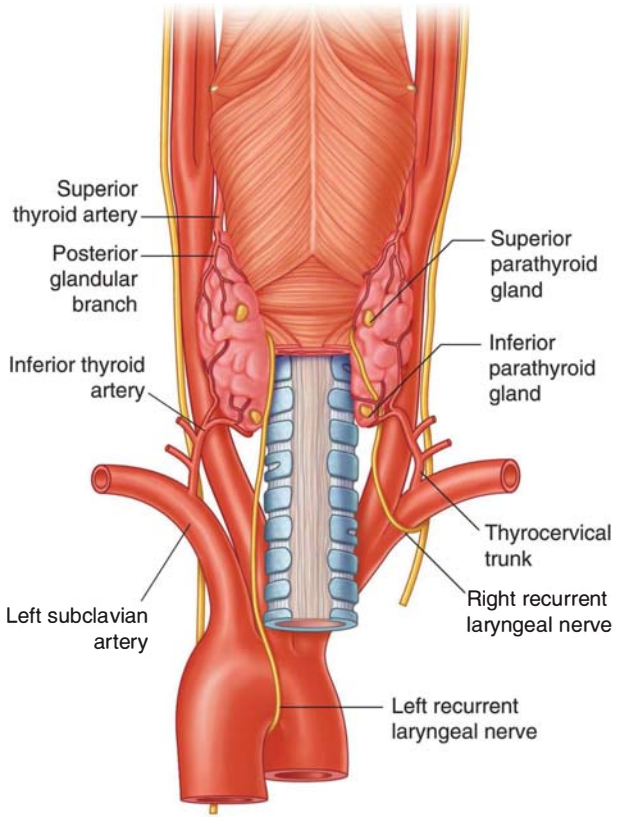


Fig. 8.176 Superior and inferior thyroid arteries and left and right recurrent laryngeal nerves: posterior view.

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empty into the right and left brachiocephalic veins, respectively.

Lymphatic drainage of the thyroid gland is to nodes beside the trachea (paratracheal nodes) and to deep cervical nodes inferior to the omohyoid muscle along the internal jugular vein.

Recurrent laryngeal nerves

The thyroid gland is closely related to the recurrent laryngeal nerves. After branching from the vagus nerve [X] and looping around the subclavian artery on the right and the arch of the aorta on the left, the **recurrent laryngeal nerves** ascend in a groove between the trachea and esophagus (Fig. 8.176). They pass deep to the posteromedial surface of the lateral lobes of the thyroid gland and enter the larynx by passing deep to the lower margin of the inferior constrictor of the pharynx.

Together with branches of the inferior thyroid arteries, the recurrent laryngeal nerves are clearly related to, and may pass through ligaments, one on each side, that bind the thyroid gland to the trachea and to the cricoid cartilage of the larynx. These relationships need to be considered when surgically removing or manipulating the thyroid gland.

Parathyroid glands

The parathyroid glands are two pairs of small, ovoid, yellowish structures on the deep surface of the lateral lobes of the thyroid gland. They are designated as the superior and inferior parathyroid glands (Fig. 8.176). However, their position is quite variable and they may be anywhere from the carotid bifurcation superiorly to the mediastinum inferiorly.

Derived from the third (the inferior parathyroid glands) and fourth (the superior parathyroid glands) pharyngeal pouches, these paired structures migrate to their final adult positions and are named accordingly.

The arteries supplying the parathyroid glands are the inferior thyroid arteries, and venous and lymphatic drainage follows that described for the thyroid gland.

In the clinic

Thyroid gland

The thyroid gland develops from a small region of tissue near the base of the tongue. This tissue descends as the thyroglossal duct from the foramen cecum in the posterior aspect of the tongue to pass adjacent to the anterior aspect of the middle of the hyoid bone. The thyroid tissue continues to migrate inferiorly and eventually comes to rest at the anterior aspect of the trachea in the root of the neck.

Consequently, the migration of thyroid tissue may be arrested anywhere along the embryological descent of the gland. Ectopic thyroid tissue is relatively rare. More frequently seen is the cystic change that arises from the thyroglossal duct. The usual symptom of a thyroglossal duct cyst is a midline mass. Ultrasound easily demonstrates its nature and position, and treatment is by surgical excision. The whole of the duct as well as a small part of the anterior aspect of the hyoid bone must be excised to prevent recurrence.

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In the clinic

Thyroidectomy

A thyroidectomy is a common surgical procedure. In most cases it involves excision of part or most of the thyroid gland. This surgical procedure is usually carried out for benign diseases, such as multinodular goiter and thyroid cancer.

Given the location of the thyroid gland, there is a possibility of damaging other structures when carrying out a thyroidectomy, namely the parathyroid glands and the recurrent laryngeal nerve (Fig. 8.177). Assessment of the vocal folds is necessary before and after thyroid surgery because the recurrent laryngeal nerves are closely related to ligaments that bind the gland to the larynx and can be easily traumatized during surgical procedures.

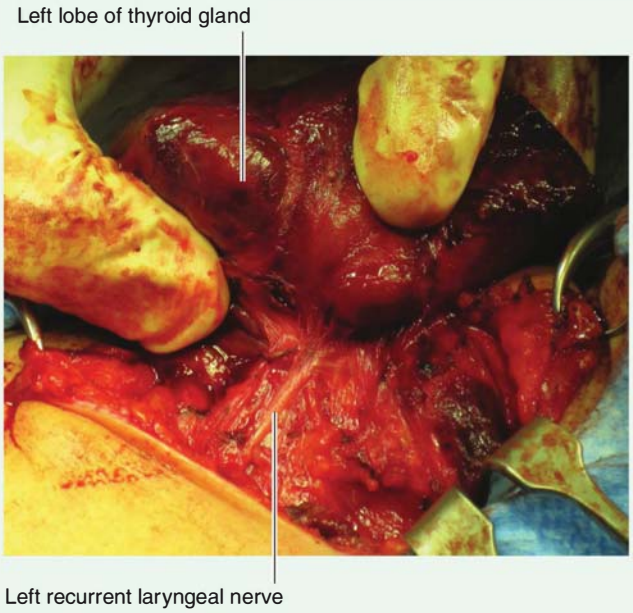


Fig. 8.177 Surgical view of left lobe of enlarged thyroid (goiter) retracted to show close association with recurrent laryngeal nerve.

In the clinic

Thyroid gland pathology

Thyroid gland pathology is extremely complex. In essence, thyroid gland pathology should be assessed from two points of view. First, the thyroid gland may be diffusely or focally enlarged, for which there are numerous causes. Second, the thyroid gland may undersecrete or oversecrete the hormone thyroxine.

One of the commonest disorders of the thyroid gland is a **multinodular goiter,** which is a diffuse irregular enlargement of the thyroid gland with areas of thyroid hypertrophy and colloid cyst formation. Most patients are euthyroid (i.e., have normal serum thyroxine levels). The typical symptom is a diffuse mass in the neck, which may be managed medically or may need surgical excision if the mass is large enough to affect the patient's life or cause respiratory problems.

Isolated nodules in the thyroid gland may be a dominant nodule in a multinodular gland or possibly an isolated tumor of the thyroid gland. Isolated tumors may or may not secrete thyroxine depending on their cellular morphology. Treatment is usually by excision.

Immunological diseases may affect the thyroid gland and may overstimulate it to produce excessive thyroxine. These diseases may be associated with other extrathyroid manifestations, which include exophthalmos, pretibial myxedema, and nail changes. Other causes of diffuse thyroid stimulation include viral thyroiditis. Some diseases may cause atrophy of the thyroid gland, leading to undersecretion of thyroxine (**myxedema**).

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In the clinic

Ectopic parathyroid glands

The parathyroid glands develop from the third and fourth pharyngeal pouches and translocate to their more adult locations during development. The position of the glands can be highly variable, sometimes being situated high in the neck or in the thorax. Tumors develop in any of these locations (Fig. 8.178).

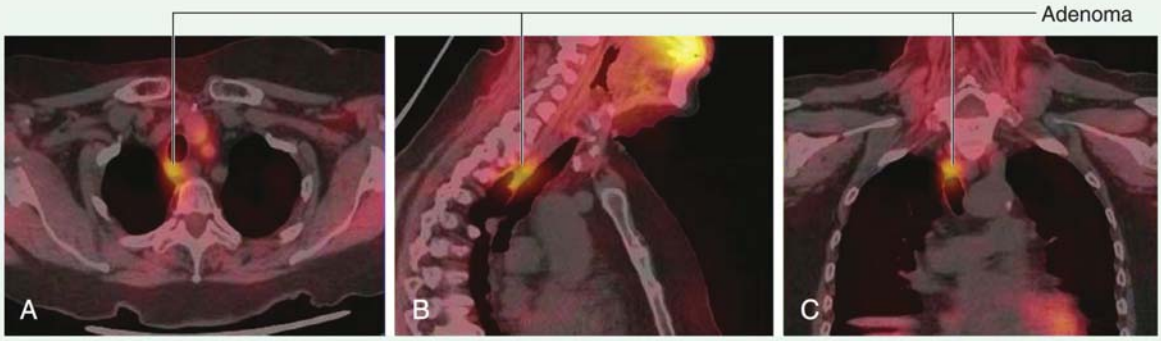


Fig. 8.178 Ectopic parathyroid adenoma in superior mediastinum. Noncontrast hybrid single photon emission computed tomography/computed tomography (SPECT/CT). **A.** Transverse view. **B.** Sagittal view. **C.** Coronal view.

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Location of structures in different regions of the anterior triangle of the neck

The regional location of major structures in the anterior triangle of the neck is summarized in Table 8.14. Structures can be identified as being within a specific subdivision, passing into a specific subdivision from outside the area, originating in one subdivision and passing to another subdivision, or passing through several subdivisions while traversing the region.

Table 8.14 Subdivisions of the anterior triangle of the neck—a regional approach

Subdivision	Boundaries	Contents
Submental triangle (unpaired)	Mandibular symphysis; anterior belly of digastric muscle; body of hyoid bone	Submental lymph nodes; tributaries forming the anterior jugular vein
Submandibular triangle (paired)	Lower border of mandible; anterior belly of digastric muscle; posterior belly of digastric muscle	Submandibular gland; submandibular lymph nodes; hypoglossal nerve [XII]; mylohyoid nerve; facial artery and vein
Carotid triangle (paired)	Posterior belly of digastric muscle; superior belly of omohyoid muscle; anterior border of sternocleidomastoid muscle	Tributaries to common facial vein; cervical branch of facial nerve [VII]; common carotid artery; external and internal carotid arteries; superior thyroid; ascending pharyngeal; lingual, facial, and occipital arteries; internal jugular vein; vagus [X], accessory [XI], and hypoglossal [XII] nerves; superior and inferior roots of ansa cervicalis; transverse cervical nerve
Muscular triangle (paired)	Midline of neck; superior belly of omohyoid muscle; anterior border of sternocleidomastoid muscle	Sternohyoid, omohyoid, sternohyoid, and thyrohyoid muscles; thyroid and parathyroid glands; pharynx

Posterior triangle of the neck

The posterior triangle of the neck is on the lateral aspect of the neck in direct continuity with the upper limb (Fig. 8.179). It is bordered:

- anteriorly by the posterior edge of the sternocleidomastoid muscle,
- posteriorly by the anterior edge of the trapezius muscle,
- basally by the middle one-third of the clavicle, and
- apically by the occipital bone just posterior to the mastoid process where the attachments of the trapezius and sternocleidomastoid come together.

The roof of the posterior triangle consists of an investing layer of cervical fascia that surrounds the sternocleidomastoid and trapezius muscles as it passes through the region.

The muscular floor of the posterior triangle is covered by the prevertebral layer of cervical fascia; and from superior to inferior consists of the splenius capitis, levator scapulae, and the posterior, middle, and anterior scalene muscles.

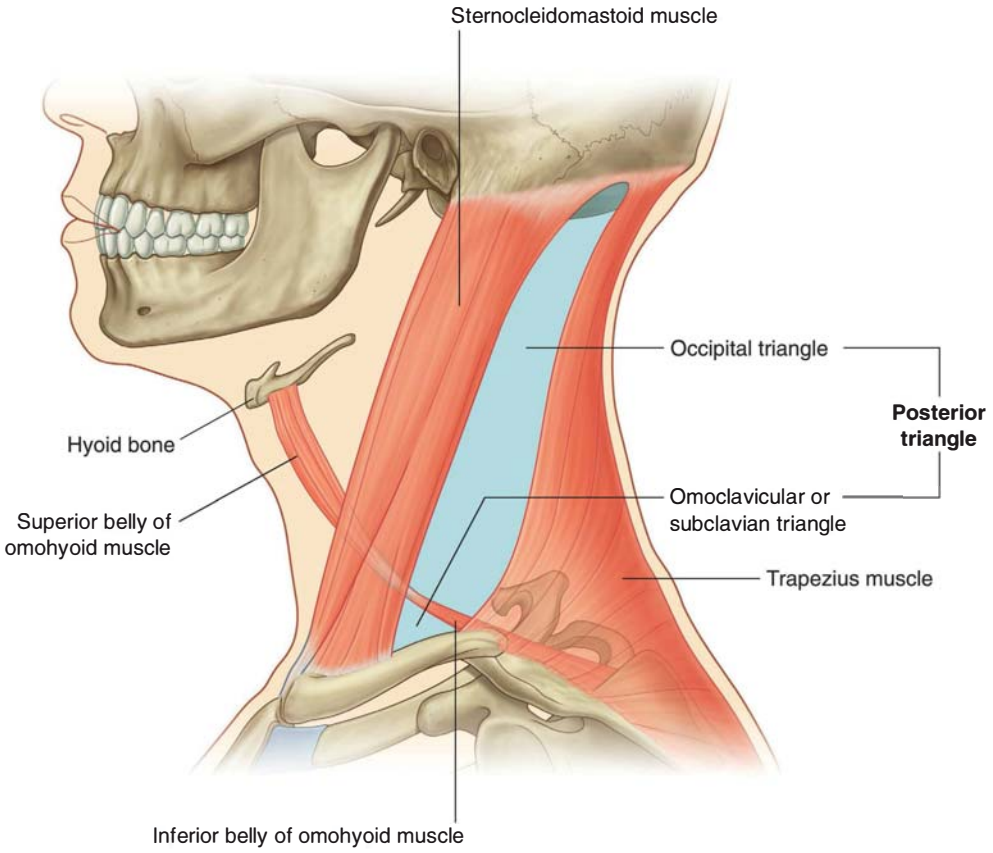


Fig. 8.179 Borders of the posterior triangle of the neck.

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Muscles

Numerous muscles participate in forming the borders and floor of the posterior triangle of the neck (Table 8.15).

In addition, the **omohyoid** muscle passes across the inferior part of the posterior triangle before disappearing under the sternocleidomastoid muscle and emerging in the anterior triangle (Fig. 8.180). It is enclosed in the investing layer of cervical fascia and crosses the posterior triangle from lateral to medial as it continues in a superior direction. It originates on the superior border of the scapula, just medial to the scapular notch and eventually inserts into the inferior border of the body of the hyoid bone. It has two bellies connected by a tendon, which is anchored by a fascial sling to the clavicle:

- The **superior belly** is in the anterior triangle.
- The **inferior belly** crosses the posterior triangle, subdividing it into a small, **omoclavicular or subclavian triangle** inferiorly and a much larger **occipital triangle** superiorly.

The omohyoid is innervated by branches of the ansa cervicalis (anterior rami from C1 to C3) and it depresses the hyoid bone.

Vessels

External jugular vein

One of the most superficial structures passing through the posterior triangle of the neck is the external jugular vein (Fig. 8.181). This large vein forms near the angle of the mandible, when the posterior branch of the retromandibular and posterior auricular veins join, and descends through the neck in the superficial fascia.

After crossing the sternocleidomastoid muscle, the external jugular vein enters the posterior triangle and continues its vertical descent.

Table 8.15 Muscles associated with the posterior triangle of the neck; parentheses indicate possible involvement

Muscle	Origin	Insertion	Innervation	Function
Sternocleidomastoid —Sternal head	Upper part of anterior surface of manubrium of sternum	Lateral one-half of superior nuchal line	Accessory nerve [XI] and branches from anterior rami of C2 to C3 (C4)	Individually—will tilt head toward shoulder on same side rotating head to turn face to opposite side; acting together, draw head forward
—Clavicular head	Superior surface of medial one-third of clavicle	Lateral surface of mastoid process		
Trapezius	Superior nuchal line; external occipital protuberance; ligamentum nuchae; spinous processes of vertebrae CVII to TXII	Lateral one-third of clavicle; acromion; spine of scapula	Motor—accessory nerve [XI]; proprioception—C3 and C4	Assists in rotating the scapula during abduction of humerus above horizontal; upper fibers—elevate, middle fibers—adduct, lower fibers—depress scapula
Splenius capitis	Lower half of ligamentum nuchae; spinous processes of vertebrae CVII to TIV	Mastoid process, skull below lateral one-third of superior nuchal line	Posterior rami of middle cervical nerves	Together, draw head backward; individually, draw and rotate head to one side (turn face to same side)
Levator scapulae	Transverse processes of CI to CIV	Upper part of medial border of scapula	C3, C4; and dorsal scapular nerve (C4, C5)	Elevates scapula
Posterior scalene	Posterior tubercles of transverse processes of vertebrae CIV to CVI	Upper surface of rib II	Anterior rami of C5 to C7	Elevation of rib II
Middle scalene	Transverse processes of vertebrae CII to CVII	Upper surface of rib I posterior to the groove for the subclavian artery	Anterior rami of C3 to C7	Elevation of rib I
Anterior scalene	Anterior tubercles of the transverse processes of vertebrae CIII to CVI	Scalene tubercle and upper surface of rib I	Anterior rami of C4 to C7	Elevation of rib I
Omohyoid	Superior border of scapula medial to scapular notch	Inferior border of body of hyoid bone	Ansa cervicalis; anterior rami of C1 to C3	Depress the hyoid bone

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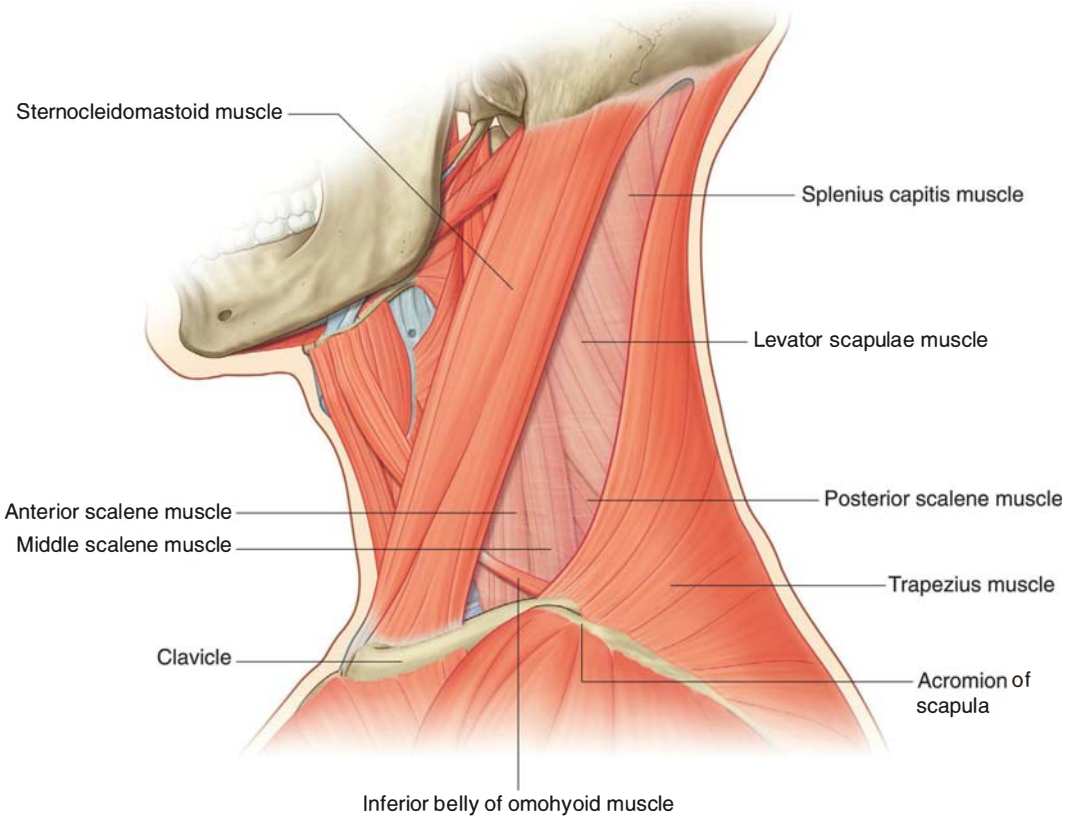


Fig. 8.180 Muscles of the posterior triangle of the neck.

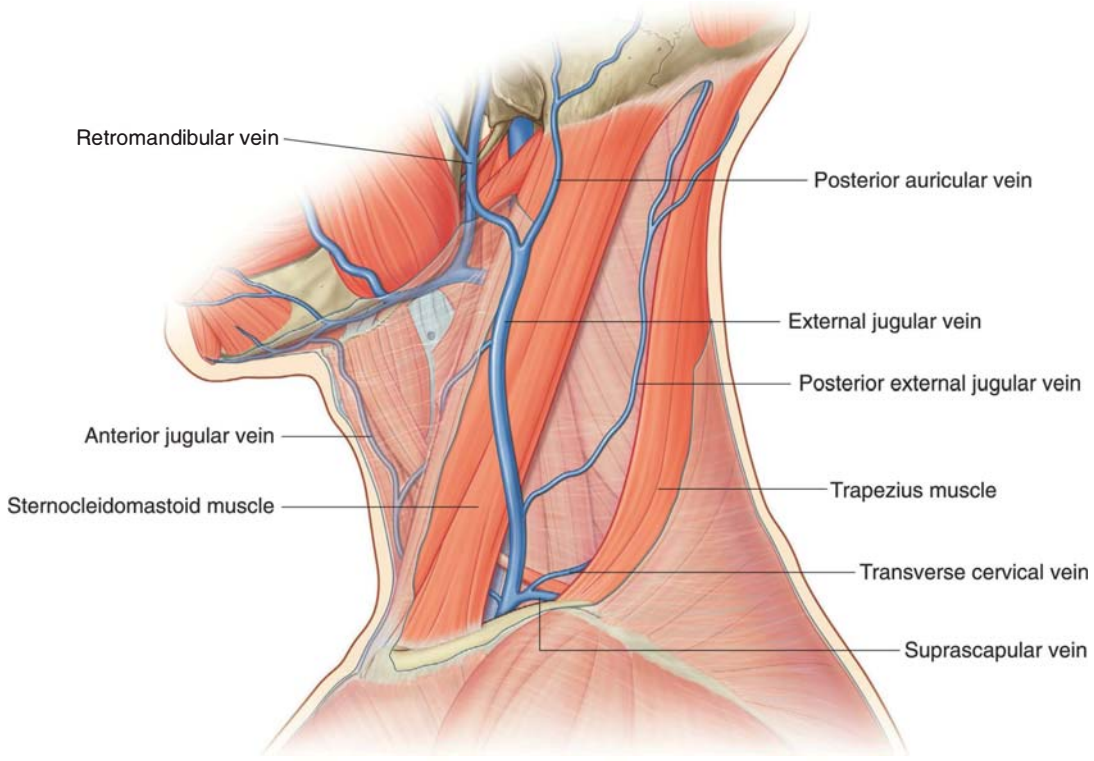


Fig. 8.181 External jugular vein in the posterior triangle of the neck.

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In the lower part of the posterior triangle, the external jugular vein pierces the investing layer of cervical fascia and ends in the subclavian vein.

Tributaries to the external jugular vein while it traverses the posterior triangle of the neck include the transverse cervical, suprascapular, and anterior jugular veins.

Subclavian artery and its branches

Several arteries are found within the boundaries of the posterior triangle of the neck. The largest is the third part of the subclavian artery as it crosses the base of the posterior triangle (Fig. 8.182).

The **first part of the subclavian artery** ascends to the medial border of the anterior scalene muscle from either the brachiocephalic trunk on the right side or directly from the arch of the aorta on the left side. It has numerous branches.

The **second part of the subclavian artery** passes laterally between the anterior and middle scalene muscles, and one branch may arise from it.

The **third part of the subclavian artery** emerges from between the anterior and middle scalene muscles to cross the base of the posterior triangle (Fig. 8.182). It extends from the lateral border of the anterior scalene muscle to the lateral border of rib I where it becomes the **axillary artery** and continues into the upper limb.

A single branch (the **dorsal scapular artery**) may arise from the third part of the subclavian artery. This branch passes posterolaterally to reach the superior angle of the scapula where it descends along the medial border of the scapula posterior to the rhomboid muscles.

Transverse cervical and suprascapular arteries

Two other small arteries also cross the base of the posterior triangle. These are the transverse cervical and the

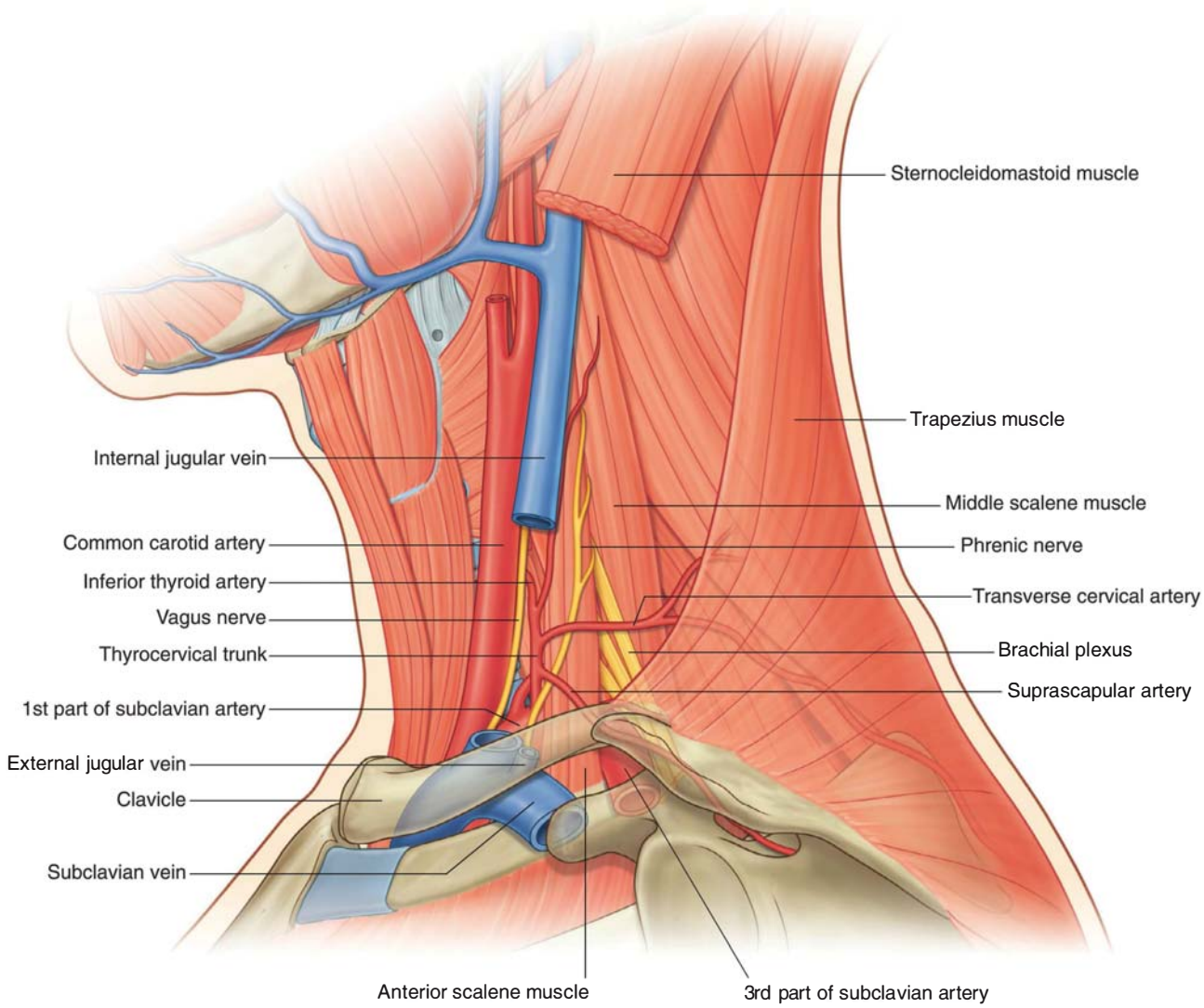


Fig. 8.182 Arteries in the posterior triangle of the neck.

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suprascapular arteries (Fig. 8.182). They are both branches of the thyrocervical trunk, which arises from the first part of the subclavian artery.

After branching from the thyrocervical trunk, the **transverse cervical artery** passes laterally and slightly posteriorly across the base of the posterior triangle anterior to the anterior scalene muscle and the brachial plexus. Reaching the deep surface of the trapezius muscle, it divides into superficial and deep branches:

- The **superficial branch** continues on the deep surface of the trapezius muscle.
- The **deep branch** continues on the deep surface of the rhomboid muscles near the medial border of the scapula.

The **suprascapular artery**, also a branch of the thyrocervical trunk, passes laterally, in a slightly downward direction across the lowest part of the posterior triangle, and ends up posterior to the clavicle (Fig. 8.182). Approaching the scapula, it passes over the superior transverse scapular ligament and distributes branches to muscles on the posterior surface of the scapula.

Veins

Veins accompany all the arteries described previously.

The **subclavian vein** is a continuation of the axillary vein and begins at the lateral border of rib I. As it crosses the base of the posterior triangle, the external jugular, and, possibly, the suprascapular and transverse cervical veins enter it (Fig. 8.181). It ends by joining with the internal jugular vein to form the brachiocephalic vein near the sternoclavicular joint. In the posterior triangle it is anterior to, and slightly lower than, the subclavian artery and passes anterior to the anterior scalene muscle.

Transverse cervical and suprascapular veins travel with each of the similarly named arteries. These veins become tributaries to either the external jugular vein or the initial part of the subclavian vein.

Nerves

A variety of nerves pass through or are within the posterior triangle. These include the accessory nerve [XI], branches of the cervical plexus, components forming the brachial plexus, and branches of the brachial plexus.

Accessory nerve

The accessory nerve [XI] exits the cranial cavity through the jugular foramen. It descends through the neck in a posterior direction, to reach the anterior border of the sternocleidomastoid muscle. Passing either deep to or through and innervating the sternocleidomastoid muscle, the accessory nerve [XI] continues to descend and enters the posterior triangle (Fig. 8.183). It crosses the posterior triangle, still in an obliquely downward direction, within

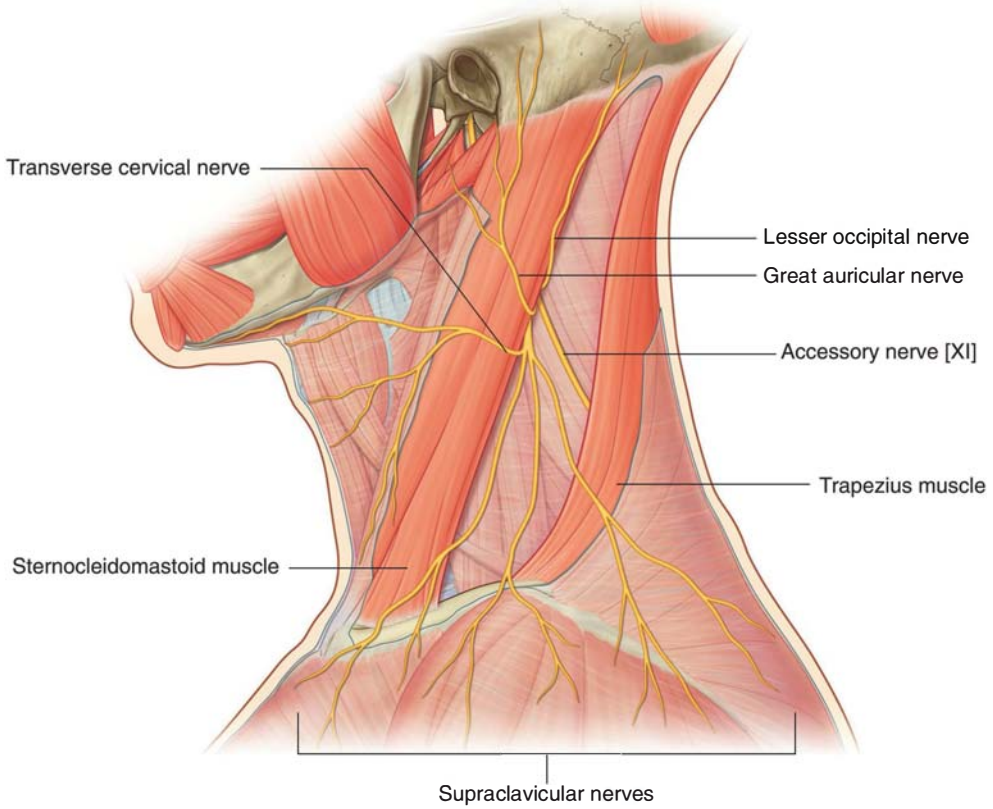


Fig. 8.183 Accessory nerve and cutaneous branches of the cervical plexus in the posterior triangle of the neck.

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the investing layer of cervical fascia as this fascia crosses between the sternocleidomastoid and trapezius muscles. When the accessory nerve [XI] reaches the anterior border of the trapezius muscle, it continues on the deep surface of the trapezius and innervates it. The superficial location of the accessory nerve as it crosses the posterior triangle makes it susceptible to injury.

Cervical plexus

The cervical plexus is formed by the anterior rami of cervical nerves C1 to C4 (Fig. 8.184).

The cervical plexus forms in the substance of the muscles making up the floor of the posterior triangle within the prevertebral layer of cervical fascia, and consists of:

- muscular (or deep) branches, and
- cutaneous (or superficial) branches.

The cutaneous branches are visible in the posterior triangle emerging from beneath the posterior border of the sternocleidomastoid muscle (Fig. 8.183).

Muscular branches

Muscular (deep) branches of the cervical plexus distribute to several groups of muscles. A major branch is the **phrenic nerve**, which supplies the diaphragm with both sensory and motor innervation (Fig. 8.184). It arises from the anterior rami of cervical nerves C3 to C5. Hooking around the upper lateral border of the anterior scalene muscle, the nerve continues inferiorly across the anterior surface of the anterior scalene within the prevertebral fascia to enter the thorax (Fig. 8.185). As the nerve descends in the neck, it is "pinned" to the anterior scalene muscle by the transverse cervical and suprascapular arteries.

Several muscular branches of the cervical plexus supply prevertebral and lateral vertebral muscles, including the rectus capitis anterior, rectus capitis lateralis, longus colli, and longus capitis (Fig. 8.185 and Table 8.16).

The cervical plexus also contributes to the formation of the superior and inferior roots of the ansa cervicalis (Fig. 8.184). This loop of nerves receives contributions from the anterior rami of the cervical nerves C1 to C3 and innervates the infrahyoid muscles.

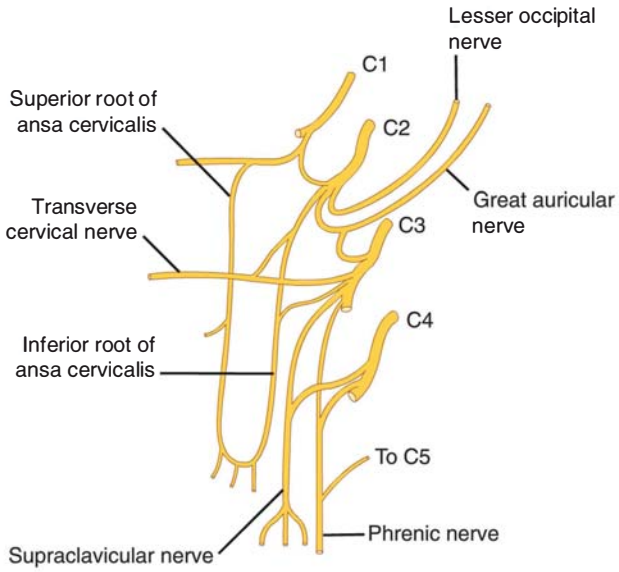


Fig. 8.184 Cervical plexus.

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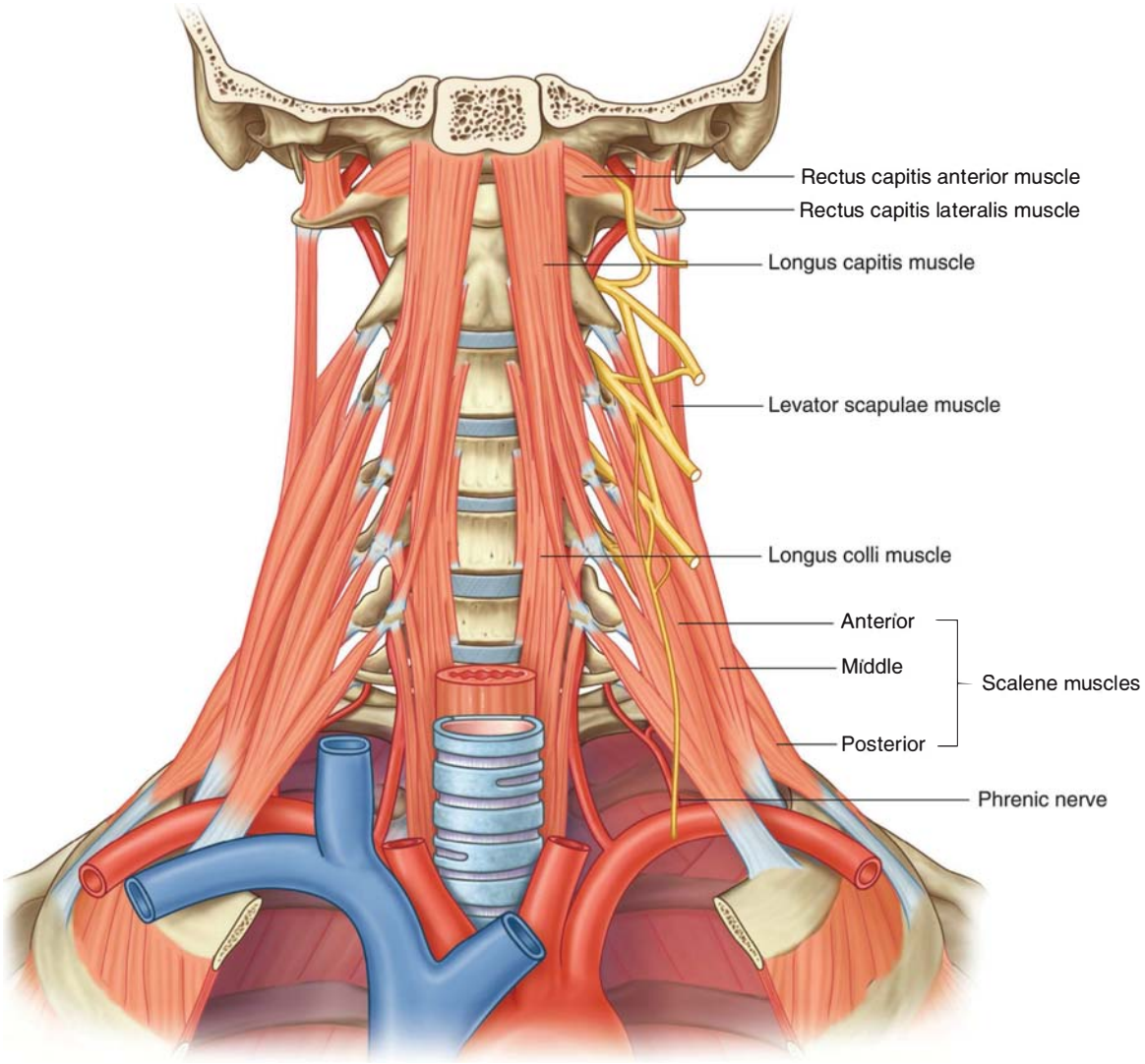


Fig. 8.185 Prevertebral and lateral vertebral muscles supplied by cervical plexus.

Table 8.16 Prevertebral and lateral vertebral muscles

Muscle	Origin	Insertion	Innervation	Function
Rectus capitis anterior	Anterior surface of lateral part of atlas and its transverse process	Inferior surface of basilar part of occipital bone	Branches from anterior rami of C1, C2	Flexes head at atlanto-occipital joint
Rectus capitis lateralis	Superior surface of transverse process of atlas	Inferior surface of jugular process of occipital bone	Branches from anterior rami of C1, C2	Flexes head laterally to same side
Longus colli —Superior oblique part	Anterior tubercles of transverse processes of vertebrae CIII to CV	Tubercle of anterior arch of atlas	Branches from anterior rami of C2 to C6	Flexes neck anteriorly and laterally and slight rotation to opposite side
—Inferior oblique part	Anterior surface of bodies of vertebrae TI, TII, and maybe TIII	Anterior tubercles of transverse processes of vertebrae CV and CVI		
—Vertical part	Anterior surface of bodies of TI to TIII and CV to CVII	Anterior surface of bodies of vertebrae CII to CIV		
Longus capitis	Tendinous slips to transverse processes of vertebrae CIII to CVI	Inferior surface of basilar part of occipital bone	Branches from anterior rami of C1 to C3	Flexes the head

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Cutaneous branches

Cutaneous (superficial) branches of the cervical plexus are visible in the posterior triangle as they pass outward from the posterior border of the sternocleidomastoid muscle (Figs. 8.183 and 8.184):

- The **lesser occipital nerve** consists of contributions from cervical nerve C2 (Fig. 8.184), ascends along the posterior border of the sternocleidomastoid muscle, and distributes to the skin of the neck and scalp posterior to the ear.
- The **great auricular nerve** consists of branches from cervical nerves C2 and C3, emerges from the posterior border of the sternocleidomastoid muscle, and ascends across the muscle to the base of the ear, supplying the skin of the parotid region, the ear, and the mastoid area.
- The **transverse cervical nerve** consists of branches from the cervical nerves C2 and C3, passes around the midpart of the sternocleidomastoid muscle, and continues horizontally across the muscle to supply the lateral and anterior parts of the neck.
- The **supraclavicular nerves** are a group of cutaneous nerves from cervical nerves C3 and C4 that, after emerging from beneath the posterior border of the sternocleidomastoid muscle, descend and supply the skin over the clavicle and shoulder as far inferiorly as rib II.

Brachial plexus

The brachial plexus forms from the anterior rami of cervical nerves C5 to C8 and thoracic nerve T1. The contributions of each of these nerves, which are between the anterior and middle scalene muscles, are the **roots** of the brachial plexus. As the roots emerge from between these muscles, they form the next component of the brachial plexus (the **trunks**) as follows:

- the anterior rami of C5 and C6 form the upper trunk,
- the anterior ramus of C7 forms the middle trunk,
- the anterior rami of C8 and T1 form the lower trunk.

The trunks cross the base of the posterior triangle (see Fig. 8.182). Several branches of the brachial plexus may be visible in the posterior triangle (see Fig. 7.54 on pg. 741). These include the:

- **dorsal scapular nerve** to the rhomboid muscles,
- **long thoracic nerve** to the serratus anterior muscle,
- nerve to the subclavius muscle, and
- **suprascapular nerve** to the supraspinatus and infraspinatus muscles.

Root of the neck

The root of the neck (Fig. 8.186) is the area immediately superior to the superior thoracic aperture and axillary inlets. It is bounded by:

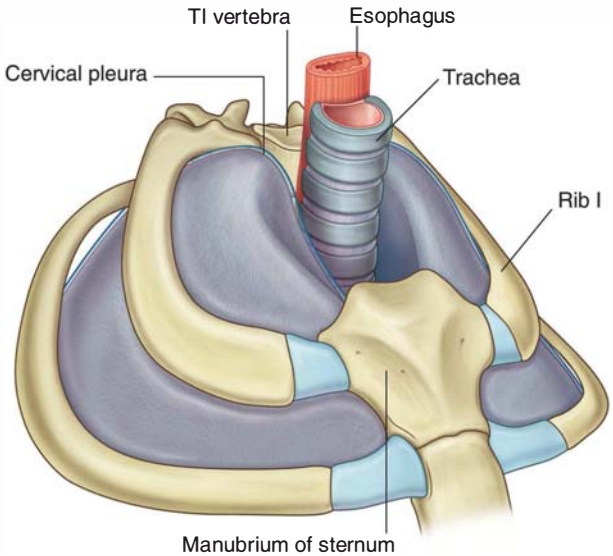


Fig. 8.186 Root of the neck.

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- the top of the manubrium of the sternum and superior margin of the clavicle anteriorly, and
- the top of the thoracic vertebra TI and the superior margin of the scapula to the coracoid process posteriorly.

It contains structures passing between the neck, thorax, and upper limb. There is also an extension of the thoracic cavity projecting into the root of the neck (Fig. 8.186). This consists of an upward projection of the pleural cavity, on both sides, and includes the cervical part of the parietal pleura (cupula), and the apical part of the superior lobe of each lung.

Anteriorly, the pleural cavity extends above the top of the manubrium of the sternum and superior border of rib I, while posteriorly, due to the downward slope of the superior thoracic aperture, the pleural cavity remains below the top of vertebra TI.

Vessels

Subclavian arteries

The subclavian arteries on both sides arch upward out of the thorax to enter the root of the neck (Fig. 8.187).

The **right subclavian artery** begins posterior to the sternoclavicular joint as one of two terminal branches of the brachiocephalic trunk. It arches superiorly and laterally to pass anterior to the extension of the pleural cavity in the root of the neck and posterior to the anterior scalene muscle. Continuing laterally across rib I, it becomes the **axillary artery** as it crosses its lateral border.

The **left subclavian artery** begins lower in the thorax than the right subclavian artery as a direct branch of the arch of the aorta. Lying posterior to the left common carotid artery and lateral to the trachea, it ascends and arches laterally, passing anterior to the extension of the pleural cavity and posterior to the anterior scalene muscle.

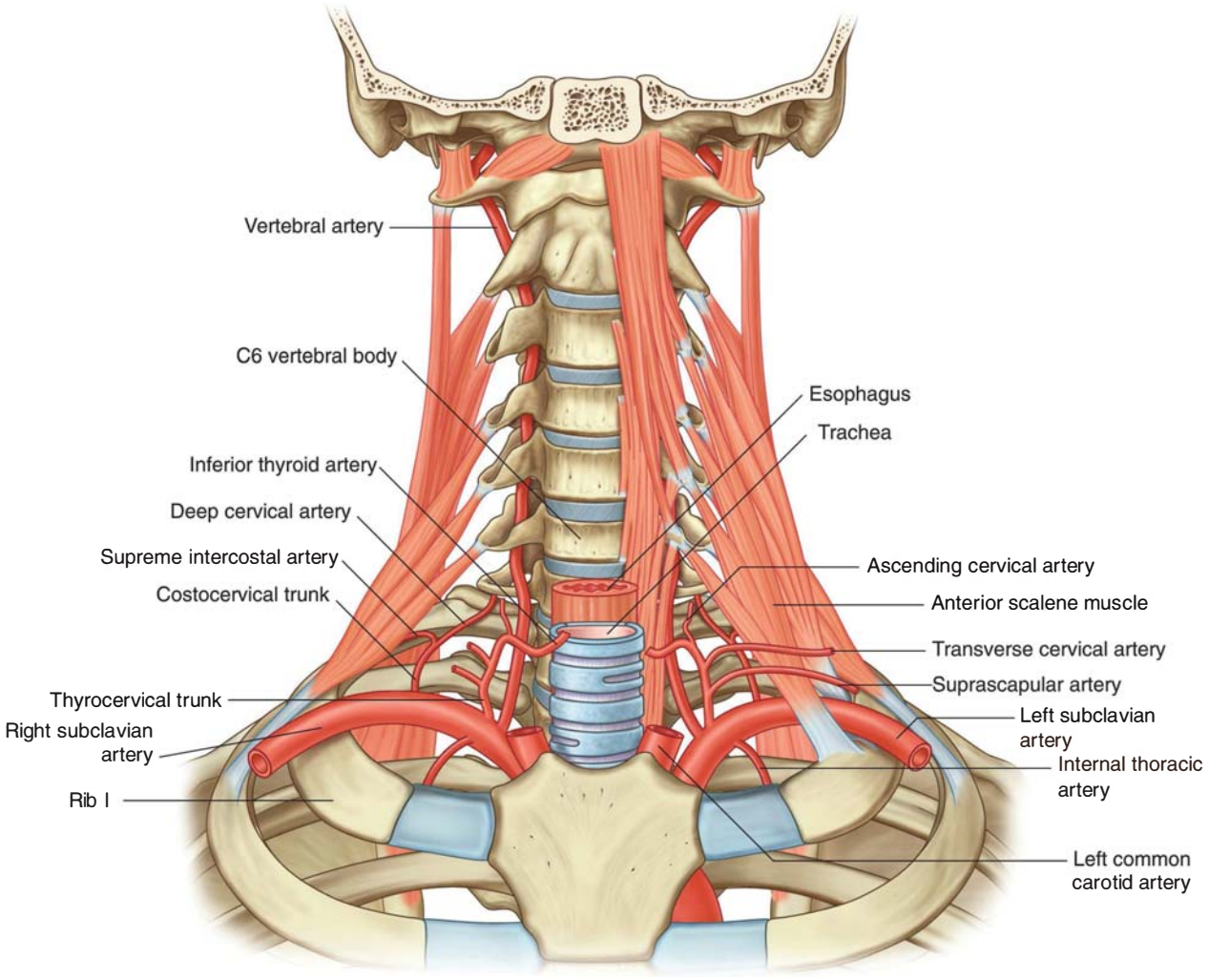


Fig. 8.187 Vasculature of the root of the neck.

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It continues laterally over rib I, and becomes the axillary artery as it crosses the lateral border of rib I.

Both subclavian arteries are divided into three parts by the anterior scalene muscle (Fig. 8.187):

- The first part extends from the origin of the artery to the anterior scalene muscle.
- The second part is the part of the artery posterior to the anterior scalene muscle.
- The third part is the part lateral to the anterior scalene muscle before the artery reaches the lateral border of rib I.

All branches from the right and left subclavian arteries arise from the first part of the artery, except in the case of one branch (the costocervical trunk) on the right side (Fig. 8.187). The branches include the vertebral artery, the thyrocervical trunk, the internal thoracic artery, and the costocervical trunk.

Vertebral artery

The **vertebral artery** is the first branch of the subclavian artery as it enters the root of the neck (Fig. 8.187). A large branch, arising from the first part of the subclavian artery medial to the anterior scalene muscle, it ascends and enters the foramen in the transverse process of vertebra CVI. Continuing to pass superiorly, the vertebral artery passes through the foramina of vertebrae CV to CI. At the superior border of vertebra CI, the artery turns medially and crosses the posterior arch of vertebra CI. From here it passes through the foramen magnum to enter the posterior cranial fossa.

Thyrocervical trunk

The second branch of the subclavian artery is the **thyrocervical trunk** (Fig. 8.187). It arises from the first part of the subclavian artery medial to the anterior scalene muscle, and divides into three branches—the inferior thyroid, the transverse cervical, and the suprascapular arteries.

Inferior thyroid artery. The inferior thyroid artery (Fig. 8.187) is the superior continuation of the thyrocervical trunk. It ascends, anterior to the anterior scalene muscle, and eventually turns medially, crossing posterior to the carotid sheath and its contents and anterior to the vertebral artery. Reaching the posterior surface of the thyroid gland it supplies the thyroid gland.

When the inferior thyroid artery turns medially, it gives off an important branch (the **ascending cervical artery**), which continues to ascend on the anterior surface of the prevertebral muscles, supplying these muscles and sending branches to the spinal cord.

Transverse cervical artery. The middle branch of the thyrocervical trunk is the **transverse cervical artery** (Fig. 8.187). This branch passes laterally, across the anterior surface of the anterior scalene muscle and the phrenic nerve, and enters and crosses the base of the posterior triangle of the neck. It continues to the deep surface of the trapezius muscle, where it divides into superficial and deep branches:

- The **superficial branch** continues on the deep surface of the trapezius muscle.
- The **deep branch** continues on the deep surface of the rhomboid muscles near the medial border of the scapula.

Suprascapular artery. The lowest branch of the thyrocervical trunk is the **suprascapular artery** (Fig. 8.187). This branch passes laterally, crossing anterior to the anterior scalene muscle, the phrenic nerve, the third part of the subclavian artery, and the trunks of the brachial plexus. At the superior border of the scapula, it crosses over the superior transverse scapular ligament and enters the supraspinatus fossa.

Internal thoracic artery

The third branch of the subclavian artery is the **internal thoracic artery** (Fig. 8.187). This artery branches from the inferior edge of the subclavian artery and descends.

It passes posterior to the clavicle and the large veins in the region and anterior to the pleural cavity. It enters the thoracic cavity posterior to the ribs and anterior to the transversus thoracis muscle and continues to descend giving off numerous branches.

Costocervical trunk

The final branch of the subclavian artery in the root of the neck is the **costocervical trunk** (Fig. 8.187). It arises in a slightly different position, depending on the side:

- On the left, it arises from the first part of the subclavian artery, just medial to the anterior scalene muscle.
- On the right, it arises from the second part of the subclavian artery.

On both sides, the costocervical trunk ascends and passes posteriorly over the dome of the pleural cavity and continues in a posterior direction behind the anterior scalene muscle. Eventually it divides into two branches—the deep cervical and the supreme intercostal arteries:

- The **deep cervical artery** ascends in the back of the neck and anastomoses with the descending branch of the occipital artery.

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- The **supreme intercostal artery** descends anterior to rib I and divides to form the posterior intercostal arteries for the first two intercostal spaces.

Veins

Numerous veins pass through the root of the neck. Small veins accompany each of the arteries described above, and large veins form major drainage channels.

The **subclavian veins** begin at the lateral margin of rib I as continuations of the **axillary veins**. Passing medially on each side, just anterior to the anterior scalene muscles, each subclavian vein is joined by the internal jugular vein to form the brachiocephalic veins.

The only tributary to each subclavian vein is an external jugular vein.

The veins accompanying the numerous arteries in this region empty into other veins.

Nerves

Several nerves and components of the nervous system pass through the root of the neck.

Phrenic nerves

The phrenic nerves are branches of the cervical plexus and arise on each side as contributions from the anterior rami of cervical nerves C3 to C5 come together. Passing around the upper lateral border of each anterior scalene muscle, the phrenic nerves continue inferiorly across the anterior surface of each anterior scalene muscle within the prevertebral layer of cervical fascia (Fig. 8.188). Leaving the lower edge of the anterior scalene muscle each phrenic nerve passes between the subclavian vein and artery to enter the thorax and continue to the diaphragm.

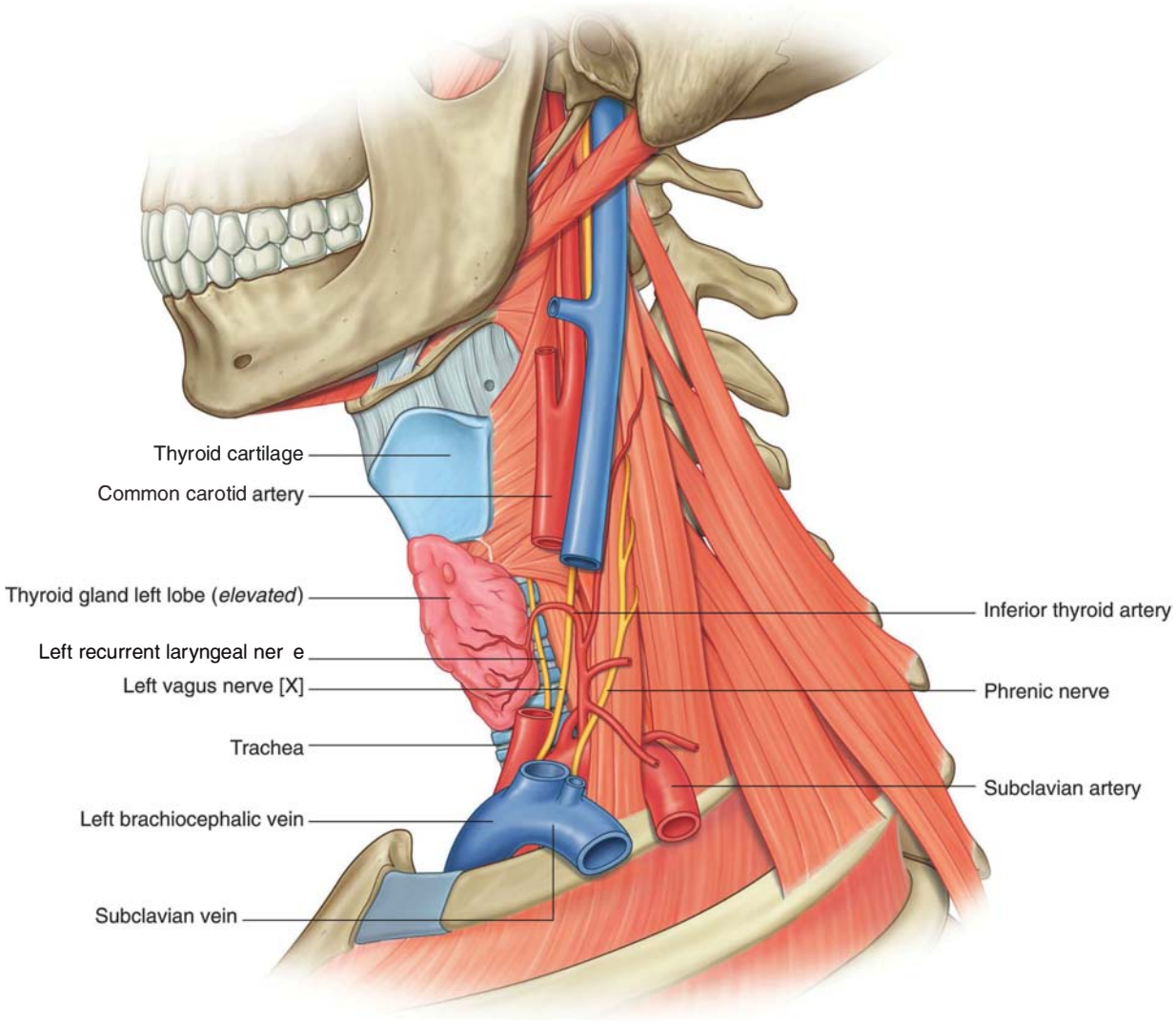


Fig. 8.188 Nerves in the root of the neck.

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Vagus nerves [X]

The vagus nerves [X] descend through the neck within the carotid sheath, posterior to and just between the common carotid artery and the internal jugular vein.

In the lower part of the neck, the vagus nerves [X] give off cardiac branches, which continue downward and medially, passing posterior to the subclavian arteries to disappear into the thorax.

In the root of the neck, each vagus nerve [X] passes anterior to the subclavian artery and posterior to the subclavian vein as it enters the thorax (Fig. 8.188).

Recurrent laryngeal nerves

The right and left recurrent laryngeal nerves are visible as they originate in (the right recurrent laryngeal nerve), or pass through (the left recurrent laryngeal nerve), the root of the neck.

The **right recurrent laryngeal nerve** is a branch of the right vagus nerve [X] as it reaches the lower edge of the first part of the subclavian artery in the root of the neck. It passes around the subclavian artery and upward and medially in a groove between the trachea and the esophagus as it heads to the larynx.

The **left recurrent laryngeal nerve** is a branch of the left vagus nerve [X] as it crosses the arch of the aorta in the superior mediastinum. It passes below and behind the arch of the aorta and ascends beside the trachea to the larynx (Fig. 8.188).

Sympathetic nervous system

Various components of the sympathetic nervous system are visible as they pass through the root of the neck (Fig. 8.189). These include:

- the cervical part of the sympathetic trunk,
- the ganglia associated with the cervical part of the sympathetic trunk, and
- cardiac nerves branching from the cervical part of the sympathetic trunk.

The sympathetic trunks are two parallel cords that run from the base of the skull to the coccyx. Along the way they are punctuated by ganglia, which are collections of neuronal cell bodies outside the CNS.

In the clinic

Recurrent laryngeal nerve palsy

Damage to either the right or left recurrent laryngeal nerve may lead initially to a hoarse voice and finally to an inability to speak. Recurrent laryngeal nerve palsy can occur from disruption of the nerves anywhere along their course. Furthermore, interruption of the vagus nerves before the division of the recurrent laryngeal nerves can also produce vocal symptoms.

Lung cancer in the apex of the right lung can affect the right recurrent laryngeal nerve, whereas cancers that infiltrate into the area between the pulmonary artery and aorta, an area known clinically as the "aortopulmonary window," can affect the left recurrent laryngeal nerve. Thyroid surgery also can traumatize the recurrent laryngeal nerves.

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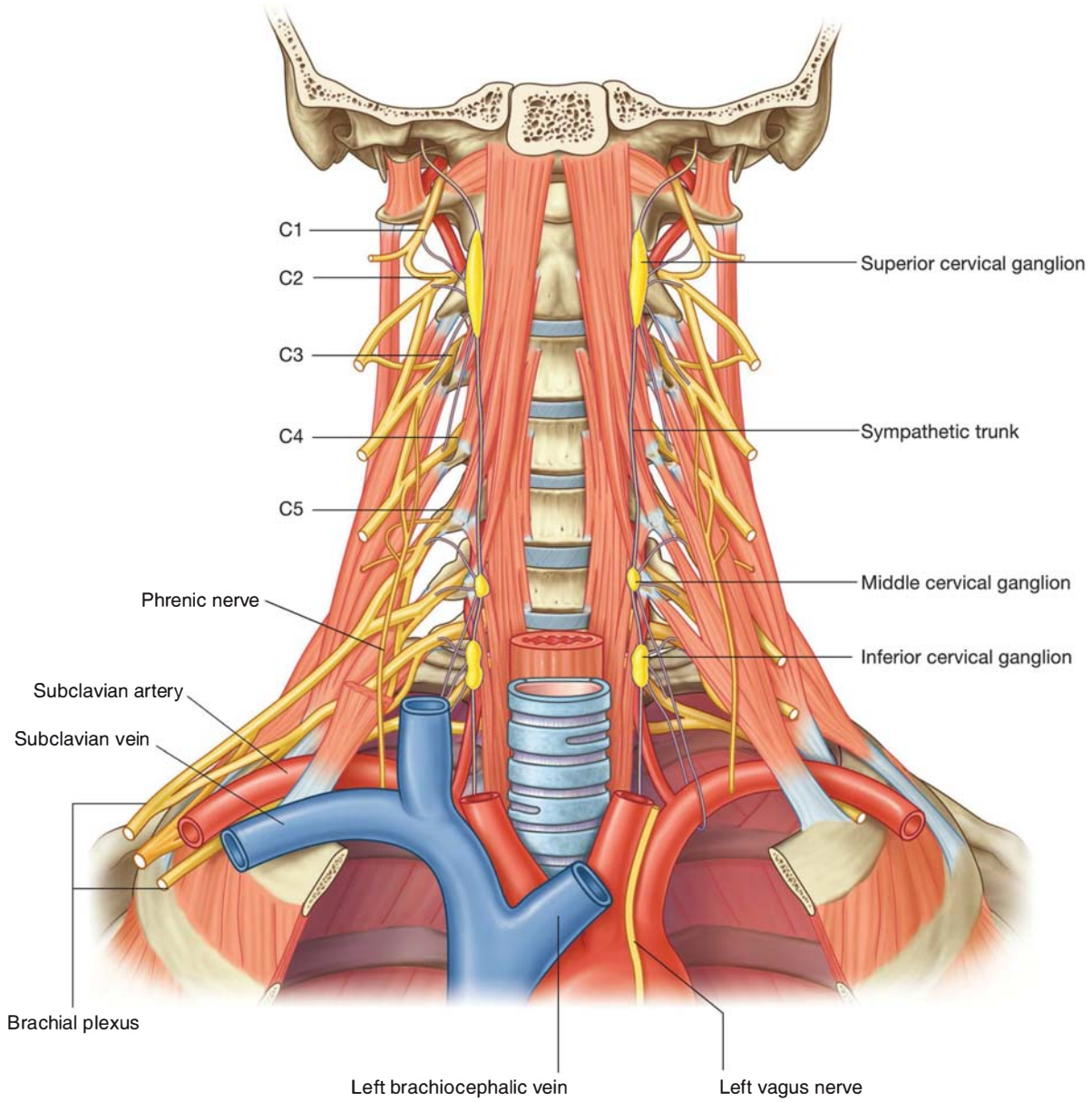


Fig. 8.189 Components of the sympathetic nervous system in the root of the neck.

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Cervical part of the sympathetic trunk

The **cervical part of the sympathetic trunk** is anterior to the longus colli and longus capitis muscles, and posterior to the common carotid artery in the carotid sheath and the internal carotid artery. It is connected to each cervical spinal nerve by a gray ramus communicans (Fig. 8.190). There are no white rami communicantes in the cervical region.

Ganglia

Three ganglia are usually described along the course of the sympathetic trunk in the cervical region, and in these ganglia ascending preganglionic sympathetic fibers from upper thoracic spinal cord levels synapse with postganglionic sympathetic fibers. The postganglionic sympathetic fibers are distributed in branches from these ganglia.

Superior cervical ganglion. A very large **superior cervical ganglion** in the area of cervical vertebrae CI and CII marks the superior extent of the sympathetic trunk (Figs. 8.189 and 8.190). Its branches pass to:

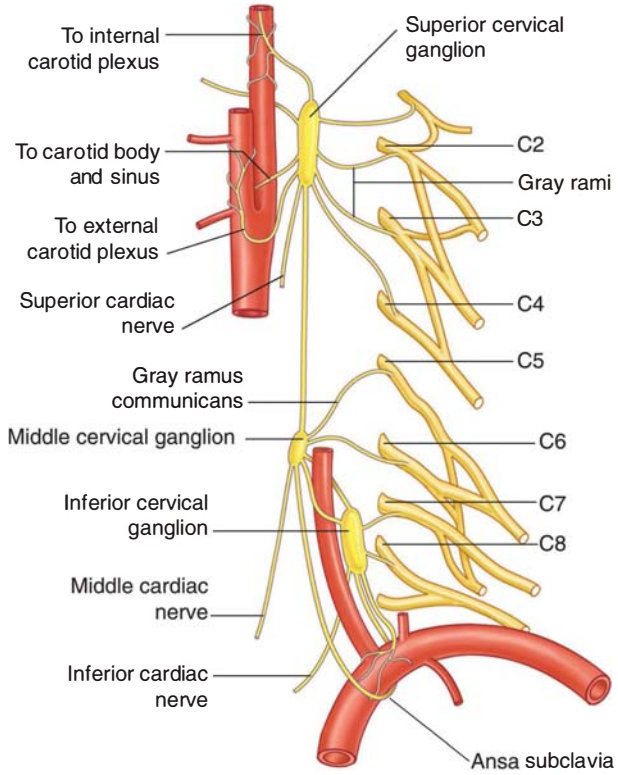


Fig. 8.190 Cervical part of the sympathetic trunk.

- the internal carotid and external carotid arteries, forming plexuses around these vessels,
- cervical spinal nerves C1 to C4 through gray rami communicantes,
- the pharynx, and
- the heart as **superior cardiac nerves**.

Middle cervical ganglion. A second ganglion inferior to the superior cervical ganglion along the course of the sympathetic trunk (the **middle cervical ganglion**) is encountered at about the level of cervical vertebra CVI (Figs. 8.189 and 8.190). Branches from this ganglion pass to:

- cervical spinal nerves C5 and C6 through gray rami communicantes, and
- the heart as **middle cardiac nerves**.

Inferior cervical ganglion. At the lower end of the cervical part of the sympathetic trunk is another ganglion (the inferior cervical ganglion), which becomes very large when it combines with the first thoracic ganglion and forms the **cervicothoracic ganglion** (**stellate ganglion**). The inferior cervical ganglion (Figs. 8.189 and 8.190) is anterior to the neck of rib I and the transverse process of cervical vertebra CVII, and posterior to the first part of the subclavian artery and the origin of the vertebral artery.

Branches from this ganglion pass to:

- spinal nerves C7 to T1 through gray rami communicantes,
- the vertebral artery, forming a plexus associated with this vessel, and
- the heart as **inferior cardiac nerves**.

This ganglion may also receive white rami communicantes from thoracic spinal nerve T1, and occasionally, from T2.

Lymphatics

Thoracic duct

The **thoracic duct** is a major lymphatic channel that begins in the abdomen, passes superiorly through the thorax, and ends in the venous channels in the neck. It passes through the lower thoracic cavity in the midline with:

- the thoracic aorta on the left,
- the azygos vein on the right, and
- the esophagus anteriorly.

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At about the level of thoracic vertebra TV the thoracic duct passes to the left and continues to ascend just to the left of the esophagus. It passes through the superior mediastinum and enters the root of the neck to the left of the esophagus (Fig. 8.191). Arching laterally, it passes posterior to the carotid sheath and turns inferiorly in front of the thyrocervical trunk, the phrenic nerve, and the vertebral artery.

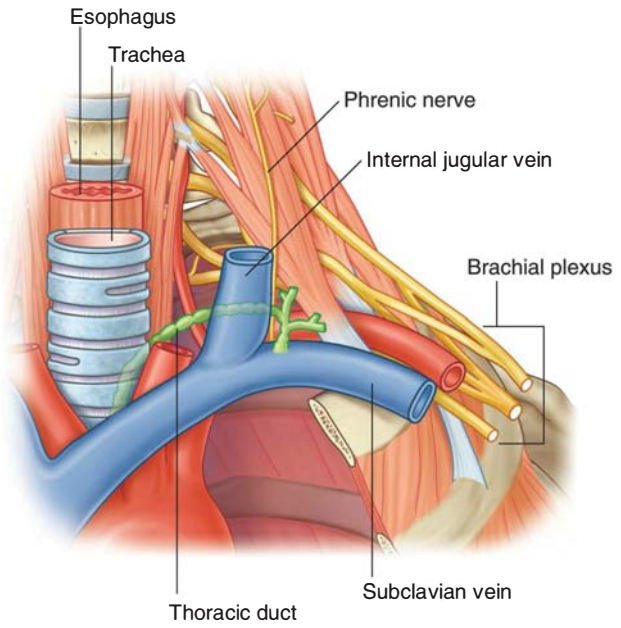


Fig. 8.191 Thoracic duct in the root of the neck.

The thoracic duct terminates in the junction between the left internal jugular and the left subclavian veins (Fig. 8.191). Near its junction with the venous system it is joined by:

- the **left jugular trunk**, which drains lymph from the left side of the head and neck,
- the **left subclavian trunk**, which drains lymph from the left upper limb, and
- occasionally, the **left bronchomediastinal trunk**, which drains lymph from the left half of the thoracic structures (Fig. 8.192).

A similar confluence of three lymphatic trunks occurs on the right side of the body. Emptying into the junction between the right internal jugular and right subclavian veins are:

- the **right jugular trunk** from the head and neck,
- the **right subclavian trunk** from the right upper limb, and
- occasionally, the **right bronchomediastinal trunk** carrying lymph from the structures in the right half of the thoracic cavity and the right upper intercostal spaces (Fig. 8.192).

There is variability in how these trunks enter the veins. They may combine into a single right lymphatic duct to enter the venous system or enter as three separate trunks.

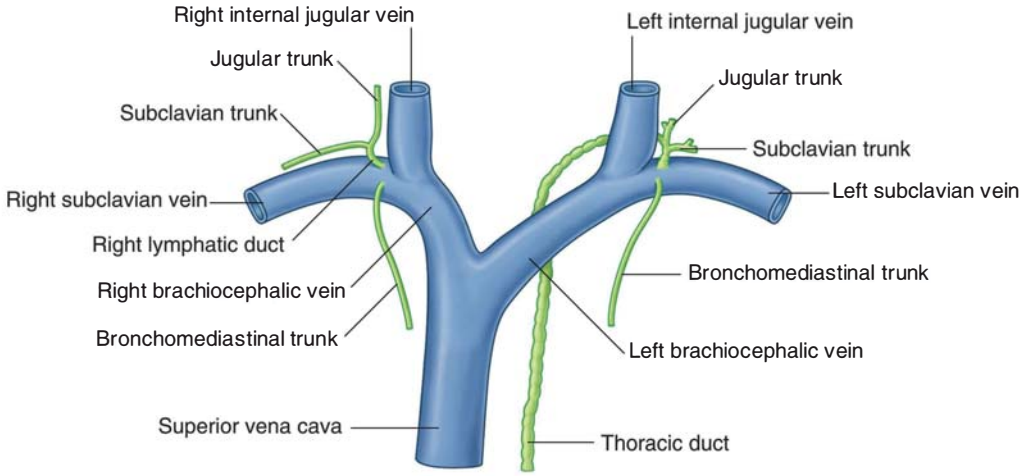


Fig. 8.192 Termination of lymphatic trunks in the root of the neck.

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Lymphatics of the neck

A description of the organization of the lymphatic system in the neck becomes a summary of the lymphatic system in the head and neck. It is impossible to separate the two regions. The components of this system include superficial nodes around the head, superficial cervical nodes along the external jugular vein, and deep cervical nodes forming a chain along the internal jugular vein (Fig. 8.193).

The basic pattern of drainage is for superficial lymphatic vessels to drain to the superficial nodes. Some of these drain to the superficial cervical nodes on their way to the deep cervical nodes and others drain directly to the deep cervical nodes.

Superficial lymph nodes

Five groups of superficial lymph nodes form a ring around the head and are primarily responsible for the lymphatic

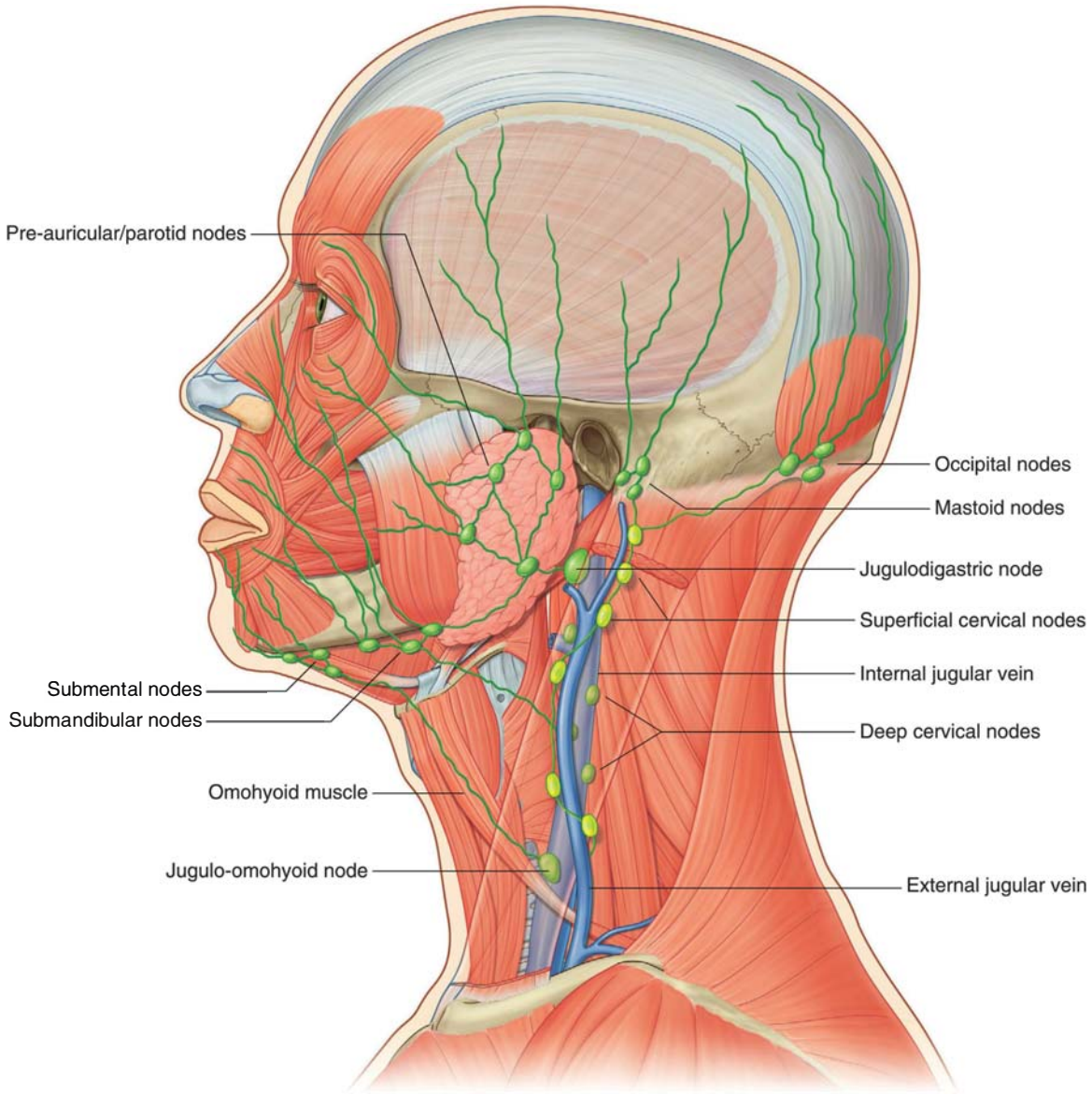


Fig. 8.193 Lymphatic system in the neck.

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drainage of the face and scalp. Their pattern of drainage is very similar to the area of distribution of the arteries near their location.

Beginning posteriorly these groups (Fig. 8.193) are:

- **occipital nodes** near the attachment of the trapezius muscle to the skull and associated with the occipital artery—lymphatic drainage is from the posterior scalp and neck;
- **mastoid nodes** (**retro-auricular/posterior auricular nodes**) posterior to the ear near the attachment of the sternocleidomastoid muscle and associated with the posterior auricular artery—lymphatic drainage is from the posterolateral half of the scalp;
- **pre-auricular and parotid nodes** anterior to the ear and associated with the superficial temporal and transverse facial arteries—lymphatic drainage is from the anterior surface of the auricle, the anterolateral scalp, the upper half of the face, the eyelids, and the cheeks;
- **submandibular nodes** inferior to the body of the mandible and associated with the facial artery—lymphatic drainage is from structures along the path of the facial artery as high as the forehead, as well as the gingivae, the teeth, and the tongue;
- **submental nodes** inferior and posterior to the chin—lymphatic drainage is from the center part of the lower lip, the chin, the floor of the mouth, the tip of the tongue, and the lower incisor teeth.

Lymphatic flow from these superficial lymph nodes passes in several directions:

- Drainage from the occipital and mastoid nodes passes to the superficial cervical nodes along the external jugular vein.
- Drainage from the pre-auricular and parotid nodes, the submandibular nodes, and the submental nodes passes to the deep cervical nodes.

Superficial cervical lymph nodes

The **superficial cervical nodes** are a collection of lymph nodes along the external jugular vein on the superficial surface of the sternocleidomastoid muscle (Fig. 8.193). They primarily receive lymphatic drainage from the posterior and posterolateral regions of the scalp through the occipital and mastoid nodes, and send lymphatic vessels in the direction of the deep cervical nodes.

Deep cervical lymph nodes

The **deep cervical nodes** are a collection of lymph nodes that form a chain along the internal jugular vein (Fig. 8.193). They are divided into upper and lower groups where the intermediate tendon of the omohyoid muscle crosses the common carotid artery and the internal jugular vein.

The most superior node in the upper deep cervical group is the **jugulodigastric node** (Fig. 8.193). This large node is where the posterior belly of the digastric muscle crosses the internal jugular vein and receives lymphatic drainage from the tonsils and tonsillar region.

Another large node, usually associated with the lower deep cervical group because it is at or just inferior to the intermediate tendon of the omohyoid muscle, is the **jugulo-omohyoid node** (Fig. 8.193). This node receives lymphatic drainage from the tongue.

The deep cervical nodes eventually receive all lymphatic drainage from the head and neck either directly or through regional groups of nodes.

From the deep cervical nodes, lymphatic vessels form the right and left jugular trunks, which empty into the right lymphatic duct on the right side or the thoracic duct on the left side.

In the clinic

Clinical lymphatic drainage of the head and neck

Enlargement of the neck lymph nodes (cervical lymphadenopathy) is a common manifestation of disease processes that occur in the head and neck. It is also a common manifestation of diffuse diseases of the body, which include lymphoma, sarcoidosis, and certain types of viral infection such as glandular fever and human immunodeficiency virus (HIV) infection.

Evaluation of cervical lymph nodes is extremely important in determining the nature and etiology of the primary disease process that has produced nodal enlargement.

Clinical evaluation includes a general health assessment, particularly relating to symptoms from the head and neck. Examination of the nodes themselves often gives the clinician a clue as to the nature of the pathological process.

- Soft, tender, and inflamed lymph nodes suggest an acute inflammatory process, which is most likely to be infective.

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In the clinic—cont'd

- Firm multinodular large-volume rubbery nodes often suggest a diagnosis of lymphoma.

Examination should also include careful assessment of other nodal regions, including the supraclavicular fossae, the axillae, the retroperitoneum, and the inguinal regions.

Further examination may include digestive tract endoscopy, chest radiography, and body CT scanning.

Most cervical lymph nodes are easily palpable and suitable for biopsy to establish a tissue diagnosis. Biopsy can be performed using ultrasound for guidance and good samples of lymph nodes may be obtained.

The lymphatic drainage of the neck is somewhat complex, clinically. A relatively simple "level" system of nodal enlargement has been designed that is extremely helpful in evaluating lymph node spread of primary head and neck tumors. Once the number of levels of nodes are determined, and the size of the lymph nodes, the best mode of treatment can be instituted. This may include surgery, radiotherapy, and chemotherapy. The lymph node level also enables a prognosis to be made.

The levels are as follows (Fig. 8.195):

- Level I—from the midline of the submental triangle up to the level of the submandibular gland.
- Level II—from the skull base to the level of the hyoid bone anteriorly from the posterior border of the sternocleidomastoid muscle.
- Level III—the inferior aspect of the hyoid bone to the bottom cricoid arch and anterior to the posterior border of the sternocleidomastoid up to the midline.
- Level IV—from the inferior aspect of the cricoid to the top of the manubrium of the sternum and anterior to the posterior border of the sternocleidomastoid muscle.

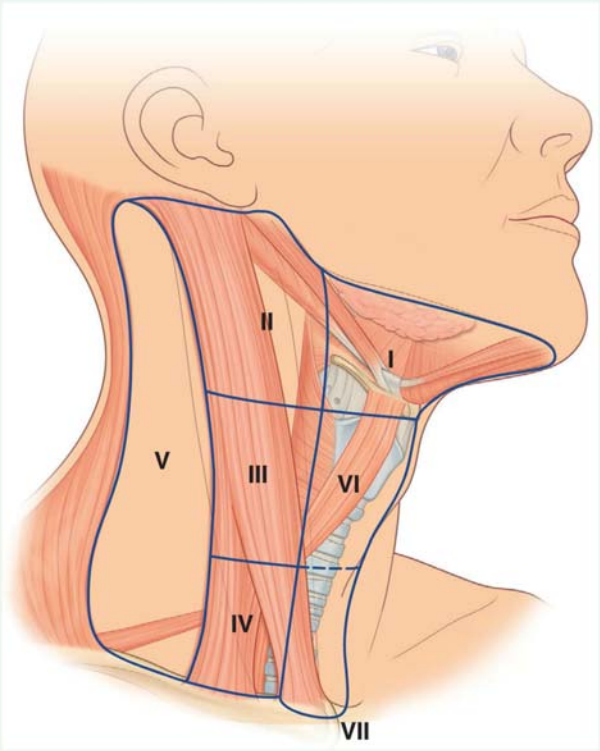


Fig. 8.195 Neck regions (levels) that are used clinically to evaluate lymph nodes.

- Level V—posterior to the sternocleidomastoid muscle and anterior to the trapezius muscle above the level of the clavicle.
- Level VI—below the hyoid bone and above the jugular (sternal) notch in the midline.
- Level VII—below the level of the jugular (sternal) notch.

PHARYNX

The pharynx is a musculofascial half-cylinder that links the oral and nasal cavities in the head to the larynx and esophagus in the neck (Fig. 8.194). The pharyngeal cavity is a common pathway for air and food.

The pharynx is attached above to the base of the skull and is continuous below, approximately at the level of vertebra CVI, with the top of the esophagus. The walls of the pharynx are attached anteriorly to the margins of the nasal cavities, oral cavity, and larynx. Based on these anterior relationships the pharynx is subdivided into three regions, the nasopharynx, oropharynx, and laryngopharynx:

- The posterior apertures (choanae) of the nasal cavities open into the nasopharynx.

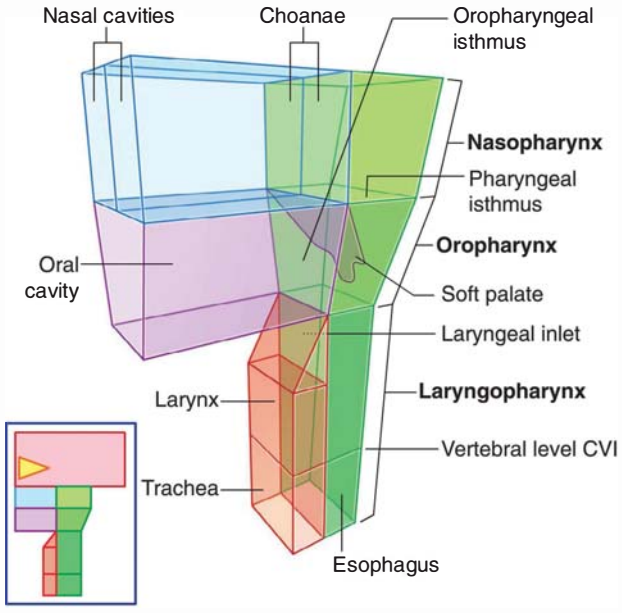


Fig. 8.194 Pharynx.

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- The posterior opening of the oral cavity (oropharyngeal isthmus) opens into the oropharynx.
- The superior aperture of the larynx (laryngeal inlet) opens into the laryngopharynx.

In addition to these openings, the pharyngeal cavity is related anteriorly to the posterior one-third of the tongue and to the posterior aspect of the larynx. The pharyngotympanic tubes open into the lateral walls of the nasopharynx.

Lingual, pharyngeal, and palatine tonsils are on the deep surface of the pharyngeal walls.

The pharynx is separated from the posteriorly positioned vertebral column by a thin retropharyngeal space containing loose connective tissue.

Although the soft palate is generally considered as part of the roof of the oral cavity, it is also related to the pharynx. The soft palate is attached to the posterior margin of the hard palate and is a type of "flutter valve" that can:

- swing up (elevate) to close the pharyngeal isthmus, and seal off the nasopharynx from the oropharynx, and
- swing down (depress) to close the oropharyngeal isthmus and seal off the oral cavity from the oropharynx.

Skeletal framework

The superior and anterior margins of the pharyngeal wall are attached to bone and cartilage, and to ligaments. The two sides of the pharyngeal wall are welded together posteriorly in the midline by a vertically oriented cord-like ligament (the pharyngeal raphe). This connective tissue structure descends from the pharyngeal tubercle on the base of the skull to the level of cervical vertebra CVI where the raphe blends with connective tissue in the posterior wall of the esophagus.

There is an irregular C-shaped line of pharyngeal wall attachment on the base of the skull (Fig. 8.196). The open part of the C faces the nasal cavities. Each arm of the C begins at the posterior margin of the medial plate of the pterygoid process of the sphenoid bone, just inferior to the cartilaginous part of the pharyngotympanic tube. The line crosses inferior to the pharyngotympanic tube and then

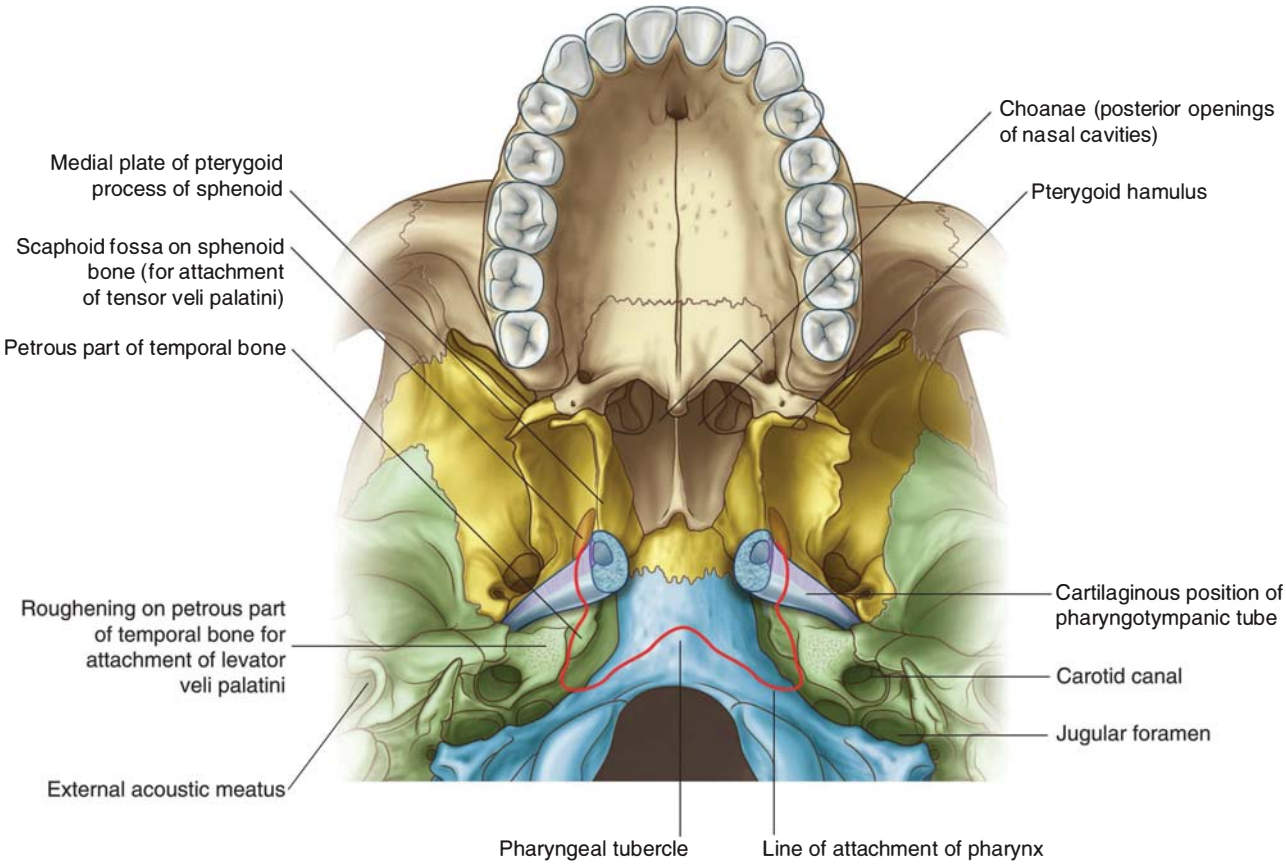


Fig. 8.196 Line of attachment of the pharynx to the base of the skull.

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passes onto the petrous part of the temporal bone where it is just medial to the roughening for the attachment of one of the muscles (levator veli palatini) of the soft palate. From here, the line swings medially onto the occipital bone and joins the line from the other side at a prominent elevation of bone in the midline (the pharyngeal tubercle).

Anterior vertical line of attachment for the lateral pharyngeal walls

The vertical line of attachment for the lateral pharyngeal walls to structures related to the nasal and oral cavities and larynx is discontinuous and in three parts (Fig. 8.197).

First part

On each side, the anterior line of attachment of the lateral pharyngeal wall begins superiorly on the posterior edge of the medial pterygoid plate of the sphenoid bone just inferior to where the pharyngotympanic tube lies against this plate. It continues inferiorly along the edge of the medial plate of the pterygoid process and onto the pterygoid hamulus. From this point, the line descends along the pterygomandibular raphe to the mandible where this part of the line terminates.

The **pterygomandibular raphe** is a linear cord-like connective tissue ligament that spans the distance between the tip of the pterygoid hamulus and a triangular roughening immediately posterior to the third molar on the mandible. It joins a muscle of the lateral pharyngeal wall (superior constrictor) with a muscle of the lateral wall of the oral cavity (buccinator).

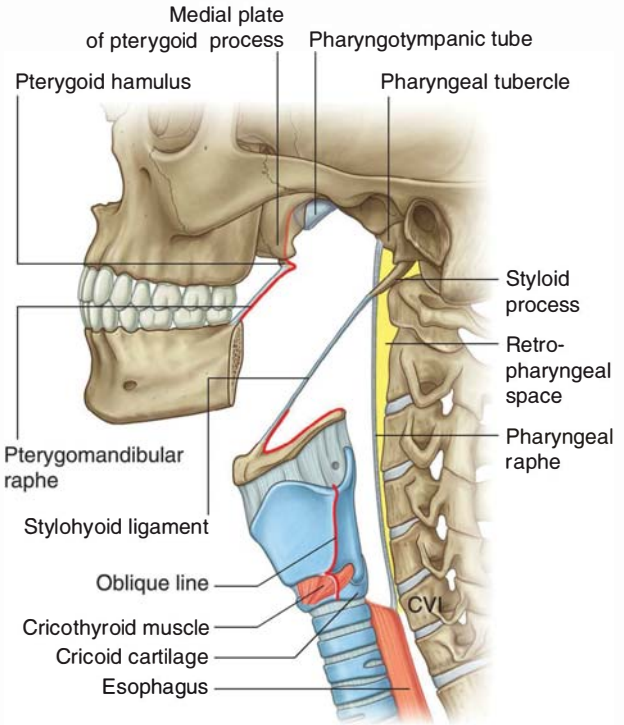


Fig. 8.197 Attachments of the lateral pharyngeal wall.

Second part

The second part of the line of attachment of the lateral pharyngeal wall is related to the hyoid bone. It begins on the lower aspect of the stylohyoid ligament, which connects the tip of the styloid process of the temporal bone to the lesser horn of the hyoid bone. The line continues onto the lesser horn and then turns and runs posteriorly along the entire upper surface of the greater horn of the hyoid where it terminates.

Third part

The most inferior and third part of the line of attachment of the lateral pharyngeal wall begins superiorly on the superior tubercle of the thyroid cartilage, and descends along the oblique line to the inferior tubercle.

From the inferior tubercle, the line of attachment continues over the cricothyroid muscle along a tendinous thickening of fascia to the cricoid cartilage where it terminates.

Pharyngeal wall

The pharyngeal wall is formed by skeletal muscles and by fascia. Gaps between the muscles are reinforced by the fascia and provide routes for structures to pass through the wall.

Muscles

The muscles of the pharynx are organized into two groups based on the orientation of muscle fibers.

The constrictor muscles have fibers oriented in a circular direction relative to the pharyngeal wall, whereas the longitudinal muscles have fibers oriented vertically.

Constrictor muscles

The three constrictor muscles on each side are major contributors to the structure of the pharyngeal wall (Fig. 8.198 and Table 8.17) and their names indicate their position—**superior, middle,** and **inferior constrictor muscles.** Posteriorly, the muscles from each side are joined together by the pharyngeal raphe. Anteriorly, these muscles attach to bones and ligaments related to the lateral margins of the nasal and oral cavities and the larynx.

The constrictor muscles overlap each other in a fashion resembling the walls of three flower pots stacked one on the other. The inferior constrictors overlap the lower margins of the middle constrictors and, in the same way, the middle constrictors overlap the superior constrictors.

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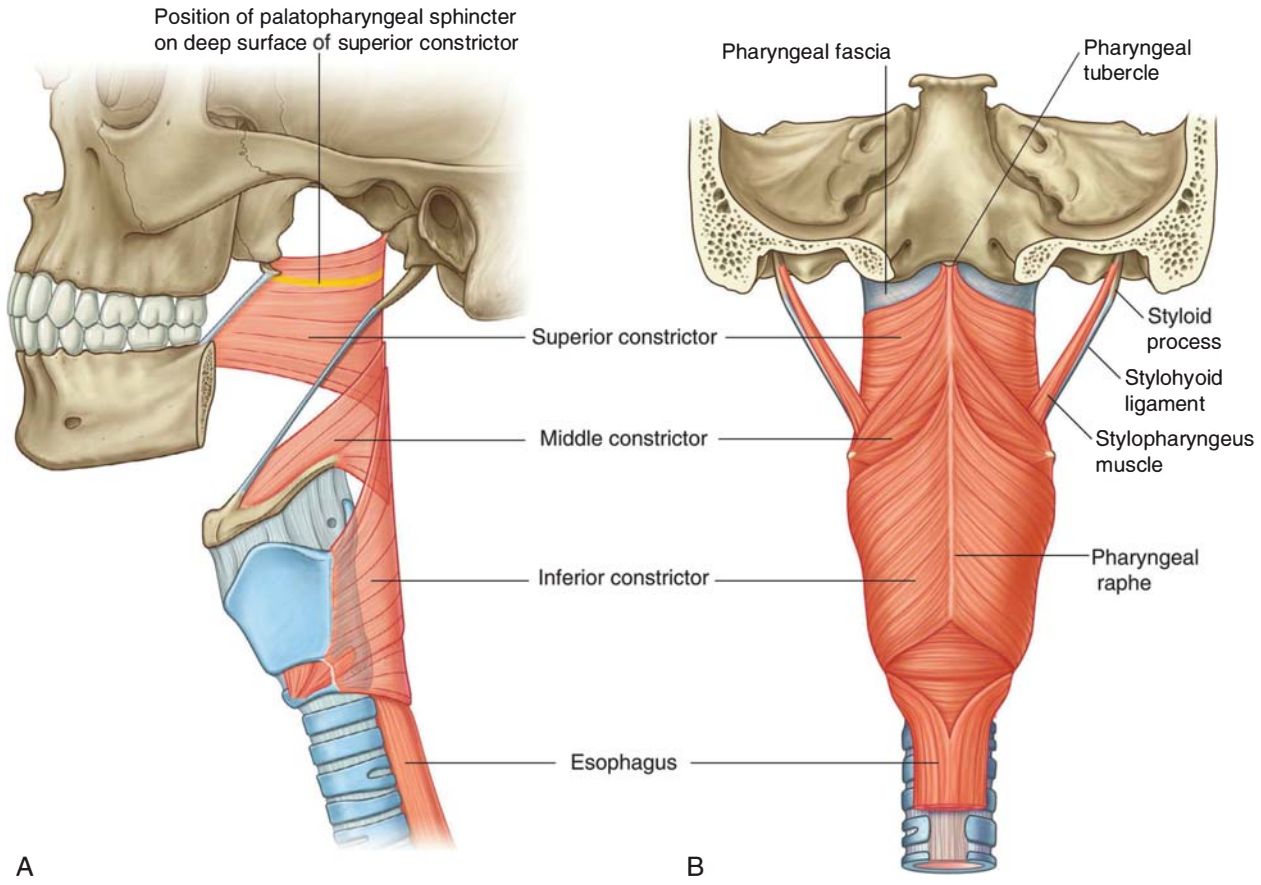


Fig. 8.198 Constrictor muscles of the pharynx. **A.** Lateral view. **B.** Posterior view.

Table 8.17 Constrictor muscles of the pharynx

Muscle	Posterior attachment	Anterior attachment	Innervation	Function
Superior constrictor	Pharyngeal raphe	Pterygomandibular raphe and adjacent bone on the mandible and pterygoid hamulus	Vagus nerve [X]	Constriction of pharynx
Middle constrictor	Pharyngeal raphe	Upper margin of greater horn of hyoid bone and adjacent margins of lesser horn and stylohyoid ligament	Vagus nerve [X]	Constriction of pharynx
Inferior constrictor	Pharyngeal raphe	Cricoid cartilage, oblique line of thyroid cartilage, and a ligament that spans between these attachments and crosses the cricothyroid muscle	Vagus nerve [X]	Constriction of pharynx

Collectively, the muscles constrict or narrow the pharyngeal cavity.

When the constrictor muscles contract sequentially from top to bottom, as in swallowing, they move a bolus of food through the pharynx and into the esophagus.

All of the constrictors are innervated by the pharyngeal branch of the vagus nerve [X].

Superior constrictors

The superior constrictor muscles together bracket the upper part of the pharyngeal cavity (Fig. 8.198).

Each muscle is attached anteriorly to the pterygoid hamulus, pterygomandibular raphe, and adjacent bone of the mandible. From these attachments, the muscle fans out posteriorly and joins with its partner muscle from the other side at the pharyngeal raphe.

A special band of muscle (the **palatopharyngeal sphincter**) originates from the anterolateral surface of the soft palate and circles the inner aspect of the pharyngeal wall, blending with the inner aspect of the superior constrictor.

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When the superior constrictor constricts during swallowing, it forms a prominent ridge on the deep aspect of the pharyngeal wall that catches the margin of the elevated soft palate, which then seals closed the pharyngeal isthmus between the nasopharynx and oropharynx.

Middle constrictors

The middle constrictor muscles are attached to the lower aspect of the stylohyoid ligament, the lesser horn of the hyoid bone, and the entire upper surface of the greater horn of the hyoid (Fig. 8.198).

Like the superior constrictors, the middle constrictor muscles fan out posteriorly and attach to the pharyngeal raphe.

The posterior part of the middle constrictors overlaps the superior constrictors.

Inferior constrictors

The inferior constrictor muscles attach anteriorly to the oblique line of the thyroid cartilage, the cricoid cartilage, and a ligament that spans between these two attachments to cartilage and crosses the cricothyroid muscle (Fig. 8.198).

Like the other constrictor muscles, the inferior constrictor muscles spread out posteriorly and attach to the pharyngeal raphe.

The posterior part of the inferior constrictors overlaps the middle constrictors. Inferiorly, the muscle fibers blend with and attach into the wall of the esophagus.

The parts of the inferior constrictors attached to the cricoid cartilage bracket the narrowest part of the pharyngeal cavity.

Longitudinal muscles

The three longitudinal muscles of the pharyngeal wall (Fig. 8.199 and Table 8.18) are named according to their origins—**stylopharyngeus** from the styloid process of the temporal bone, **salpingopharyngeus** from the cartilaginous part of the pharyngotympanic tube (*salpinx* is Greek for "tube"), and **palatopharyngeus** from the soft palate.

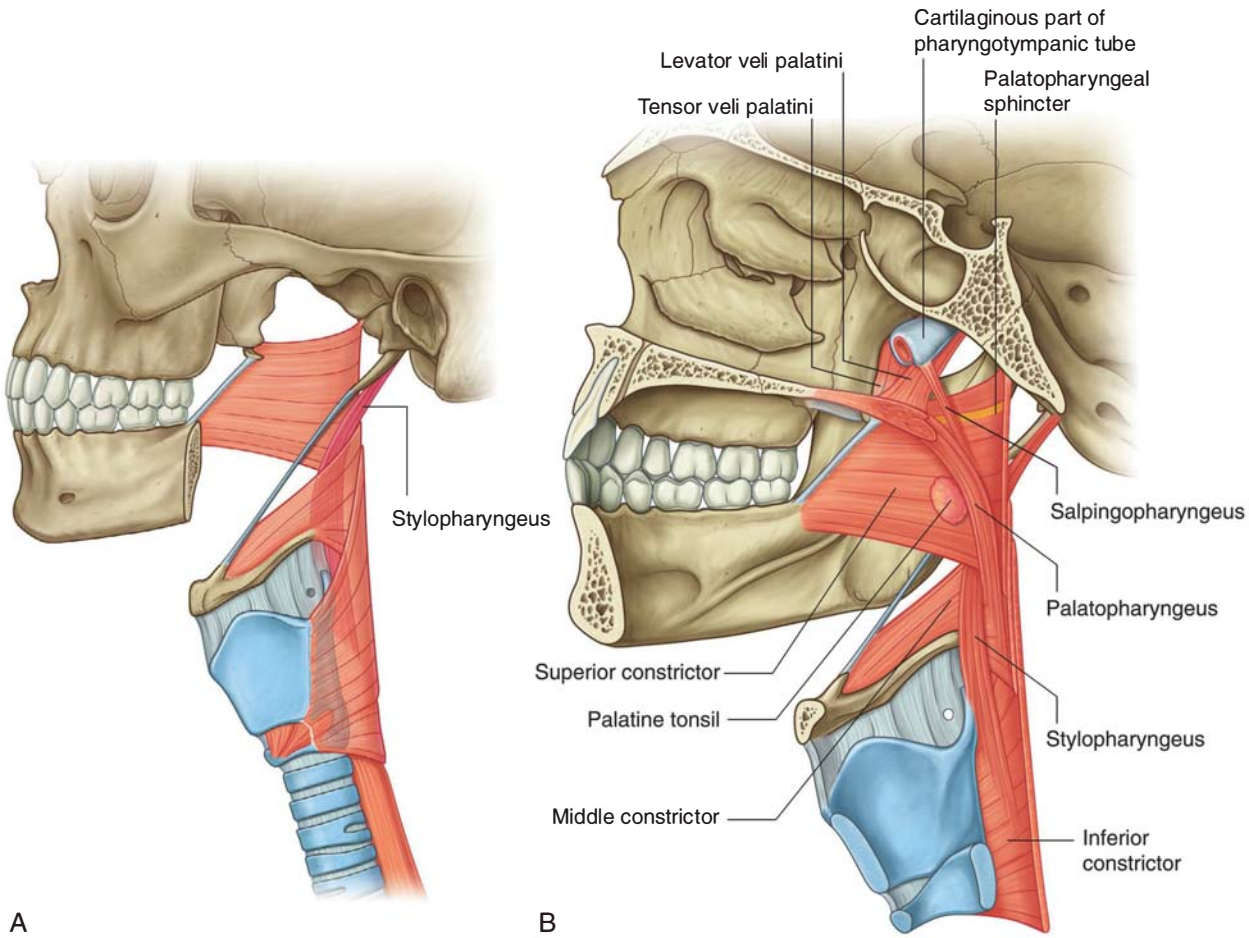


Fig. 8.199 Longitudinal muscles of the pharynx. **A.** Stylopharyngeus muscle. **B.** Medial view.

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Table 8.18 Longitudinal muscles of the pharynx

Muscle	Origin	Insertion	Innervation	Function
Stylopharyngeus	Medial side of base of styloid process	Pharyngeal wall	Glossopharyngeal nerve [IX]	Elevation of the pharynx
Salpingopharyngeus	Inferior aspect of pharyngeal end of pharyngotympanic tube	Pharyngeal wall	Vagus nerve [X]	Elevation of the pharynx
Palatopharyngeus	Upper surface of palatine aponeurosis	Pharyngeal wall	Vagus nerve [X]	Elevation of the pharynx; closure of the oropharyngeal isthmus

From their sites of origin, these muscles descend and attach into the pharyngeal wall.

The longitudinal muscles elevate the pharyngeal wall, or during swallowing, pull the pharyngeal wall up and over a bolus of food being moved through the pharynx and into the esophagus.

Stylopharyngeus

The cylindrical stylopharyngeus muscle (Fig. 8.199A) originates from the base of the medial surface of the styloid process of the temporal bone and descends between the superior and middle constrictor muscles to fan out on, and blend with, the deep surface of the pharyngeal wall. It is innervated by the glossopharyngeal nerve [IX].

Salpingopharyngeus

The salpingopharyngeus (Fig. 8.199B) is a small muscle originating from the inferior aspect of the pharyngotympanic tube, descending on, and blending into, the deep surface of the pharyngeal wall. It is innervated by the vagus nerve [X].

Palatopharyngeus

The palatopharyngeus (Fig. 8.199B), in addition to being a muscle of the pharynx, is also a muscle of the soft palate (see p. 1048). It is attached to the upper surface of the palatine aponeurosis, and passes posteriorly and inferiorly to blend with the deep surface of the pharyngeal wall.

The palatopharyngeus forms an important fold in the overlying mucosa (the **palatopharyngeal arch**). This arch is visible through the oral cavity and is a landmark for finding the **palatine tonsil**, which is immediately anterior to it on the oropharyngeal wall.

In addition to elevating the pharynx, the palatopharyngeus participates in closing the oropharyngeal isthmus by depressing the palate and moving the palatopharyngeal fold toward the midline.

The palatopharyngeus is innervated by the vagus nerve [X].

Fascia

The pharyngeal fascia is separated into two layers, which sandwich the pharyngeal muscles between them:

- A thin layer (**buccopharyngeal fascia**) coats the outside of the muscular part of the wall and is a component of the pretracheal layer of cervical fascia (see p. 1002).
- A much thicker layer (**pharyngobasilar fascia**) lines the inner surface.

The fascia reinforces the pharyngeal wall where muscle is deficient. This is particularly evident above the level of the superior constrictor where the pharyngeal wall is formed almost entirely of fascia (Fig. 8.199). This part of the wall is reinforced externally by muscles of the soft palate (tensor and levator veli palatini).

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Gaps in the pharyngeal wall and structures passing through them

Gaps between muscles of the pharyngeal wall provide important routes for muscles and neurovascular tissues (Fig. 8.200).

Above the margin of the superior constrictor, the pharyngeal wall is deficient in muscle and completed by pharyngeal fascia.

The tensor and levator veli palatini muscles of the soft palate initially descend from the base of the skull and are lateral to the pharyngeal fascia. In this position, they reinforce the pharyngeal wall:

- The levator veli palatini passes through the pharyngeal fascia inferior to the pharyngotympanic tube and enters the soft palate.
- The tendon of the tensor veli palatini turns medially around the pterygoid hamulus and passes through the origin of the buccinator muscle to enter the soft palate.

One of the largest and most important apertures in the pharyngeal wall is between the superior and middle constrictor muscles of the pharynx and the posterior border of the mylohyoid muscle, which forms the floor of the mouth (Fig. 8.200). This triangular-shaped gap (**oropharyngeal triangle**) not only enables the stylopharyngeus to slip into the pharyngeal wall, but also allows muscles, nerves, and vessels to pass between regions lateral to the pharyngeal wall and the oral cavity, particularly to the tongue.

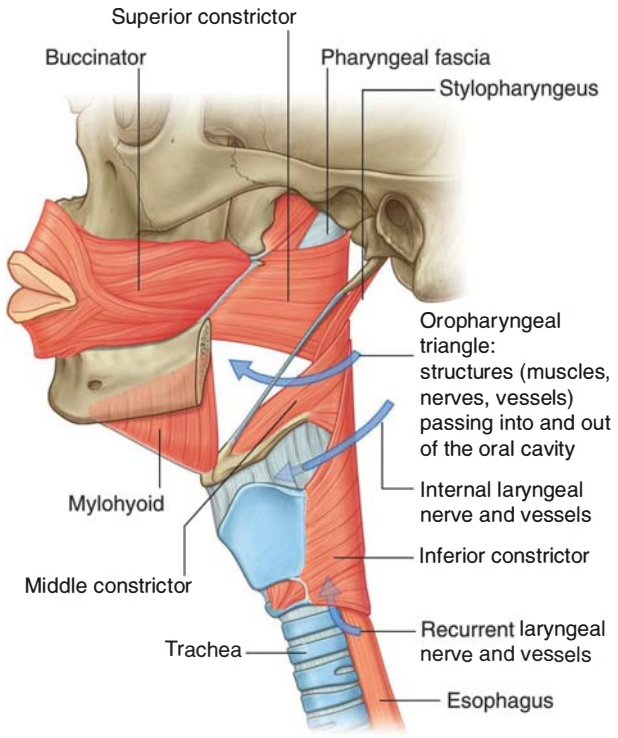


Fig. 8.200 Gaps between muscles in the pharyngeal wall.

The gap between the middle and inferior constrictor muscles allows the internal laryngeal vessels and nerve access to the aperture in the thyrohyoid membrane to enter the larynx.

The recurrent laryngeal nerves and accompanying inferior laryngeal vessels enter the larynx posterior to the inferior horn of the thyroid cartilage deep to the inferior margin of the inferior constrictor muscle.

Nasopharynx

The nasopharynx is behind the posterior apertures (choanae) of the nasal cavities and above the level of the soft palate (Fig. 8.201). Its ceiling is formed by the sloping base of the skull and consists of the posterior part of the body of the sphenoid bone and the basal part of the occipital bone. The ceiling and lateral walls of the nasopharynx form a domed vault at the top of the pharyngeal cavity that is always open.

The cavity of the nasopharynx is continuous below with the cavity of the oropharynx at the pharyngeal isthmus. The position of the pharyngeal isthmus is marked on the pharyngeal wall by a mucosal fold caused by the underlying palatopharyngeal sphincter, which is part of the superior constrictor muscle.

Elevation of the soft palate and constriction of the palatopharyngeal sphincter closes the pharyngeal isthmus during swallowing and separates the nasopharynx from the oropharynx.

There is a large collection of lymphoid tissue (the **pharyngeal tonsil**) in the mucosa covering the roof of the nasopharynx. Enlargement of this tonsil, known then as adenoids, can occlude the nasopharynx so that breathing is only possible through the oral cavity (Fig. 8.201A).

The most prominent features on each lateral wall of the nasopharynx are:

- the pharyngeal opening of the pharyngotympanic tube, and
- mucosal elevations and folds covering the end of the pharyngotympanic tube and the adjacent muscles.

The opening of the pharyngotympanic tube is posterior to and slightly above the level of the hard palate, and lateral to the top of the soft palate (Fig. 8.201A).

Because the pharyngotympanic tube projects into the nasopharynx from a posterolateral direction, its posterior

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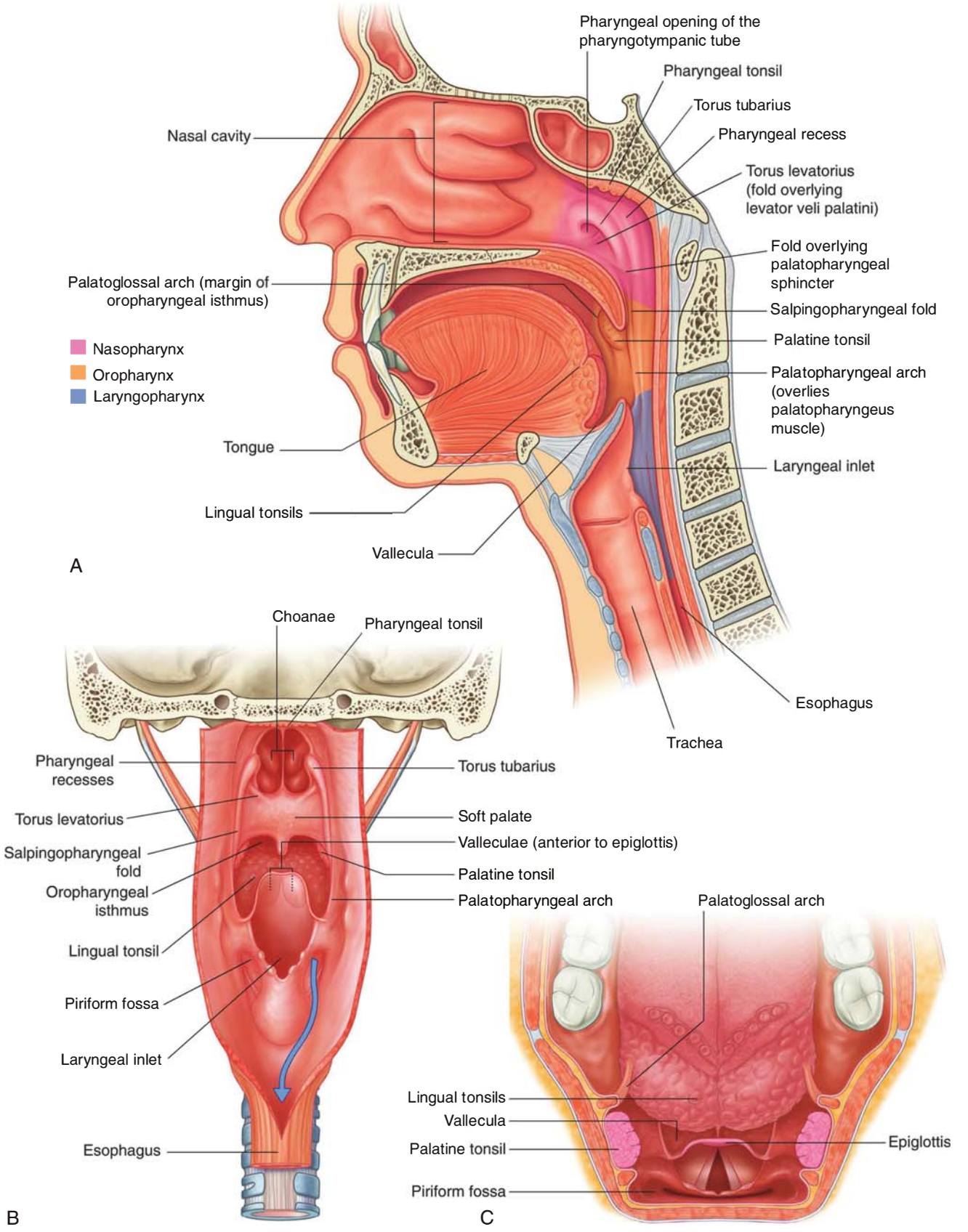


Fig. 8.201 Mucosal features of the pharynx. **A.** Lateral view. **B.** Posterior view with the pharyngeal wall opened. **C.** Superior view.

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rim forms an elevation or bulge on the pharyngeal wall. Posterior to this tubal elevation (**torus tubarius**) is a deep recess (**pharyngeal recess**) (Fig. 8.201A).

Mucosal folds related to the pharyngotympanic tube include:

- the small vertical **salpingopharyngeal fold**, which descends from the tubal elevation and overlies the salpingopharyngeus muscle, and
- a broad fold or elevation (**torus levatorius**) that appears to emerge from just under the opening of the pharyngotympanic tube, continues medially onto the upper surface of the soft palate, and overlies the levator veli palatini muscle.

Oropharynx

The oropharynx is posterior to the oral cavity, inferior to the level of the soft palate, and superior to the upper margin of the epiglottis (Fig. 8.201). The palatoglossal folds (arches), one on each side, that cover the palatoglossal muscles, mark the boundary between the oral cavity and the oropharynx. The arched opening between the two folds is the oropharyngeal isthmus. Just posterior and medial to these folds are another pair of folds (arches), the palatopharyngeal folds, one on each side, that overlie the palatopharyngeus muscles.

The anterior wall of the oropharynx inferior to the oropharyngeal isthmus is formed by the upper part of the posterior one-third or pharyngeal part of the tongue. Large collections of lymphoid tissue (the lingual tonsils) are in the mucosa covering this part of the tongue. A pair of mucosal pouches (**valleculae**), one on each side of the midline, between the base of the tongue and epiglottis, are depressions formed between a midline mucosal fold and two lateral folds that connect the tongue to the epiglottis.

The palatine tonsils are on the lateral walls of the oropharynx. On each side, there is a large ovoid collection of lymphoid tissue in the mucosa lining the superior constrictor muscle and between the palatoglossal and palatopharyngeal arches. The palatine tonsils are visible through the oral cavity just posterior to the palatoglossal folds.

When holding liquid or solids in the oral cavity, the oropharyngeal isthmus is closed by depression of the soft palate, elevation of the back of the tongue, and movement toward the midline of the palatoglossal and palatopharyngeal folds. This allows a person to breathe while chewing or manipulating material in the oral cavity.

On swallowing, the oropharyngeal isthmus is opened, the palate is elevated, the laryngeal cavity is closed, and the food or liquid is directed into the esophagus. A person cannot breathe and swallow at the same time because the airway is closed at two sites, the pharyngeal isthmus and the larynx.

Laryngopharynx

The laryngopharynx extends from the superior margin of the epiglottis to the top of the esophagus at the level of vertebra CVI (Fig. 8.201).

The laryngeal inlet opens into the anterior wall of the laryngopharynx. Inferior to the laryngeal inlet, the anterior wall consists of the posterior aspect of the larynx.

There is another pair of mucosal recesses (**piriform fossae**) between the central part of the larynx and the more lateral lamina of the thyroid cartilage. The piriform fossae form channels that direct solids and liquids from the oral cavity around the raised laryngeal inlet and into the esophagus.

Tonsils

Collections of lymphoid tissue in the mucosa of the pharynx surrounding the openings of the nasal and oral cavities are part of the body's defense system. The largest of these collections form distinct masses (**tonsils**). Tonsils occur mainly in three areas (Fig. 8.201):

- The pharyngeal tonsil, known as adenoids when enlarged, is in the midline on the roof of the nasopharynx.
- The palatine tonsils are on each side of the oropharynx between the palatoglossal and palatopharyngeal arches just posterior to the oropharyngeal isthmus. (The palatine tonsils are visible through the open mouth of a patient when the tongue is depressed.)
- The lingual tonsils refer collectively to numerous lymphoid nodules on the posterior one-third of the tongue.

Small lymphoid nodules also occur in the pharyngotympanic tube near its opening into the nasopharynx, and on the upper surface of the soft palate.

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Vessels

Arteries

Numerous vessels supply the pharyngeal wall (Fig. 8.202).

Arteries that supply upper parts of the pharynx include:

- the ascending pharyngeal artery,
- the ascending palatine and tonsillar branches of the facial artery, and
- numerous branches of the maxillary and the lingual arteries.

All these vessels are from the external carotid artery.

Arteries that supply the lower parts of the pharynx include pharyngeal branches from the inferior thyroid artery, which originates from the thyrocervical trunk of the subclavian artery.

The major blood supply to the palatine tonsil is from the tonsillar branch of the facial artery, which penetrates the superior constrictor muscle.

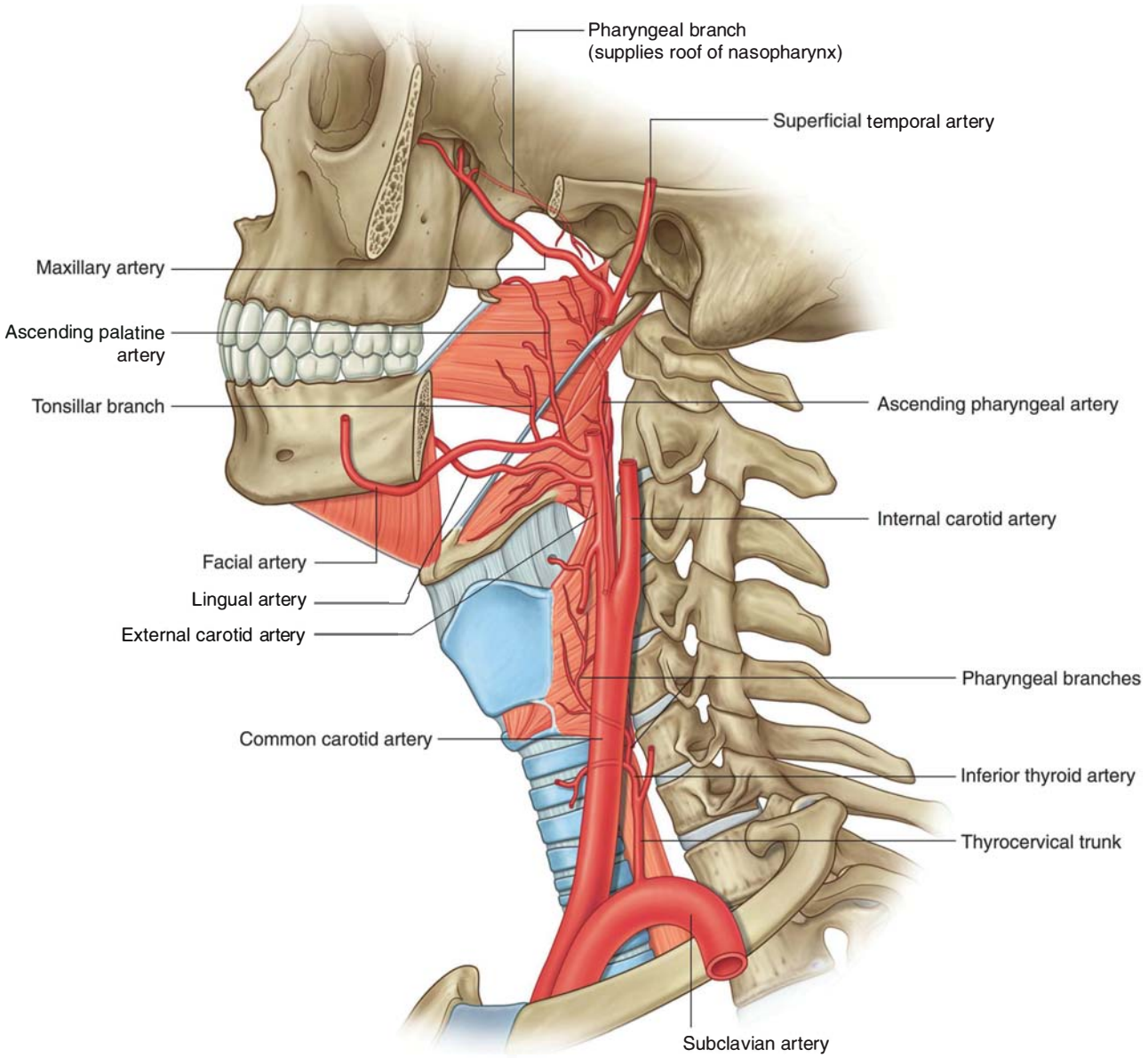


Fig. 8.202 Arterial supply of the pharynx.

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Veins

Veins of the pharynx form a plexus, which drains superiorly into the pterygoid plexus in the infratemporal fossa, and inferiorly into the facial and internal jugular veins (Fig. 8.203).

Lymphatics

Lymphatic vessels from the pharynx drain into the deep cervical nodes and include **retropharyngeal** (between the nasopharynx and vertebral column), **paratracheal**, and **infrahyoid nodes** (Fig. 8.203).

The palatine tonsils drain through the pharyngeal wall into the jugulodigastric nodes in the region where the facial vein drains into the internal jugular vein (and inferior to the posterior belly of the digastric muscle).

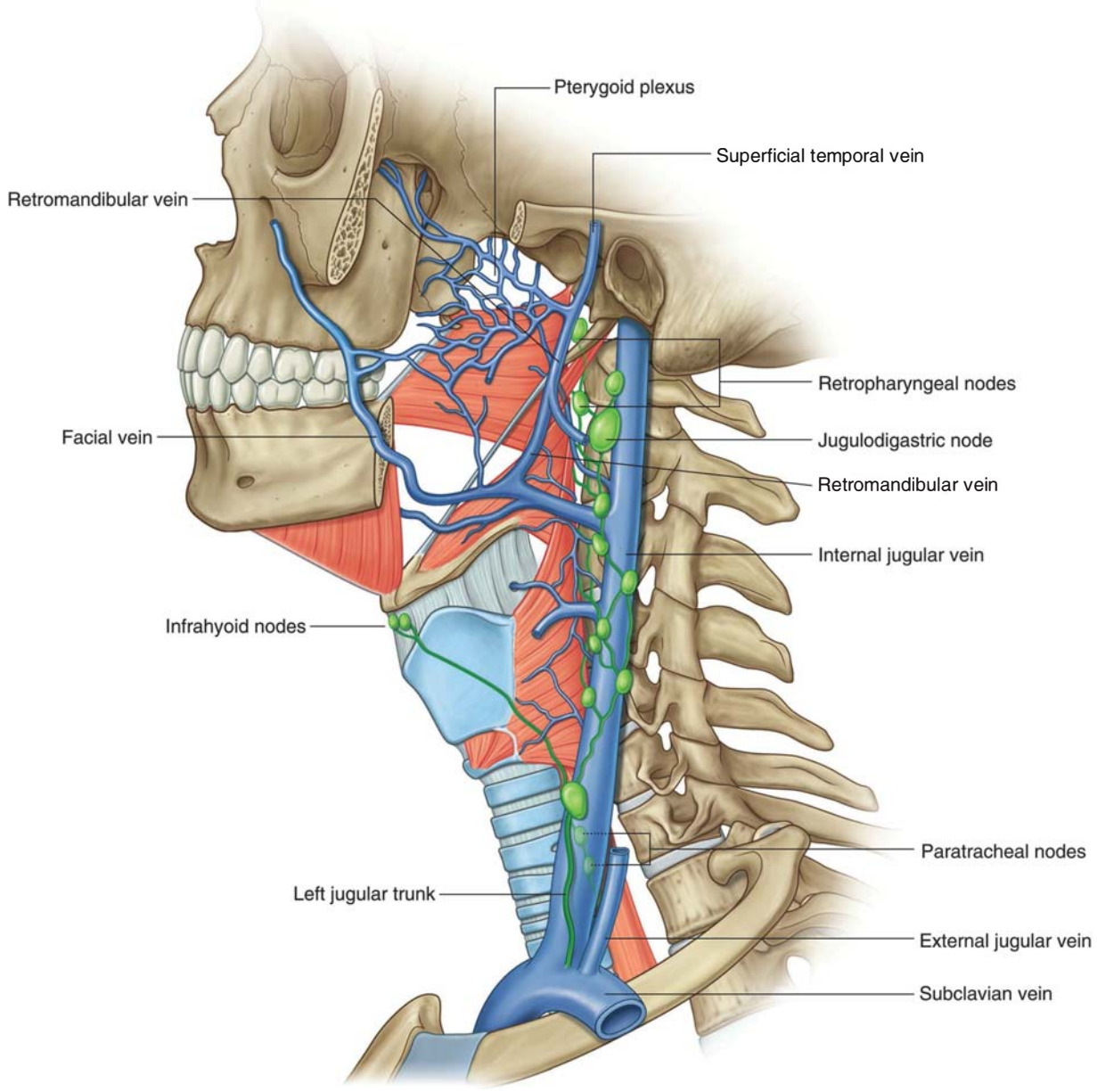


Fig. 8.203 Venous and lymphatic drainage of the pharynx.

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Nerves

Motor and most sensory innervation (except for the nasal region) of the pharynx is mainly through branches of the vagus [X] and glossopharyngeal [IX] nerves, which form a plexus in the outer fascia of the pharyngeal wall (Fig. 8.204A).

The **pharyngeal plexus** is formed by:

- the pharyngeal branch of the vagus nerve [X],
- branches from the **external laryngeal nerve** from the **superior laryngeal branch** of the vagus nerve [X], and
- pharyngeal branches of the glossopharyngeal nerve [IX].

The **pharyngeal branch of the vagus nerve [X]** originates from the upper part of its **inferior ganglion** above the origin of the superior laryngeal nerve and is the major motor nerve of the pharynx.

All muscles of the pharynx are innervated by the vagus nerve [X] mainly through the pharyngeal plexus, except for the stylopharyngeus, which is innervated directly by a branch of the glossopharyngeal nerve [IX] (Fig. 8.204B).

Each subdivision of the pharynx has a different sensory innervation:

- The nasopharynx is innervated by a pharyngeal branch of the maxillary nerve [V_2] that originates in the pterygopalatine fossa and passes through the palatovaginal canal in the sphenoid bone to reach the roof of the pharynx.
- The oropharynx is innervated by the glossopharyngeal nerve [IX] via the pharyngeal plexus.
- The laryngopharynx is innervated by the vagus nerve [X] via the internal branch of the superior laryngeal artery.

Glossopharyngeal nerve [IX]

The glossopharyngeal nerve [IX] is related to the pharynx throughout most of its course outside the cranial cavity.

After exiting the skull through the jugular foramen, the glossopharyngeal nerve [IX] descends on the posterior surface of the stylopharyngeus muscle (Fig. 8.204B),

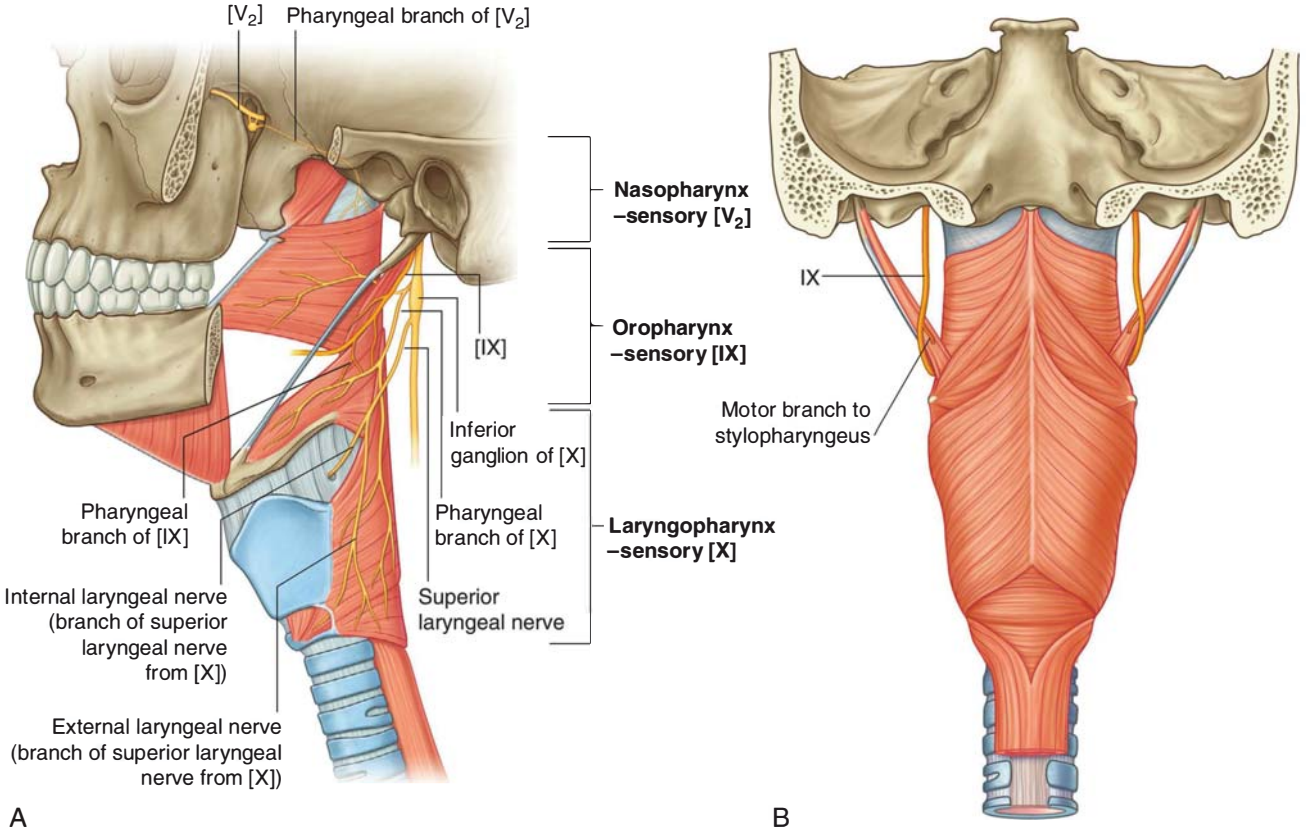


Fig. 8.204 Innervation of the pharynx. **A.** Lateral view. **B.** Posterior view showing innervation of stylopharyngeus muscle.

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passes onto the lateral surface of the stylopharyngeus, and then passes anteriorly through the gap (oropharyngeal triangle) between the superior constrictor, middle constrictor, and mylohyoid muscles to eventually reach the posterior aspect of the tongue.

As the glossopharyngeal nerve [IX] passes under the free edge of the superior constrictor, it is just inferior to the palatine tonsil lying on the deep surface of the superior constrictor.

Pharyngeal branches to the pharyngeal plexus and a motor branch to the stylopharyngeus muscle are among branches that originate from the glossopharyngeal nerve [IX] in the neck. Because sensory innervation of the oropharynx is by the glossopharyngeal nerve [IX], this nerve carries sensory innervation from the palatine tonsil and is also the afferent limb of the gag reflex (see "In the clinic" on pg. 901).

LARYNX

The larynx is a hollow musculoligamentous structure with a cartilaginous framework that caps the lower respiratory tract.

The cavity of the larynx is continuous below with the trachea, and above opens into the pharynx immediately posterior and slightly inferior to the tongue and the posterior opening (oropharyngeal isthmus) of the oral cavity (Fig. 8.205A,B).

The larynx is both a valve (or sphincter) to close the lower respiratory tract, and an instrument to produce sound. It is composed of:

- three large unpaired cartilages (cricoid, thyroid, and epiglottis),

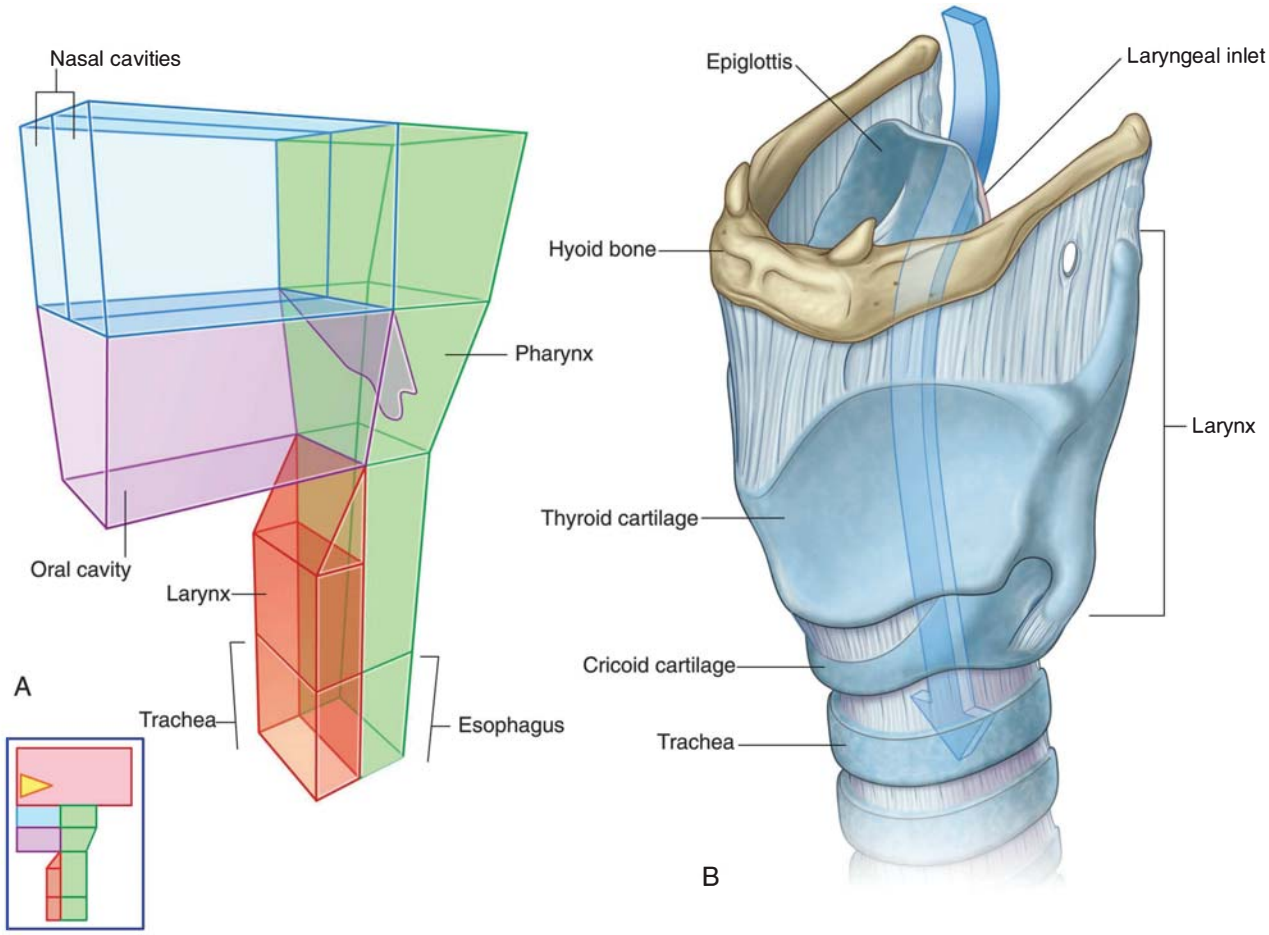


Fig. 8.205 Larynx. **A.** Relationship to other cavities. **B.** Lateral view.

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- three pairs of smaller cartilages (arytenoid, corniculate, and cuneiform), and
- a fibro-elastic membrane and numerous intrinsic muscles.

The larynx is suspended from the hyoid bone above and attached to the trachea below by membranes and ligaments. It is highly mobile in the neck and can be moved up and down and forward and backward by the action of extrinsic muscles that attach either to the larynx itself or to the hyoid bone.

During swallowing, the dramatic upward and forward movements of the larynx facilitate closing the laryngeal inlet and opening the esophagus.

Motor and sensory innervation of the larynx is provided by the vagus nerve [X].

Laryngeal cartilages

Cricoid cartilage

The cricoid cartilage is the most inferior of the laryngeal cartilages and completely encircles the airway (Fig. 8.206). It is shaped like a signet ring with a broad **lamina of cricoid cartilage** posterior to the airway and a much narrower **arch of cricoid cartilage** circling anteriorly.

The posterior surface of the lamina is characterized by two shallow oval depressions separated by a vertical ridge. The esophagus is attached to the ridge and the depressions are for attachment of the posterior cricoarytenoid muscles.

The cricoid cartilage has two articular facets on each side for articulation with other laryngeal cartilages:

- One facet is on the sloping superolateral surface of the lamina and articulates with the base of an arytenoid cartilage.
- The other facet is on the lateral surface of the lamina near its base and is for articulation with the medial surface of the inferior horn of the thyroid cartilage.

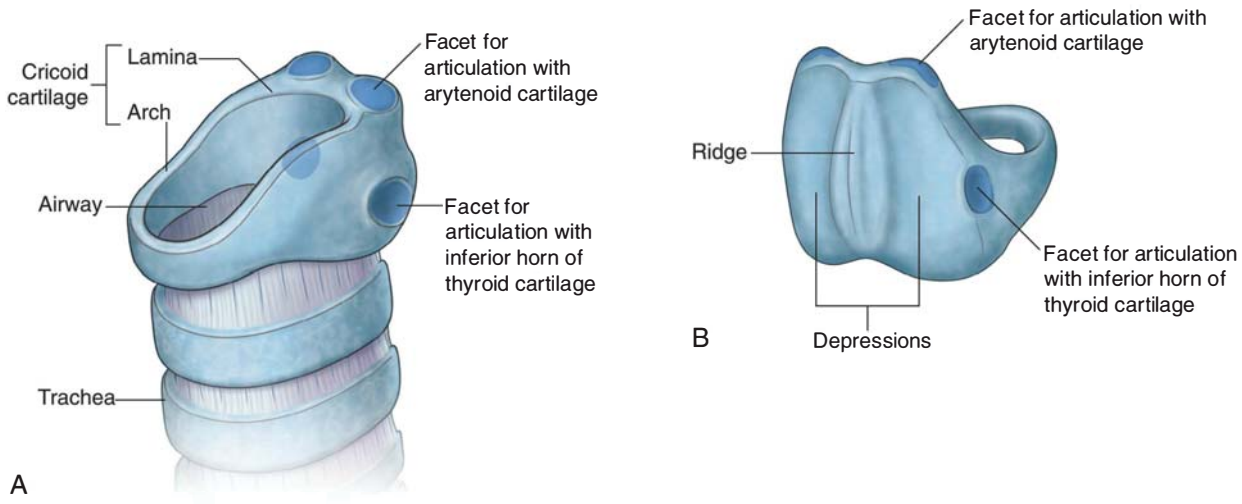


Fig. 8.206 Cricoid cartilage. **A.** Anterolateral view. **B.** Posterior view.

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Thyroid cartilage

The thyroid cartilage (Fig. 8.207) is the largest of the laryngeal cartilages. It is formed by a right and a left lamina, which are widely separated posteriorly, but converge and join anteriorly. The most superior point of the site of fusion between the two broad flat laminae projects forward as the **laryngeal prominence** (Adam's apple). The angle between the two laminae is more acute in men (90°) than in women (120°) so the laryngeal prominence is more apparent in men than women.

Just superior to the laryngeal prominence, the **superior thyroid notch** separates the two laminae as they diverge laterally. Both the superior thyroid notch and the laryngeal prominence are palpable landmarks in the neck. There is a less distinct **inferior thyroid notch** in the midline along the base of the thyroid cartilage.

The posterior margin of each lamina of the thyroid cartilage is elongated to form a **superior horn** and an **inferior horn**:

- The medial surface of the inferior horn has a facet for articulation with the cricoid cartilage.
- The superior horn is connected by a **lateral thyrohyoid ligament** to the posterior end of the greater horn of the hyoid bone.

The lateral surface of each thyroid lamina is marked by a ridge (the **oblique line**), which curves anteriorly from the base of the superior horn to a little short of midway along the inferior margin of the lamina.

The ends of the oblique line are expanded to form **superior** and **inferior thyroid tubercles**. The oblique line is a site of attachment for the extrinsic muscles of the larynx (sternothyroid, thyrohyoid, and inferior constrictor).

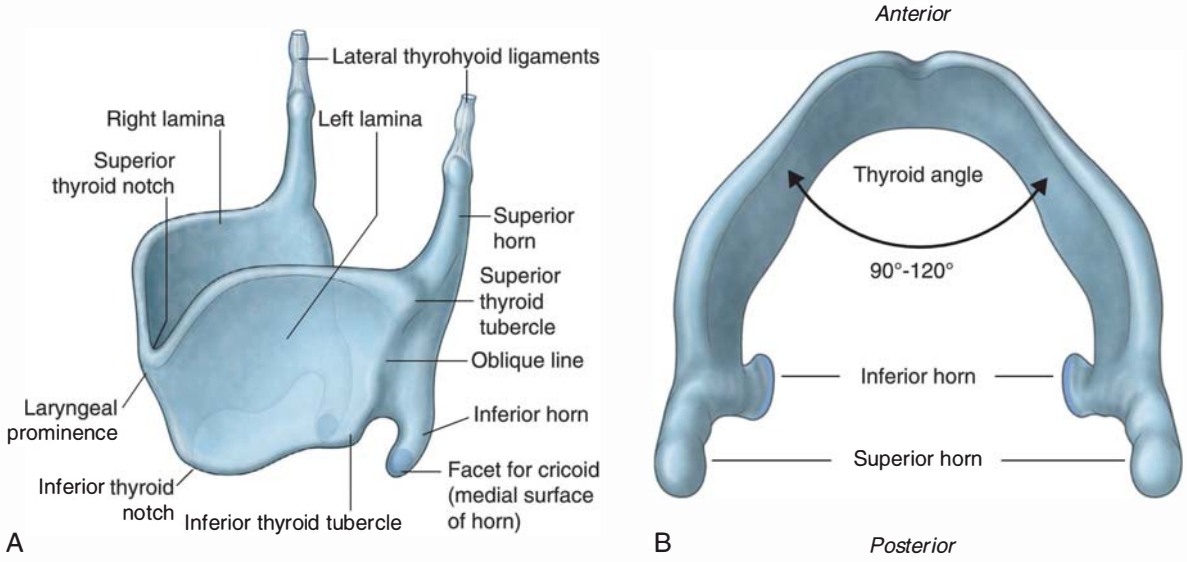


Fig. 8.207 Thyroid cartilage. **A.** Anterolateral view. **B.** Superior view.

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Epiglottis

The epiglottis is a leaf-shaped cartilage attached by its stem to the posterior aspect of the thyroid cartilage at the angle (Fig. 8.208) and projects posterosuperiorly from its attachment to the thyroid cartilage. The attachment is via the **thyro-epiglottic ligament** in the midline approximately midway between the laryngeal prominence and the inferior thyroid notch. The upper margin of the epiglottis is behind the pharyngeal part of the tongue.

The inferior half of the posterior surface of the epiglottis is raised slightly to form an epiglottic tubercle.

Arytenoid cartilages

The two arytenoid cartilages are pyramid-shaped cartilages with three surfaces, a **base of arytenoid cartilage** and an **apex of arytenoid cartilage** (Fig. 8.209):

- The base is concave and articulates with the sloping articular facet on the superolateral surface of the lamina of cricoid cartilage.
- The apex articulates with a corniculate cartilage.
- The **medial surface** of each cartilage faces the other.
- The **anterolateral surface** has two depressions, separated by a ridge, for muscle (vocalis) and ligament (vestibular ligament) attachment.

The anterior angle of the base is elongated into a **vocal process** to which the vocal ligament is attached. The lateral angle is similarly elongated into a muscular process for attachment of the posterior and lateral crico-arytenoid muscles.

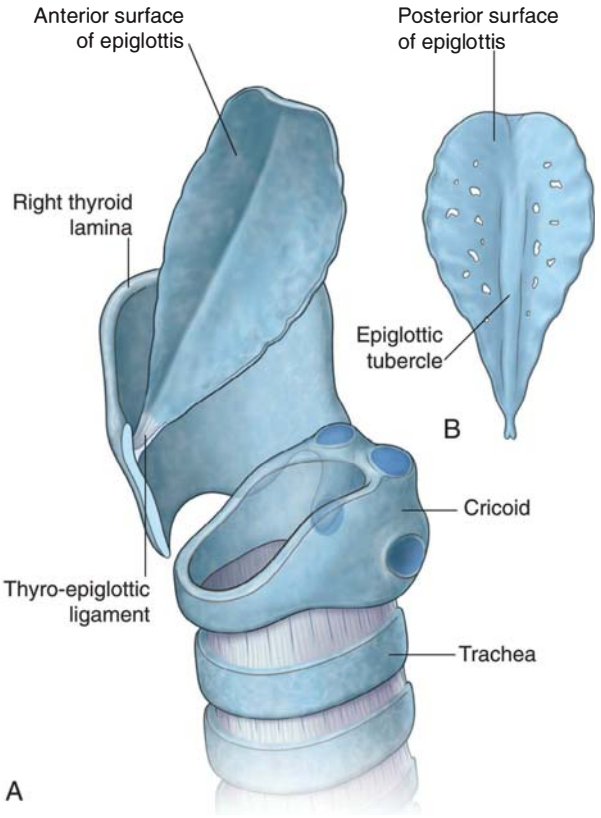


Fig. 8.208 Epiglottis. **A.** Anterolateral view. **B.** Posterior surface.

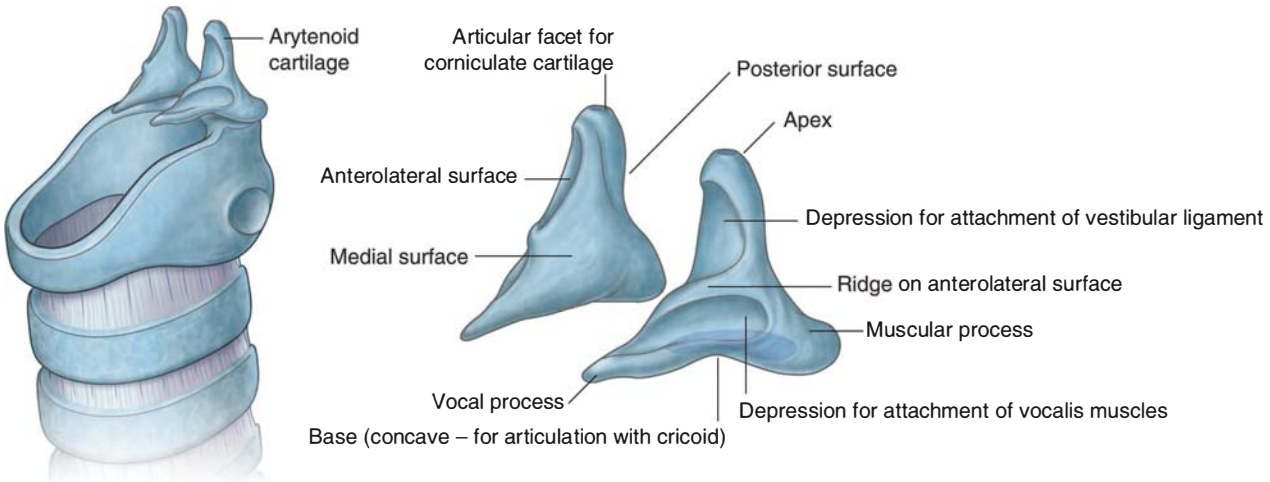


Fig. 8.209 Arytenoid cartilages.

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Corniculate

The corniculate cartilages (Fig. 8.210) are two small conical cartilages whose bases articulate with the apices of the arytenoid cartilages. Their apices project posteromedially toward each other.

Cuneiform

These two small club-shaped cartilages (Fig. 8.210) lie anterior to the corniculate cartilages and are suspended in the part of the fibro-elastic membrane of the larynx that attaches the arytenoid cartilages to the lateral margin of the epiglottis.

Extrinsic ligaments

Thyrohyoid membrane

The thyrohyoid membrane is a tough fibro-elastic ligament that spans between the superior margin of the thyroid cartilage below and the hyoid bone above (Fig. 8.211). It is attached to the superior margin of the thyroid laminae and adjacent anterior margins of the superior horns, and ascends medial to the greater horns and posterior to the body of the hyoid bone to attach to the superior margins of these structures.

An aperture in the lateral part of the thyrohyoid membrane on each side is for the superior laryngeal artery, the internal branch of the superior laryngeal nerve, and lymphatics.

The posterior borders of the thyrohyoid membrane are thickened to form the **lateral thyrohyoid ligaments**. The membrane is also thickened anteriorly in the midline to form the **median thyrohyoid ligament**.

Occasionally, there is a small cartilage (**triticeal cartilage**) in each lateral thyrohyoid ligament.

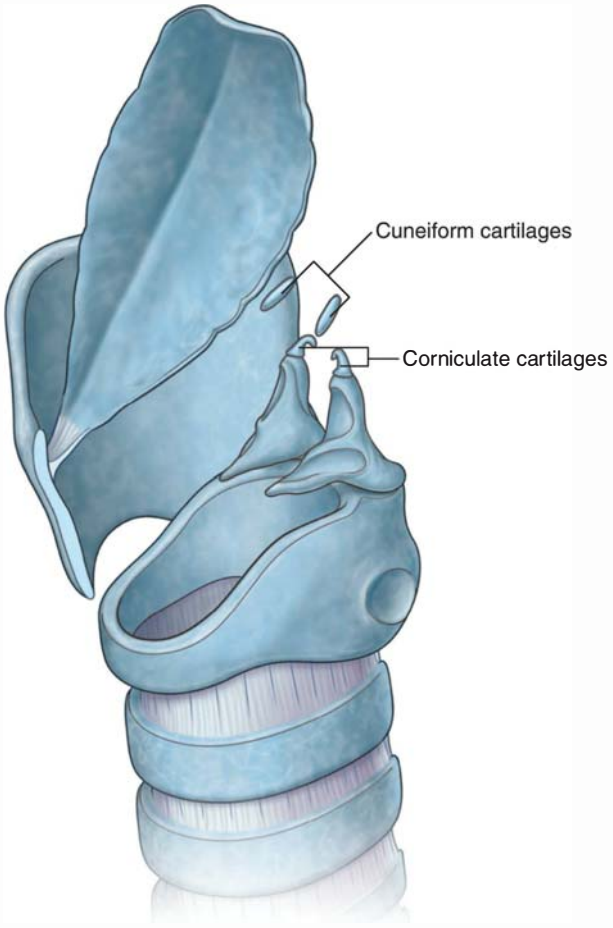


Fig. 8.210 Corniculate and cuneiform cartilages.

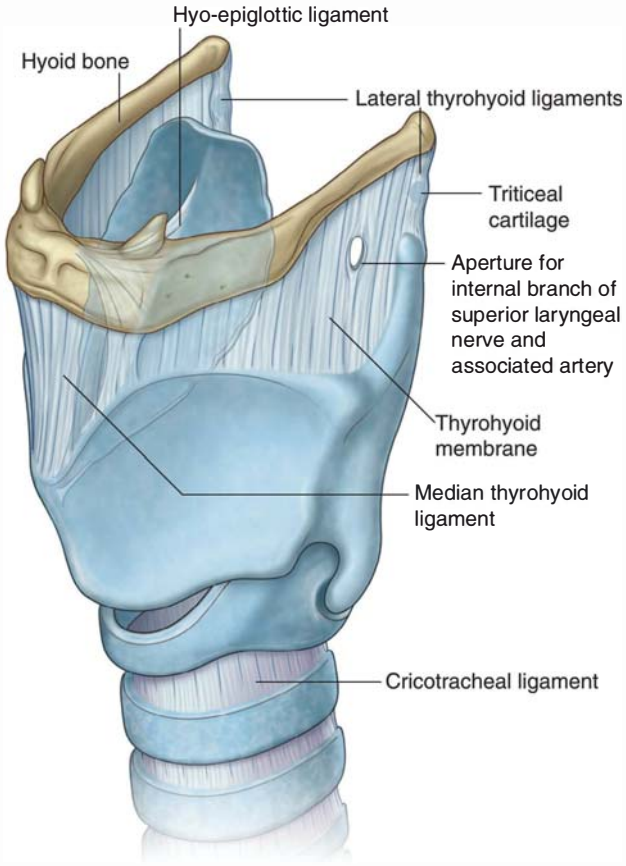


Fig. 8.211 Extrinsic ligaments of the larynx.

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Hyo-epiglottic ligament

The hyo-epiglottic ligament (Fig. 8.211) extends from the midline of the epiglottis, anterosuperiorly to the body of the hyoid bone.

Cricotracheal ligament

The cricotracheal ligament (Fig. 8.211) runs from the lower border of the cricoid cartilage to the adjacent upper border of the first tracheal cartilage.

Intrinsic ligaments

Fibro-elastic membrane of the larynx

The fibro-elastic membrane of the larynx links together the laryngeal cartilages and completes the architectural framework of the laryngeal cavity. It is composed of two parts—a lower cricothyroid ligament and an upper quadrangular membrane.

Cricothyroid ligament (cricovocal membrane, cricothyroid membrane)

The cricothyroid ligament (Fig. 8.212) is attached to the arch of cricoid cartilage and extends superiorly to end in a free upper margin within the space enclosed by the thyroid cartilage. On each side, this upper free margin attaches:

- anteriorly to the thyroid cartilage, and
- posteriorly to the vocal processes of the arytenoid cartilages.

The free margin between these two points of attachment is thickened to form the **vocal ligament**, which is under the **vocal fold** (**true vocal cord**) of the larynx.

The cricothyroid ligament is also thickened anteriorly in the midline to form a distinct **median cricothyroid ligament**, which spans the distance between the arch of cricoid cartilage and the inferior thyroid notch and adjacent deep surface of the thyroid cartilage up to the attachment of the vocal ligaments.

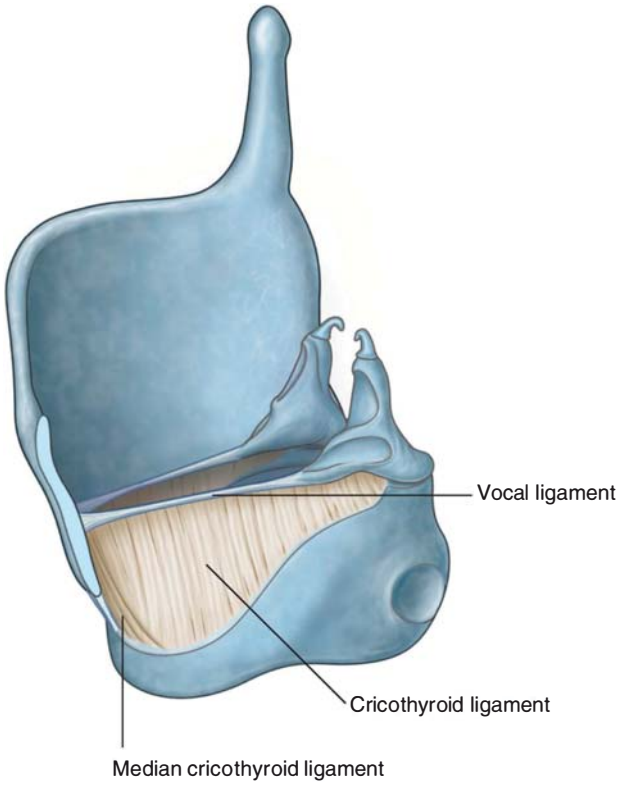


Fig. 8.212 Cricothyroid ligament.

In emergency situations, when the airway is blocked above the level of the vocal folds, the median cricothyroid ligament can be perforated to establish an airway. Except for small vessels and the occasional presence of a pyramidal lobe of the thyroid gland, normally there are few structures between the median cricothyroid ligament and skin.

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Quadrangular membrane

The quadrangular membrane on each side runs between the lateral margin of the epiglottis and the anterolateral surface of the arytenoid cartilage on the same side (Fig. 8.213). It is also attached to the corniculate cartilage, which articulates with the apex of arytenoid cartilage.

Each quadrangular membrane has a free upper margin, between the top of the epiglottis and the corniculate cartilage, and a free lower margin. The free lower margin is thickened to form the **vestibular ligament** under the **vestibular fold** (**false vocal cord**) of the larynx.

The vestibular ligament is attached posteriorly to the superior depression on the anterolateral surface of the arytenoid cartilage and anteriorly to the thyroid angle just superior to the attachment of the vocal ligament.

On each side, the vestibular ligament of the quadrangular membrane is separated from the vocal ligament of the cricothyroid ligament below by a gap. Because the vestibular ligament attaches to the anterolateral surface of the arytenoid cartilage and the vocal ligament attaches to the vocal process of the same cartilage, the vestibular ligament is lateral to the vocal ligament when viewed from above (Fig. 8.214).

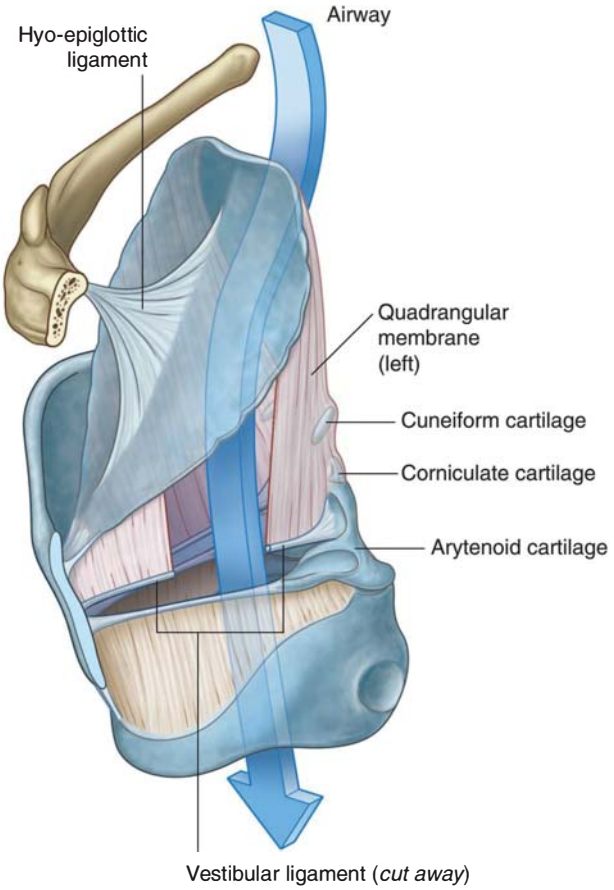


Fig. 8.213 Quadrangular membrane.

Laryngeal joints

Cricothyroid joints

The joints between the inferior horns of the thyroid cartilage and the cricoid cartilage, and between the cricoid cartilage and arytenoid cartilages are synovial. Each is surrounded by a capsule and is reinforced by associated ligaments. The cricothyroid joints enable the thyroid cartilage to move forward and tilt downward on the cricoid cartilage (Fig. 8.215).

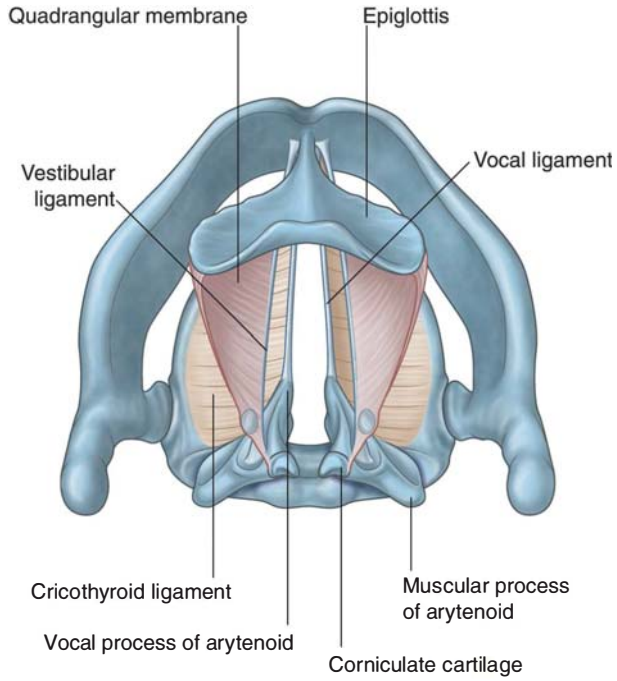


Fig. 8.214 Fibro-elastic membrane of the larynx (superior view).

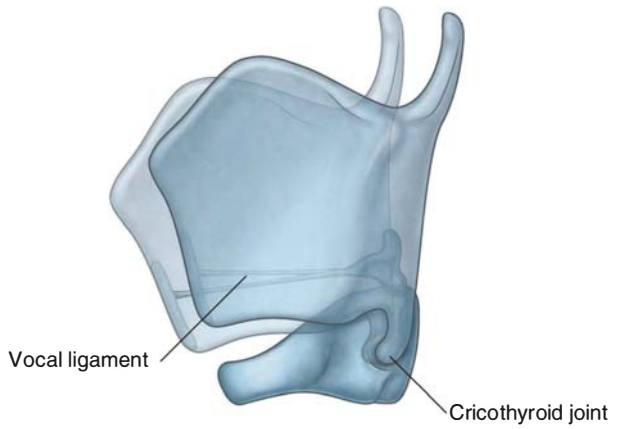


Fig. 8.215 Movements of the cricothyroid joints.

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Because the vocal ligaments pass between the posterior aspect of the thyroid angle and the arytenoid cartilages that sit on the lamina of cricoid cartilage, forward movement and downward rotation of the thyroid cartilage on the cricoid cartilage effectively lengthens and puts tension on the vocal ligaments.

Crico-arytenoid joints

The crico-arytenoid joints between articular facets on the superolateral surfaces of the cricoid cartilage and the bases of the arytenoid cartilages enable the arytenoid cartilages to slide away or toward each other and to rotate so that the vocal processes pivot either toward or away from the midline. These movements abduct and adduct the vocal ligaments (Fig. 8.216).

Cavity of the larynx

The central cavity of the larynx (Fig. 8.217) is tubular and lined by mucosa. Its architectural support is provided by the fibro-elastic membrane of the larynx and by the laryngeal cartilages to which it is attached.

The superior aperture of the cavity (laryngeal inlet) opens into the anterior aspect of the pharynx just below and posterior to the tongue (Fig. 8.217A):

- Its anterior border is formed by mucosa covering the superior margin of the epiglottis.
- Its lateral borders are formed by mucosal folds (**aryepiglottic folds**), which enclose the superior margins of the quadrangular membranes and adjacent soft tissues, and two tubercles on the more posterolateral margin of the laryngeal inlet on each side mark the positions of the underlying cuneiform and corniculate cartilages.
- Its posterior border in the midline is formed by a mucosal fold that forms a depression (**interarytenoid notch**) between the two corniculate tubercles.

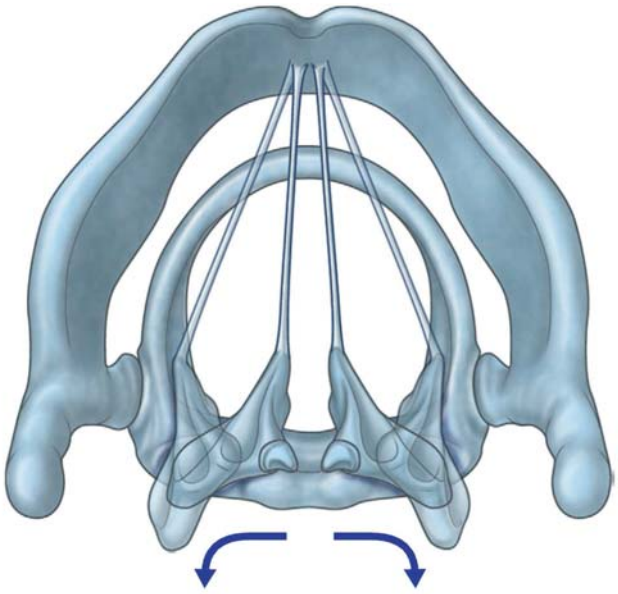


Fig. 8.216 Movements of the crico-arytenoid joints.

The inferior opening of the laryngeal cavity is continuous with the lumen of the trachea, is completely encircled by the cricoid cartilage, and is horizontal in position unlike the laryngeal inlet, which is oblique and points posterosuperiorly into the pharynx. In addition, the inferior opening is continuously open, whereas the laryngeal inlet can be closed by downward movement of the epiglottis.

Division into three major regions

Two pairs of mucosal folds, the vestibular and vocal folds, which project medially from the lateral walls of the laryngeal cavity, constrict it and divide it into three major regions—the vestibule, a middle chamber, and the infraglottic cavity (Fig. 8.217B):

- The **vestibule** is the upper chamber of the laryngeal cavity between the laryngeal inlet and the vestibular folds, which enclose the vestibular ligaments and associated soft tissues.
- The middle part of the laryngeal cavity is very thin and is between the vestibular folds above and the vocal folds below.
- The **infraglottic space** is the most inferior chamber of the laryngeal cavity and is between the vocal folds (which enclose the vocal ligaments and related soft tissues) and the inferior opening of the larynx.

Laryngeal ventricles and saccules

On each side, the mucosa of the middle cavity bulges laterally through the gap between the vestibular and vocal ligaments to produce an expanded trough-shaped space (a **laryngeal ventricle**) (Fig. 8.217A). An elongate tubular extension of each ventricle (laryngeal saccule) projects anterosuperiorly between the vestibular fold and thyroid cartilage and may reach as high as the top of the thyroid cartilage. Within the walls of these laryngeal saccules are numerous mucous glands. Mucus secreted into the saccules lubricates the vocal folds.

Rima vestibuli and rima glottidis

When viewed from above (Fig. 8.217C,D), there is a triangular opening (the **rima vestibuli**) between the two

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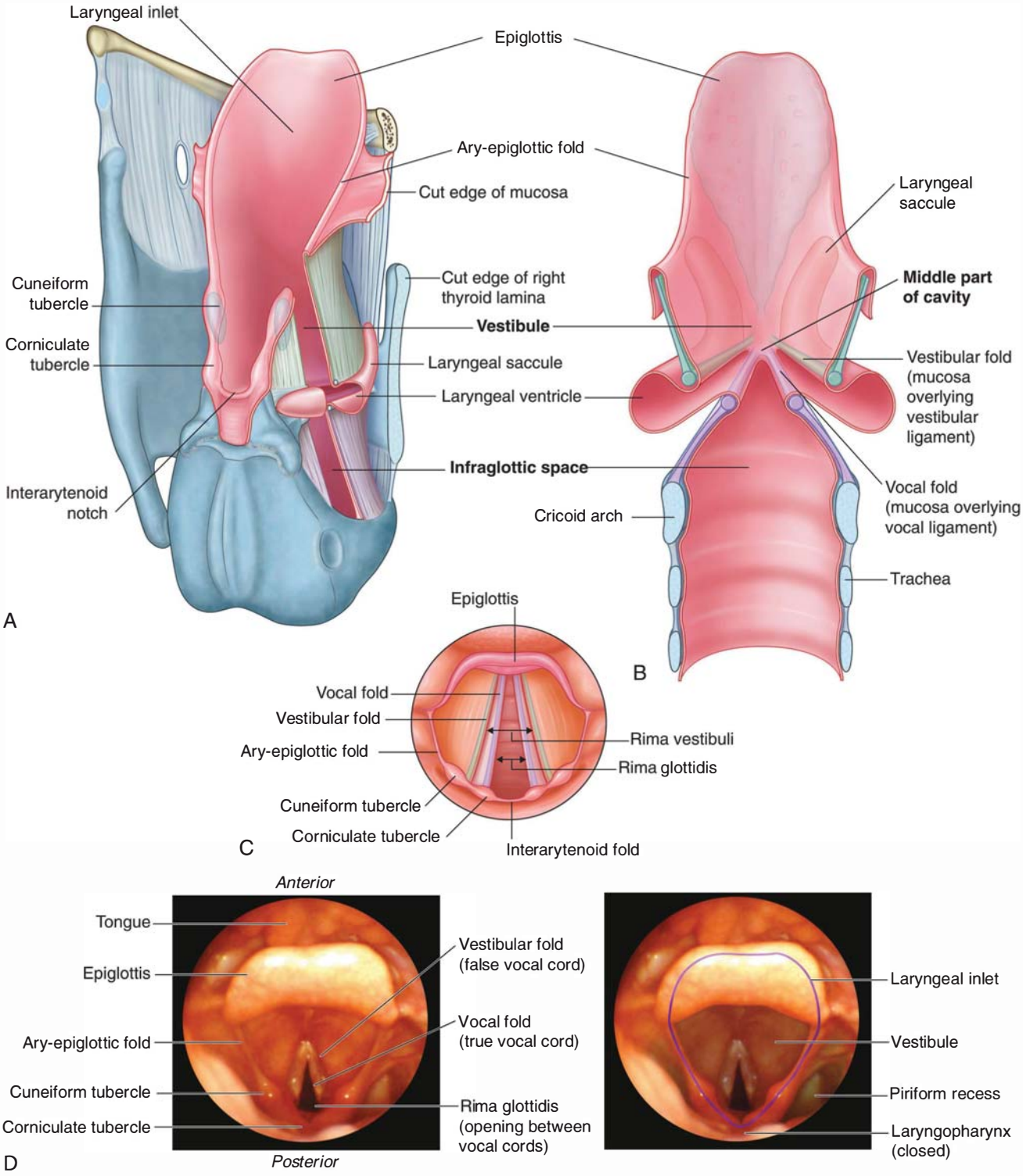


Fig. 8.217 Laryngeal cavity. **A.** Posterolateral view. **B.** Posterior view (cut away). **C.** Superior view through the laryngeal inlet. **D.** Labeled photograph of the larynx, superior view.

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adjacent vestibular folds at the entrance to the middle chamber of the laryngeal cavity. The apex of the opening is anterior and its base is formed by the posterior wall of the laryngeal cavity.

Inferior to the vestibular folds, the vocal folds (true vocal cords) and adjacent mucosa-covered parts of the arytenoid cartilages form the lateral walls of a similar, but narrower, triangular opening (the **rima glottidis** between the two adjacent vocal folds). This opening separates the middle chamber above from the infraglottic cavity below. The base of this triangular opening is formed by the fold of mucosa (**interarytenoid fold**) at the bottom of the interarytenoid notch.

Both the rima glottidis and the rima vestibuli can be opened and closed by movement of the arytenoid cartilages and associated fibro-elastic membranes.

Intrinsic muscles

The intrinsic muscles of the larynx (Table 8.19) adjust tension in the vocal ligaments, open and close the rima glottidis, control the inner dimensions of the vestibule, close the rima vestibuli, and facilitate closing of the laryngeal inlet. They do this mainly by:

- acting on the cricothyroid and crico-arytenoid joints,
- adjusting the distance between the epiglottis and arytenoid cartilages,
- pulling directly on the vocal ligaments, and
- forcing soft tissues associated with the quadrangular membranes and vestibular ligaments toward the midline.

Table 8.19 Intrinsic muscles of the larynx

Muscle	Origin	Insertion	Innervation	Function
Cricothyroid	Anterolateral aspect of arch of cricoid cartilage	Oblique part—inferior horn of the thyroid cartilage; straight part—inferior margin of thyroid cartilage	External branch of superior laryngeal nerve from the vagus nerve [X]	Forward and downward rotation of the thyroid cartilage at the cricothyroid joint
Posterior crico-arytenoid	Oval depression on posterior surface of lamina of cricoid cartilage	Posterior surface of muscular process of arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	Abduction and external rotation of the arytenoid cartilage. The posterior crico-arytenoid muscles are the primary abductors of the vocal folds. In other words, they are the primary openers of the rima glottidis.
Lateral crico-arytenoid	Superior surface of arch of cricoid cartilage	Anterior surface of muscular process of arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	Internal rotation of the arytenoid cartilage and adduction of vocal folds
Transverse arytenoid	Lateral border of posterior surface of arytenoid cartilage	Lateral border of posterior surface of opposite arytenoid cartilage	Recurrent laryngeal branch of the vagus nerve [X]	Adduction of arytenoid cartilages
Oblique arytenoid	Posterior surface of muscular process of arytenoid cartilage	Posterior surface of apex of adjacent arytenoid cartilage; extends into ary-epiglottic fold	Recurrent laryngeal branch of the vagus nerve [X]	Sphincter of the laryngeal inlet
Thyro-arytenoid	Thyroid angle and adjacent cricothyroid ligament	Anterolateral surface of arytenoid cartilage; some fibers continue in ary-epiglottic folds to the lateral margin of the epiglottis	Recurrent laryngeal branch of the vagus nerve [X]	Sphincter of vestibule and of laryngeal inlet
Vocalis	Lateral surface of vocal process of arytenoid cartilage	Vocal ligament and thyroid angle	Recurrent laryngeal branch of the vagus nerve [X]	Adjusts tension in vocal folds

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Cricothyroid muscles

The fan-shaped **cricothyroid muscles** are attached to the anterolateral surfaces of the arch of the cricoid cartilage and expand superiorly and posteriorly to attach to the thyroid cartilage (Fig. 8.218).

Each muscle has an oblique part and a straight part:

- The **oblique part** runs in a posterior direction from the arch of the cricoid cartilage to the inferior horn of the thyroid cartilage.
- The **straight part** runs more vertically from the arch of the cricoid cartilage to the posteroinferior margin of the thyroid lamina.

The cricothyroid muscles move the cricothyroid joints. They pull the thyroid cartilage forward and rotate it down relative to the cricoid cartilage. These actions lengthen the vocal folds.

The cricothyroid muscles are the only intrinsic muscles of the larynx innervated by the superior laryngeal branches of the vagus nerves [X]. All other intrinsic muscles are innervated by the recurrent laryngeal branches of the vagus nerves [X].

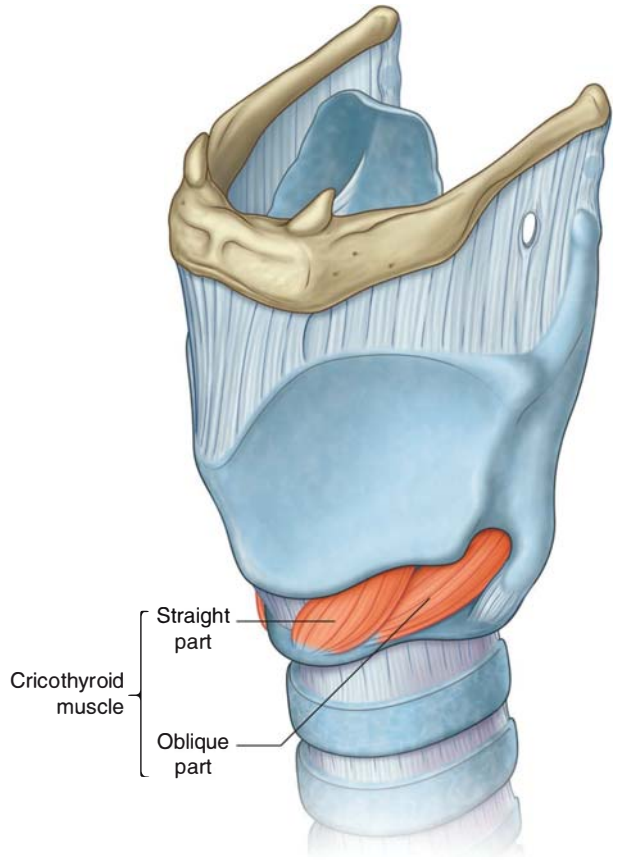


Fig. 8.218 Cricothyroid muscle.

Posterior crico-arytenoid muscles

There is a right and a left **posterior crico-arytenoid muscle** (Fig. 8.219). The fibers of each muscle originate from a large shallow depression on the posterior surface of the lamina of the cricoid cartilage, and run superiorly and laterally to converge on the muscular processes of the arytenoid cartilage.

The posterior crico-arytenoid muscles abduct and externally (laterally) rotate the arytenoid cartilages, thereby opening the rima glottidis. These muscles are the primary abductors of the vocal folds. They are innervated by the recurrent laryngeal branches of the vagus nerves [X].

Lateral crico-arytenoid muscles

The **lateral crico-arytenoid muscle** on each side originates from the upper surface of the arch of the cricoid cartilage, and runs posteriorly and superiorly to insert on the muscular process of the arytenoid cartilage (Fig. 8.219).

The lateral crico-arytenoid muscles internally rotate the arytenoid cartilages. These movements result in adducted (closed) vocal folds.

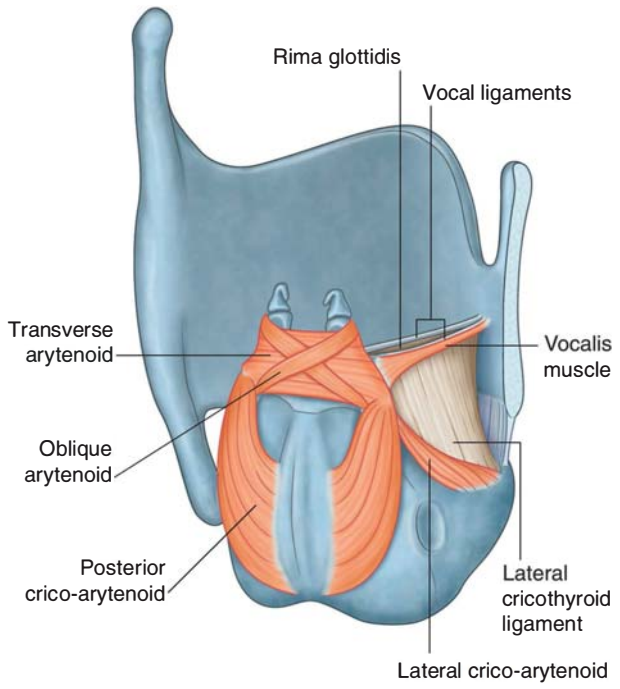


Fig. 8.219 Crico-arytenoid, oblique and transverse arytenoid, and vocalis muscles.

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The lateral crico-arytenoids are innervated by the recurrent laryngeal branches of the vagus nerves [X].

Transverse arytenoid muscle

The single **transverse arytenoid muscle** spans the distance between adjacent lateral margins of the arytenoid cartilages and covers the posterior surfaces of these cartilages (Fig. 8.219). It adducts the arytenoid cartilages and is innervated by the recurrent laryngeal branches of the vagus nerves [X].

Oblique arytenoid muscles

Each of the two **oblique arytenoid muscles** runs from the posterior surface of the muscular process of one arytenoid cartilage to the apex of the arytenoid cartilage on the other side (Fig. 8.219). Some fibers of the muscle continue laterally around the margin of the arytenoid cartilage and into the ary-epiglottic fold where they continue as the **ary-epiglottic part** of the muscle (Fig. 8.220).

The oblique arytenoids can narrow the laryngeal inlet by constricting the distance between the arytenoid cartilages and the epiglottis. They are innervated by the recurrent laryngeal branches of the vagus nerves [X].

Vocalis

The **vocalis muscles** are elongate muscles lateral to and running parallel with each vocal ligament (Fig. 8.219). The fibers in each muscle are attached posteriorly to the lateral surface of the vocal process and adjacent depression on the anterolateral surface of the arytenoid cartilage, and anteriorly insert along the length of the vocal ligament to the thyroid angle.

The vocalis muscles adjust tension in the vocal folds and are innervated by the recurrent laryngeal branches of the vagus nerves [X].

Thyro-arytenoid muscles

The two **thyro-arytenoid muscles** are broad flat muscles lateral to the fibro-elastic membrane of the larynx and the laryngeal ventricles and saccules (Fig. 8.220). Each muscle runs from a vertical line of origin on the lower half of the thyroid angle and adjacent external surface of the cricothyroid ligament to the anterolateral surface of the arytenoid cartilage. Some of the fibers may continue into the ary-epiglottic fold and reach the margin of the epiglottis. These fibers are the **thyro-epiglottic part** of the muscle.

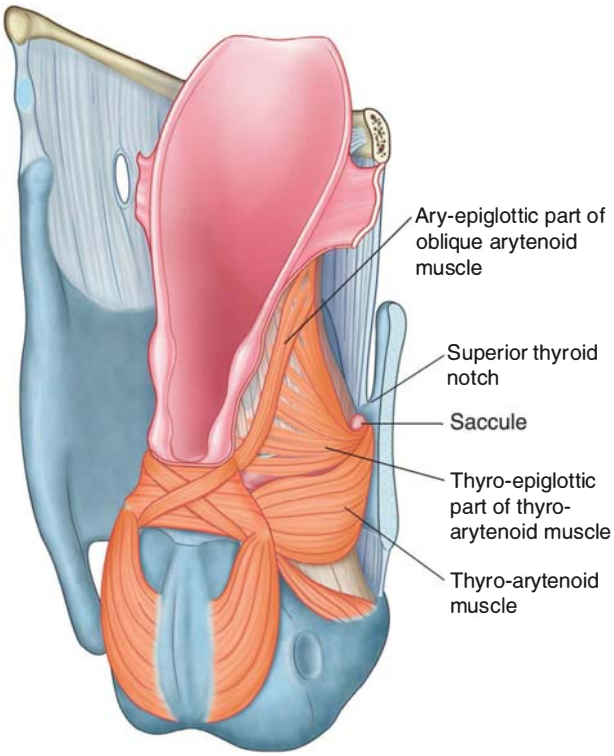


Fig. 8.220 Thyro-arytenoid muscle.

Because the thyro-arytenoid muscles are broad and lateral to the quadrangular membrane, they act as a sphincter of the vestibule by pushing soft tissues medial to the muscles toward the midline. The muscles also narrow the laryngeal inlet by pulling the arytenoid cartilages forward while simultaneously pulling the epiglottis toward the arytenoid cartilages.

The thyro-arytenoid muscles are innervated by the recurrent laryngeal branches of the vagus nerves [X].

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Function of the larynx

The larynx is an elaborate sphincter for the lower respiratory tract and provides a mechanism for producing sounds. Adjustments of the size of the central cavity of the larynx result from changes in the dimensions of the rima glottidis, the rima vestibuli, the vestibule, and the laryngeal inlet (Fig. 8.221). These changes result from muscle actions and laryngeal mechanics.

Respiration

During quiet respiration, the laryngeal inlet, vestibule, rima vestibuli, and rima glottidis are open. The arytenoid cartilages are abducted and the rima glottidis is triangular shaped (Fig. 8.221A). During forced inspiration (Fig. 8.221B), the arytenoid cartilages are rotated laterally, mainly by the action of the posterior crico-arytenoid muscles. As a result, the vocal folds are abducted and the

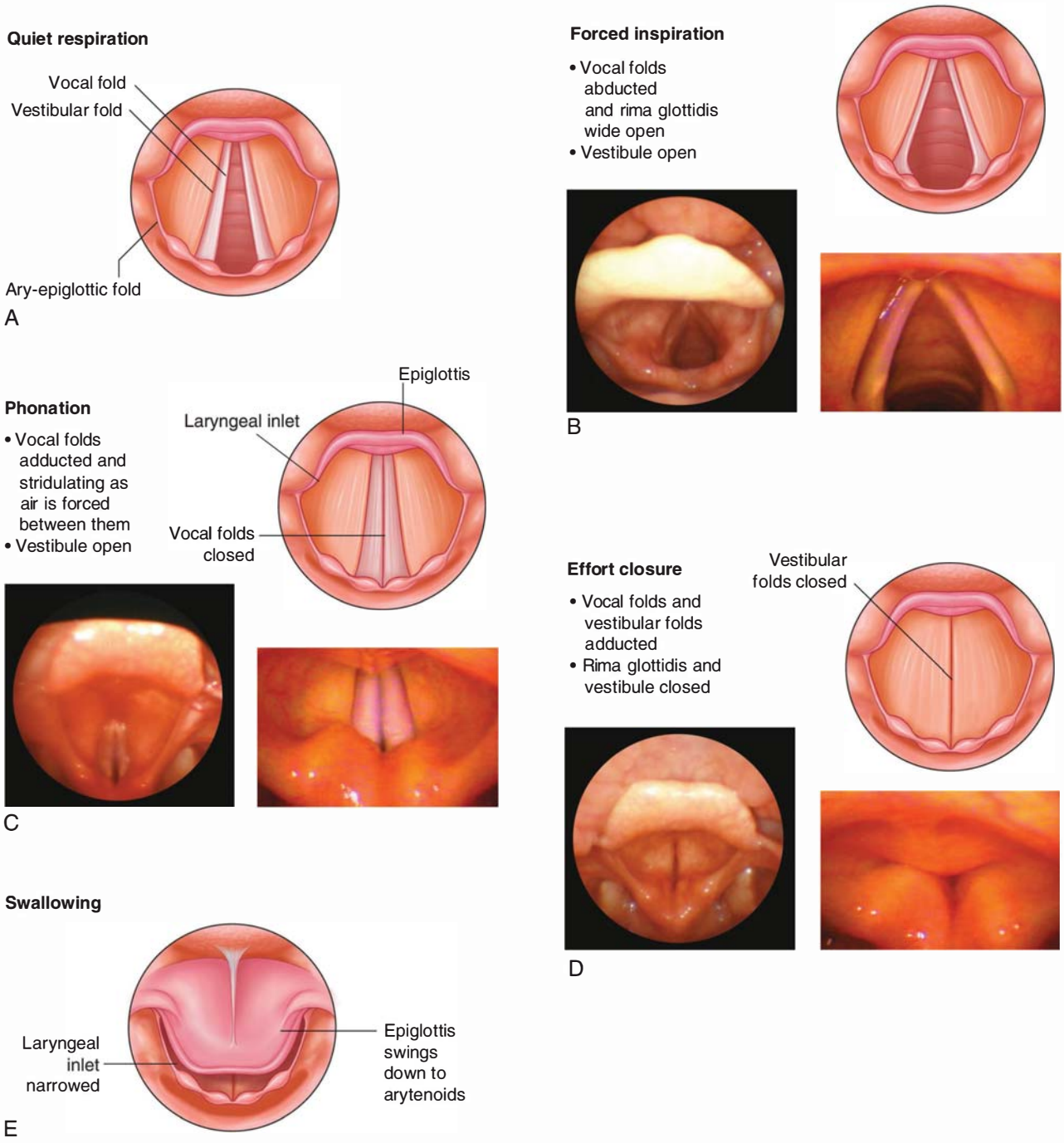


Fig. 8.221 Laryngeal function. **A.** Quiet respiration. **B.** Forced inspiration. **C.** Phonation. **D.** Effort closure. **E.** Swallowing.

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rima glottidis widens into a rhomboid shape, which effectively increases the diameter of the laryngeal airway.

Phonation

When phonating, the arytenoid cartilages and vocal folds are adducted and air is forced through the closed rima glottidis (Fig. 8.221C). This action causes the vocal folds to vibrate against each other and produce sounds, which can then be modified by the upper parts of the airway and oral cavity. Tension in the vocal folds can be adjusted by the vocalis and cricothyroid muscles.

Effort closure

Effort closure of the larynx (Fig. 8.221D) occurs when air is retained in the thoracic cavity to stabilize the trunk, for example during heavy lifting, or as part of the mechanism for increasing intra-abdominal pressure. During effort closure, the rima glottidis is completely closed, as is the rima vestibuli and lower parts of the vestibule. The result is to completely and forcefully shut the airway.

Swallowing

During swallowing, the rima glottidis, rima vestibuli, and vestibule are closed and the laryngeal inlet is narrowed. In addition, the larynx moves up and forward. This action causes the epiglottis to swing downward toward the arytenoid cartilages and to effectively narrow or close the laryngeal inlet (Fig. 8.221E). The up and forward movement of the larynx also opens the esophagus, which is attached to the posterior aspect of the lamina of the cricoid cartilage. All these actions together prevent solids and liquids from entry into the airway and facilitate their movement through the piriform fossae into the esophagus.

In the clinic

Tracheostomy

A tracheostomy is a procedure in which a hole is made in the trachea and a tube is inserted to enable ventilation.

In the emergency situation a tracheostomy is typically performed when there is obstruction to the larynx as a result of inhalation of a foreign body, severe edema secondary to anaphylactic reaction, or severe head and neck trauma.

In the emergency situation the cricothyroid ligament can be identified by simple palpation and a small needle can be inserted to establish an airway. This procedure is called a "cricothyrotomy."

The typical situation in which a tracheostomy is performed is in the calm atmosphere of an operating theater. A small transverse incision is placed in the lower third of the neck anteriorly. The strap muscles are deviated laterally and the trachea can be easily visualized. Occasionally it is necessary to divide the isthmus of the thyroid gland. An incision is made in the second and third tracheal rings and a small tracheostomy tube inserted.

After the tracheostomy has been in situ for the required length of time, it is simply removed. The hole through which it was inserted almost inevitably closes without any intervention.

Patients with long-term tracheostomies are unable to vocalize because no air is passing through the vocal cords.

In the clinic

Laryngoscopy

Laryngoscopy is a medical procedure that is used to inspect the larynx. The functions of laryngoscopy include the evaluation of patients with difficulty swallowing, assessment of the vocal cords, and assessment of the larynx for tumors, masses, and weak voice.

The larynx is typically visualized using two methods. Indirect laryngoscopy involves passage of a small rod-mounted mirror (not dissimilar to a dental mirror) into the oropharynx permitting indirect visualization of the larynx. Direct laryngoscopy can be performed using a device with a curved metal tip that holds the tongue and epiglottis forward, allowing direct inspection of the larynx. This procedure can be performed only in the unconscious patient or in a patient in whom the gag reflex is not intact. Other methods of inspection include the passage of fiberoptic endoscopes through either the oral cavity or nasal cavity.

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Vessels

Arteries

The major blood supply to the larynx is by the superior and inferior laryngeal arteries (Fig. 8.222):

- The **superior laryngeal artery** originates near the upper margin of the thyroid cartilage from the superior thyroid branch of the external carotid artery, and accompanies the internal branch of the superior laryngeal nerve through the thyrohyoid membrane to reach the larynx.
- The **inferior laryngeal artery** originates from the inferior thyroid branch of the thyrocervical trunk of the subclavian artery low in the neck and, together with the recurrent laryngeal nerve, ascends in the groove between the esophagus and trachea—it enters the larynx by passing deep to the margin of the inferior constrictor muscle of the pharynx.

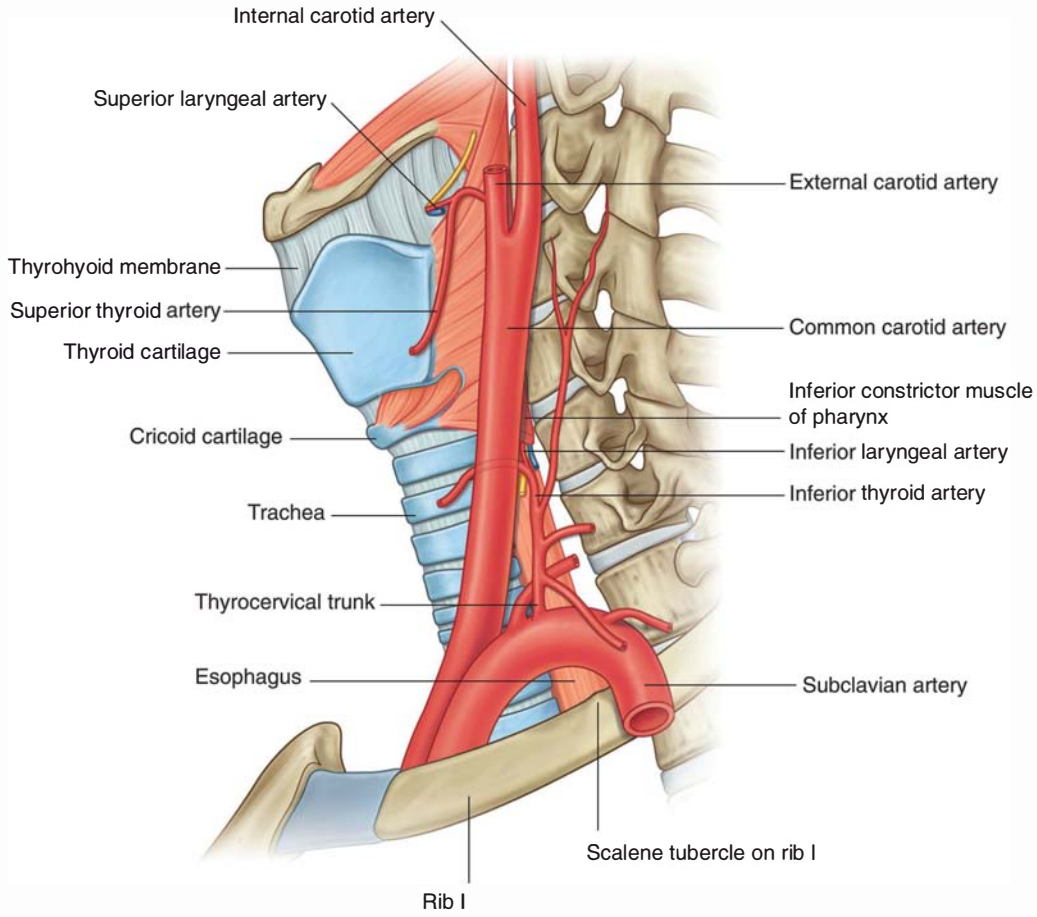


Fig. 8.222 Arterial supply of the larynx, left lateral view.

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Veins

Veins draining the larynx accompany the arteries:

- **Superior laryngeal veins** drain into superior thyroid veins, which in turn drain into the internal jugular veins (Fig. 8.223).
- **Inferior laryngeal veins** drain into inferior thyroid veins, which drain into the left brachiocephalic vein.

Lymphatics

Lymphatics drain regions above and below the vocal folds:

- Those above the vocal folds follow the superior laryngeal artery and terminate in deep cervical nodes associated with the bifurcation of the common carotid artery.
- Those below the vocal folds drain into deep nodes associated with the inferior thyroid artery or with nodes associated with the front of the cricothyroid ligament or upper trachea.

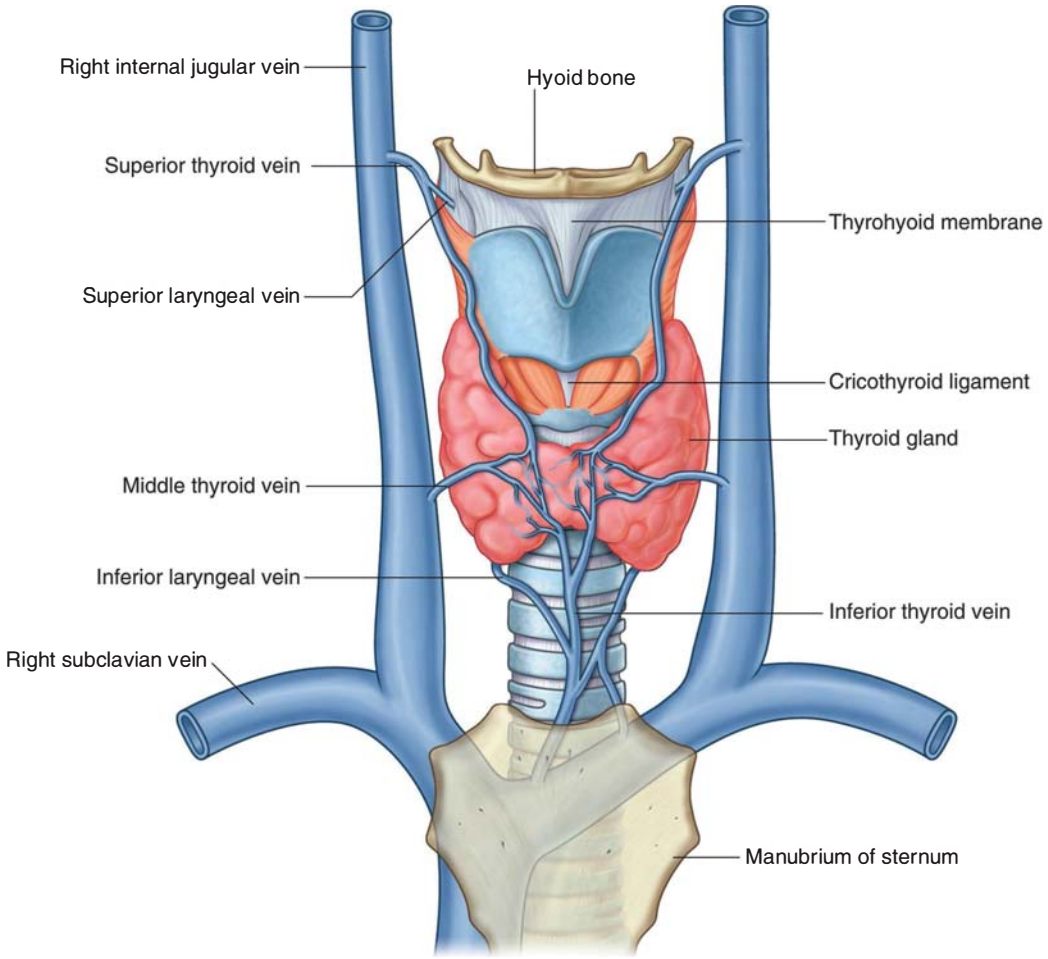


Fig. 8.223 Venous drainage of the larynx, anterior view.

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Nerves

Sensory and motor innervation of the larynx is by two branches of the vagus nerves [X]—the superior laryngeal nerves and the recurrent laryngeal nerves (Fig. 8.224).

Superior laryngeal nerves

The **superior laryngeal nerves** originate from the inferior vagal ganglia high in the neck (Fig. 8.224). On each side, the nerve descends medial to the internal carotid artery and divides into **internal** and **external branches** just above the level of the superior horn of the hyoid bone:

- The external branch (external laryngeal nerve) descends along the lateral wall of the pharynx to supply and penetrate the inferior constrictor of the pharynx and ends by supplying the cricothyroid muscle.
- The internal branch (internal laryngeal nerve) passes anteroinferiorly to penetrate the thyrohyoid membrane—it is mainly sensory and supplies the laryngeal cavity down to the level of the vocal folds.

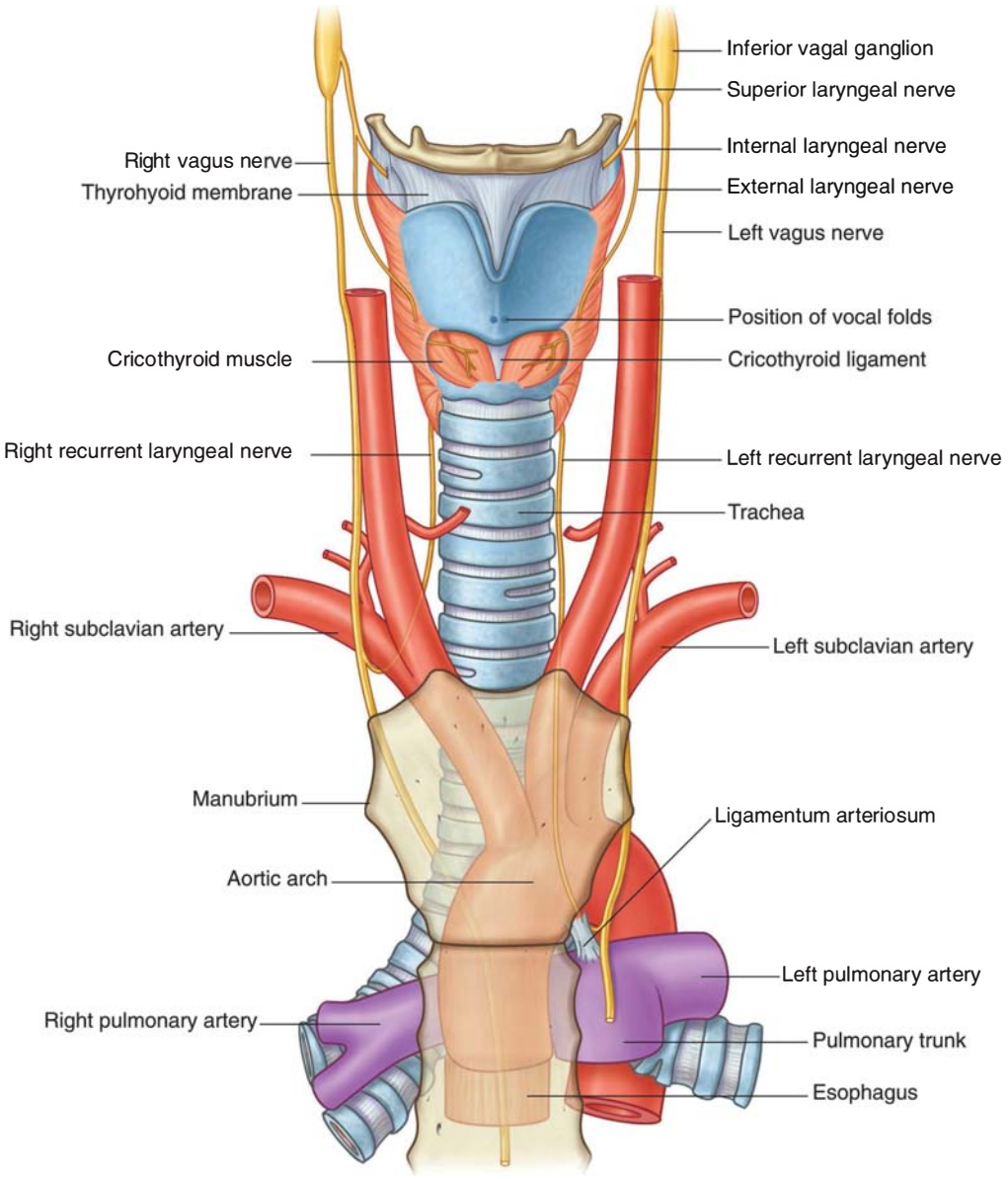


Fig. 8.224 Innervation of the larynx.

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Recurrent laryngeal nerves

The recurrent laryngeal nerves are (Fig. 8.224):

- sensory to the laryngeal cavity below the level of the vocal folds, and
- motor to all intrinsic muscles of the larynx except for the cricothyroid.

The left recurrent laryngeal nerve originates in the thorax, whereas the right recurrent laryngeal nerve originates in the root of the neck. Both nerves generally ascend in the neck in the groove between the esophagus and trachea and enter the larynx deep to the margin of the inferior constrictor. They may pass medial to, lateral to, or through the lateral ligament of the thyroid gland, which attaches the thyroid gland to the trachea and lower part of the cricoid cartilage on each side.

NASAL CAVITIES

The two nasal cavities are the uppermost parts of the respiratory tract and contain the olfactory receptors. They are elongated wedge-shaped spaces with a large inferior base and a narrow superior apex (Figs. 8.225 and 8.226) and are held open by a skeletal framework consisting mainly of bone and cartilage.

The smaller anterior regions of the cavities are enclosed by the external nose, whereas the larger posterior regions are more central within the skull. The anterior apertures of the nasal cavities are the nares, which open onto the inferior surface of the nose. The posterior apertures are the choanae, which open into the nasopharynx.

The nasal cavities are separated:

- from each other by a midline nasal septum,
- from the oral cavity below by the hard palate, and

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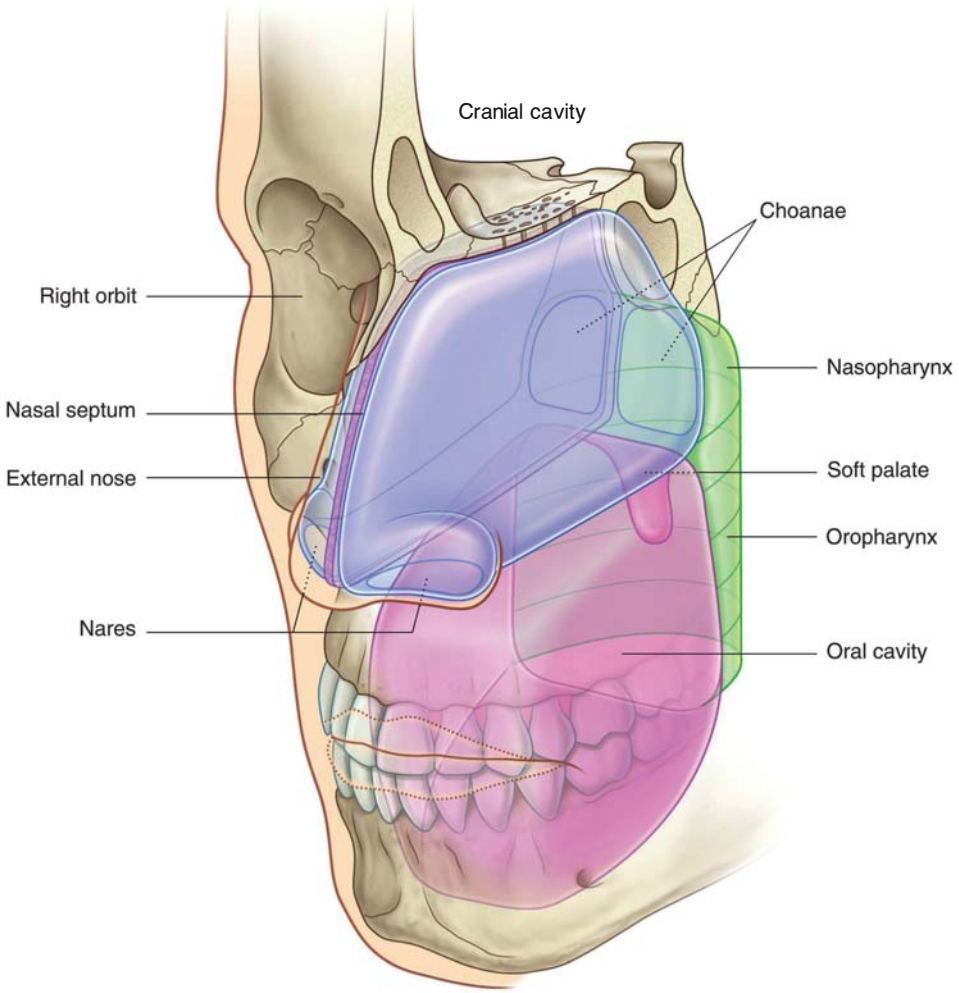


Fig. 8.225 Nasal cavities (anterolateral view). Relationship to other cavities.

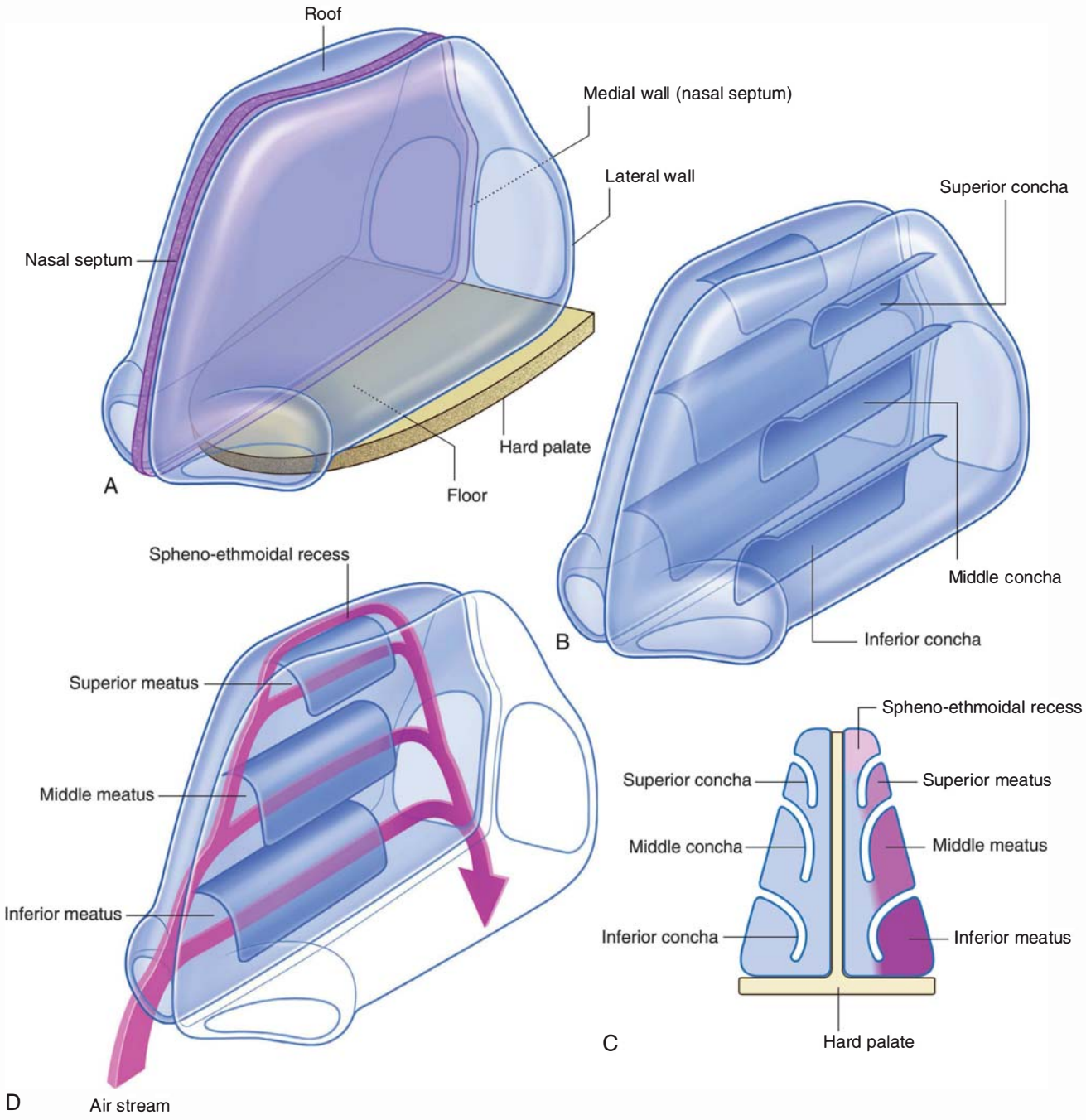


Fig. 8.226 Nasal cavities. **A.** Floor, roof, and lateral walls. **B.** Conchae on lateral walls. **C.** Coronal section. **D.** Air channels in right nasal cavity.

- from the cranial cavity above by parts of the frontal, ethmoid, and sphenoid bones.

Lateral to the nasal cavities are the orbits.

Each nasal cavity has a floor, roof, medial wall, and lateral wall (Fig. 8.226A).

Lateral wall

The lateral wall is characterized by three curved shelves of bone (conchae), which are one above the other and project medially and inferiorly across the nasal cavity (Fig. 8.226B). The medial, anterior, and posterior margins of the conchae are free.

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The conchae divide each nasal cavity into four air channels (Fig. 8.226C,D):

- an **inferior nasal meatus** between the **inferior concha** and the nasal floor,
- a **middle nasal meatus** between the inferior and **middle concha**,
- a **superior nasal meatus** between the middle and **superior concha**, and
- a **spheno-ethmoidal recess** between the superior concha and the nasal roof.

These conchae increase the surface area of contact between tissues of the lateral wall and the respired air.

The openings of the paranasal sinuses, which are extensions of the nasal cavity that erode into the surrounding bones during childhood and early adulthood, are on the lateral wall and roof of the nasal cavities (Fig. 8.227). In addition, the lateral wall also contains the opening of the nasolacrimal duct, which drains tears from the eye into the nasal cavity.

Regions

Each nasal cavity consists of three general regions—the nasal vestibule, the respiratory region, and the olfactory region (Fig. 8.228):

- The **nasal vestibule** is a small dilated space just internal to the naris that is lined by skin and contains hair follicles.
- The **respiratory region** is the largest part of the nasal cavity, has a rich neurovascular supply, and is lined by respiratory epithelium composed mainly of ciliated and mucous cells.
- The **olfactory region** is small, is at the apex of each nasal cavity, is lined by olfactory epithelium, and contains the olfactory receptors.

In addition to housing receptors for the sense of smell (olfaction), the nasal cavities adjust the temperature and humidity of respired air by the action of a rich blood supply, and trap and remove particulate matter from the airway by

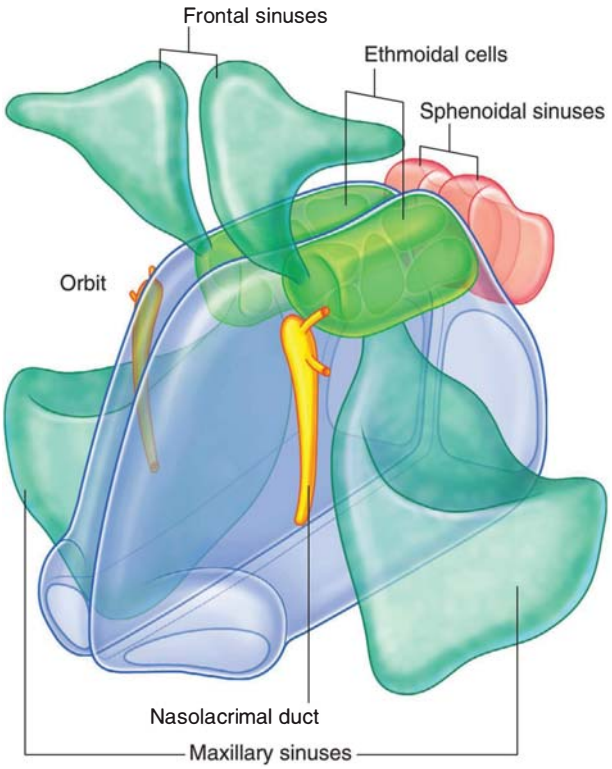


Fig. 8.227 Paranasal sinuses and nasolacrimal duct.

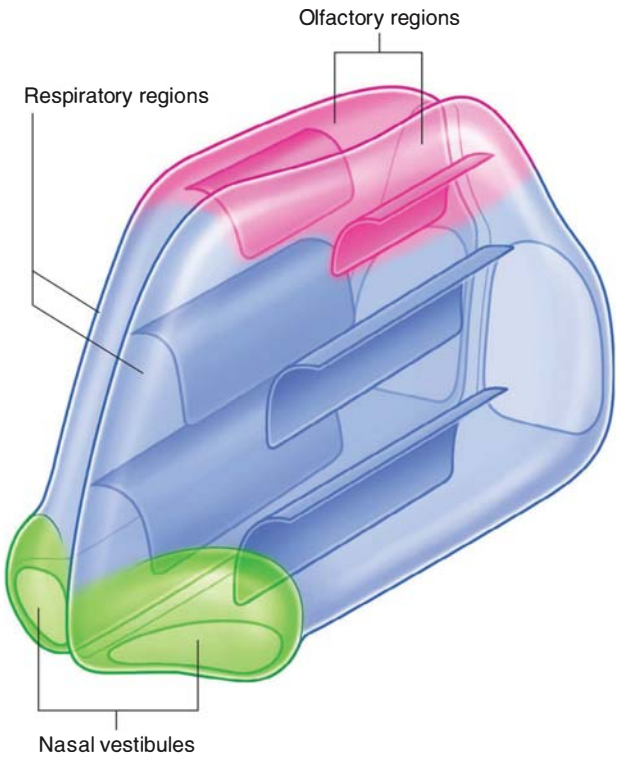


Fig. 8.228 Regions of the nasal cavities.

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filtering the air through hair in the vestibule and by capturing foreign material in abundant mucus. The mucus normally is moved posteriorly by cilia on epithelial cells in the nasal cavities and is swallowed.

Innervation and blood supply

Innervation of the nasal cavities is by three cranial nerves:

- Olfaction is carried by the olfactory nerve [I].
- General sensation is carried by the trigeminal nerve [V], the anterior region by the ophthalmic nerve [V_1], and the posterior region by the maxillary nerve [V_2].
- All glands are innervated by parasympathetic fibers in the facial nerve [VII] (greater petrosal nerve), which join branches of the maxillary nerve [V_2] in the pterygopalatine fossa.

Sympathetic fibers are ultimately derived from the T1 spinal cord level. They synapse mainly in the superior cervical sympathetic ganglion, and postganglionic fibers reach the nasal cavities along blood vessels, or by joining branches of the maxillary nerve [V_2] in the pterygopalatine fossa.

Blood supply to the nasal cavities is by:

- terminal branches of the maxillary and facial arteries, which originate from the external carotid artery, and
- ethmoidal branches of the ophthalmic artery, which originates from the internal carotid artery.

Skeletal framework

Bones that contribute to the skeletal framework of the nasal cavities include:

- the unpaired ethmoid, sphenoid, frontal, and vomer bones, and
- the paired nasal, maxillary, palatine, and lacrimal bones and inferior conchae.

Of all the bones associated with the nasal cavities, the ethmoid is a key element.

Ethmoid bone

The single ethmoid bone is one of the most complex bones in the skull. It contributes to the roof, lateral wall, and medial wall of both nasal cavities, and contains the ethmoidal cells (ethmoidal sinuses).

The ethmoid bone is cuboidal in overall shape (Fig. 8.229A) and is composed of two rectangular box-shaped **ethmoidal labyrinths**, one on each side, united superiorly across the midline by a perforated sheet of bone (the **cribriform plate**). A second sheet of bone (the **perpendicular plate**) descends vertically in the median sagittal plane from the cribriform plate to form part of the nasal septum.

Each ethmoidal labyrinth is composed of two delicate sheets of bone, which sandwich between them the ethmoidal cells.

- The lateral sheet of bone (the **orbital plate**) is flat and forms part of the medial wall of the orbit.
- The medial sheet of bone forms the upper part of the lateral wall of the nasal cavity and is characterized by two processes and a swelling (Fig. 8.229B)—the two processes are curved shelves of bone (the superior and middle conchae), which project across the nasal cavity and curve downward ending in free medial margins, while inferior to the origin of the middle concha, the middle ethmoidal cells form a prominent bulge (the **ethmoidal bulla**), on the medial wall of the labyrinth.

Extending anterosuperiorly from just under the bulla is a groove (the **ethmoidal infundibulum**), which continues upward, and narrows to form a channel that penetrates the ethmoidal labyrinth and opens into the frontal sinus. This channel is for the frontonasal duct, which drains the frontal sinus.

The superior surface of the ethmoidal labyrinth articulates with the frontal bone, which usually completes the roof of the ethmoidal cells, while the anterior surface articulates with the frontal process of the maxilla and with the lacrimal bone. The inferior surface articulates with the upper medial margin of the maxilla.

A delicate irregularly shaped projection (the **uncinate process**) on the anterior aspect of the inferior surface of the ethmoidal labyrinth extends posteroinferiorly across a large defect (**maxillary hiatus**) in the medial wall of the maxilla to articulate with the inferior concha.

The cribriform plate is at the apex of the nasal cavities and fills the **ethmoidal notch** in the frontal bone (Fig. 8.229) and separates the nasal cavities below from the cranial cavity above. Small perforations in the bone allow the fibers of the olfactory nerve [I] to pass between the two regions.

A large triangular process (the **crista galli**) at the midline on the superior surface of the cribriform plate anchors a fold (falx cerebri) of dura mater in the cranial cavity.

The perpendicular plate of the ethmoid bone is quadrangular in shape, descends in the midline from the cribriform plate, and forms the upper part of the median nasal septum (Fig. 8.229). It articulates:

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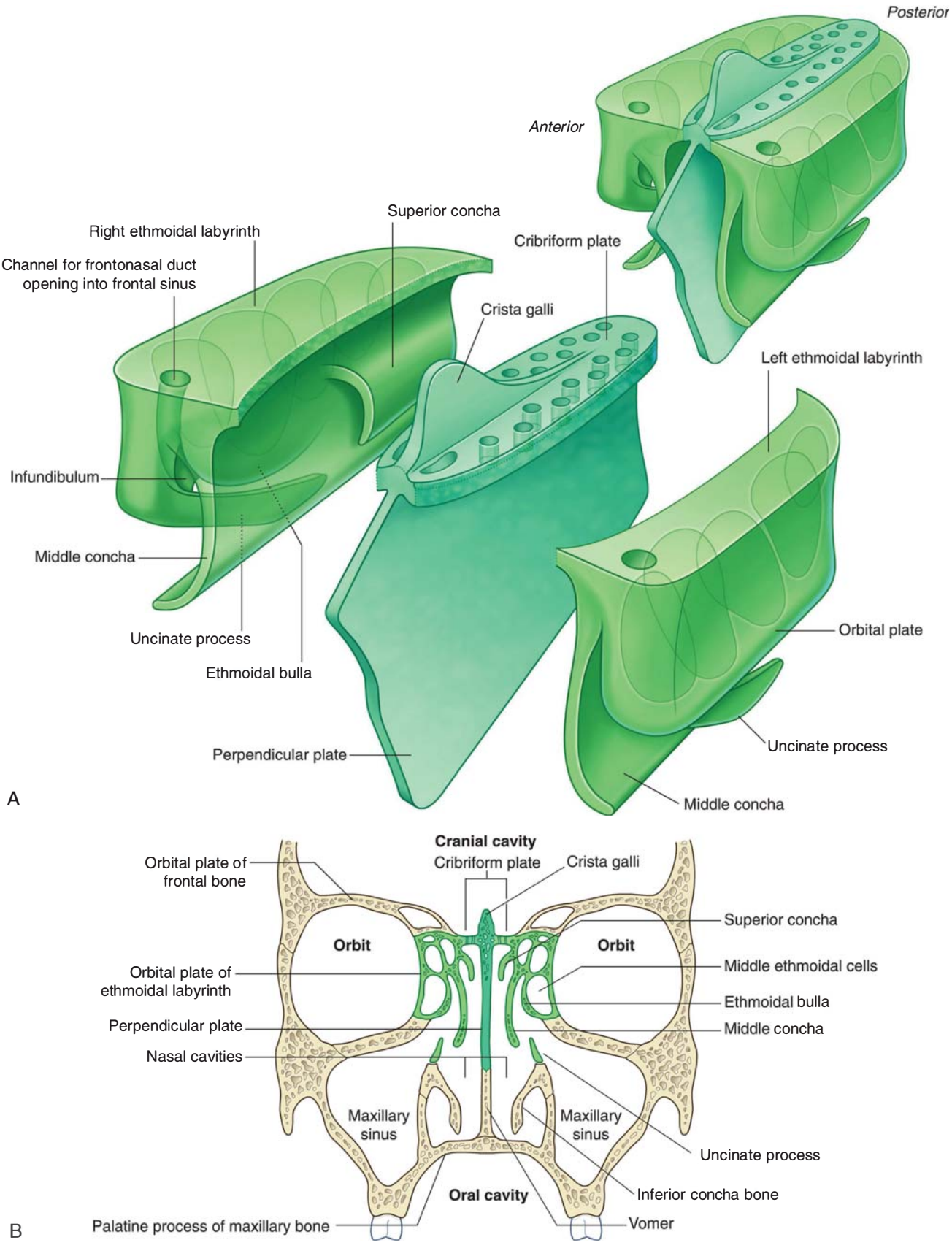


Fig. 8.229 Ethmoid bone. **A.** Overall shape. **B.** Coronal section through skull.

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- posteriorly with the sphenoidal crest on the body of the sphenoid bone,
- anteriorly with the nasal spine on the frontal bone and with the site of articulation at the midline between the two nasal bones, and
- inferiorly and anteriorly with the septal cartilage and posteriorly with the vomer.

External nose

The external nose extends the nasal cavities onto the front of the face and positions the nares so that they point downward (Fig. 8.230). It is pyramidal in shape with its apex anterior in position. The upper angle of the nose between the openings of the orbits is continuous with the forehead.

Like posterior regions, the anterior parts of the nasal cavities found within the nose are held open by a skeletal framework, which is composed partly of bone and mainly of cartilage:

- The bony parts are where the nose is continuous with the skull—here the nasal bones and parts of the maxillae and frontal bones provide support.
- Anteriorly, and on each side, support is provided by **lateral processes** of the septal cartilage, **major alar** and three or four **minor alar cartilages**, and a single septal cartilage in the midline that forms the anterior part of the nasal septum.

Paranasal sinuses

There are four paranasal air sinuses—the ethmoidal cells, and the sphenoidal, maxillary, and frontal sinuses (Fig. 8.231A,B). Each is named according to the bone in which it is found.

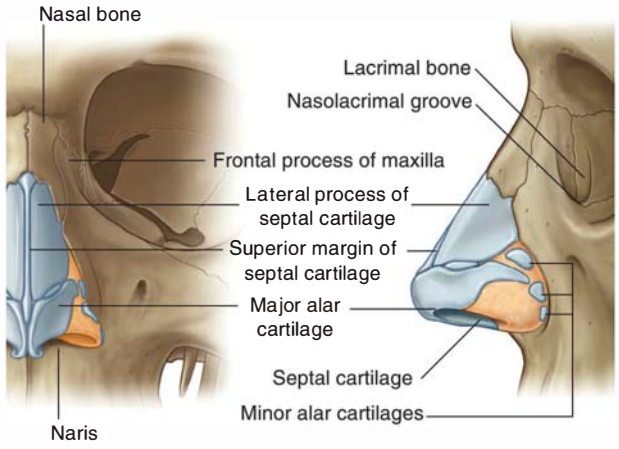


Fig. 8.230 External nose.

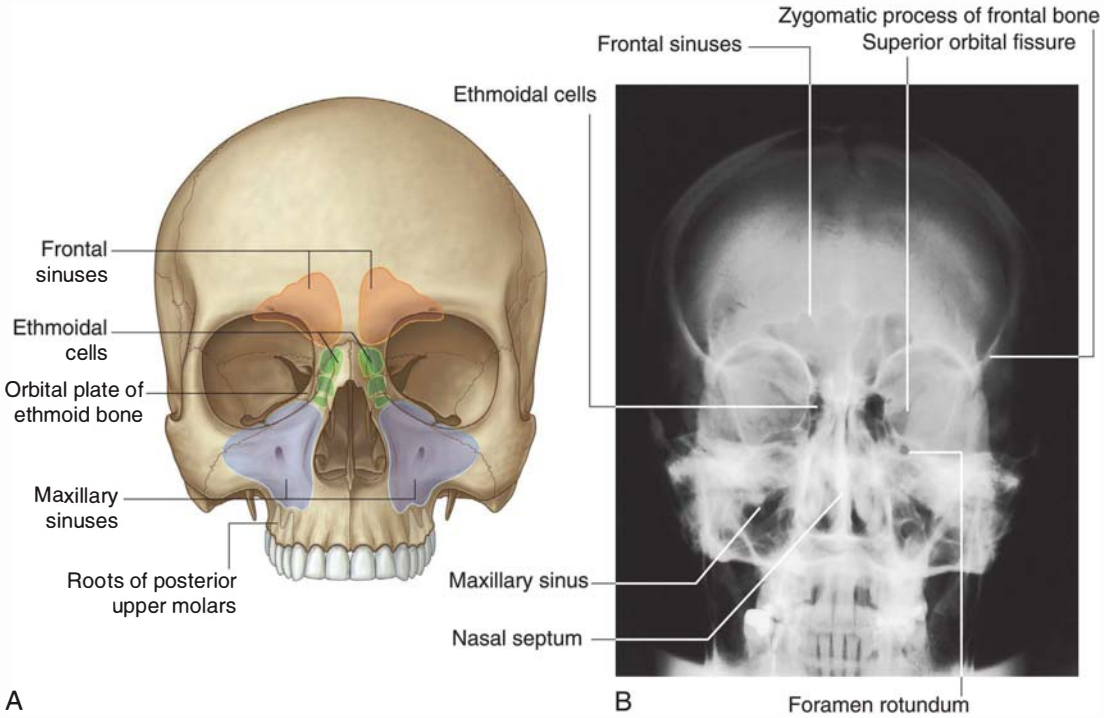


Fig. 8.231 Paranasal sinuses. **A.** Anterior view. **B.** Posteroanterior skull radiograph.

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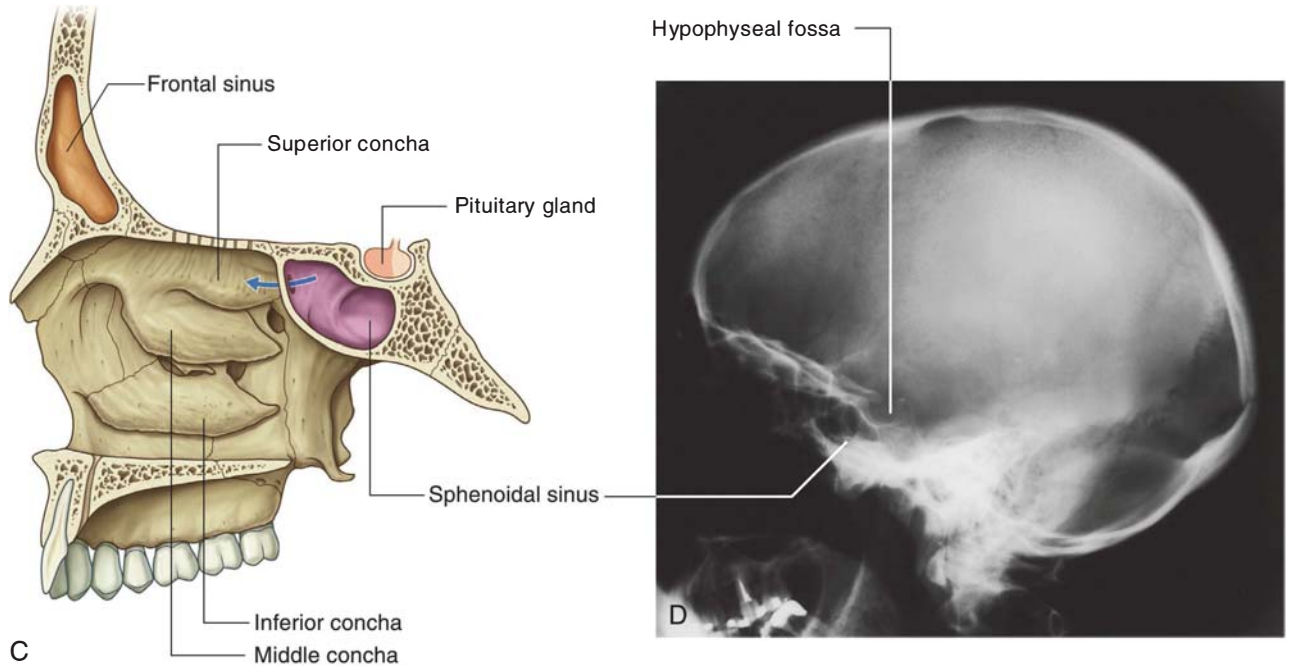


Fig. 8.231, cont'd **C.** Paramedian view of right nasal cavity. **D.** Lateral skull radiograph.

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The paranasal sinuses develop as outgrowths from the nasal cavities and erode into the surrounding bones. All of the paranasal sinuses:

- are lined by respiratory mucosa, which is ciliated and mucus secreting,
- open into the nasal cavities, and
- are innervated by branches of the trigeminal nerve [V].

Frontal sinuses

The frontal sinuses, one on each side, are variable in size and are the most superior of the sinuses (Fig. 8.231A–C). Each is triangular in shape and is in the part of the frontal bone under the forehead. The base of each triangular sinus is oriented vertically in the bone at the midline above the bridge of the nose and the apex is laterally approximately one-third of the way along the upper margin of the orbit.

Each frontal sinus drains onto the lateral wall of the middle meatus via the frontonasal duct, which penetrates the ethmoidal labyrinth and continues as the ethmoidal infundibulum at the front end of the **semilunar hiatus**.

The frontal sinuses are innervated by branches of the supra-orbital nerve from the ophthalmic nerve [V_1]. Their blood supply is from branches of the anterior ethmoidal arteries.

Ethmoidal cells

The ethmoidal cells on each side fill the ethmoidal labyrinth (Fig. 8.231A,B). Each cluster of cells is separated from the orbit by the thin orbital plate of the ethmoidal labyrinth, and from the nasal cavity by the medial wall of the ethmoidal labyrinth.

The ethmoidal cells are formed by a variable number of individual air chambers, which are divided into anterior, middle, and posterior ethmoidal cells based on the location of their apertures on the lateral wall of the nasal cavity:

- The anterior ethmoidal cells open into the ethmoidal infundibulum or the frontonasal duct.
- The middle ethmoidal cells open onto the ethmoidal bulla, or onto the lateral wall just above this structure.
- The posterior ethmoidal cells open onto the lateral wall of the superior nasal meatus.

Because the ethmoidal cells often erode into bones beyond the boundaries of the ethmoidal labyrinth, their walls may be completed by the frontal, maxillary, lacrimal, sphenoid, and palatine bones.

The ethmoidal cells are innervated by:

- the **anterior** and **posterior ethmoidal branches** of the nasociliary nerve from the ophthalmic nerve [V_1], and

- the maxillary nerve [V_2] via orbital branches from the pterygopalatine ganglion.

The ethmoidal cells receive their blood supply through branches of the anterior and posterior ethmoidal arteries.

Maxillary sinuses

The maxillary sinuses, one on each side, are the largest of the paranasal sinuses and completely fill the bodies of the maxillae (Fig. 8.231A,B). Each is pyramidal in shape with the apex directed laterally and the base deep to the lateral wall of the adjacent nasal cavity. The medial wall or base of the maxillary sinus is formed by the maxilla, and by parts of the inferior concha and palatine bone that overlie the maxillary hiatus.

The opening of the maxillary sinus is near the top of the base, in the center of the semilunar hiatus, which grooves the lateral wall of the middle nasal meatus.

Relationships of the maxillary sinus are as follows:

- The superolateral surface (roof) is related above to the orbit.
- The anterolateral surface is related below to the roots of the upper molar and premolar teeth and in front to the face.
- The posterior wall is related behind to the infratemporal fossa.

The maxillary sinuses are innervated by infra-orbital and alveolar branches of the maxillary nerve [V_2], and receive their blood through branches from the infra-orbital and superior alveolar branches of the maxillary arteries.

Sphenoidal sinuses

The sphenoidal sinuses, one on either side within the body of the sphenoid, open into the roof of the nasal cavity via apertures on the posterior wall of the spheno-ethmoidal recess (Fig. 8.231C,D). The apertures are high on the anterior walls of the sphenoid sinuses.

The sphenoidal sinuses are related:

- above to the cranial cavity, particularly to the pituitary gland and to the optic chiasm,
- laterally, to the cranial cavity, particularly to the cavernous sinuses, and
- below and in front, to the nasal cavities.

Because only thin shelves of bone separate the sphenoidal sinuses from the nasal cavities below and hypophyseal fossa above, the pituitary gland can be surgically approached through the roof of the nasal cavities by passing first through the anteroinferior aspect of the sphenoid bone and into the sphenoidal sinuses and then through the top of the sphenoid bone into the hypophyseal fossa.

Innervation of the sphenoidal sinuses is provided by:

- the posterior ethmoidal branch of the ophthalmic nerve [V_1], and
- the maxillary nerve [V_2] via orbital branches from the pterygopalatine ganglion.

The sphenoidal sinuses are supplied by branches of the pharyngeal arteries from the maxillary arteries.

Walls, floor, and roof

Medial wall

The medial wall of each nasal cavity is the mucosa-covered surface of the thin nasal septum, which is oriented vertically in the median sagittal plane and separates the right and left nasal cavities from each other.

The nasal septum (Fig. 8.232) consists of:

- the **septal nasal cartilage** anteriorly,
- posteriorly, mainly the vomer and the perpendicular plate of the ethmoid bone,
- small contributions by the nasal bones where they meet in the midline, and the nasal spine of the frontal bone, and
- contributions by the nasal crests of the maxillary and palatine bones, rostrum of the sphenoid bone, and the incisor crest of the maxilla.

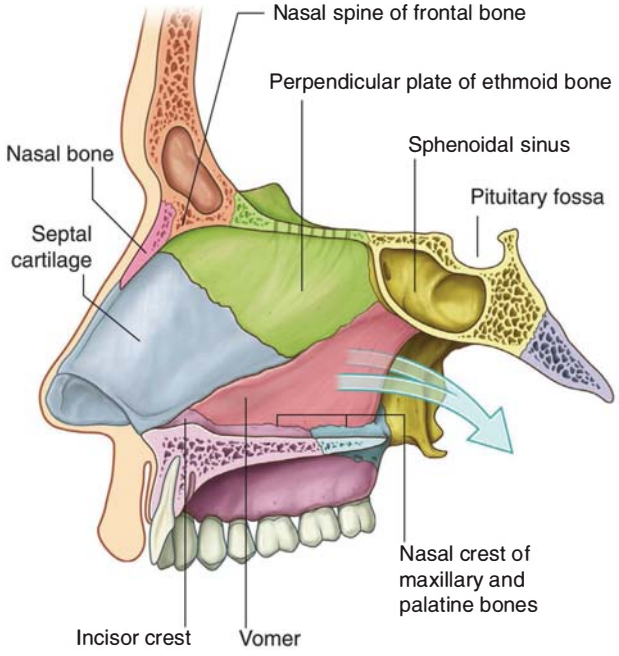


Fig. 8.232 Medial wall of the nasal cavity—the nasal septum.

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In the clinic

Deviated nasal septum

The nasal septum is typically situated in the midline; however, septal deviation to one side or the other is not uncommon, and in many cases is secondary to direct trauma. Extreme septal deviation can produce nasal occlusion. The deviation can be corrected surgically.

Floor

The floor of each nasal cavity (Fig. 8.233) is smooth, concave, and much wider than the roof. It consists of:

- soft tissues of the external nose, and
- the upper surface of the palatine process of the maxilla and the horizontal plate of the palatine bone, which together form the hard palate.

The naris opens anteriorly into the floor, and the superior aperture of the incisive canal is deep to the mucosa immediately lateral to the nasal septum near the front of the hard palate.

Roof

The roof of the nasal cavity is narrow and is highest in central regions where it is formed by the cribriform plate of the ethmoid bone (Fig. 8.234).

Anterior to the cribriform plate the roof slopes inferiorly to the nares and is formed by:

- the nasal spine of the frontal bone and the nasal bones, and
- the lateral processes of the septal cartilage and major alar cartilages of the external nose.

Posteriorly, the roof of each cavity slopes inferiorly to the choana and is formed by:

- the anterior surface of the sphenoid bone,
- the ala of the vomer and adjacent sphenoidal process of the palatine bone, and
- the vaginal process of the medial plate of the pterygoid process.

Underlying the mucosa, the roof is perforated superiorly by openings in the cribriform plate, and anterior to these openings by a separate foramen for the anterior ethmoidal nerve and vessels.

The opening between the sphenoidal sinus and the spheno-ethmoidal recess is on the posterior slope of the roof.

Lateral wall

The lateral wall of each nasal cavity is complex and is formed by bone, cartilage, and soft tissues.

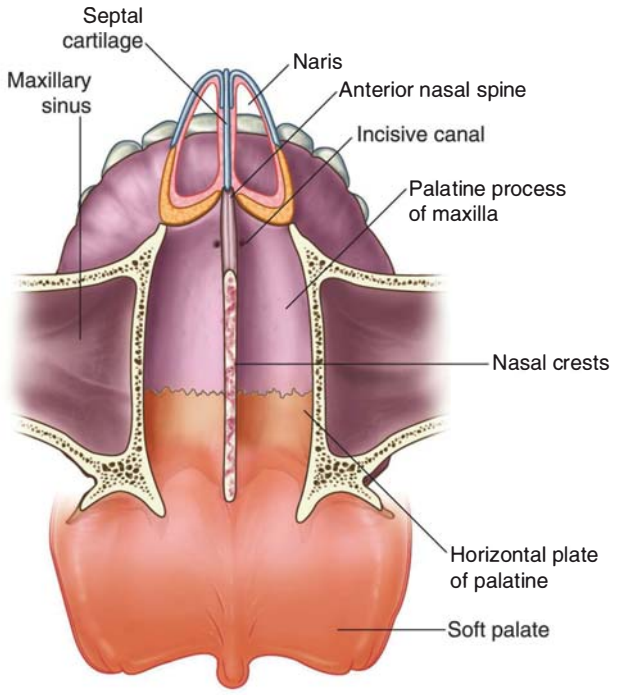


Fig. 8.233 Floor of the nasal cavity (superior view).

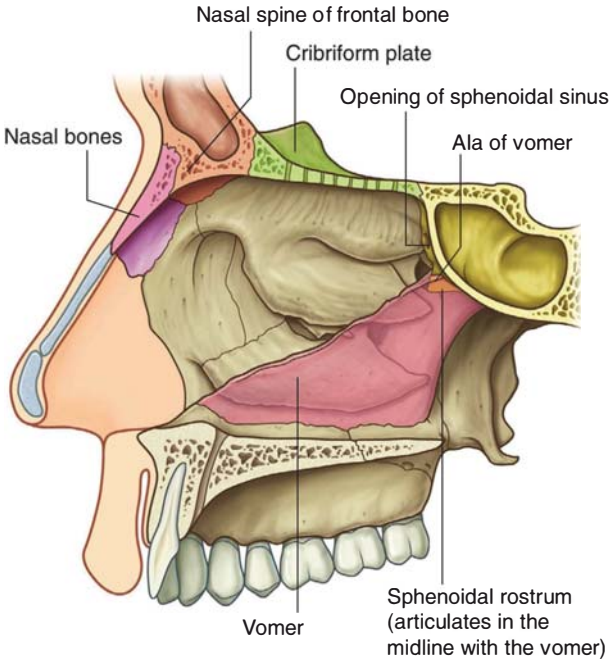


Fig. 8.234 Roof of the nasal cavity.

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Bony support for the lateral wall (Fig. 8.235A) is provided by:

- the ethmoidal labyrinth, superior concha, middle concha and uncinate process,
- the perpendicular plate of the palatine bone,
- the medial pterygoid plate of the sphenoid bone,
- the medial surfaces of the lacrimal bones and maxillae, and
- the inferior concha.

In the external nose, the lateral wall of the cavity is supported by cartilage (lateral process of the septal cartilage and major and minor alar cartilages) and by soft tissues. The surface of the lateral wall is irregular in contour and is interrupted by the three nasal conchae.

The inferior, middle, and superior conchae (Fig. 8.235B) extend medially across the nasal cavity, separating it into four air channels, an inferior, middle, and superior meatus and a spheno-ethmoidal recess. The conchae do not extend forward into the external nose. The anterior end of each concha curves inferiorly to form a lip that overlies the end of the related meatus.

Immediately inferior to the attachment of the middle concha and just anterior to the midpoint of the concha, the lateral wall of the middle meatus elevates to form the dome-shaped ethmoidal bulla (Fig. 8.235C). This is formed by the underlying middle ethmoidal cells, which expand the medial wall of the ethmoidal labyrinth.

Inferior to the ethmoidal bulla is a curved gutter (the semilunar hiatus), which is formed by the mucosa covering the lateral wall as it spans a defect in the bony wall between the ethmoidal bulla above and the uncinate process below.

The anterior end of the semilunar hiatus forms a channel (the ethmoidal infundibulum), which curves upward and continues as the frontonasal duct through the anterior part of the ethmoidal labyrinth to open into the frontal sinus.

The nasolacrimal duct and most of the paranasal sinuses open onto the lateral wall of the nasal cavity (Fig. 8.235C):

- The nasolacrimal duct opens onto the lateral wall of the inferior nasal meatus under the anterior lip of the inferior concha—it drains tears from the conjunctival sac of the eye into the nasal cavity and originates at the inferior end of the lacrimal sac on the anteromedial wall of the orbit.
- The frontal sinus drains via the frontonasal duct and ethmoidal infundibulum into the anterior end of the semilunar hiatus on the lateral wall of the middle nasal meatus—the anterior ethmoidal cells drain into the frontonasal duct or ethmoidal infundibulum (in some cases, the frontal sinus drains directly into the anterior

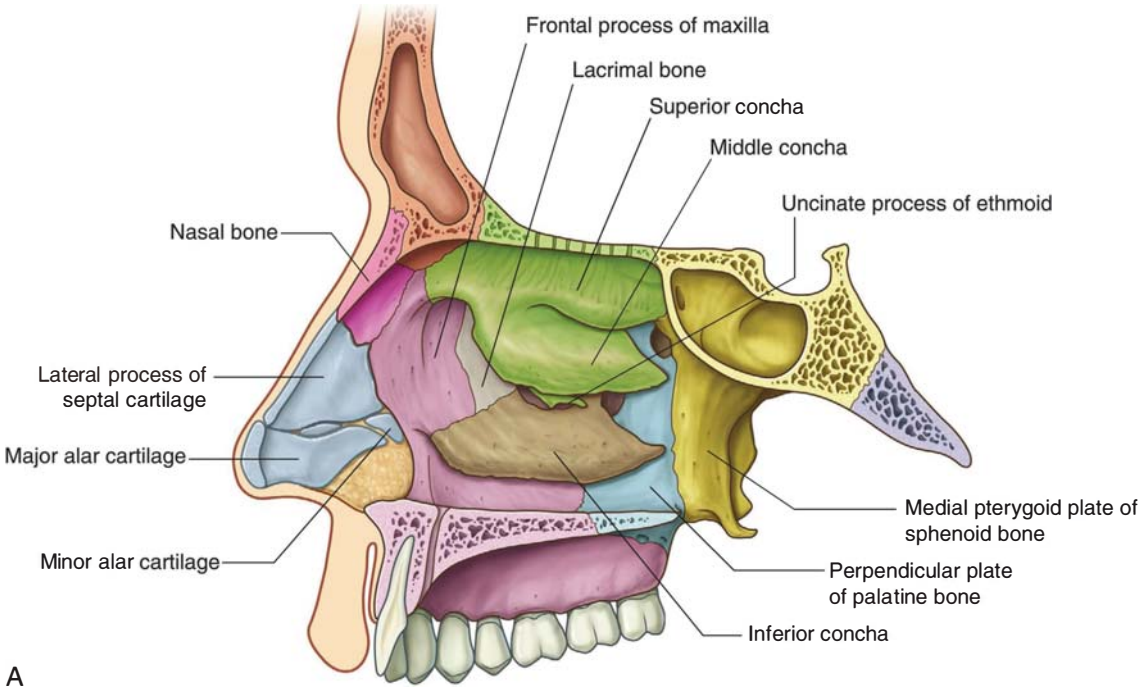
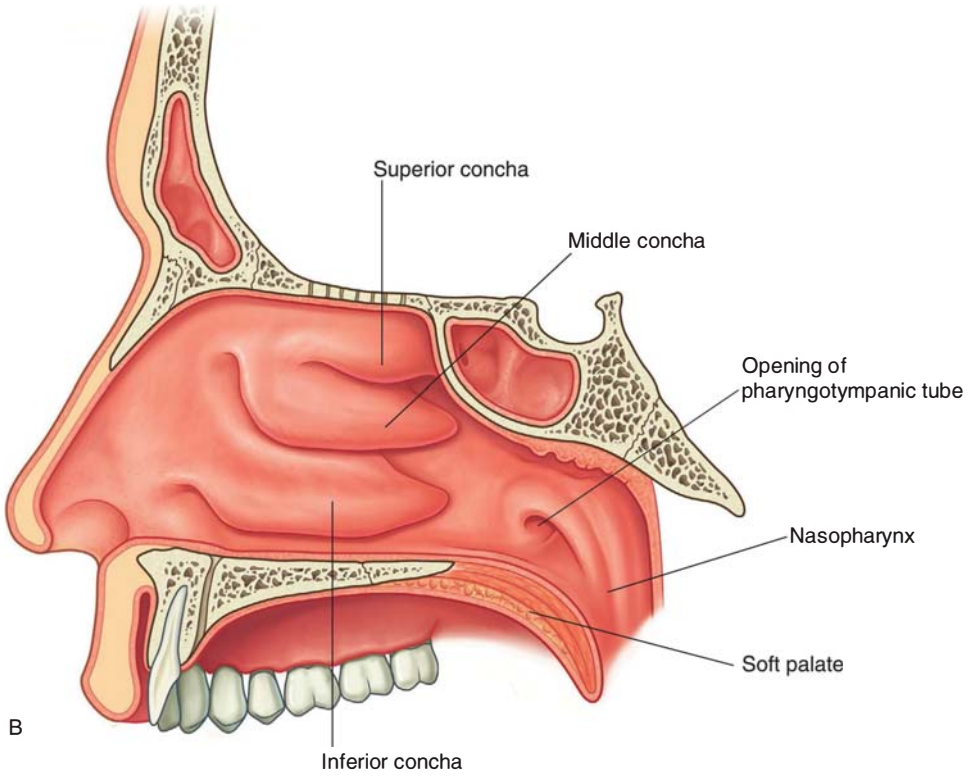


Fig. 8.235 Lateral wall of the nasal cavity. **A.** Bones.

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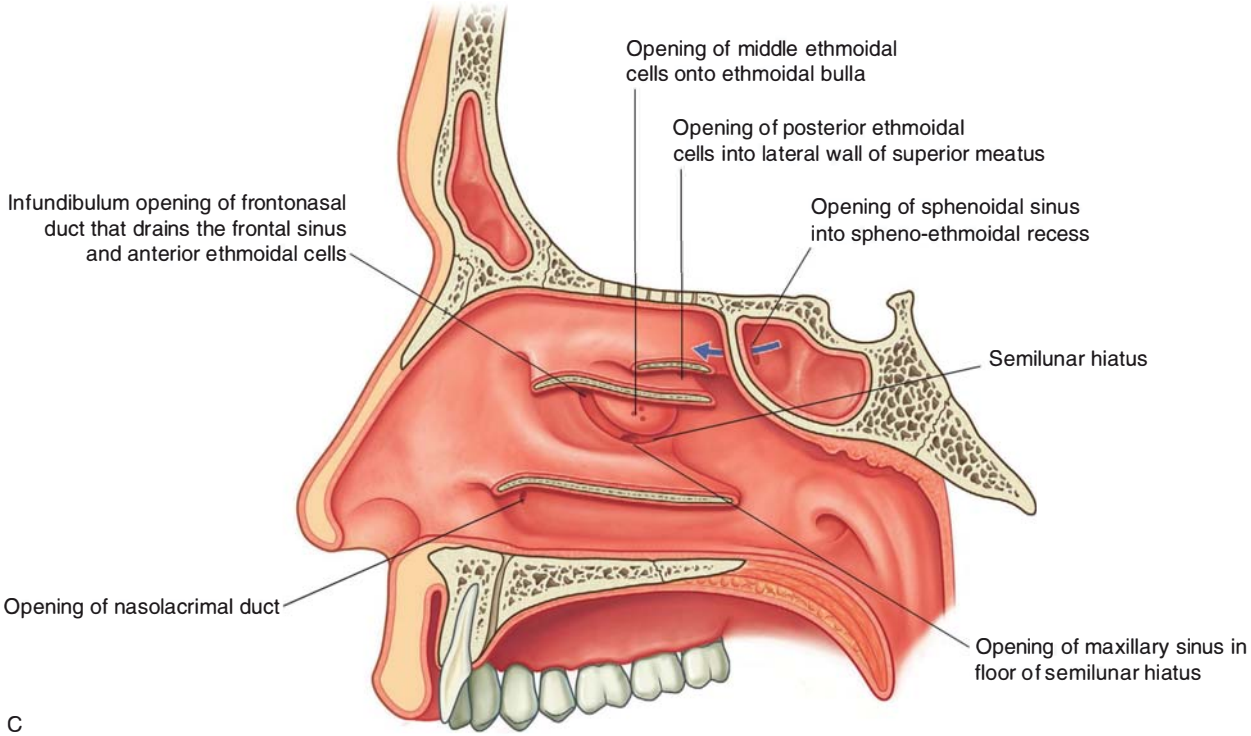


Fig. 8.235, cont'd **B.** Covered with mucosa. **C.** Conchae broken away at attachment to lateral wall.

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end of the middle nasal meatus and the frontonasal duct ends blindly in the anterior ethmoidal cells).
- The middle ethmoidal cells open onto or just above the ethmoidal bulla.
- The posterior ethmoidal cells usually open onto the lateral wall of the superior nasal meatus.
- The large maxillary sinus opens into the semilunar hiatus, usually just inferior to the center of the ethmoidal bulla—this opening is near the roof of the maxillary sinus.

The only paranasal sinus that does not drain onto the lateral wall of the nasal cavity is the sphenoidal sinus, which usually opens onto the sloping posterior roof of the nasal cavity.

Nares

The nares are oval apertures on the inferior aspect of the external nose and are the anterior openings of the nasal cavities (Fig. 8.236A). They are held open by the surrounding alar cartilages and septal cartilage, and by the inferior nasal spine and adjacent margins of the maxillae.

Although the nares are continuously open, they can be widened further by the action of the related muscles of

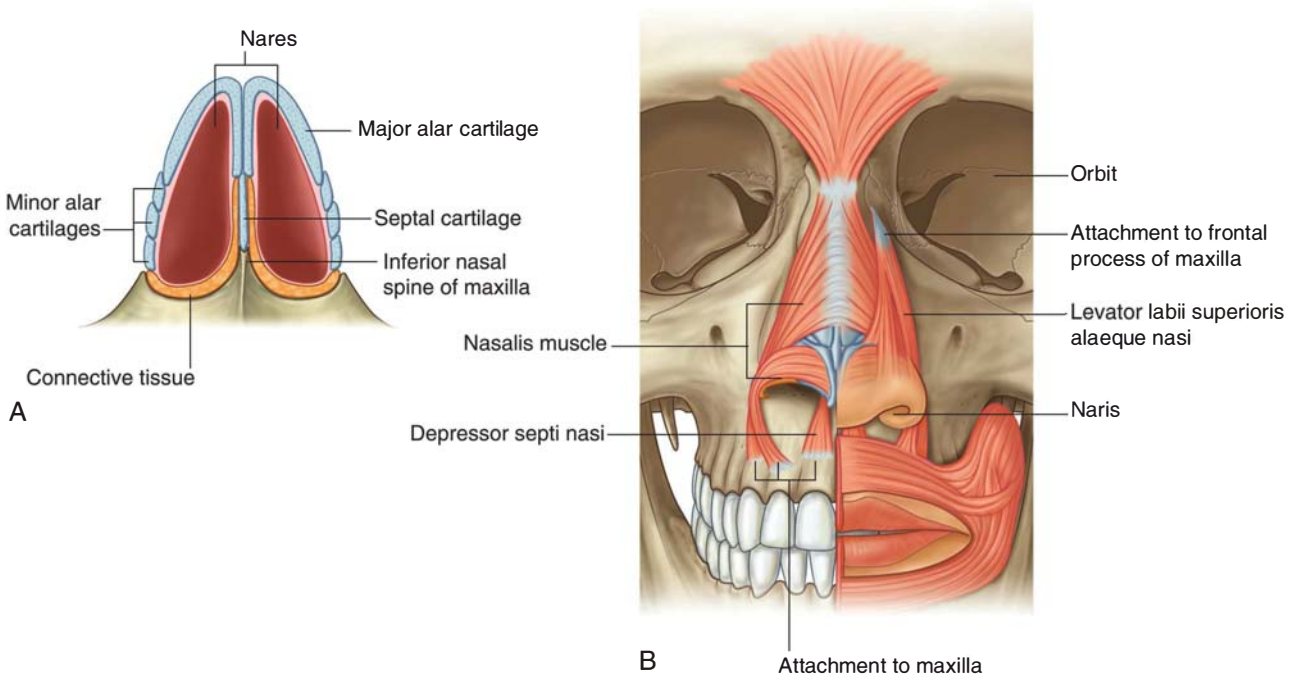


Fig. 8.236 Nares. **A.** Inferior view. **B.** Associated muscles.

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facial expression (nasalis, depressor septi nasi, and levator labii superioris alaeque nasi muscles; Fig. 8.236B).

Choanae

The choanae are the oval-shaped openings between the nasal cavities and the nasopharynx (Fig. 8.237). Unlike the nares, which have flexible borders of cartilage and soft tissues, the choanae are rigid openings completely surrounded by bone, and their margins are formed:

- inferiorly by the posterior border of the horizontal plate of the palatine bone,
- laterally by the posterior margin of the medial plate of the pterygoid process, and
- medially by the posterior border of the vomer.

The roof of the choanae is formed:

- anteriorly by the ala of the vomer and the vaginal process of the medial plate of the pterygoid process, and
- posteriorly by the body of the sphenoid bone.

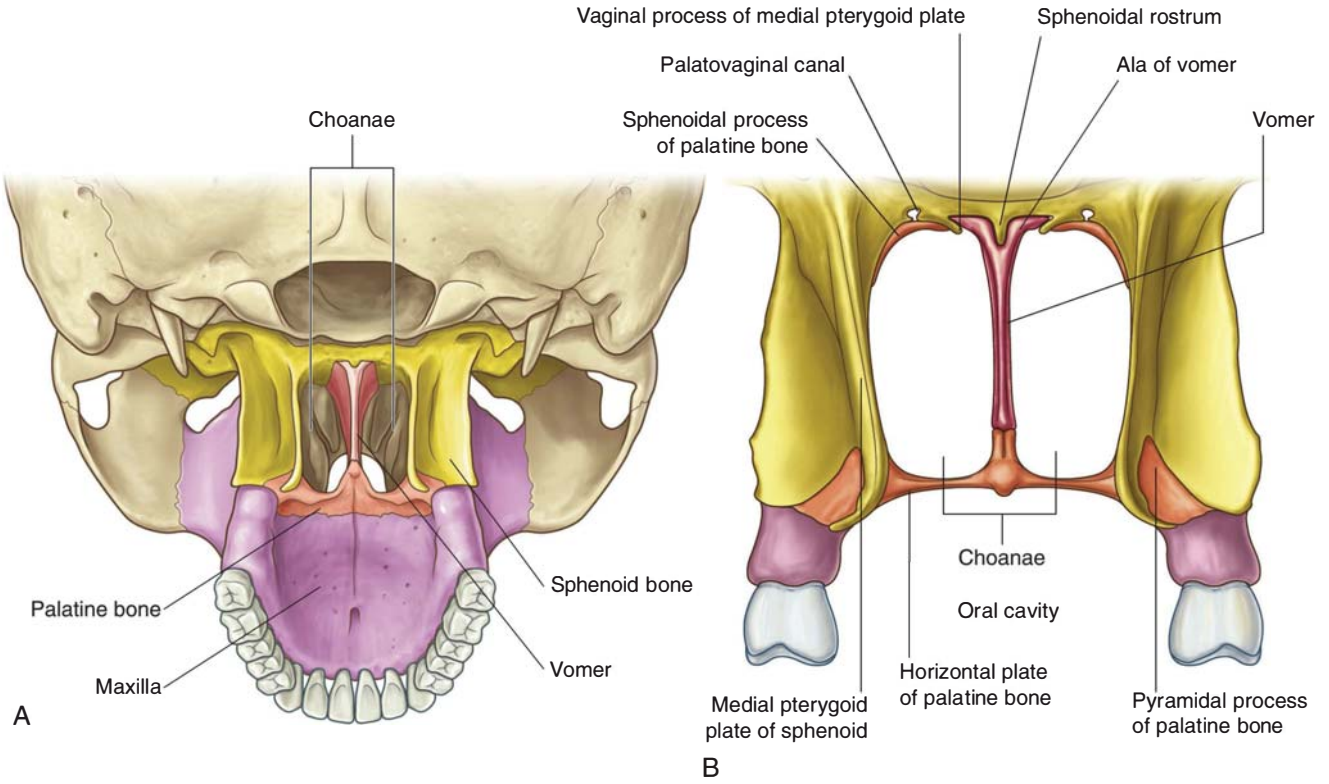


Fig. 8.237 Choanae (posterior view). **A.** Overview. **B.** Magnified view.

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Gateways

There are a number of routes by which nerves and vessels enter and leave the soft tissues lining each nasal cavity (Fig. 8.238), and these include the cribriform plate, sphenopalatine foramen, incisive canal, and small foramina in the lateral wall, and around the margin of the nares.

Cribriform plate

The fibers of the olfactory nerve [I] exit the nasal cavity and enter the cranial cavity through perforations in the cribriform plate. In addition, small foramina between the cribriform plate and surrounding bone allow the anterior ethmoidal nerve, a branch of the ophthalmic nerve [V_1], and accompanying vessels to pass from the orbit into the cranial cavity and then down into the nasal cavity.

In addition, there is a connection in some individuals between nasal veins and the superior sagittal sinus of the cranial cavity through a prominent foramen (the foramen cecum) in the midline between the crista galli and frontal bone.

Sphenopalatine foramen

One of the most important routes by which nerves and vessels enter and leave the nasal cavity is the sphenopalatine foramen in the posterolateral wall of the superior nasal meatus. This foramen is just superior to the attachment of the posterior end of the middle nasal concha and is formed by the sphenopalatine notch in the palatine bone and the body of the sphenoid bone.

The sphenopalatine foramen is a route of communication between the nasal cavity and the pterygopalatine fossa. Major structures passing through the foramen are:

- the sphenopalatine branch of the maxillary artery,
- the nasopalatine branch of the maxillary nerve [V_2], and
- superior nasal branches of the maxillary nerve [V_2].

Incisive canal

Another route by which structures enter and leave the nasal cavities is through the **incisive canal** in the floor of each nasal cavity. This canal is immediately lateral to the nasal septum and just posterosuperior to the root of the central incisor in the maxilla. The two incisive canals, one on each side, both open into the single unpaired incisive fossa in the roof of the oral cavity and transmit:

- the nasopalatine nerve from the nasal cavity into the oral cavity, and
- the terminal end of the greater palatine artery from the oral cavity into the nasal cavity.

Small foramina in the lateral wall

Other routes by which vessels and nerves get into and out of the nasal cavity include the nares and small foramina in the lateral wall:

- Internal nasal branches of the infra-orbital nerve of the maxillary nerve [V_2] and alar branches of the nasal artery from the facial artery loop around the margin of the naris to gain entry to the lateral wall of the nasal cavity from the face.
- Inferior nasal branches from the greater palatine branch of the maxillary nerve [V_2] enter the lateral wall of the nasal cavity from the palatine canal by passing through small foramina on the lateral wall.

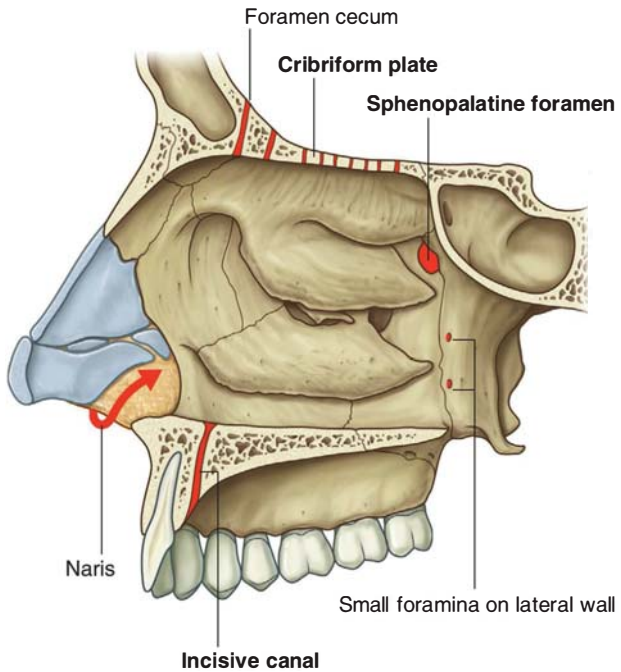


Fig. 8.238 Gateways to the nasal cavities.

Vessels

The nasal cavities have a rich vascular supply for altering the humidity and temperature of respired air. In fact, the submucosa of the respiratory region, particularly that related to the conchae and septum, is often described as "erectile" or "cavernous" because the tissue enlarges or shrinks depending on the amount of blood flowing into the system.

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Arteries

Arteries that supply the nasal cavity include vessels that originate from both the internal and external carotid arteries (Fig. 8.239):

- Vessels that originate from branches of the external carotid artery include the sphenopalatine, greater palatine, superior labial, and lateral nasal arteries.
- Vessels that originate from branches of the internal carotid artery are the anterior and posterior ethmoidal arteries.

Sphenopalatine artery

The largest vessel supplying the nasal cavity is the **sphenopalatine artery** (Fig. 8.239), which is the terminal branch of the maxillary artery in the pterygopalatine fossa. It leaves the pterygopalatine fossa and enters the nasal cavity by passing medially through the sphenopalatine foramen and onto the lateral wall of the nasal cavity.

Posterior lateral nasal branches supply a large part of the lateral wall and anastomose anteriorly with branches from the anterior and posterior ethmoidal arteries, and with lateral nasal branches of the facial artery.

Posterior septal branches of the sphenopalatine artery pass over the roof of the cavity and onto the nasal septum where they contribute to the blood supply of the medial wall. One of these latter branches continues forward down the nasal septum to anastomose with the terminal end of the greater palatine artery and septal branches of the superior labial artery.

Greater palatine artery

The terminal end of the **greater palatine artery** enters the anterior aspect of the floor of the nasal cavity by passing up through the incisive canal from the roof of the oral cavity (Fig. 8.239).

Like the sphenopalatine artery, the greater palatine artery arises in the pterygopalatine fossa as a branch of the maxillary artery. It passes first onto the roof of the oral cavity by passing down through the palatine canal and greater palatine foramen to the posterior aspect of the palate, then passes forward on the undersurface of the palate, and up through the incisive fossa and canal to reach the floor of the nasal cavity. The greater palatine artery supplies anterior regions of the medial wall and adjacent floor of the nasal cavity, and anastomoses with the septal branch of the sphenopalatine artery.

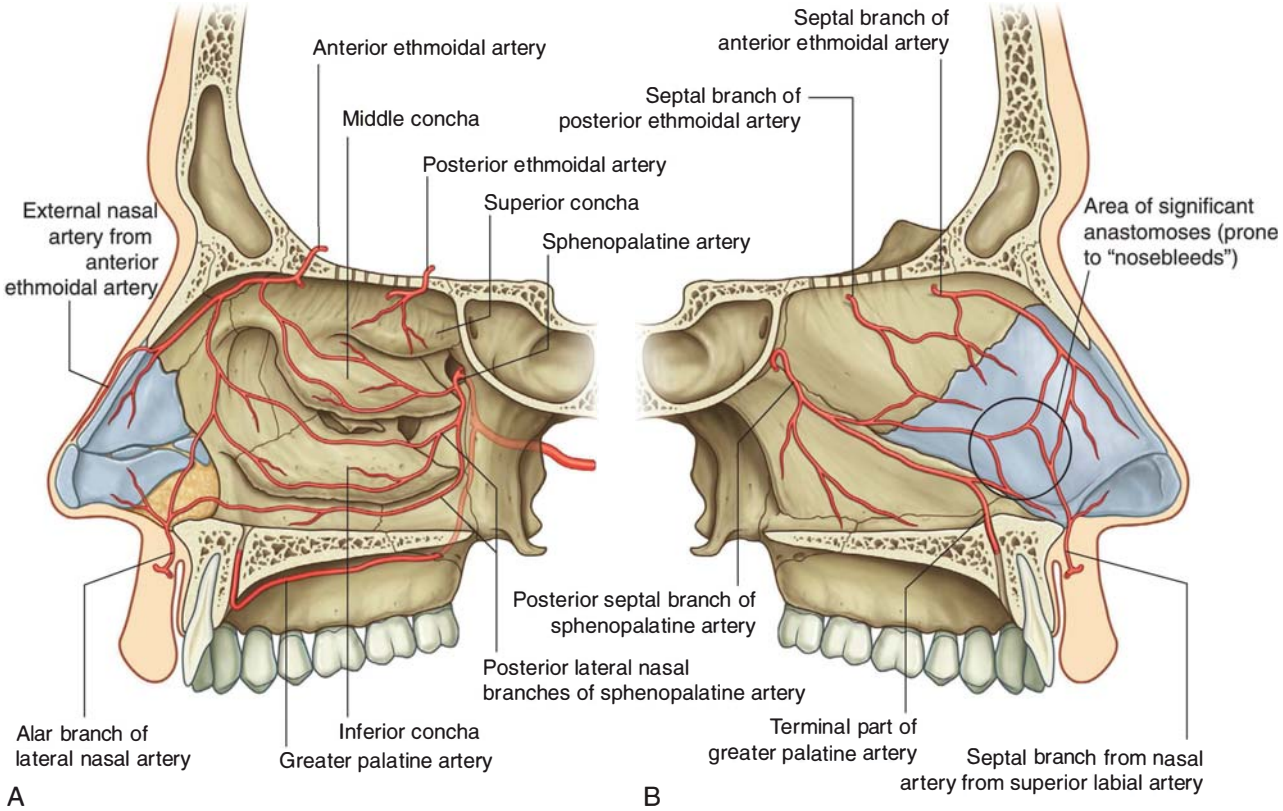


Fig. 8.239 Arterial supply of the nasal cavities. **A.** Lateral wall of the right nasal cavity. **B.** Septum (medial wall of right nasal cavity).

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Superior labial and lateral nasal arteries

The superior labial artery and the lateral nasal artery originate from the facial artery on the front of the face.

The **superior labial artery** originates from the facial artery near the lateral end of the oral fissure and passes medially in the lip, supplying the lip and giving rise to branches that supply the nose and nasal cavity. An alar branch supplies the region around the lateral aspect of the naris and a septal branch passes into the nasal cavity and supplies anterior regions of the nasal septum.

The **lateral nasal artery** originates from the facial artery in association with the margin of the external nose and contributes to the blood supply of the external nose. Alar branches pass around the lateral margin of the naris and supply the nasal vestibule.

Anterior and posterior ethmoidal arteries

The anterior and posterior ethmoidal arteries (Fig. 8.239) originate in the orbit from the ophthalmic artery, which originates in the cranial cavity as a major branch of the internal carotid artery. They pass through canals in the medial wall of the orbit between the ethmoidal labyrinth and frontal bone, supply the adjacent paranasal sinuses, and then enter the cranial cavity immediately lateral and superior to the cribriform plate.

The **posterior ethmoidal artery** descends into the nasal cavity through the cribriform plate and has branches to the upper parts of the medial and lateral walls.

The **anterior ethmoidal artery** passes forward, with the accompanying anterior ethmoidal nerve, in a groove on the cribriform plate and enters the nasal cavity by descending through a slit-like foramen immediately lateral to the crista galli. It gives rise to branches that supply the medial (septal) and lateral wall of the nasal cavity and then continues forward on the deep surface of the nasal bone, and terminates by passing between the nasal bone and lateral nasal cartilage to emerge on the external nose as the external nasal branch to supply skin and adjacent tissues.

Vessels that supply the nasal cavities form extensive anastomoses with each other. This is particularly evident in the anterior region of the medial wall where there are anastomoses between branches of the greater palatine, sphenopalatine, superior labial, and anterior ethmoidal arteries, and where the vessels are relatively close to the surface (Fig. 8.239B). This area is the major site of nosebleeds, or epistaxis.

Veins

Veins draining the nasal cavities generally follow the arteries (Fig. 8.240):

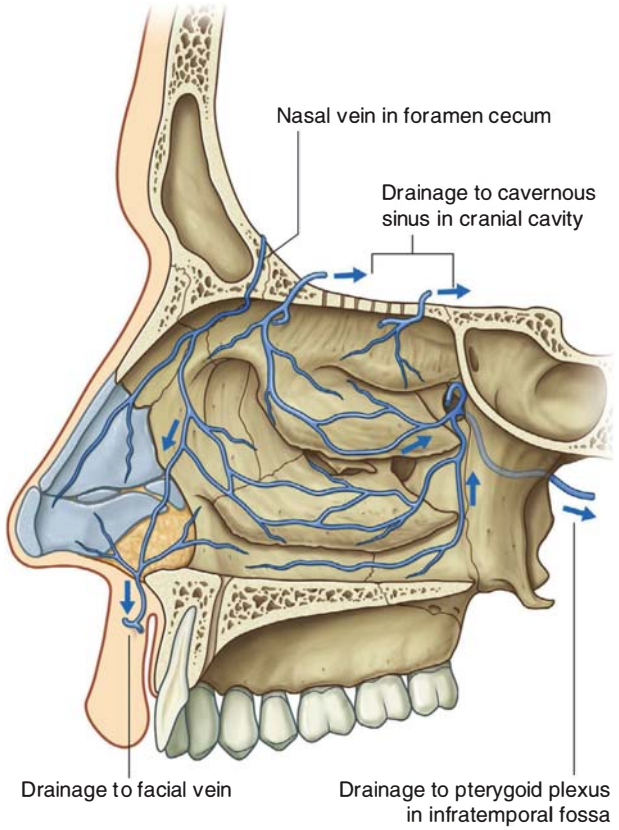


Fig. 8.240 Venous drainage of the nasal cavities.

- Veins that pass with branches that ultimately originate from the maxillary artery drain into the pterygoid plexus of veins in the infratemporal fossa.
- Veins from anterior regions of the nasal cavities join the facial vein.

In some individuals, an additional nasal vein passes superiorly through a midline aperture (the foramen cecum), in the frontal bone anterior to the crista galli, and joins with the anterior end of the superior sagittal sinus. Because this nasal vein connects an intracranial venous sinus with extracranial veins, it is classified as an emissary vein. Emissary veins in general are routes by which infections can track from peripheral regions into the cranial cavity.

Veins that accompany the anterior and posterior ethmoidal arteries are tributaries of the superior ophthalmic vein, which is one of the largest emissary veins and drains into the cavernous sinus on either side of the hypophyseal fossa.

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Innervation

Nerves that innervate the nasal cavities (Fig. 8.241) are:

- the olfactory nerve [I] for olfaction, and
- branches of the ophthalmic [V_1] and maxillary [V_2] nerves for general sensation.

Secretomotor innervation of mucous glands in the nasal cavities and paranasal sinuses is by parasympathetic fibers from the facial nerve [VII], which mainly join branches of the maxillary nerve [V_2] in the pterygopalatine fossa.

Olfactory nerve [I]

The olfactory nerve [I] is composed of axons from receptors in the olfactory epithelium at the top of each nasal cavity. Bundles of these axons pass superiorly through perforations in the cribriform plate to synapse with neurons in the olfactory bulb of the brain.

Branches from the ophthalmic nerve [V_1]

Branches from the ophthalmic nerve [V_1] that innervate the nasal cavity are the anterior and posterior ethmoidal nerves, which originate from the nasociliary nerve in the orbit.

Anterior and posterior ethmoidal nerves

The anterior ethmoidal nerve (Fig. 8.241) travels with the anterior ethmoidal artery and leaves the orbit through a canal between the ethmoidal labyrinth and the frontal bone. It passes through and supplies the adjacent ethmoidal cells and frontal sinus, and then enters the cranial cavity immediately lateral and superior to the cribriform plate. It then travels forward in a groove on the cribriform plate and enters the nasal cavity by descending through a slit-like foramen immediately lateral to the crista galli. It has branches to the medial and lateral wall of the nasal cavity and then continues forward on the undersurface of the nasal bone. It passes onto the external surface

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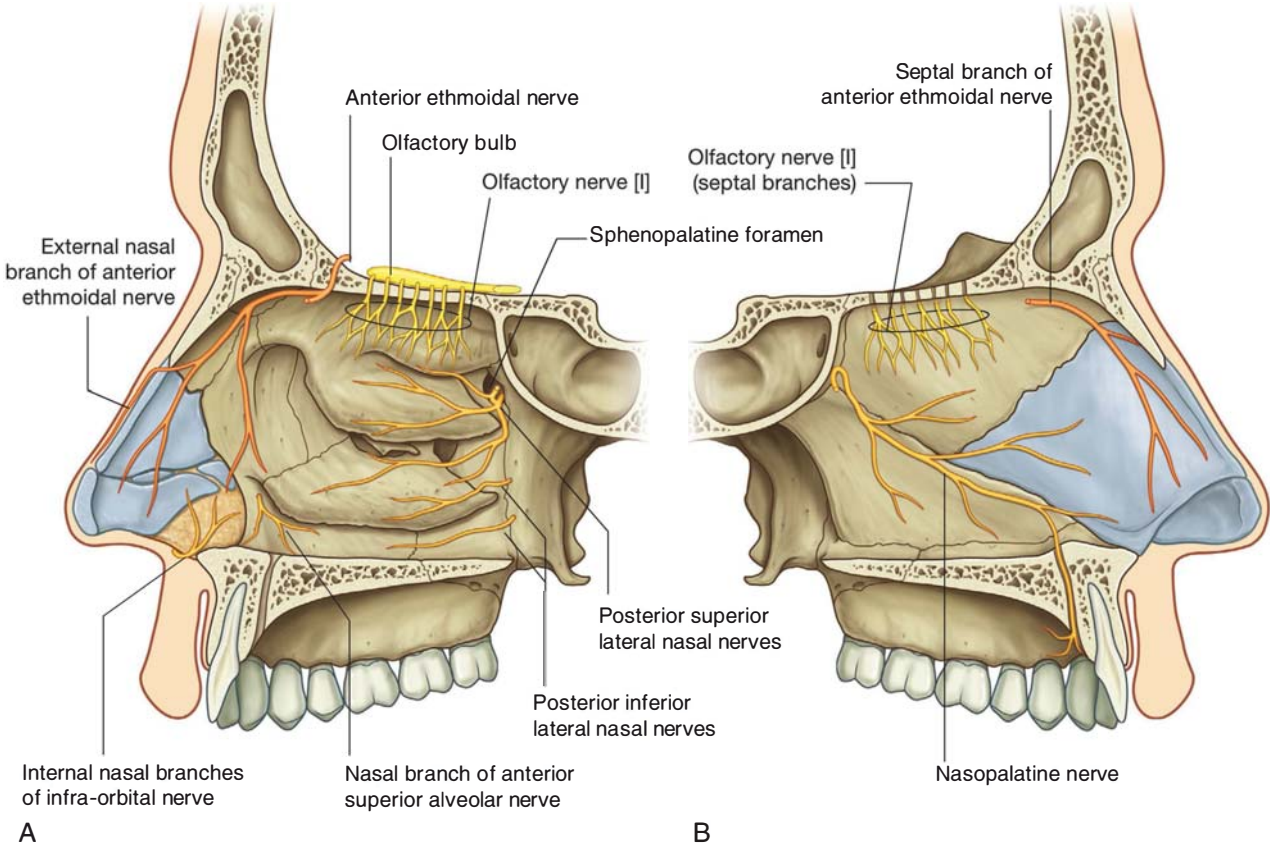


Fig. 8.241 Innervation of the nasal cavities. **A.** Lateral wall of right nasal cavity. **B.** Medial wall of right nasal cavity.

of the nose by traveling between the nasal bone and lateral nasal cartilage, and then terminates as the **external nasal nerve**, which supplies skin around the naris, in the nasal vestibule, and on the tip of the nose.

Like the anterior ethmoidal nerve, the posterior ethmoidal nerve leaves the orbit through a similar canal in the medial wall of the orbit. It terminates by supplying the mucosa of the ethmoidal cells and sphenoidal sinus and normally does not extend into the nasal cavity itself.

Branches from the maxillary nerve [V_2]

A number of nasal branches from the maxillary nerve [V_2] innervate the nasal cavity. Many of these nasal branches (Fig. 8.241) originate in the pterygopalatine fossa, which is just lateral to the lateral wall of the nasal cavity, and leave the fossa to enter the nasal cavity by passing medially through the sphenopalatine foramen or through smaller foramina in the lateral wall:

- A number of these nerves (**posterior superior lateral nasal nerves**) pass forward on and supply the lateral wall of the nasal cavity.
- Others (**posterior superior medial nasal nerves**) cross the roof to the nasal septum and supply both these regions.
- The largest of these nerves is the **nasopalatine nerve**, which passes forward and down the medial wall of the nasal cavity to pass through the incisive canal onto the roof of the oral cavity, and terminates by supplying the oral mucosa posterior to the incisor teeth.
- Other nasal nerves (**posterior inferior nasal nerves**) originate from the greater palatine nerve, descending from the pterygopalatine fossa in the palatine canal just lateral to the nasal cavity, and pass through small bony foramina to innervate the lateral wall of the nasal cavity.
- A small nasal nerve also originates from the anterior superior alveolar branch of the infra-orbital nerve and passes medially through the maxilla to supply the lateral wall near the anterior end of the inferior concha.

Parasympathetic innervation

Secretomotor innervation of glands in the mucosa of the nasal cavity and paranasal sinuses is by preganglionic parasympathetic fibers carried in the greater petrosal branch of the facial nerve [VII]. These fibers enter the pterygopalatine fossa and synapse in the pterygopalatine ganglion (see Fig. 8.153 and pp. 997–998). Postganglionic parasympathetic fibers then join branches of the maxillary nerve [V_2] to leave the fossa and ultimately reach target glands.

Sympathetic innervation

Sympathetic innervation, mainly involved with regulating blood flow in the nasal mucosa, is from spinal cord level T1. Preganglionic sympathetic fibers enter the sympathetic trunk and ascend to synapse in the superior cervical sympathetic ganglion. Postganglionic sympathetic fibers pass onto the internal carotid artery, enter the cranial cavity, and then leave the internal carotid artery to form the deep petrosal nerve, which joins the greater petrosal nerve of the facial nerve [VII] and enters the pterygopalatine fossa (see Figs. 8.152 and 8.153 and pp. 997–998).

Like the parasympathetic fibers, the sympathetic fibers follow branches of the maxillary nerve [V_2] into the nasal cavity.

Lymphatics

Lymph from anterior regions of the nasal cavities drains forward onto the face by passing around the margins of the nares (Fig. 8.242). These lymphatics ultimately connect with the submandibular nodes.

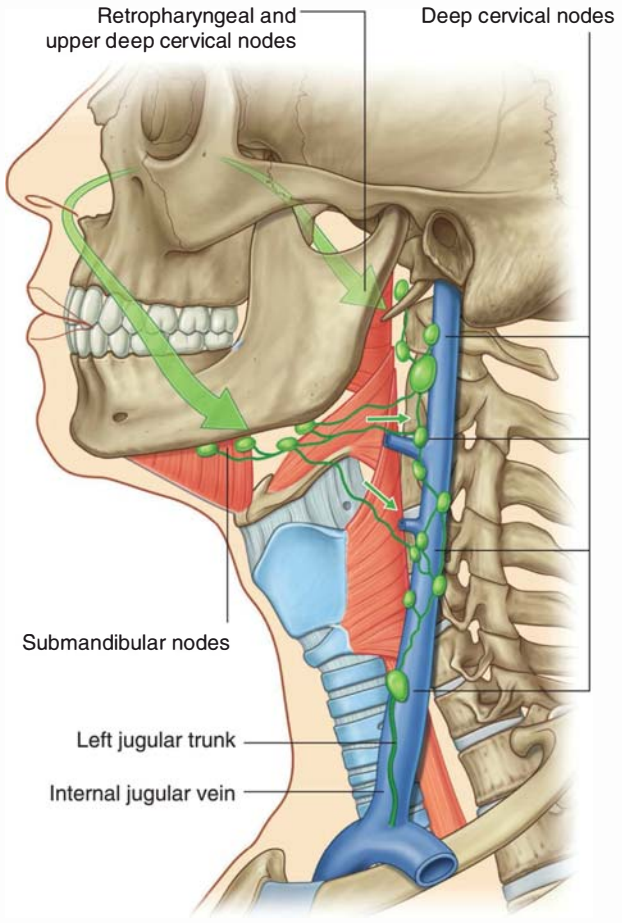


Fig. 8.242 Lymphatic drainage of the nasal cavities.

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Lymph from posterior regions of the nasal cavity and the paranasal sinuses drains into upper deep cervical nodes. Some of this lymph passes first through the retropharyngeal nodes.

ORAL CAVITY

The oral cavity is inferior to the nasal cavities (Fig. 8.243A). It has a roof and floor and lateral walls, opens onto the face through the oral fissure, and is continuous with the cavity of the pharynx at the oropharyngeal isthmus.

The roof of the oral cavity consists of the hard and soft palates. The floor is formed mainly of soft tissues, which include a muscular diaphragm and the tongue. The lateral walls (cheeks) are muscular and merge anteriorly with the lips surrounding the **oral fissure** (the anterior opening of the oral cavity).

The posterior aperture of the oral cavity is the oropharyngeal isthmus, which opens into the oral part of the pharynx.

The oral cavity is separated into two regions by the upper and lower dental arches consisting of the teeth and alveolar bone that supports them (Fig. 8.243B):

- The outer **oral vestibule**, which is horseshoe shaped, is between the dental arches and the deep surfaces of the cheeks and lips—the oral fissure opens into it and can be opened and closed by muscles of facial expression, and by movements of the lower jaw.
- The inner **oral cavity proper** is enclosed by the dental arches.

The degree of separation between the upper and lower arches is established by elevating or depressing the lower jaw (mandible) at the temporomandibular joint.

The oropharyngeal isthmus at the back of the oral cavity proper can be opened and closed by surrounding soft tissues, which include the soft palate and tongue.

The oral cavity has multiple functions:

- It is the inlet for the digestive system involved with the initial processing of food, which is aided by secretions from salivary glands.
- It manipulates sounds produced by the larynx and one outcome of this is speech.

It can be used for breathing because it opens into the pharynx, which is a common pathway for food and air. For this reason, the oral cavity can be used by physicians to access the lower airway, and dentists use "rubber dams" to prevent debris such as tooth fragments from passing through the oropharyngeal isthmus and pharynx into either the esophagus or the lower airway.

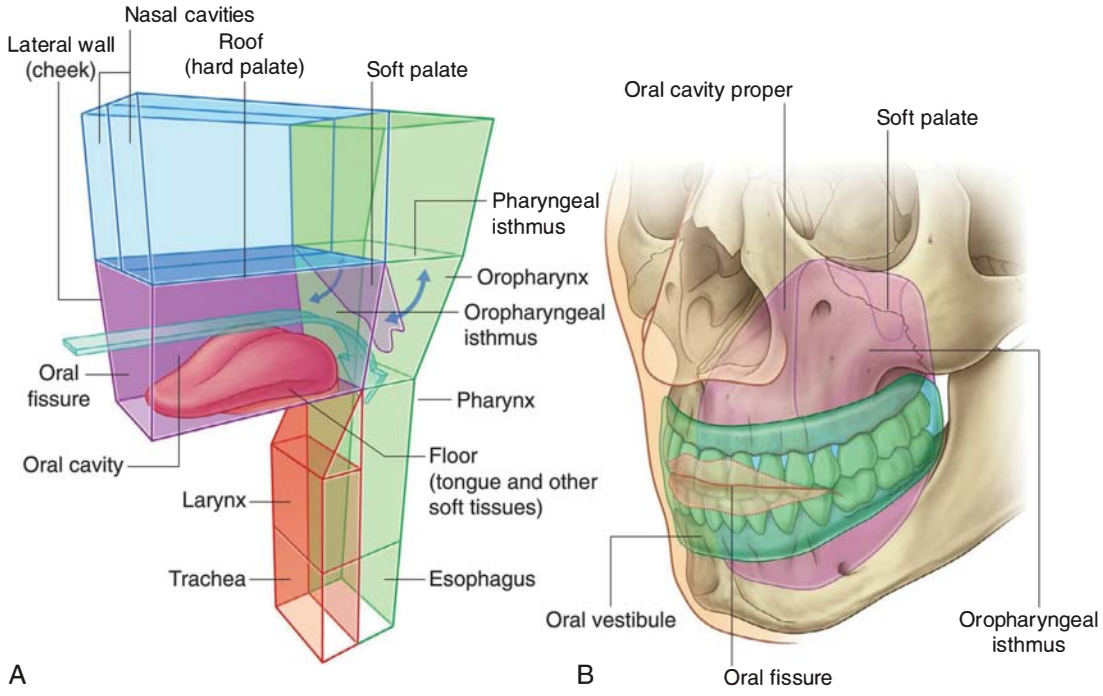


Fig. 8.243 Oral cavity. **A.** Relationship to other cavities. **B.** Oral vestibule and oral cavity proper.

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Multiple nerves innervate the oral cavity

General sensory innervation is carried predominantly by branches of the trigeminal nerve [V]:

- The upper parts of the cavity, including the palate and the upper teeth, are innervated by branches of the maxillary nerve [V_2].
- The lower parts, including the teeth and oral part of the tongue, are innervated by branches of the mandibular nerve [V_3].
- Taste (special afferent [SA]) from the oral part or anterior two-thirds of the tongue is carried by branches of the facial nerve [VII], which join and are distributed with branches of the trigeminal nerve [V].
- Parasympathetic fibers to the glands within the oral cavity are also carried by branches of the facial nerve [VII], which are distributed with branches of the trigeminal nerve [V].
- Sympathetic fibers in the oral cavity ultimately come from spinal cord level T1, synapse in the superior cervical sympathetic ganglion, and are eventually distributed to the oral cavity along branches of the trigeminal nerve [V] or directly along blood vessels.

All muscles of the tongue are innervated by the hypoglossal nerve [XII], except the palatoglossus, which is innervated by the vagus nerve [X].

All muscles of the soft palate are innervated by the vagus nerve [X], except for the tensor veli palatini, which is innervated by a branch from the mandibular nerve [V_3]. The muscle (mylohyoid) that forms the floor of the oral cavity is also innervated by the mandibular nerve [V_3].

Skeletal framework

Bones that contribute to the skeletal framework of the oral cavity or are related to the anatomy of structures in the oral cavity include:

- the paired maxillae, palatine, and temporal bones, and
- the unpaired mandible, sphenoid, and hyoid bones.

In addition, the cartilaginous parts of the pharyngotympanic tubes on the inferior aspect of the base of the skull are related to the attachment of muscles of the soft palate.

Maxillae

The two maxillae contribute substantially to the architecture of the roof of the oral cavity. The parts involved are the alveolar and palatine processes (Fig. 8.244A).

The palatine process is a horizontal shelf that projects from the medial surface of each maxilla. It originates just superior to the medial aspect of the alveolar process and extends to the midline where it is joined, at a suture, with the palatine process from the other side. Together, the two palatine processes form the anterior two-thirds of the hard palate.

In the midline on the inferior surface of the hard palate and at the anterior end of the intermaxillary suture is a single small fossa (incisive fossa) just behind the incisor teeth. Two incisive canals, one on each side, extend posterosuperiorly from the roof of this fossa to open onto the floor of the nasal cavity. The canals and fossae allow passage of the greater palatine vessels and the nasopalatine nerves.

Palatine bones

The parts of each L-shaped palatine bone that contribute to the roof of the oral cavity are the horizontal plate and the pyramidal process (Fig. 8.244A).

The horizontal plate projects medially from the inferior aspect of the palatine bone and is joined by sutures to its partner in the midline and, on the same side, with the palatine process of the maxilla anteriorly.

A single **posterior nasal spine** is formed at the midline where the two horizontal plates join and projects backward from the margin of the hard palate. The posterior margin of the horizontal plates and the posterior nasal spine are associated with attachment of the soft palate.

The greater palatine foramen, formed mainly by the horizontal plate of the palatine bone and completed laterally by the adjacent part of the maxilla, opens onto the posterolateral aspect of the horizontal plate. This foramen is the inferior opening of the palatine canal, which continues superiorly into the pterygopalatine fossa and transmits the greater palatine nerve and vessels to the palate.

Also opening onto the palatine bone is the lesser palatine foramen. This foramen is the inferior opening of the lesser palatine canal, which branches from the greater palatine canal and transmits the lesser palatine nerve and vessels to the soft palate.

The pyramidal process projects posteriorly and fills the space between the inferior ends of the medial and lateral plates of the pterygoid process of the sphenoid bone.

Sphenoid bone

The pterygoid processes and spines of the sphenoid bone are associated with structures related to the soft palate, which forms part of the roof of the oral cavity (Fig. 8.244A).

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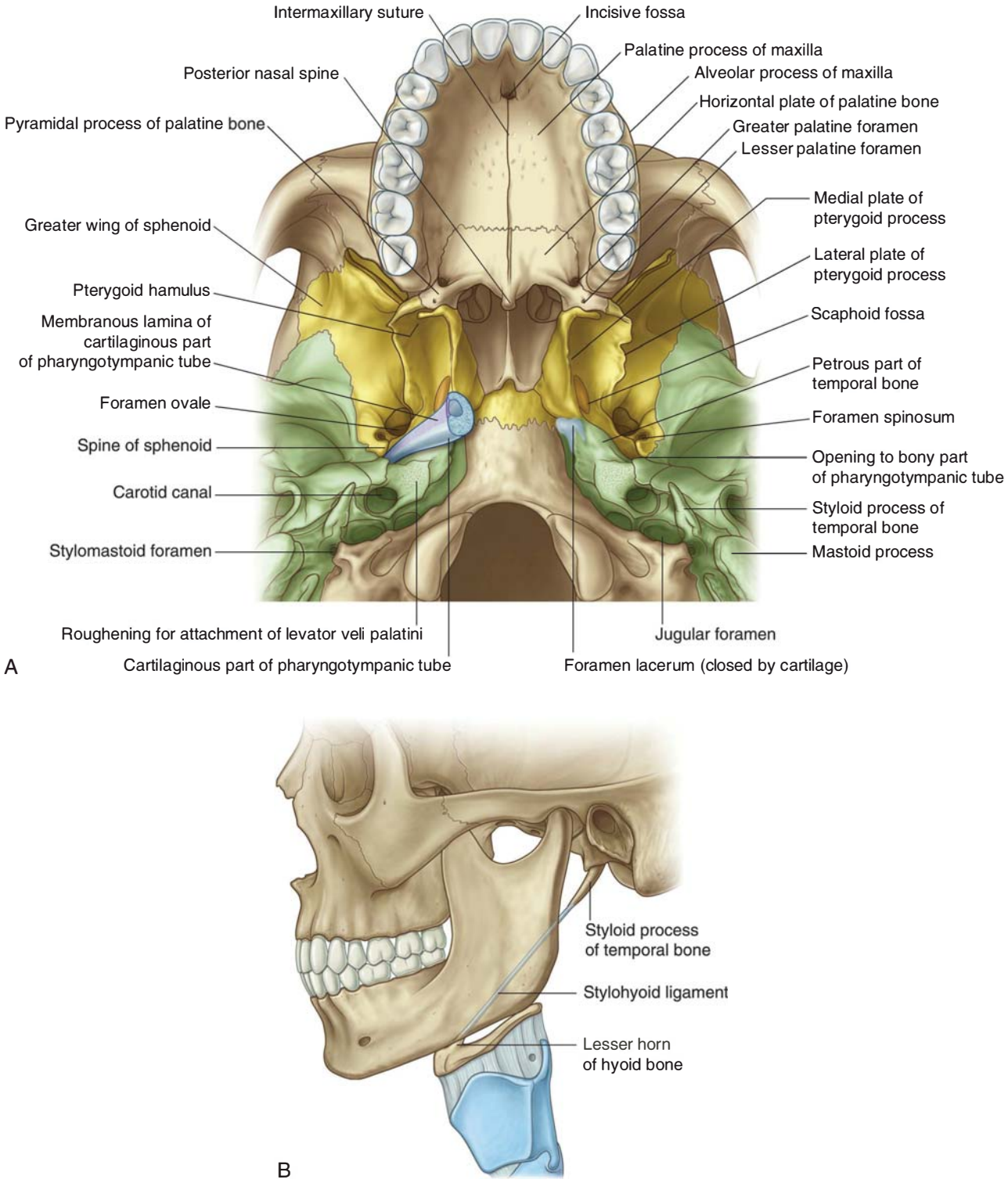


Fig. 8.244 Base and lateral aspects of the skull. **A.** Features in the base of the skull related to structures associated with the oral cavity. **B.** Styloid process of the temporal bone.

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The pterygoid processes descend, one on each side, from the lateral aspect of the body of the sphenoid bone. Each process has a medial and a lateral plate. These two vertically oriented plates project from the posterior aspect of the process. The V-shaped gap that occurs inferiorly between the two plates is filled by the pyramidal process of the palatine bone.

Projecting posterolaterally from the inferior margin of the medial plate of the pterygoid process is an elongate hook-shaped structure (the pterygoid hamulus). This hamulus is immediately behind the alveolar arch and inferior to the posterior margin of the hard palate. It is:

- a "pulley" for one of the muscles (tensor veli palatini) of the soft palate, and
- the attachment site for the upper end of the pterygomandibular raphe, which is attached below to the mandible and joins together the superior constrictor of the pharynx and the buccinator muscle of the cheek.

At the root of the medial plate of the pterygoid process on the base of the skull is a small canoe-shaped fossa (**scaphoid fossa**), which begins just medial to the foramen ovale and descends anteriorly and medially to the root of the medial plate of the pterygoid process (Fig. 8.244A). This fossa is for the attachment of one of the muscles of the soft palate (tensor veli palatini).

The spines of the sphenoid, one on each side, are vertical projections from the inferior surfaces of the greater wings of the sphenoid bone (Fig. 8.244A). Each spine is immediately posteromedial to the foramen spinosum.

The medial aspect of the spine provides attachment for the most lateral part of the tensor veli palatini muscle of the soft palate.

Temporal bone

The styloid process and inferior aspect of the petrous part of the temporal bone provide attachment for muscles associated with the tongue and soft palate, respectively.

The styloid process projects anteroinferiorly from the underside of the temporal bone. It can be as long as 1 inch (2.5 cm) and points toward the lesser horn of the hyoid bone to which it is attached by the stylohyoid ligament (Fig. 8.244B). The root of the styloid process is immediately anterior to the stylomastoid foramen and lateral to the jugular foramen. The styloglossus muscle of the tongue attaches to the anterolateral surface of the styloid process.

The inferior aspect of the temporal bone has a triangular roughened area immediately anteromedial to the opening of the carotid canal (Fig. 8.244A). The levator veli palatini muscle of the soft palate is attached here.

Cartilaginous part of the pharyngotympanic tube

The trumpet-shaped cartilaginous part of the pharyngotympanic tube is in a groove between the anterior margin of the petrous part of the temporal bone and the posterior margin of the greater wing of the sphenoid (Fig. 8.244A).

The medial and lateral walls of the cartilaginous part of the pharyngotympanic tube are formed mainly of cartilage, whereas the more inferolateral wall is more fibrous and is known as the **membranous lamina**.

The apex of the cartilaginous part of the pharyngotympanic tube connects laterally to the opening of the bony part in the temporal bone.

The expanded medial end of the cartilaginous part of the pharyngotympanic tube is immediately posterior to the upper margin of the medial plate of the pterygoid process and opens into the nasopharynx.

The cartilaginous part of the pharyngotympanic tube is lateral to the attachment of the levator veli palatini muscle to the petrous part of the temporal bone and medial to the spine of the sphenoid. The tensor veli palatini muscle is attached, in part, to the membranous lamina.

Mandible

The mandible is the bone of the lower jaw (Fig. 8.245). It consists of a body of right and left parts, which are fused anteriorly in the midline (**mandibular symphysis**), and two rami. The site of fusion is particularly visible on the external surface of the bone as a small vertical ridge in the midline.

The upper surface of the body of the mandible bears the alveolar arch (Fig. 8.245B), which anchors the lower teeth, and on its external surface on each side is a small mental foramen (Fig. 8.245B).

Posterior to the mandibular symphysis on the internal surface of the mandible are two pairs of small spines, one pair immediately above the other pair. These are the **superior** and **inferior mental spines** (**superior** and **inferior genial spines**) (Fig. 8.245A,C), and are attachment sites for a pair of muscles that pass into the tongue and a pair of muscles that connect the mandible to the hyoid bone.

Extending from the midline and originating inferior to the mental spines is a raised line or ridge (the **mylohyoid line**) (Fig. 8.245C), which runs posteriorly and superiorly along the internal surface of each side of the body of the mandible to end just below the level of the last molar tooth.

Above the anterior one-third of the mylohyoid line is a shallow depression (the **sublingual fossa**) (Fig. 8.245C), and below the posterior two-thirds of the mylohyoid line is another depression (the **submandibular fossa**) (Fig. 8.245C).

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Between the last molar tooth and the mylohyoid line is a shallow groove for the lingual nerve.

Immediately posterior to the last molar tooth on the medial upper surface of the body of the mandible is a small triangular depression (**retromolar triangle**) (Fig. 8.245A,C). The pterygomandibular raphe attaches just medial to the apex of this triangle and extends from here to the tip of the pterygoid hamulus above.

The ramus of the mandible, one on each side, is quadrangular shaped and oriented in the sagittal plane. On the medial surface of the ramus is a large **mandibular foramen** for transmission of the inferior alveolar nerve and vessels (Fig. 8.245C).

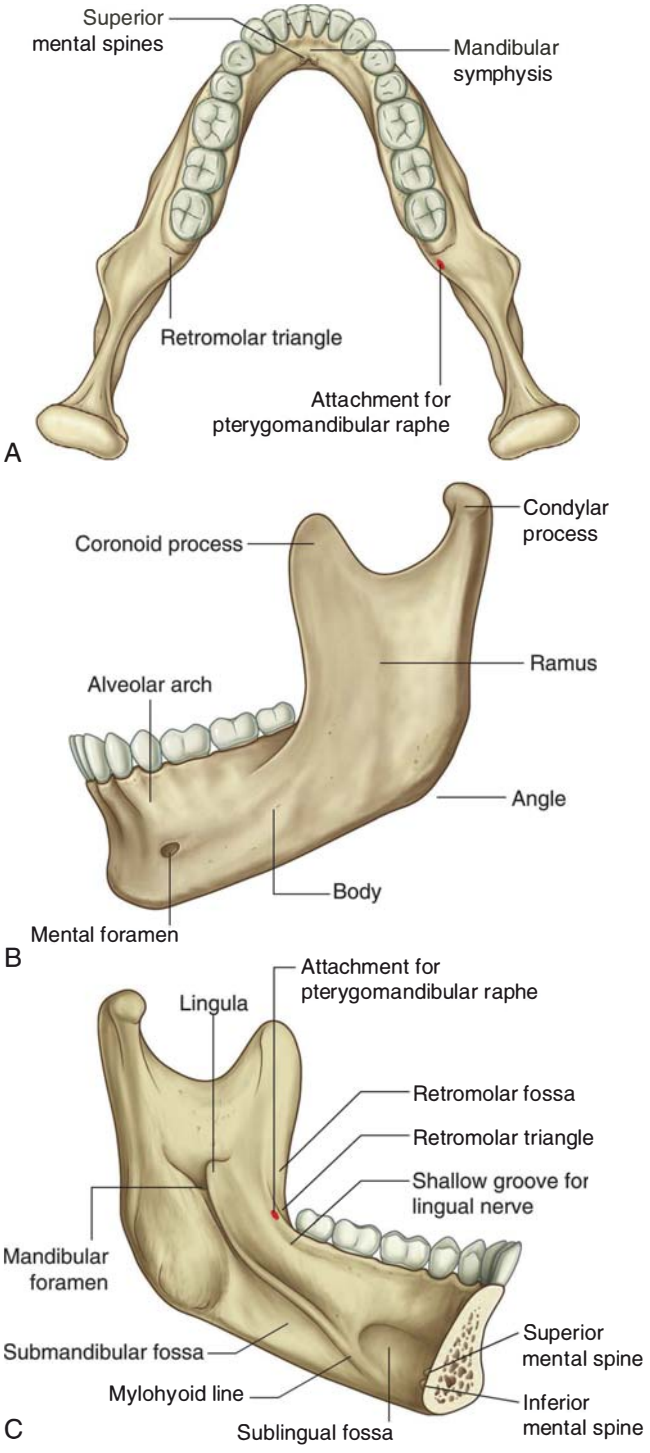


Fig. 8.245 Mandible. **A.** Superior view. **B.** Lateral view. **C.** Medial view.

Hyoid bone

The hyoid bone is a small U-shaped bone in the neck between the larynx and the mandible. It has an anterior body of hyoid bone and two large greater horns, one on each side, which project posteriorly and superiorly from the body (Fig. 8.246). There are two small conical lesser horns on the superior surface where the greater horns join with the body. The stylohyoid ligaments attach to the apices of the lesser horns.

The hyoid bone is a key bone in the neck because it connects the floor of the oral cavity in front with the pharynx behind and the larynx below.

Walls: the cheeks

The walls of the oral cavity are formed by the cheeks.

Each cheek consists of fascia and a layer of skeletal muscle sandwiched between skin externally and oral mucosa internally. The thin layer of skeletal muscle within the cheeks is principally the buccinator muscle.

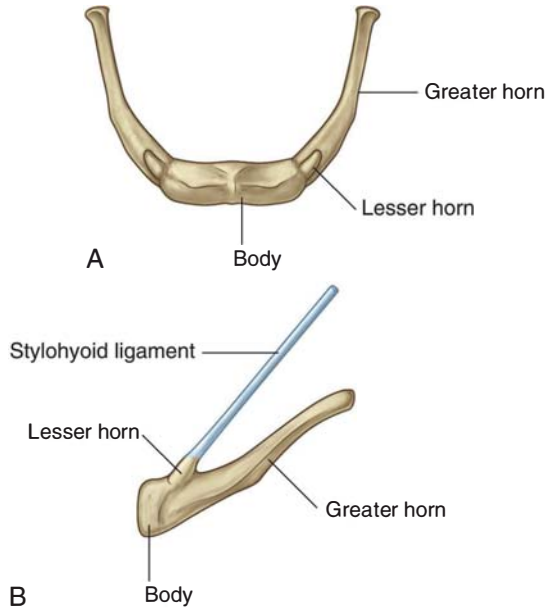


Fig. 8.246 Hyoid bone. **A.** Anterior view. **B.** Lateral view.

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Buccinator

The buccinator muscle is one of the muscles of facial expression (Fig. 8.247; also see p. 909). It is in the same plane as the superior constrictor muscle of the pharynx. In fact, the posterior margin of the buccinator muscle is joined to the anterior margin of the superior constrictor muscle by the pterygomandibular raphe, which runs between the tip of the pterygoid hamulus of the sphenoid bone above and a roughened area of bone immediately behind the last molar tooth on the mandible below.

The buccinator and superior constrictor muscles therefore provide continuity between the walls of the oral and pharyngeal cavities.

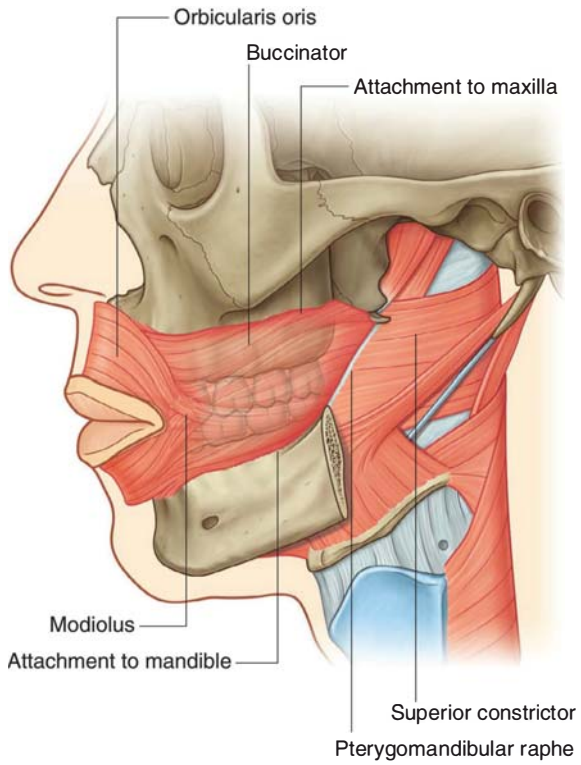


Fig. 8.247 Buccinator muscle.

The buccinator muscle, in addition to originating from the pterygomandibular raphe, also originates directly from the alveolar part of the mandible and alveolar process of the maxilla.

From its three sites of origin, the muscle fibers of the buccinator run forward to blend with those of the orbicularis oris muscle and to insert into the modiolus, which is a small button-shaped nodule of connective tissue at the interface between the muscles of the lips and cheeks on each side.

The buccinator muscle holds the cheeks against the alveolar arches and keeps food between the teeth when chewing.

The buccinator is innervated by the buccal branch of the facial nerve [VII]. General sensation from the skin and oral mucosa of the cheeks is carried by the buccal branch of the mandibular nerve [V_3].

Floor

The floor of the oral cavity proper is formed mainly by three structures:

- a muscular diaphragm, which fills the U-shaped gap between the left and right sides of the body of the mandible and is composed of the paired mylohyoid muscles;
- two cord-like geniohyoid muscles above the diaphragm, which run from the mandible in front to the hyoid bone behind; and
- the tongue, which is superior to the geniohyoid muscles.

Also present in the floor of the oral cavity proper are salivary glands and their ducts. The largest of these glands, on each side, are the sublingual gland and the oral part of the submandibular gland.

Mylohyoid muscles

The two thin mylohyoid muscles (Table 8.20), one on each side, together form a muscular diaphragm that defines the

Table 8.20 Muscles in the floor of the oral cavity

Muscle	Origin	Insertion	Innervation	Function
Mylohyoid	Mylohyoid line of mandible	Median fibrous raphe and adjacent part of hyoid bone	Nerve to mylohyoid from the inferior alveolar branch of mandibular nerve [V_3]	Supports and elevates floor of oral cavity; depresses mandible when hyoid is fixed; elevates and pulls hyoid forward when mandible is fixed
Geniohyoid	Inferior mental spines of mandible	Body of hyoid bone	C1	Elevates and pulls hyoid bone forward; depresses mandible when hyoid is fixed

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inferior limit of the floor of the oral cavity (Fig. 8.248A). Each muscle is triangular in shape with its apex pointed forward.

The lateral margin of each triangular muscle is attached to the mylohyoid line on the medial side of the body of the mandible. From here, the muscle fibers run slightly downward to the medial margin at the midline where the fibers are joined together with those of their partner muscle on the other side by a raphe. The raphe extends from the posterior aspect of the mandibular symphysis in front to the body of the hyoid bone behind.

The posterior margin of each mylohyoid muscle is free except for a small medial attachment to the hyoid bone.

The mylohyoid muscles:

- contribute structural support to the floor of the oral cavity,
- participate in elevating and pulling forward the hyoid bone, and therefore the attached larynx, during the initial stages of swallowing, and
- when the hyoid bone is fixed in position, depress the mandible and open the mouth.

Like the muscles of mastication, the mylohyoid muscles are innervated by the mandibular nerve [V_3]. The specific branch that innervates the mylohyoid muscles is the nerve to the mylohyoid from the inferior alveolar nerve.

Geniohyoid muscles

The geniohyoid muscles (Table 8.20) are paired cord-like muscles that run, one on either side of the midline, from the inferior mental spines on the posterior surface of the mandibular symphysis to the anterior surface of the body of the hyoid bone (Fig. 8.248B,C). They are immediately

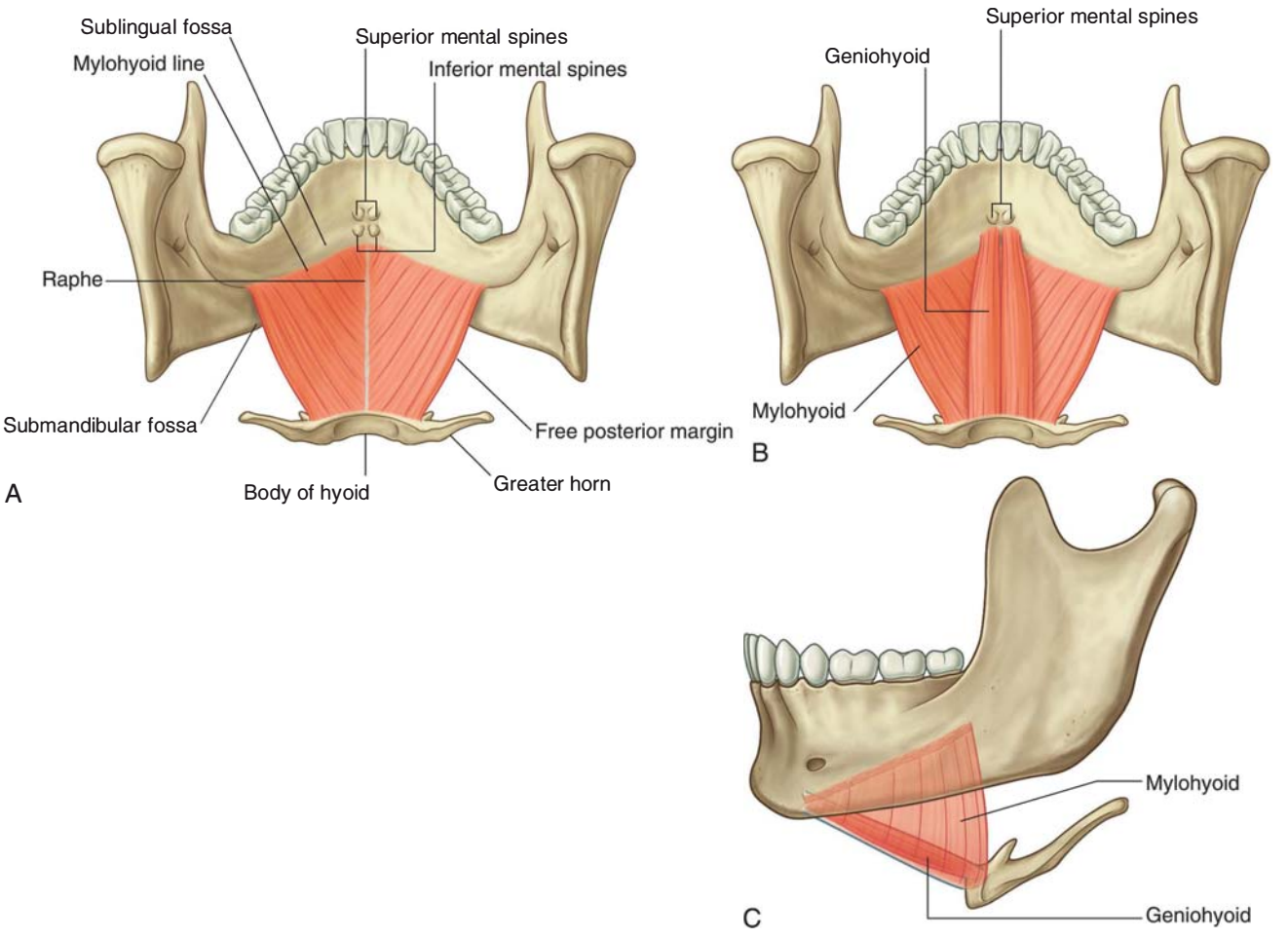


Fig. 8.248 A. Mylohyoid muscles. **B.** Geniohyoid muscles. **C.** Lateral view.

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superior to the mylohyoid muscles in the floor of the mouth and inferior to the genioglossus muscles that form part of the root of the tongue.

The geniohyoid muscles:

- mainly pull the hyoid bone, and therefore the attached larynx, up and forward during swallowing; and
- because they pass posteroinferiorly from the mandible to the hyoid bone, when the hyoid bone is fixed, they can act with the mylohyoid muscles to depress the mandible and open the mouth.

Unlike other muscles that move the mandible at the temporomandibular joint, the geniohyoid muscles are innervated by a branch of cervical nerve C1, which "hitchhikes" from the neck along the hypoglossal nerve [XII] into the floor of the oral cavity.

Gateway into the floor of the oral cavity

In addition to defining the lower limit of the floor of the oral cavity, the free posterior border of the mylohyoid muscle on each side forms one of the three margins of a large triangular aperture **(oropharyngeal triangle)**, which is a major route by which structures in the upper neck and infratemporal fossa of the head pass to and from structures in the floor of the oral cavity (Fig. 8.249). The other two muscles that complete the margins of the aperture are the superior and middle constrictor muscles of the pharynx.

Most structures that pass through the aperture are associated with the tongue and include muscles (hyoglossus, styloglossus), vessels (lingual artery and vein), nerves (lingual, hypoglossal [XII], glossopharyngeal [IX]), and lymphatics.

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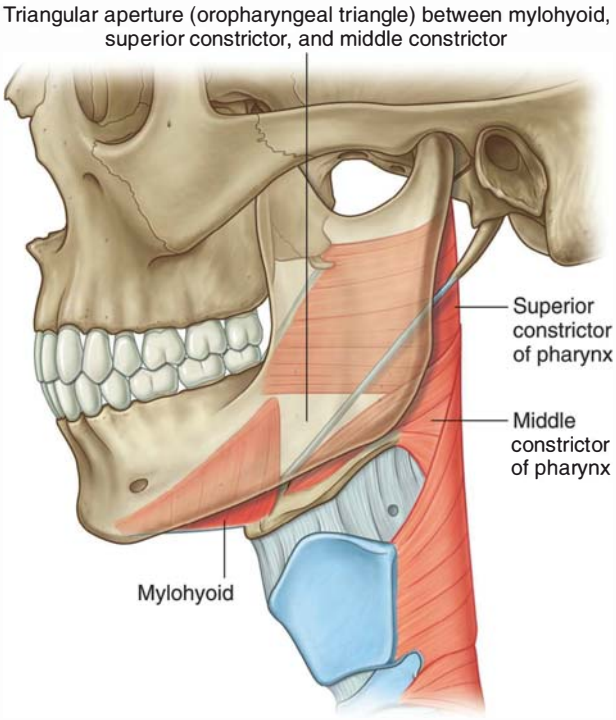


Fig. 8.249 Gateway into the floor of the oral cavity.

A large salivary gland (the submandibular gland) is "hooked" around the free posterior margin of the mylohyoid muscle and therefore also passes through the opening.

Tongue

The tongue is a muscular structure that forms part of the floor of the oral cavity and part of the anterior wall of the oropharynx (Fig. 8.250A). Its anterior part is in the oral cavity and is somewhat triangular in shape with a blunt **apex of the tongue**. The apex is directed anteriorly and sits immediately behind the incisor teeth. The **root of the tongue** is attached to the mandible and the hyoid bone.

The superior surface of the oral or anterior two-thirds of the tongue is oriented in the horizontal plane.

The pharyngeal surface or posterior one-third of the tongue curves inferiorly and becomes oriented more in the vertical plane. The oral and pharyngeal surfaces are separated by a V-shaped **terminal sulcus of the tongue**. This terminal sulcus forms the inferior margin of the oropharyngeal isthmus between the oral and pharyngeal cavities. At the apex of the V-shaped sulcus is a small depression (the **foramen cecum of the tongue**), which marks the site in the embryo where the epithelium invaginated to form the thyroid gland. In some people a thyroglossal duct persists and connects the foramen cecum on the tongue with the thyroid gland in the neck.

Papillae

The superior surface of the oral part of the tongue is covered by hundreds of papillae (Fig. 8.250B):

- **Filiform papillae** are small cone-shaped projections of the mucosa that end in one or more points.
- **Fungiform papillae** are rounder in shape and larger than the filiform papillae, and tend to be concentrated along the margins of the tongue.
- The largest of the papillae are the vallate papillae, which are blunt-ended cylindrical papillae invaginations in the tongue's surface—there are only about 8 to 12 vallate papillae in a single V-shaped line immediately anterior to the terminal sulcus of the tongue.
- **Foliate papillae** are linear folds of mucosa on the sides of the tongue near the terminal sulcus of tongue.

The papillae in general increase the area of contact between the surface of the tongue and the contents of the oral cavity. All except the filiform papillae have taste buds on their surfaces.

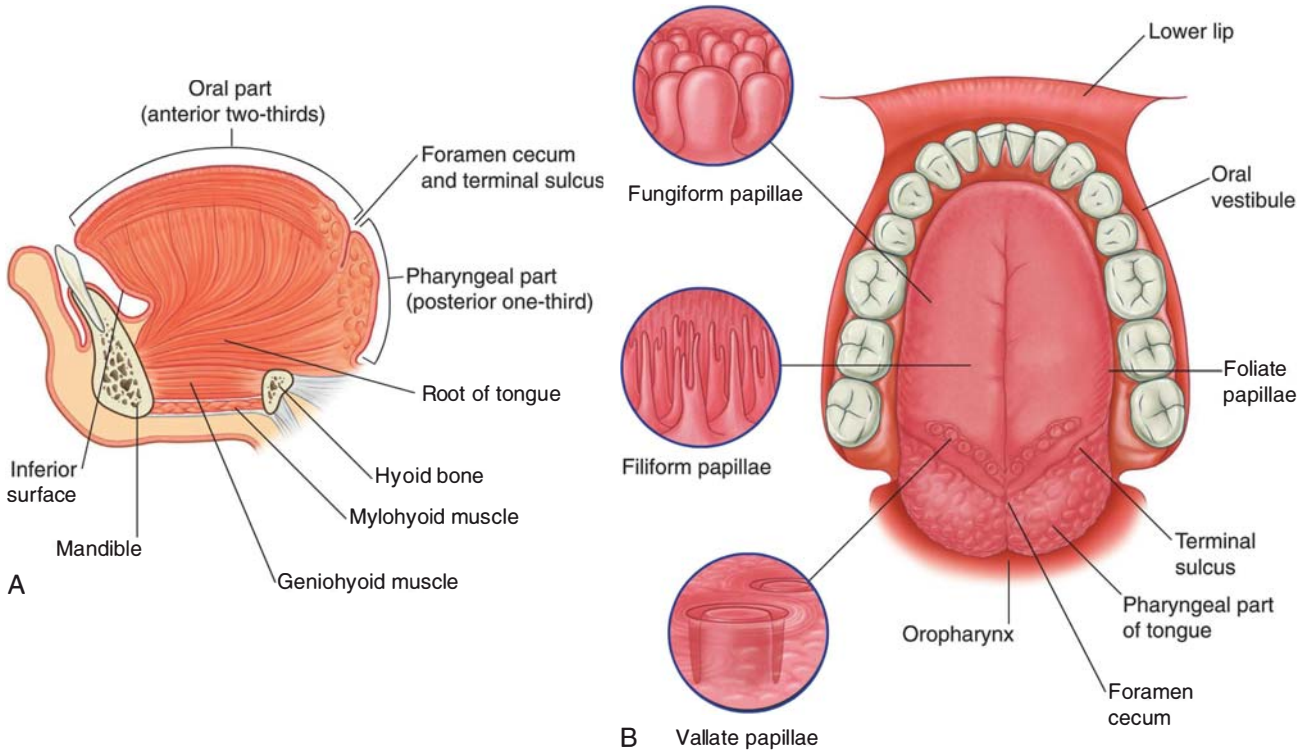


Fig. 8.250 Tongue. **A.** Paramedian sagittal section. **B.** Superior view.

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Inferior surface of tongue

The undersurface of the oral part of the tongue lacks papillae, but does have a number of linear mucosal folds (see Fig. 8.261). A single median fold (the **frenulum of the tongue**) is continuous with the mucosa covering the floor of the oral cavity, and overlies the lower margin of a midline sagittal septum, which internally separates the right and left sides of the tongue. On each side of the frenulum is a lingual vein, and lateral to each vein is a rough **fimbriated fold**.

Pharyngeal surface

The mucosa covering the pharyngeal surface of the tongue is irregular in contour because of the many small nodules of lymphoid tissue in the submucosa. These nodules are collectively the **lingual tonsil**.

There are no papillae on the pharyngeal surface.

Muscles

The bulk of the tongue is composed of muscle (Fig. 8.250 and Table 8.21).

The tongue is completely divided into left and right halves by a median sagittal septum composed of connective tissue. This means that all muscles of the tongue are paired. There are intrinsic and extrinsic lingual muscles.

Except for the palatoglossus, which is innervated by the vagus nerve [X], all muscles of the tongue are innervated by the hypoglossal nerve [XII].

Table 8.21 Muscles of the tongue

Muscle	Origin	Insertion	Innervation	Function
Intrinsic				
Superior longitudinal (just deep to surface of tongue)	Submucosal connective tissue at the back of the tongue and from the median septum of the tongue	Muscle fibers pass forward and obliquely to submucosal connective tissue and mucosa on margins of tongue	Hypoglossal nerve [XII]	Shortens tongue; curls apex and sides of tongue
Inferior longitudinal (between genioglossus and hyoglossus muscles)	Root of tongue (some fibers from hyoid)	Apex of tongue	Hypoglossal nerve [XII]	Shortens tongue; uncurls apex and turns it downward
Transverse	Median septum of the tongue	Submucosal connective tissue on lateral margins of tongue	Hypoglossal nerve [XII]	Narrows and elongates tongue
Vertical	Submucosal connective tissue on dorsum of tongue	Connective tissue in more ventral regions of tongue	Hypoglossal nerve [XII]	Flattens and widens tongue
Extrinsic				
Genioglossus	Superior mental spines	Body of hyoid; entire length of tongue	Hypoglossal nerve [XII]	Protrudes tongue; depresses center of tongue
Hyoglossus	Greater horn and adjacent part of body of hyoid bone	Lateral surface of tongue	Hypoglossal nerve [XII]	Depresses tongue
Styloglossus	Styloid process (anterolateral surface)	Lateral surface of tongue	Hypoglossal nerve [XII]	Elevates and retracts tongue
Palatoglossus	Inferior surface of palatine aponeurosis	Lateral margin of tongue	Vagus nerve [X] (via pharyngeal branch to pharyngeal plexus)	Depresses palate; moves palatoglossal fold toward midline; elevates back of the tongue

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Intrinsic muscles

The intrinsic muscles of the tongue (Fig. 8.251) originate and insert within the substance of the tongue. They are divided into **superior longitudinal**, **inferior longitudinal**, **transverse**, and **vertical muscles**, and they alter the shape of the tongue by:

- lengthening and shortening it,
- curling and uncurling its apex and edges, and
- flattening and rounding its surface.

Working in pairs or one side at a time the intrinsic muscles of the tongue contribute to precision movements of the tongue required for speech, eating, and swallowing.

Extrinsic muscles

Extrinsic muscles of the tongue (Fig. 8.251 and Table 8.21) originate from structures outside the tongue and insert into the tongue. There are four major extrinsic muscles on each side, the genioglossus, hyoglossus, styloglossus, and palatoglossus. These muscles protrude, retract, depress, and elevate the tongue.

Genioglossus

The thick fan-shaped **genioglossus muscles** make a substantial contribution to the structure of the tongue. They occur on either side of the midline septum that separates left and right halves of the tongue.

The genioglossus muscles originate from the superior mental spines on the posterior surface of the mandibular

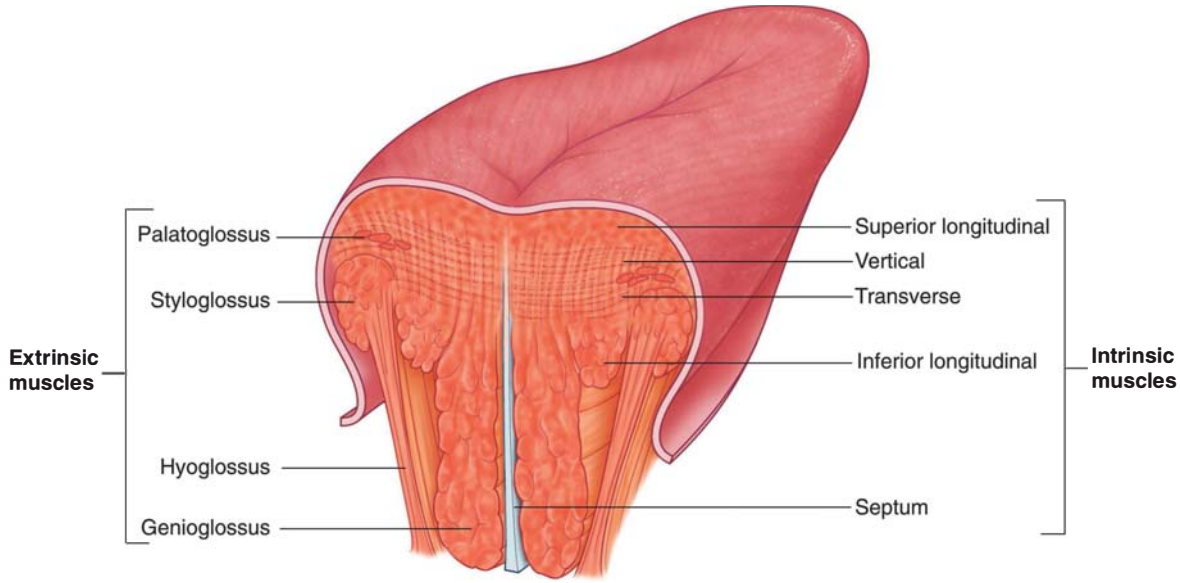


Fig. 8.251 Muscles of the tongue.

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symphysis immediately superior to the origin of the geniohyoid muscles from the inferior mental spines (Fig. 8.252). From this small site of origin, each muscle expands posteriorly and superiorly. The most inferior fibers attach to the hyoid bone. The remaining fibers spread out superiorly to blend with the intrinsic muscles along virtually the entire length of the tongue.

The genioglossus muscles:

- depress the central part of the tongue, and
- protrude the anterior part of the tongue out of the oral fissure (i.e., stick the tongue out).

Like most muscles of the tongue, the genioglossus muscles are innervated by the hypoglossal nerves [XII].

Asking a patient to "stick your tongue out" can be used as a test for the hypoglossal nerves [XII]. If the nerves are functioning normally, the tongue should protrude evenly in the midline. If the nerve on one side is not fully functional, the tip of the tongue will point to that side.

Hyoglossus

The hyoglossus muscles are thin quadrangular muscles lateral to the genioglossus muscles (Fig. 8.253).

Each hyoglossus muscle originates from the entire length of the greater horn and the adjacent part of the body of the hyoid bone. At its origin from the hyoid bone, the hyoglossus muscle is lateral to the attachment of the middle constrictor muscle of the pharynx. The muscle passes superiorly and anteriorly through the gap

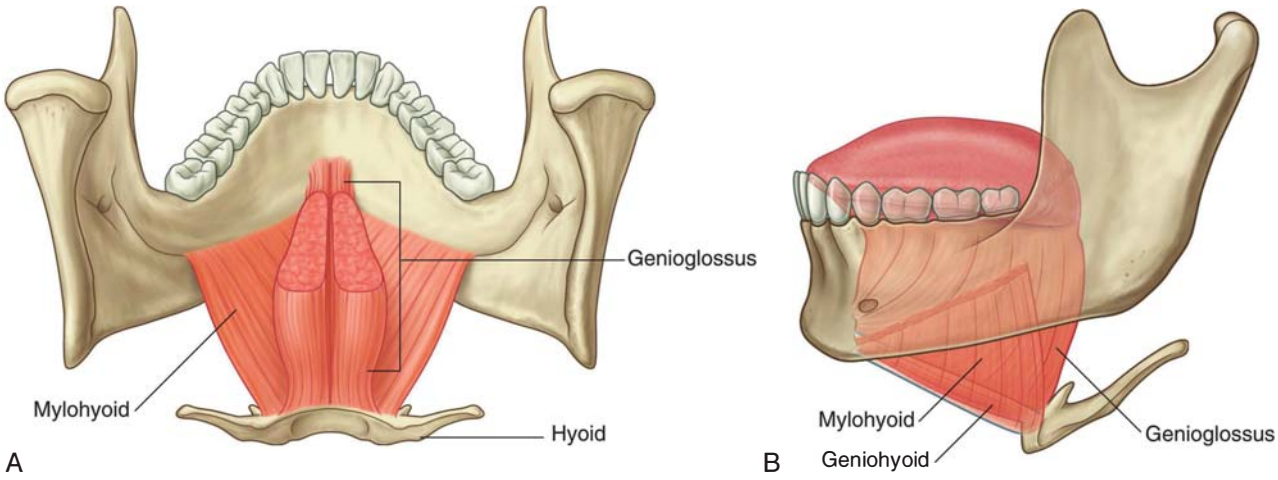


Fig. 8.252 Genioglossus muscles. **A.** Posterior view. **B.** Lateral (left) view.

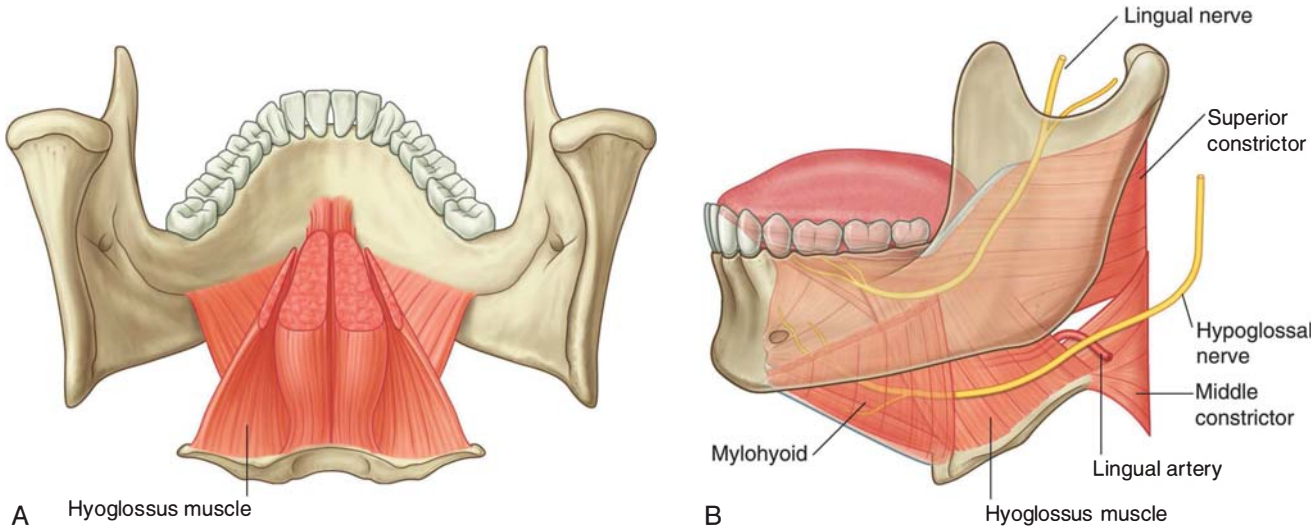


Fig. 8.253 Hyoglossus muscles. **A.** Posterior view. **B.** Lateral (left) view.

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(oropharyngeal triangle) between the superior constrictor, middle constrictor, and mylohyoid to insert into the tongue lateral to the genioglossus and medial to the styloglossus.

The hyoglossus muscle depresses the tongue and is innervated by the hypoglossal nerve [XII].

An important landmark. The hyoglossus muscle is an important landmark in the floor of the oral cavity:

- The lingual artery from the external carotid artery in the neck enters the tongue deep to the hyoglossus, between the hyoglossus and genioglossus.
- The hypoglossal nerve [XII] and lingual nerve (branch of the mandibular nerve [V_3]), from the neck and infratemporal fossa of the head, respectively, enter the tongue on the external surface of the hyoglossus.

Styloglossus

The styloglossus muscles originate from the anterior surface of the styloid processes of the temporal bones. From here, each muscle passes inferiorly and medially through the gap (oropharyngeal triangle) between the middle constrictor, superior constrictor, and mylohyoid muscles to enter the lateral surface of the tongue where they blend with the superior margin of the hyoglossus and with the intrinsic muscles (Fig. 8.254).

The styloglossus muscles retract the tongue and pull the back of the tongue superiorly. They are innervated by the hypoglossal nerves [XII].

Palatoglossus

The palatoglossus muscles are muscles of the soft palate and the tongue. Each originates from the undersurface of the palatine aponeurosis and passes anteroinferiorly to the lateral side of the tongue (Fig. 8.255).

The palatoglossus muscles:

- elevate the back of the tongue,
- move the palatoglossal arches of mucosa toward the midline, and
- depress the soft palate.

These movements facilitate closing of the oropharyngeal isthmus and as a result separate the oral cavity from the oropharynx.

Unlike other muscles of the tongue, but similar to most other muscles of the soft palate, the palatoglossus muscles are innervated by the vagus nerves [X].

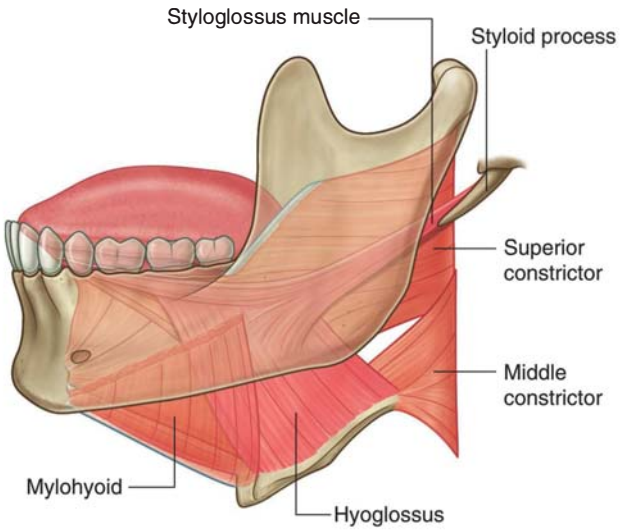


Fig. 8.254 Styloglossus muscles.

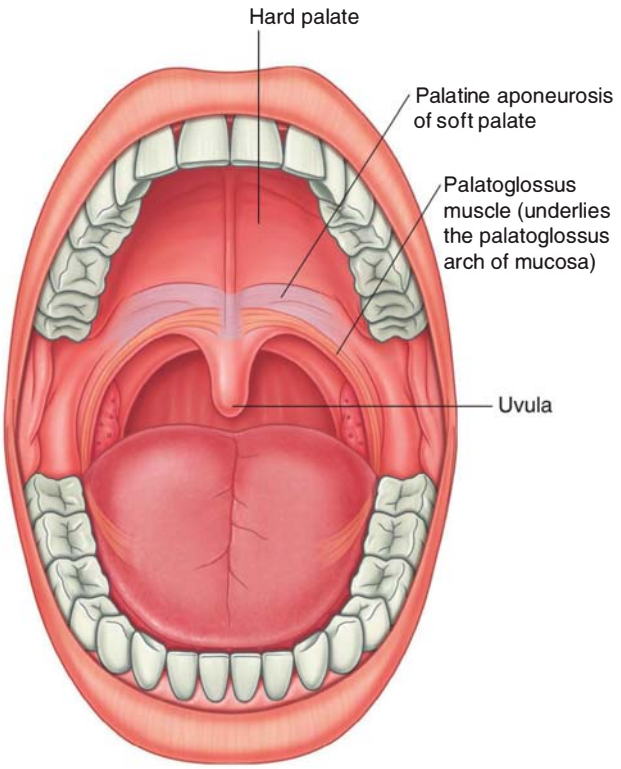


Fig. 8.255 Palatoglossus muscles.

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Vessels

Arteries

The major artery of the tongue is the **lingual artery** (Fig. 8.256).

On each side, the lingual artery originates from the external carotid artery in the neck adjacent to the tip of the greater horn of the hyoid bone. It forms an upward bend and then loops downward and forward to pass deep to the hyoglossus muscle, and accompanies the muscle through the aperture (oropharyngeal triangle) formed by the margins of the mylohyoid, superior constrictor, and middle constrictor muscles, and enters the floor of the oral cavity.

The lingual artery then travels forward in the plane between the hyoglossus and genioglossus muscles to the apex of the tongue.

In addition to the tongue, the lingual artery supplies the sublingual gland, gingiva, and oral mucosa in the floor of the oral cavity.

Veins

The tongue is drained by dorsal lingual and deep lingual veins (Fig. 8.256).

The **deep lingual veins** are visible through the mucosa on the undersurface of the tongue. Although they accompany the lingual arteries in anterior parts of the tongue, they become separated from the arteries posteriorly by the hyoglossus muscles. On each side, the deep lingual vein travels with the hypoglossal nerve [XII] on the external surface of the hyoglossus muscle and passes out of the floor of the oral cavity through the aperture formed by the margins of the mylohyoid, superior constrictor, and middle constrictor muscles. It joins the internal jugular vein in the neck.

The **dorsal lingual vein** follows the lingual artery between the hyoglossus and genioglossus muscles and, like the deep lingual vein, drains into the internal jugular vein in the neck.

Innervation

Innervation of the tongue is complex and involves a number of nerves (Figs. 8.256 and 8.257).

Glossopharyngeal nerve [IX]

Taste (SA) and general sensation from the pharyngeal part of the tongue are carried by the glossopharyngeal nerve [IX].

The glossopharyngeal nerve [IX] leaves the skull through the jugular foramen and descends along the

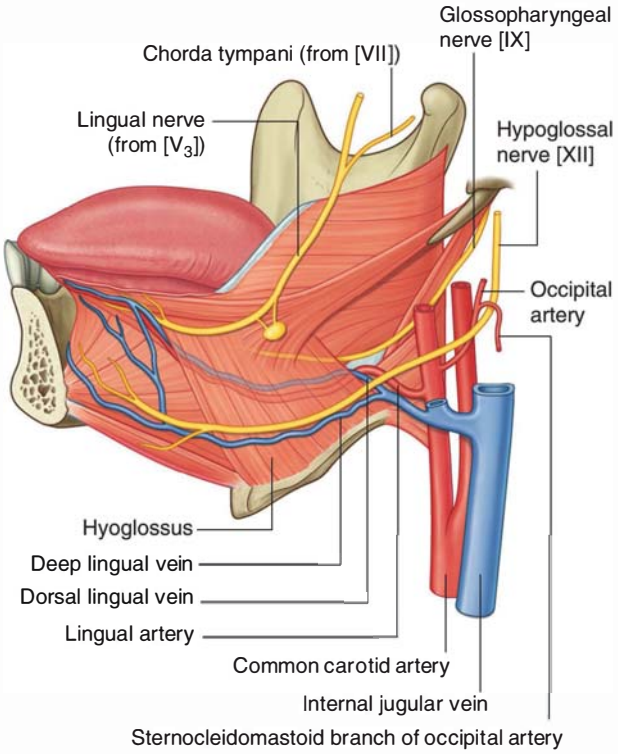


Fig. 8.256 Arteries, veins, and nerves of the tongue.

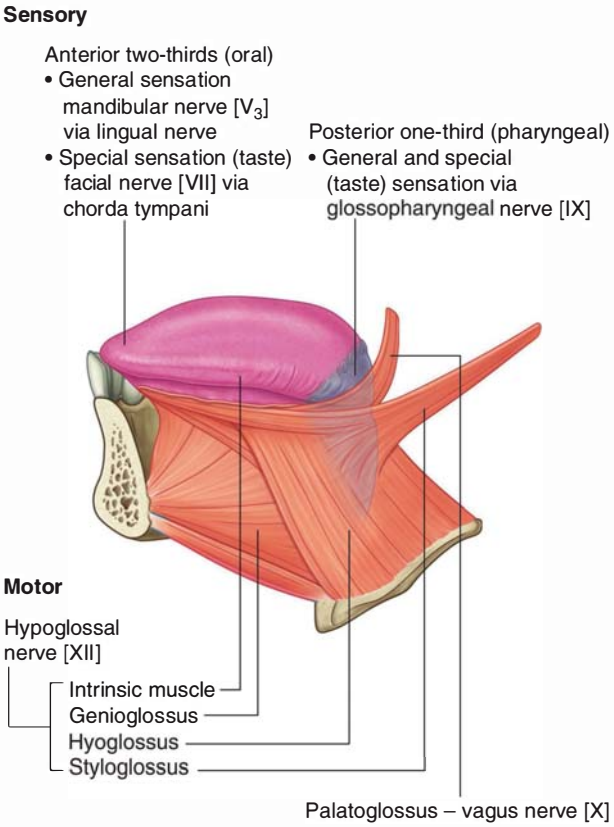


Fig. 8.257 Innervation of the tongue.

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posterior surface of the stylopharyngeus muscle. It passes around the lateral surface of the stylopharyngeus and then slips through the posterior aspect of the gap (oropharyngeal triangle) between the superior constrictor, middle constrictor, and mylohyoid muscles. The nerve then passes forward on the oropharyngeal wall just below the inferior pole of the palatine tonsil and enters the pharyngeal part of the tongue deep to the styloglossus and hyoglossus muscles. In addition to taste and general sensation on the posterior one-third of the tongue, branches creep anterior to the terminal sulcus of the tongue to carry taste (SA) and general sensation from the vallate papillae.

Lingual nerve

General sensory innervation from the anterior two-thirds or oral part of the tongue is carried by the **lingual nerve**, which is a major branch of the mandibular nerve [V_3]. It originates in the infratemporal fossa and passes anteriorly into the floor of the oral cavity by passing through the gap (oropharyngeal triangle) between the mylohyoid, superior constrictor, and middle constrictor muscles (Fig. 8.258). As it travels through the gap, it passes immediately inferior to the attachment of the superior constrictor to the mandible and continues forward on the medial surface of the mandible adjacent to the last molar tooth and deep to the gingiva. In this position, the nerve can be palpated against the bone by placing a finger into the oral cavity.

The lingual nerve then continues anteromedially across the floor of the oral cavity, loops under the submandibular duct, and ascends into the tongue on the external and superior surface of the hyoglossus muscle.

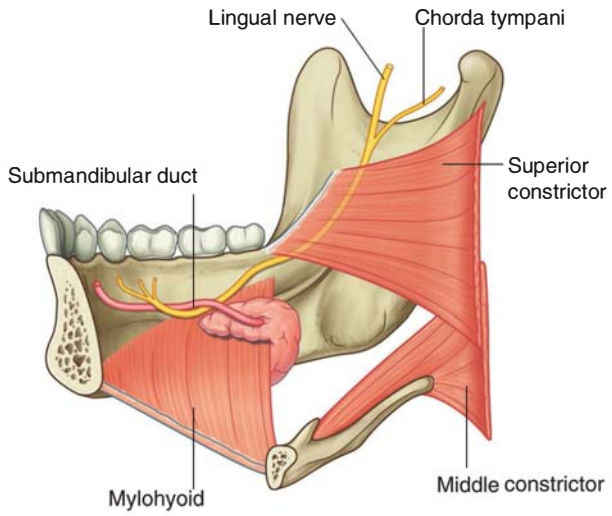


Fig. 8.258 Lingual nerve in the floor of the oral cavity (medial view).

In addition to general sensation from the oral part of the tongue, the lingual nerve also carries general sensation from the mucosa on the floor of the oral cavity and gingiva associated with the lower teeth. The lingual nerve also carries parasympathetic and taste fibers from the oral part of the tongue that are part of the facial nerve [VII].

Facial nerve [VII]

Taste (SA) from the oral part of the tongue is carried into the central nervous system by the facial nerve [VII]. Special sensory (SA) fibers of the facial nerve [VII] leave the tongue and oral cavity as part of the lingual nerve. The fibers then enter the chorda tympani nerve, which is a branch of the facial nerve [VII] that joins the lingual nerve in the infratemporal fossa (Fig. 8.258; also see p. 987).

Hypoglossal nerve [XII]

All muscles of the tongue are innervated by the hypoglossal nerve [XII] except for the palatoglossus muscle, which is innervated by the vagus nerve [X].

The hypoglossal nerve [XII] leaves the skull through the hypoglossal canal and descends almost vertically in the neck to a level just below the angle of the mandible (Fig. 8.259). Here it angles sharply forward around the

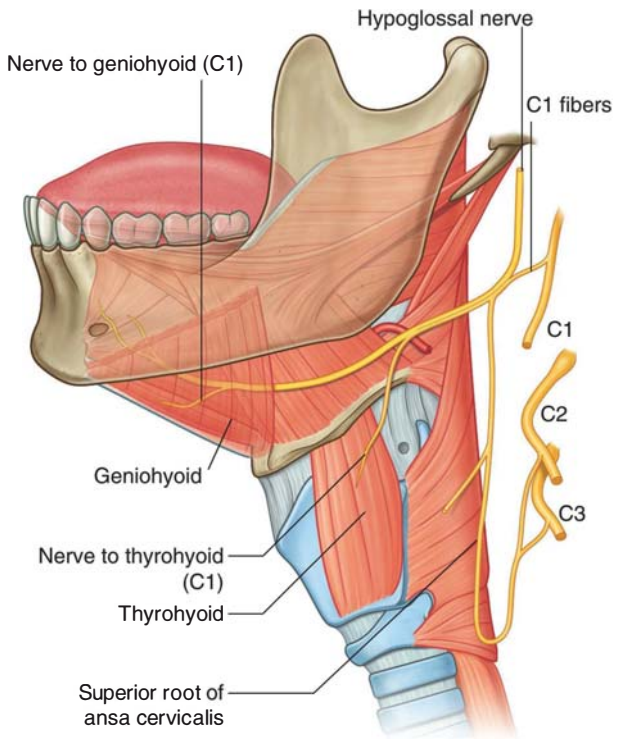


Fig. 8.259 Hypoglossal nerve and C1 fibers.

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sternocleidomastoid branch of the occipital artery, crosses the external carotid artery, and continues forward, crossing the loop of the lingual artery, to reach the external surface of the lower one-third of the hyoglossus muscle.

The hypoglossal nerve [XII] follows the hyoglossus muscle through the gap (oropharyngeal triangle) between the superior constrictor, middle constrictor, and mylohyoid muscles to reach the tongue.

In the upper neck, a branch from the anterior ramus of C1 joins the hypoglossal nerve [XII]. Most of these C1 fibers leave the hypoglossal nerve [XII] as the superior root of the ansa cervicalis (Fig. 8.259). Near the posterior border of the hyoglossus muscle, the remaining fibers leave the hypoglossal nerve [XII] and form two nerves:

- the thyrohyoid branch, which remains in the neck to innervate the thyrohyoid muscle, and
- the branch to the geniohyoid, which passes into the floor of the oral cavity to innervate the geniohyoid.

Lymphatics

All lymphatic vessels from the tongue ultimately drain into the deep cervical chain of nodes along the internal jugular vein:

- The pharyngeal part of the tongue drains through the pharyngeal wall directly into mainly the jugulodigastric node of the deep cervical chain.
- The oral part of the tongue drains both directly into the deep cervical nodes, and indirectly into these nodes by passing first through the mylohyoid muscle and into submental and submandibular nodes.

The submental nodes are inferior to the mylohyoid muscles and between the digastric muscles, while the submandibular nodes are below the floor of the oral cavity along the inner aspect of the inferior margins of the mandible.

The tip of the tongue drains through the mylohyoid muscle into the submental nodes and then into mainly the jugulo-omohyoid node of the deep cervical chain.

Salivary glands

Salivary glands are glands that open or secrete into the oral cavity. Most are small glands in the submucosa or mucosa of the oral epithelium lining the tongue, palate, cheeks, and lips, and open into the oral cavity directly or via small ducts. In addition to these small glands are much larger glands, which include the paired parotid, submandibular, and sublingual glands.

Parotid gland

The parotid gland (see pp. 911–912) on each side is entirely outside the boundaries of the oral cavity in a shallow triangular-shaped trench (Fig. 8.260) formed by:

- the sternocleidomastoid muscle behind,
- the ramus of the mandible in front, and
- superiorly, the base of the trench is formed by the external acoustic meatus and the posterior aspect of the zygomatic arch.

The gland normally extends anteriorly over the masseter muscle, and inferiorly over the posterior belly of the digastric muscle.

The parotid duct passes anteriorly across the external surface of the masseter muscle and then turns medially to penetrate the buccinator muscle of the cheek and open into the oral cavity adjacent to the crown of the second upper molar tooth.

The parotid gland encloses the external carotid artery, the retromandibular vein, and the origin of the extracranial part of the facial nerve [VII].

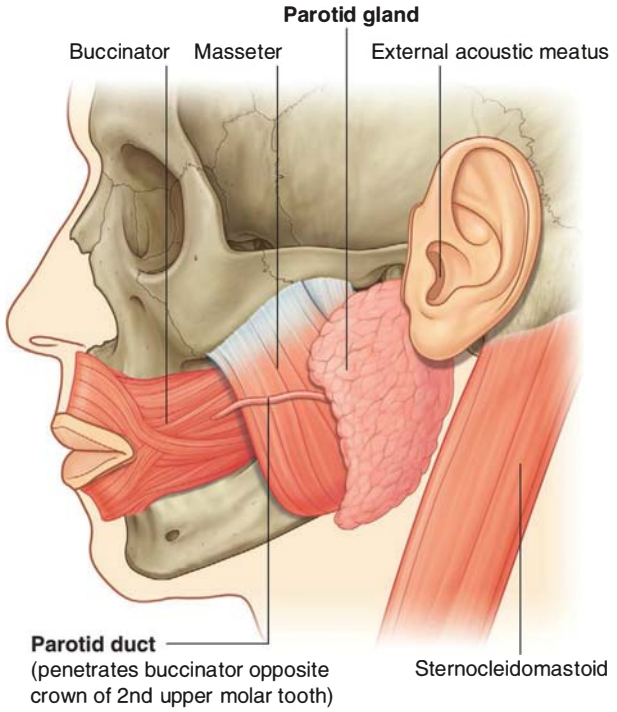


Fig. 8.260 Parotid gland.

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Submandibular glands

The elongate **submandibular glands** are smaller than the parotid glands but larger than the sublingual glands. Each is hook shaped (Fig. 8.261A,B):

- The larger arm of the hook is directed forward in the horizontal plane below the mylohyoid muscle and is therefore outside the boundaries of the oral cavity—this larger superficial part of the gland is directly against a shallow impression on the medial side of the mandible (submandibular fossa) inferior to the mylohyoid line.
- The smaller arm of the hook (or deep part) of the gland loops around the posterior margin of the mylohyoid muscle to enter and lie within the floor of the oral cavity where it is lateral to the root of the tongue on the lateral surface of the hyoglossus muscle.

The **submandibular duct** emerges from the medial side of the deep part of the gland in the oral cavity and passes forward to open on the summit of a small **sublingual caruncle** (papilla) beside the base of the frenulum of the tongue (Fig. 8.261C,D).

The lingual nerve loops under the submandibular duct, crossing first the lateral side and then the medial side of the duct, as the nerve descends anteromedially through the floor of the oral cavity and then ascends into the tongue.

Sublingual glands

The sublingual glands are the smallest of the three major paired salivary glands. Each is almond shaped and is immediately lateral to the submandibular duct and associated lingual nerve in the floor of the oral cavity (Fig. 8.261).

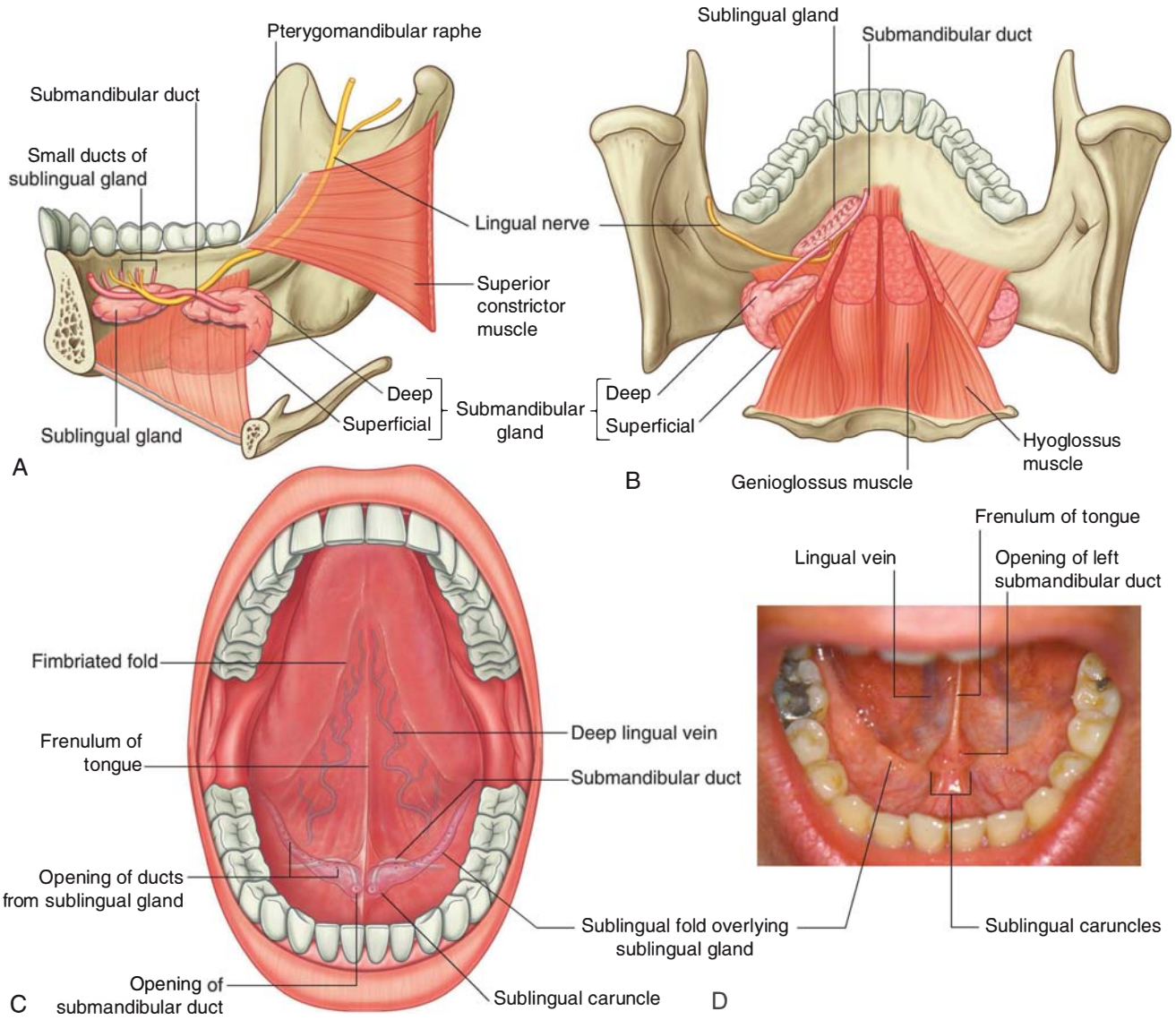


Fig. 8.261 Submandibular and sublingual glands. **A.** Medial view. **B.** Posterior view. **C.** Anterior view. **D.** Anterosuperior view.

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Each sublingual gland lies directly against the medial surface of the mandible where it forms a shallow groove (sublingual fossa) superior to the anterior one-third of the mylohyoid line.

The superior margin of the sublingual gland raises an elongate fold of mucosa (**sublingual fold**), which extends from the posterolateral aspect of the floor of the oral cavity to the sublingual papilla beside the base of the frenulum of the tongue at the midline anteriorly (Fig. 8.261D).

The sublingual gland drains into the oral cavity via numerous small ducts (minor sublingual ducts), which open onto the crest of the sublingual fold. Occasionally, the more anterior part of the gland is drained by a duct (major sublingual duct) that opens together with the submandibular duct on the sublingual caruncle.

Vessels

Vessels that supply the parotid gland originate from the external carotid artery and from its branches that are adjacent to the gland. The submandibular and sublingual glands are supplied by branches of the facial and lingual arteries.

Veins from the parotid gland drain into the external jugular vein, and those from the submandibular and sublingual glands drain into lingual and facial veins.

Lymphatic vessels from the parotid gland drain into nodes that are on or in the gland. These parotid nodes then drain into superficial and deep cervical nodes.

Lymphatics from the submandibular and sublingual glands drain mainly into submandibular nodes and then into deep cervical nodes, particularly the jugulo-omohyoid node.

Innervation

Parasympathetic

Parasympathetic innervation to all salivary glands in the oral cavity is by branches of the facial nerve [VII], which join branches of the maxillary [V_2] and mandibular [V_3] nerves to reach their target destinations.

The parotid gland, which is entirely outside the oral cavity, receives its parasympathetic innervation from fibers that initially traveled in the glossopharyngeal nerve [IX], which eventually joins a branch of the mandibular nerve [V_3] in the infratemporal fossa (Fig. 8.262).

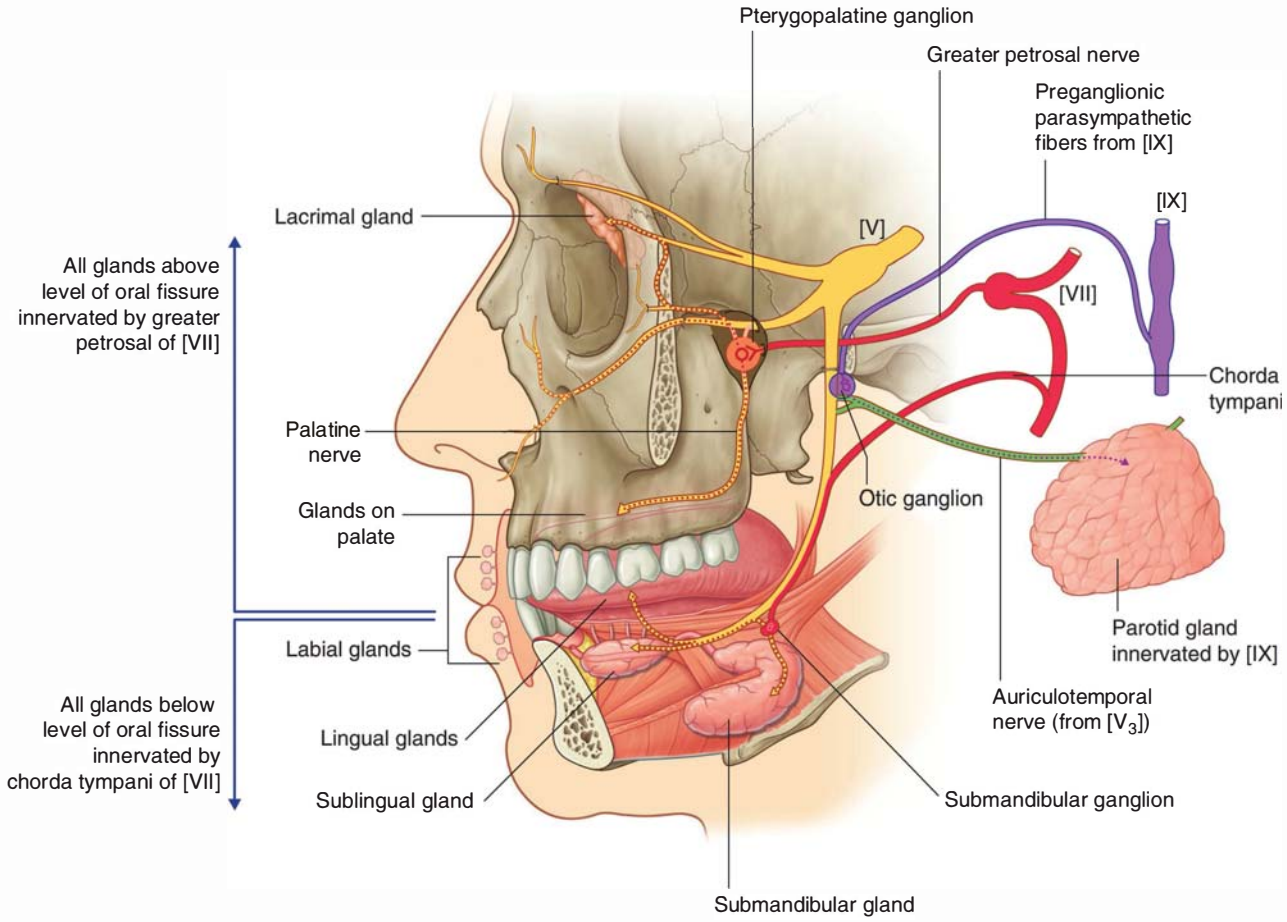


Fig. 8.262 Secretomotor (parasympathetic) innervation of the salivary gland and lacrimal gland.

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Greater petrosal nerve

All salivary glands above the level of the oral fissure, as well as all mucus glands in the nose and the lacrimal gland in the orbit, are innervated by parasympathetic fibers carried in the greater petrosal branch of the facial nerve [VII] (Fig. 8.262). Preganglionic parasympathetic fibers carried in this nerve enter the pterygopalatine fossa and synapse with postganglionic parasympathetic fibers in the pterygopalatine ganglion formed around branches of the maxillary nerve [V_2]. Postganglionic parasympathetic fibers join general sensory branches of the maxillary nerve, such as the palatine nerves, destined for the roof of the oral cavity, to reach their target glands.

Chorda tympani

All glands below the level of the oral fissure, which include those small glands in the floor of the oral cavity, in the lower lip, and in the tongue, and the larger submandibular and sublingual glands, are innervated by parasympathetic fibers carried in the chorda tympani branch of the facial nerve [VII] (Fig. 8.262).

The chorda tympani joins the lingual branch of the mandibular nerve [V_3] in the infratemporal fossa and passes with it into the oral cavity. On the external surface of the hyoglossus muscle, preganglionic parasympathetic fibers leave the inferior aspect of the lingual nerve to synapse with postganglionic parasympathetic fibers in the submandibular ganglion, which appears to hang off the lingual nerve (Fig. 8.263). Postganglionic parasympathetic fibers leave the ganglion and pass directly to the submandibular and sublingual glands while others hop back onto the lingual nerve and travel with branches of the lingual nerve to target glands.

Roof—palate

The roof of the oral cavity consists of the palate, which has two parts—an anterior hard palate and a posterior soft palate (Fig. 8.264).

Hard palate

The hard palate separates the oral cavity from the nasal cavities. It consists of a bony plate covered above and below by mucosa:

- Above, it is covered by respiratory mucosa and forms the floor of the nasal cavities.
- Below, it is covered by a tightly bound layer of oral mucosa and forms much of the roof of the oral cavity (Fig. 8.264).

The palatine processes of the maxillae form the anterior three-quarters of the hard palate. The horizontal plates of the palatine bones form the posterior one-quarter. In the oral cavity, the upper alveolar arch borders the hard palate anteriorly and laterally. Posteriorly, the hard palate is continuous with the soft palate.

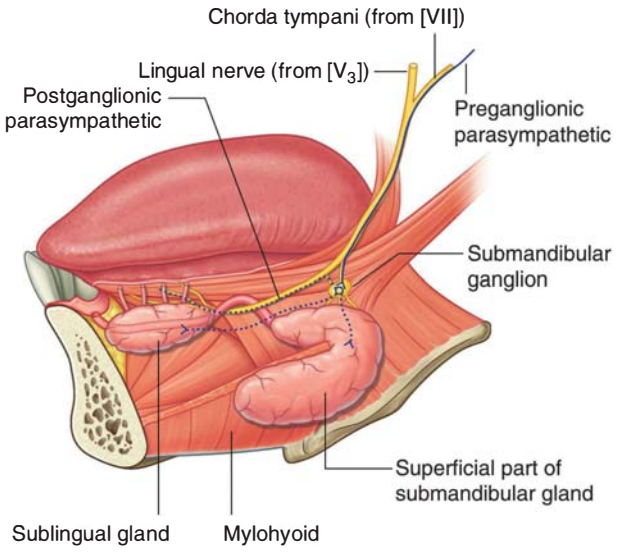


Fig. 8.263 Course of parasympathetic fibers carried in the chorda tympani nerve.

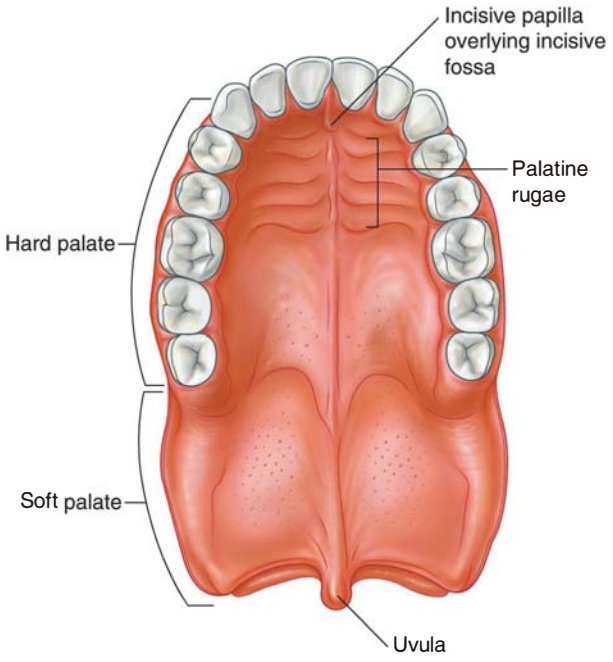


Fig. 8.264 Palate.

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The mucosa of the hard palate in the oral cavity possesses numerous **transverse palatine folds** (**palatine rugae**) and a median longitudinal ridge (**palatine raphe**), which ends anteriorly in a small oval elevation (**incisive papilla**). The incisive papilla (Fig. 8.264) overlies the incisive fossa formed between the horizontal plates of the maxillae immediately behind the incisor teeth.

Soft palate

The soft palate (Fig. 8.264) continues posteriorly from the hard palate and acts as a valve that can be:

- depressed to help close the oropharyngeal isthmus, and
- elevated to separate the nasopharynx from the oropharynx.

The soft palate is formed and moved by four muscles and is covered by mucosa that is continuous with the mucosa lining the pharynx and oral and nasal cavities.

The small tear-shaped muscular projection that hangs from the posterior free margin of the soft palate is the **uvula**.

Muscles of the soft palate

Five muscles (Table 8.22) on each side contribute to the formation and movement of the soft palate. Two of these, the tensor veli palatini and levator veli palatini, descend into the palate from the base of the skull. Two others, the palatoglossus and palatopharyngeus, ascend into the palate from the tongue and pharynx, respectively. The last muscle, the musculus uvulae, is associated with the uvula.

All muscles of the palate are innervated by the vagus nerve [X], except for the tensor veli palatini, which is innervated by the mandibular nerve [V_3] (via the nerve to the medial pterygoid).

Tensor veli palatini and the palatine aponeurosis

The **tensor veli palatini muscle** is composed of two parts—a vertical muscular part and a more horizontal fibrous part, which forms the palatine aponeurosis (Fig. 8.265A).

The vertical part of the tensor veli palatini is thin and triangular in shape with its base attached to the skull and its apex pointed inferiorly. The base is attached along an oblique line that begins medially at the scaphoid fossa near the root of the pterygoid process of the sphenoid bone and continues laterally along the membranous part of the pharyngotympanic tube to the spine of the sphenoid bone.

The tensor veli palatini descends vertically along the lateral surface of the medial plate of the pterygoid process and pharyngeal wall to the pterygoid hamulus where the fibers converge to form a small tendon (Fig. 8.265A).

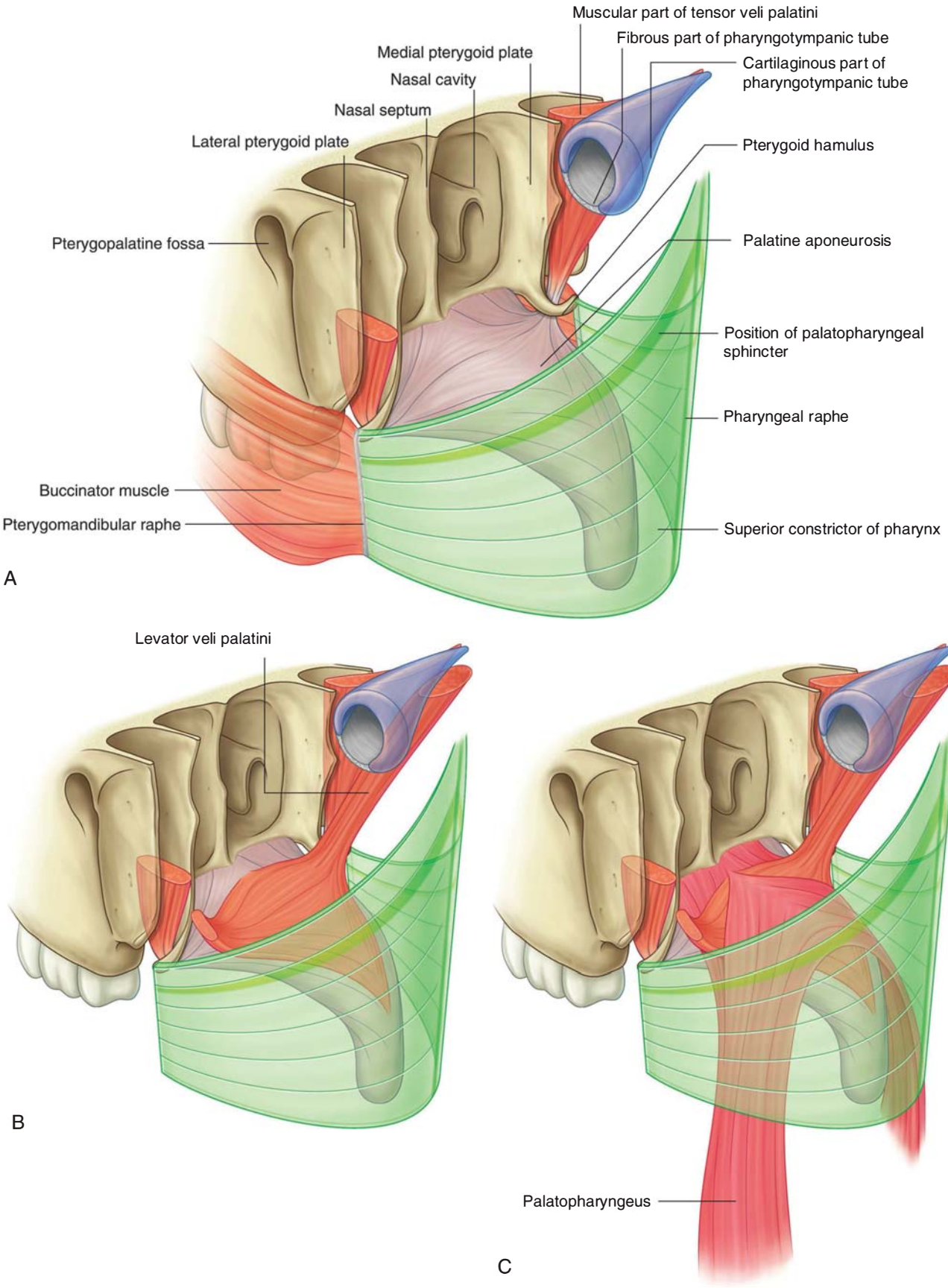
The tendon loops 90° medially around the pterygoid hamulus, penetrating the origin of the buccinator muscle as it does, and expands like a fan to form the fibrous horizontal part of the muscle. This fibrous part is continuous across the midline with its partner on the other side to form the palatine aponeurosis.

The **palatine aponeurosis** is attached anteriorly to the margin of the hard palate, but is unattached posteriorly where it ends in a free margin. This expansive aponeurosis

Table 8.22 Muscles of the soft palate

Muscle	Origin	Insertion	Innervation	Function
Tensor veli palatini	Scaphoid fossa of sphenoid bone; fibrous part of pharyngotympanic tube; spine of sphenoid	Palatine aponeurosis	Mandibular nerve [V_3] via the branch to medial pterygoid muscle	Tenses the soft palate; opens the pharyngotympanic tube
Levator veli palatini	Petrous part of temporal bone anterior to opening for carotid canal	Superior surface of palatine aponeurosis	Vagus nerve [X] via pharyngeal branch to pharyngeal plexus	Only muscle to elevate the soft palate above the neutral position
Palatopharyngeus	Superior surface of palatine aponeurosis	Pharyngeal wall	Vagus nerve [X] via pharyngeal branch to pharyngeal plexus	Depresses soft palate; moves palatopharyngeal arch toward midline; elevates pharynx
Palatoglossus	Inferior surface of palatine aponeurosis	Lateral margin of tongue	Vagus nerve [X] via pharyngeal branch to pharyngeal plexus	Depresses palate; moves palatoglossal arch toward midline; elevates back of the tongue
Musculus uvulae	Posterior nasal spine of hard palate	Connective tissue of uvula	Vagus nerve [X] via pharyngeal branch to pharyngeal plexus	Elevates and retracts uvula; thickens central region of soft palate

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Fig. 8.265 **A.** Tensor veli palatini muscles and the palatine aponeurosis. **B.** Levator veli palatini muscles. **C.** Palatopharyngeus muscles.

is the major structural element of the soft palate to which the other muscles of the palate attach.

The tensor veli palatini:

- tenses (makes firm) the soft palate so that the other muscles attached to the palate can work more effectively, and
- opens the pharyngotympanic tube when the palate moves during yawning and swallowing as a result of its attachment superiorly to the membranous part of the pharyngotympanic tube.

The tensor veli palatini is innervated by the nerve to the medial pterygoid from the mandibular nerve [V_3].

Levator veli palatini

The levator veli palatini muscle originates from the base of the skull and descends to the upper surface of the palatine aponeurosis (Fig. 8.265B). On the skull, it originates from a roughened area on the petrous part of the temporal bone immediately anterior to the opening of the carotid canal. Some fibers also originate from adjacent parts of the pharyngotympanic tube.

The levator veli palatini passes anteroinferiorly through fascia of the pharyngeal wall, passes medial to the pharyngotympanic tube, and inserts onto the palatine aponeurosis (Fig. 8.265B). Its fibers interlace at the midline with those of the levator veli palatini on the other side.

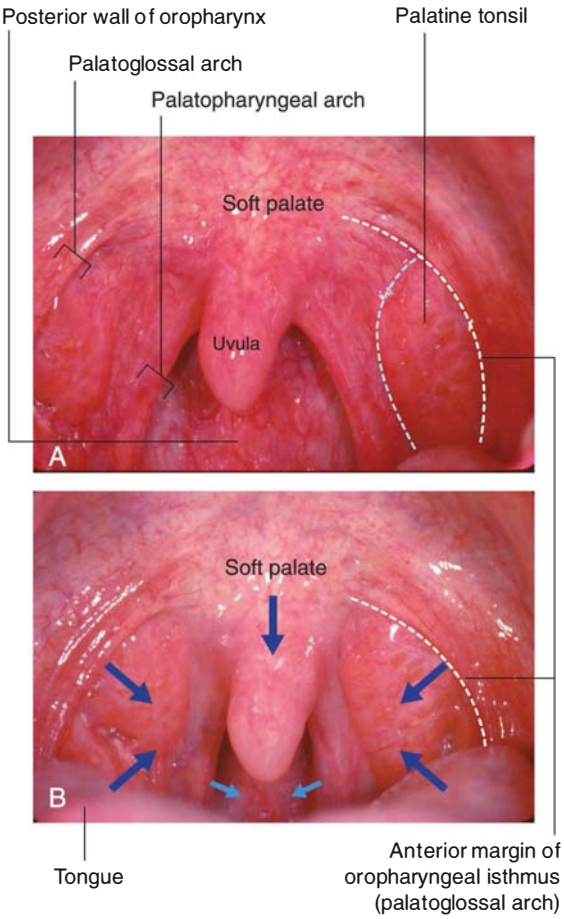
Unlike the tensor veli palatini muscles, the levator veli palatini muscles do not pass around the pterygoid hamulus, but course directly from the base of the skull to the upper surface of the palatine aponeurosis. Therefore, they are the only muscles that can elevate the palate above the neutral position and close the pharyngeal isthmus between the nasopharynx and oropharynx.

The levator veli palatini is innervated by the vagus nerve [X] through the pharyngeal branch to the pharyngeal plexus. Clinically, the levator veli palatini can be tested by asking a patient to say "ah." If the muscle on each side is functioning normally, the palate elevates evenly in the midline. If one side is not functioning, the palate deviates away from the abnormal side.

Palatopharyngeus

The palatopharyngeus muscle originates from the superior surface of the palatine aponeurosis and passes posterolaterally over its margin to descend and become one of the longitudinal muscles of the pharyngeal wall (Fig. 8.265C). It is attached to the palatine aponeurosis by two flat lamellae separated by the levator veli palatini muscle. The more anterior and lateral of these two lamellae is attached to the posterior margin of the hard palate as well as to the palatine aponeurosis.

The two palatopharyngeus muscles, one on each side, underlie the **palatopharyngeal arches** on the oropharyngeal wall. The palatopharyngeal arches lie posterior and medial to the **palatoglossal arches** when viewed anteriorly through the oral cavity (Fig. 8.266).



Closure of oropharyngeal isthmus
- Medial and downward movement of palatoglossal arches
- Medial and downward movement of palatopharyngeal arches
- Upward movement of tongue
- Downward and forward movement of soft palate

Fig. 8.266 Open mouth with soft palate. **A.** Oropharyngeal isthmus opened. **B.** Oropharyngeal isthmus closed.

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On each side, the palatine tonsil is between the palatopharyngeal and palatoglossal arches on the lateral oropharyngeal wall (Fig. 8.266A).

The palatopharyngeus muscles:

- depress the palate and move the palatopharyngeal arches toward the midline like curtains—both these actions help close the oropharyngeal isthmus; and
- elevate the pharynx during swallowing.

The palatopharyngeus is innervated by the vagus nerve [X] through the pharyngeal branch to the pharyngeal plexus.

Palatoglossus

The palatoglossus muscle attaches to the inferior (oral) surface of the palatine aponeurosis and passes inferiorly and anteriorly into the lateral surface of the tongue (Fig. 8.267).

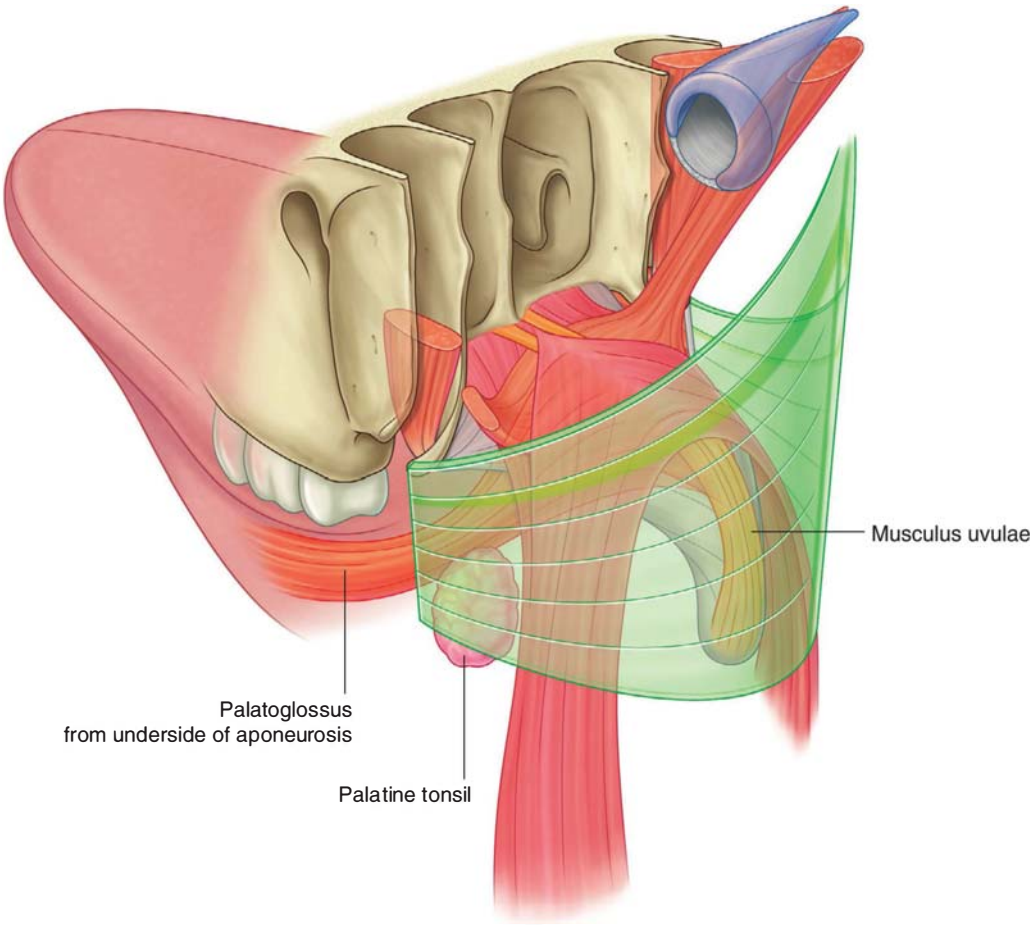


Fig. 8.267 Palatoglossus muscles and musculus uvulae.

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The palatoglossus muscle underlies a fold of mucosa that arches from the soft palate to the tongue. These palatoglossal arches, one on each side, are lateral and anterior to the palatopharyngeal arches and define the lateral margins of the oropharyngeal isthmus (Fig. 8.266A).

The palatine tonsil is between the palatoglossal and palatopharyngeal arches on the lateral oropharyngeal wall (Figs. 8.266 and 8.267).

The palatoglossus muscles depress the palate, move the palatoglossal arches toward the midline like curtains, and elevate the back of the tongue. These actions help close the oropharyngeal isthmus.

The palatoglossus is innervated by the vagus nerve [X] through the pharyngeal branch to the pharyngeal plexus.

Musculus uvulae

The musculus uvulae originates from the posterior nasal spine on the posterior margin of the hard palate and passes directly posteriorly over the dorsal aspect of the palatine aponeurosis to insert into connective tissue underlying the mucosa of the uvula (Fig. 8.267). It passes between the two lamellae of the palatopharyngeus superior to the attachment of the levator veli palatini. Along the midline, the musculus uvulae blends with its partner on the other side.

The musculus uvulae elevates and retracts the uvula. This action thickens the central part of the soft palate and helps the levator veli palatini muscles close the pharyngeal isthmus between the nasopharynx and oropharynx.

The musculus uvulae is innervated by the vagus nerve [X] through the pharyngeal branch to the pharyngeal plexus.

Vessels

Arteries

Arteries of the palate include the greater palatine branch of the maxillary artery, the ascending palatine branch of the facial artery, and the palatine branch of the ascending pharyngeal artery. The maxillary, facial, and ascending pharyngeal arteries are all branches that arise in the neck from the external carotid artery (Fig. 8.268).

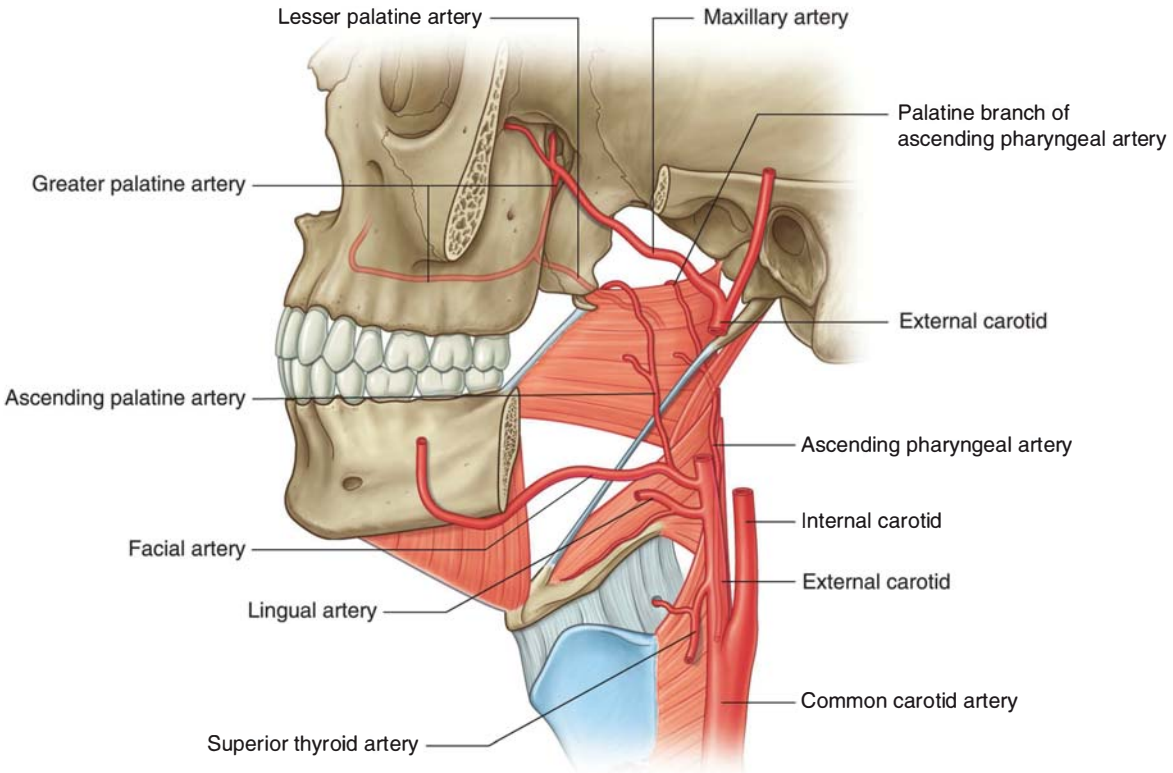


Fig. 8.268 Arteries of the palate.

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Ascending palatine artery and palatine branch

The **ascending palatine artery** of the facial artery ascends along the external surface of the pharynx. The palatine branch loops medially over the top of the superior constrictor muscle of the pharynx to penetrate the pharyngeal fascia with the levator veli palatini muscle and follow the levator veli palatini to the soft palate.

The **palatine branch** of the ascending pharyngeal artery follows the same course as the palatine branch of the ascending palatine artery from the facial artery and may replace the vessel.

Greater palatine artery

The **greater palatine artery** originates from the maxillary artery in the pterygopalatine fossa. It descends into the palatine canal where it gives origin to a small **lesser palatine branch**, and then continues through the greater palatine foramen onto the inferior surface of the hard palate (Fig. 8.269). The greater palatine artery passes forward on the hard palate and then leaves the palate superiorly through the incisive canal to enter the medial wall of the nasal cavity where it terminates. The greater palatine artery is the major artery of the hard palate. It also supplies palatal gingiva. The lesser palatine branch passes through the lesser palatine foramen just posterior to the greater palatine foramen, and contributes to the vascular supply of the soft palate.

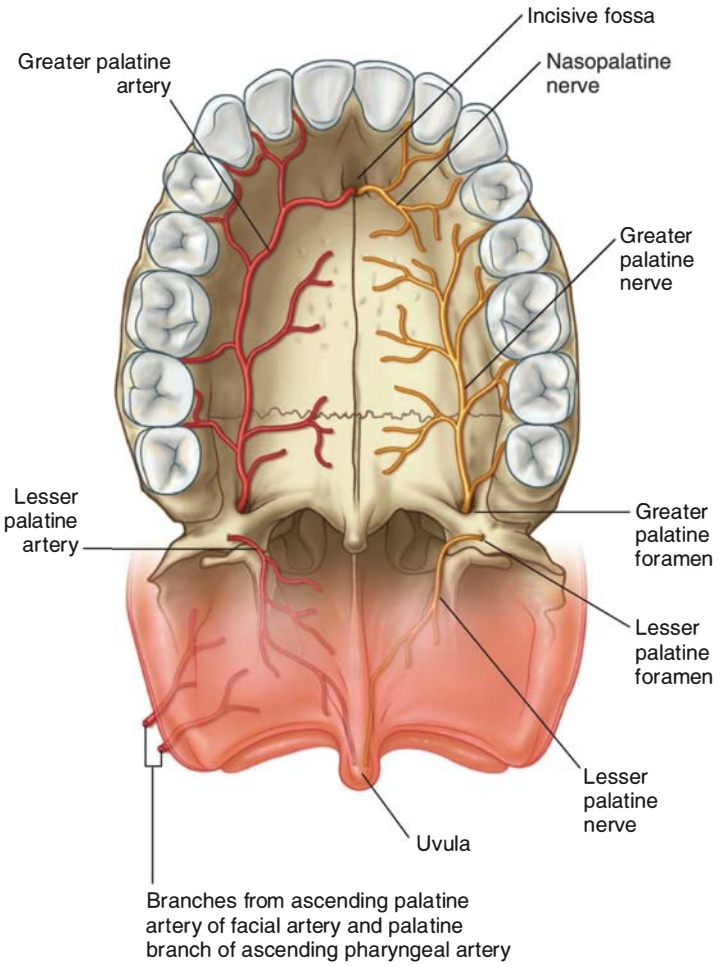


Fig. 8.269 Palatine nerves and arteries.

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Veins

Veins from the palate generally follow the arteries and ultimately drain into the pterygoid plexus of veins in the infratemporal fossa (Fig. 8.270; also see pp. 991–992), or into a network of veins associated with the palatine tonsil, which drain into the pharyngeal plexus of veins or directly into the facial vein.

Lymphatics

Lymphatic vessels from the palate drain into deep cervical nodes (Fig. 8.270).

Innervation

The palate is supplied by the greater and lesser palatine nerves and the nasopalatine nerve (Figs. 8.269 and 8.271).

General sensory fibers carried in all these nerves originate in the pterygopalatine fossa from the maxillary nerve [V_2].

Parasympathetic (to glands) and SA (taste on soft palate) fibers from a branch of the facial nerve [VII] join the nerves in the pterygopalatine fossa, as do the sympathetics (mainly to blood vessels) ultimately derived from the T1 spinal cord level.

Greater and lesser palatine nerves

The greater and lesser palatine nerves descend through the pterygopalatine fossa and palatine canal to reach the palate (Fig. 8.271):

- The greater palatine nerve travels through the greater palatine foramen and turns anteriorly to supply the hard palate and gingiva as far as the first premolar.

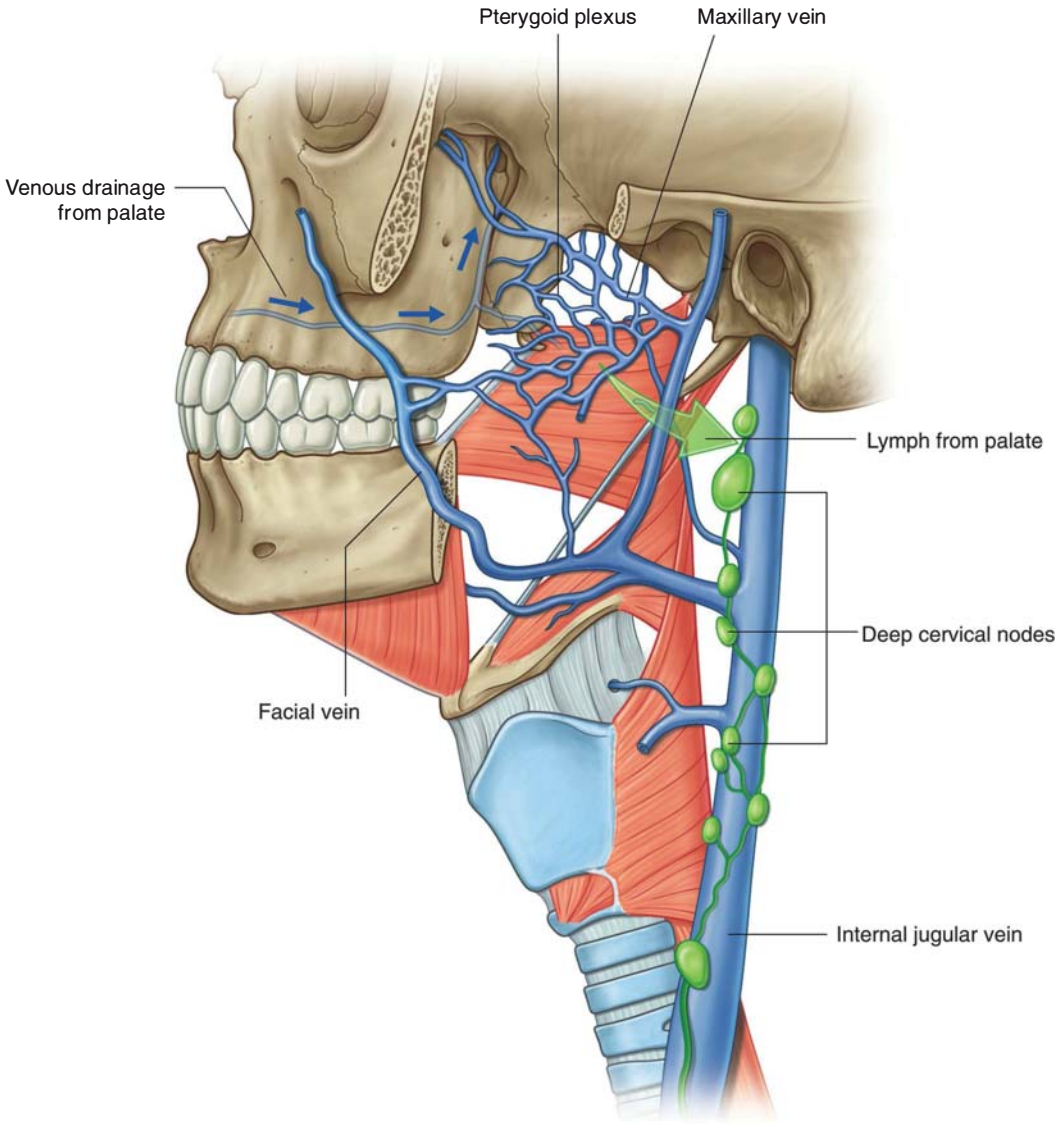


Fig. 8.270 Venous and lymphatic drainage of the palate.

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- The lesser palatine nerve passes posteromedially to supply the soft palate.

Nasopalatine nerve

The nasopalatine nerve also originates in the pterygopalatine fossa, but passes medially into the nasal cavity. It continues medially over the roof of the nasal cavity to reach the medial wall, then anteriorly and obliquely down the wall to reach the incisive canal in the anterior floor, and descends through the incisive canal and fossa to reach the inferior surface of the hard palate (Fig. 8.271).

The nasopalatine nerve supplies gingiva and mucosa adjacent to the incisors and canine.

Oral fissure and lips

The oral fissure is the slit-like opening between the lips that connects the oral vestibule to the outside (Fig. 8.272). It

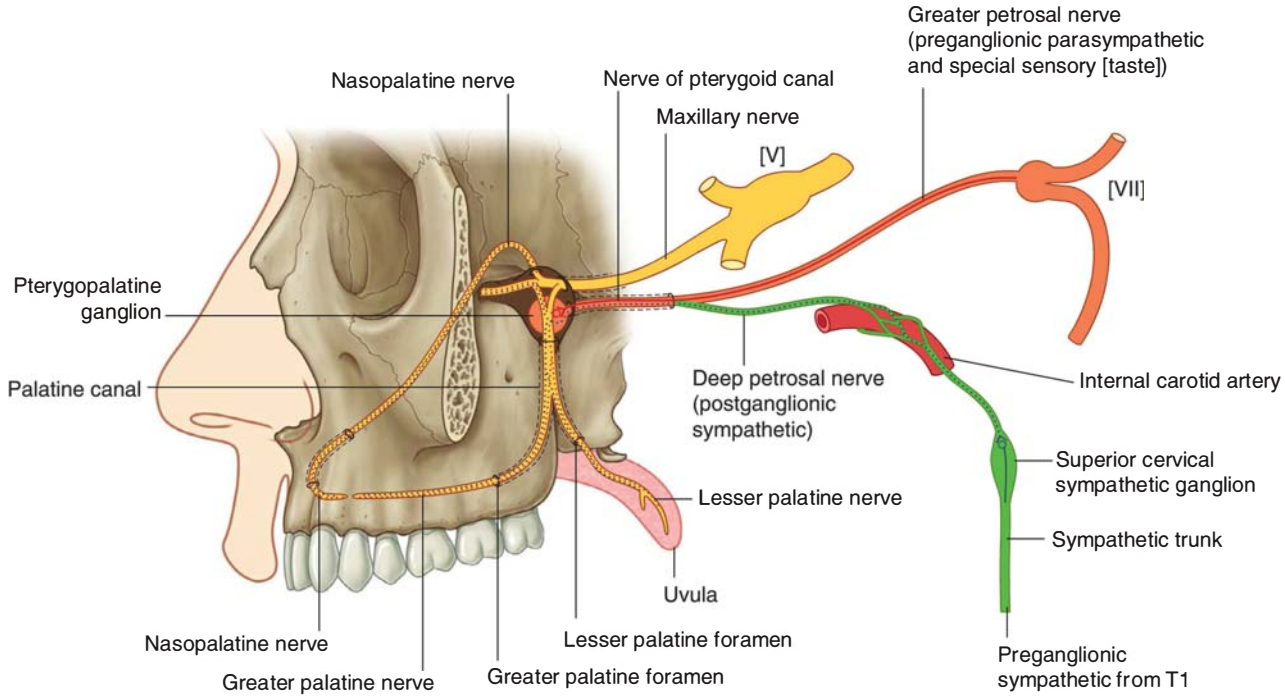


Fig. 8.271 Innervation of the palate.

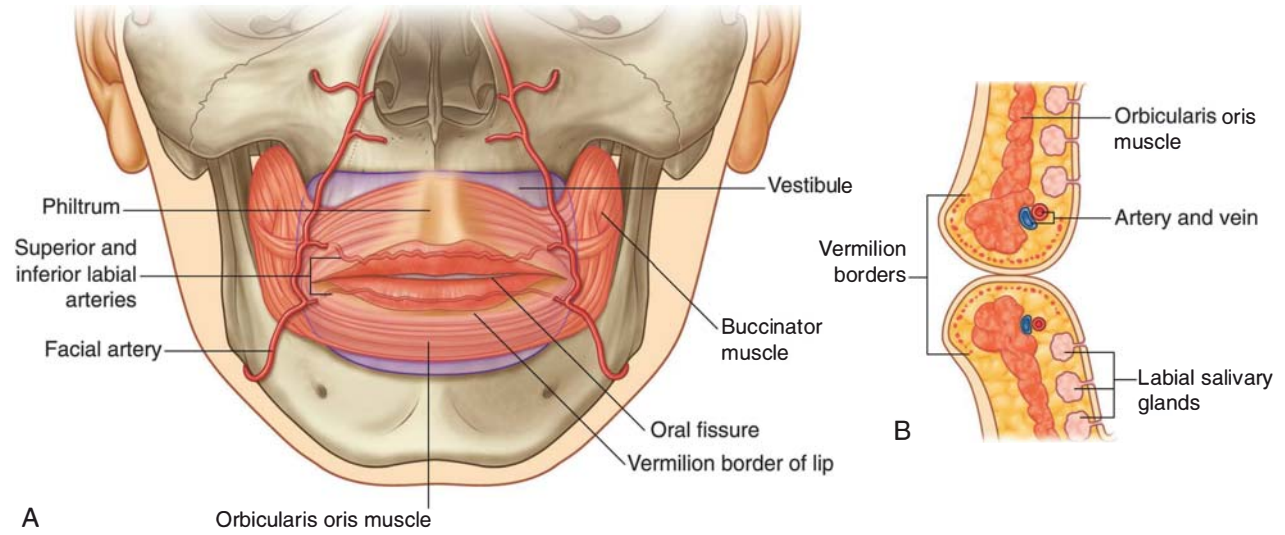


Fig. 8.272 Oral fissure and lips. **A.** Anterior view. **B.** Sagittal section.

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can be opened and closed, and altered in shape by the movements of the muscles of facial expression associated with the lips and surrounding regions, and by movements of the lower jaw (mandible).

The **lips** are entirely composed of soft tissues (Fig. 8.272B). They are lined internally by oral mucosa and covered externally by skin. Externally, there is an area of transition from the thicker skin that covers the face to the thinner skin that overlies the margins of the lips and continues as oral mucosa onto the deep surfaces of the lips.

Blood vessels are closer to the surface in areas where the skin is thin and as a consequence there is a vermilion border that covers the margins of the lips.

The upper lip has a shallow vertical groove on its external surface (the **philtrum**) sandwiched between two elevated ridges of skin (Fig. 8.272A). The philtrum and ridges are formed embryologically by fusion of the medial nasal processes.

On the inner surface of both lips, a fold of mucosa (the **median labial frenulum**) connects the lip to the adjacent gum.

The lips enclose the orbicularis oris muscle, neurovascular tissues, and labial glands (Fig. 8.272B). The small pea-shaped labial glands are between the muscle tissue and the oral mucosa and open into the oral vestibule.

A number of muscles of facial expression control the shape and size of the oral fissure. The most important of these is the orbicularis oris muscle, which encircles the orifice and acts as a sphincter. A number of other muscles of facial expression blend into the orbicularis oris or other tissues of the lips and open or adjust the contours of the oral fissure. These include the buccinator, levator labii superioris, zygomaticus major and minor, levator anguli oris, depressor labii inferioris, depressor anguli oris, and platysma (see pp. 908–910).

Oropharyngeal isthmus

The oropharyngeal isthmus is the opening between the oral cavity and the oropharynx (see Fig. 8.266). It is formed:

- laterally by the palatoglossal arches;
- superiorly by the soft palate; and
- inferiorly by the sulcus terminalis of the tongue that divides the oral surface of the tongue (anterior two-thirds) from the pharyngeal surface (posterior one-third).

The oropharyngeal isthmus can be closed by elevation of the posterior aspect of the tongue, depression of the palate, and medial movement of the palatoglossal arches toward the midline.

Medial movement of the palatopharyngeal arches medial and posterior to the palatoglossal arches is also involved in closing the oropharyngeal isthmus. By closing the oropharyngeal isthmus, food or liquid can be held in the oral cavity while breathing.

Teeth and gingivae

The **teeth** are attached to sockets (alveoli) in two elevated arches of bone on the mandible below and the maxillae above (alveolar arches). If the teeth are removed, the alveolar bone is resorbed and the arches disappear.

The **gingivae** (**gums**) are specialized regions of the oral mucosa that surround the teeth and cover adjacent regions of the alveolar bone.

The different types of teeth are distinguished on the basis of morphology, position, and function (Fig. 8.273A).

In adults, there are 32 teeth, 16 in the upper jaw and 16 in the lower jaw. On each side in both maxillary and mandibular arches are two incisor, one canine, two premolar, and three molar teeth.

- The **incisor teeth** are the "front teeth" and have one root and a chisel-shaped crown, which "cuts."
- The **canine teeth** are posterior to the incisors, are the longest teeth, have a crown with a single pointed cusp, and "grasp."
- The **premolar teeth** (bicuspids) have a crown with two pointed cusps, one on the buccal (cheek) side of the tooth and the other on the lingual (tongue) or palatal (palate) side, generally have one root (but the upper first premolar next to the canine may have two), and "grind."
- The **molar teeth** are behind the premolar teeth, have three roots and crowns with three to five cusps, and "grind."

Two successive sets of teeth develop in humans, deciduous teeth ("baby" teeth) (Fig. 8.273B) and permanent teeth ("adult" teeth). The deciduous teeth emerge from the gingivae at between six months and two years of age. Permanent teeth begin to emerge and replace the deciduous teeth at around age six years, and can continue to emerge into adulthood.

The 20 deciduous teeth consist of two incisor, one canine, and two molar teeth on each side of the upper and lower jaws. These teeth are replaced by the incisor, canine, and premolar teeth of the permanent teeth. The permanent molar teeth erupt posterior to the deciduous molars and require the jaws to elongate forward to accommodate them.

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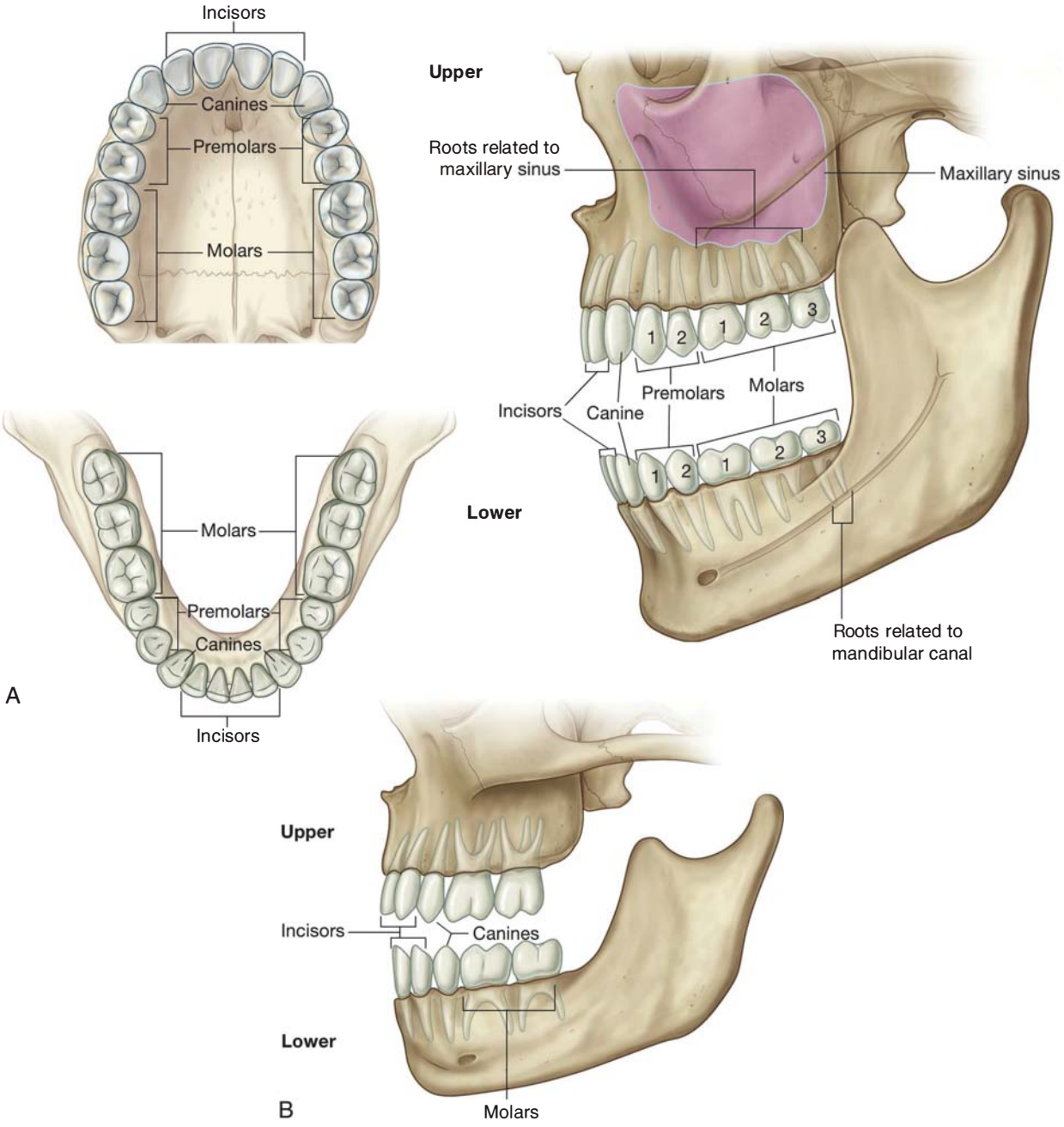


Fig. 8.273 Teeth. **A.** Adult upper and lower permanent teeth. **B.** Deciduous ("baby") teeth.

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Vessels

Arteries

All teeth are supplied by vessels that branch either directly or indirectly from the maxillary artery (Fig. 8.274).

Inferior alveolar artery

All lower teeth are supplied by the **inferior alveolar artery**, which originates from the maxillary artery in the infratemporal fossa. The vessel enters the mandibular canal of the mandible, passes anteriorly in bone supplying vessels to the more posterior teeth, and divides opposite the first premolar into **incisor** and **mental branches**. The mental branch leaves the mental foramen to supply the chin, while the incisor branch continues in bone to supply the anterior teeth and adjacent structures.

Anterior and posterior superior alveolar arteries

All upper teeth are supplied by anterior and posterior superior alveolar arteries.

The **posterior superior alveolar artery** originates from the maxillary artery just after the maxillary artery enters the pterygopalatine fossa and it leaves the fossa through the pterygomaxillary fissure. It descends on the posterolateral surface of the maxilla, branches, and enters small canals in the bone to supply the molar and premolar teeth.

The **anterior superior alveolar artery** originates from the infra-orbital artery, which arises from the maxillary artery in the pterygopalatine fossa. The infra-orbital artery leaves the pterygopalatine fossa through the inferior orbital fissure and enters the inferior orbital groove and canal in the floor of the orbit. The anterior superior alveolar artery originates from the infra-orbital artery in the infra-orbital canal. It passes through bone and branches to supply the incisor and canine teeth.

Gingival supply

The gingivae are supplied by multiple vessels and the source depends on which side of each tooth the gingiva is—the side facing the oral vestibule or cheek (vestibular or buccal side), or the side facing the tongue or palate (lingual or palatal side):

- Buccal gingiva of the lower teeth is supplied by branches from the inferior alveolar artery, whereas the lingual side is supplied by branches from the lingual artery of the tongue.

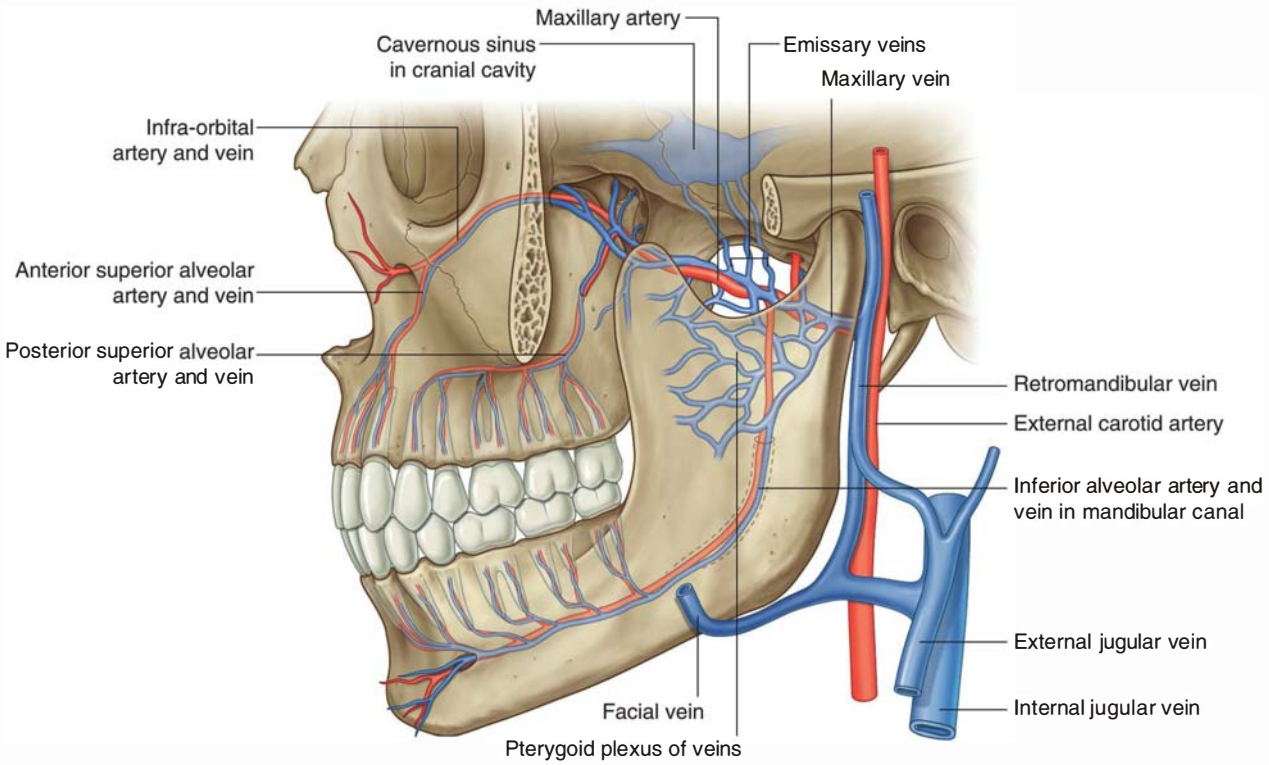


Fig. 8.274 Arteries and veins of the teeth.

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- Buccal gingiva of the upper teeth is supplied by branches of the anterior and posterior superior alveolar arteries.
- Palatal gingiva is supplied by branches from the nasopalatine (incisor and canine teeth) and greater palatine (premolar and molar teeth) arteries.

Veins

Veins from the upper and lower teeth generally follow the arteries (Fig. 8.274).

Inferior alveolar veins from the lower teeth, and superior alveolar veins from the upper teeth drain mainly into the pterygoid plexus of veins in the infratemporal fossa, although some drainage from the anterior teeth may be via tributaries of the facial vein.

The pterygoid plexus drains mainly into the maxillary vein and ultimately into the retromandibular vein and jugular system of veins. In addition, small communicating vessels pass superiorly, from the plexus, and pass through small emissary foramina in the base of the skull to connect with the cavernous sinus in the cranial cavity. Infection originating in the teeth can track into the cranial cavity through these small emissary veins.

Venous drainage from the teeth can also be via vessels that pass through the mental foramen to connect with the facial vein.

Veins from the gingivae also follow the arteries and ultimately drain into the facial vein or into the pterygoid plexus of veins.

Lymphatics

Lymphatic vessels from the teeth and gingivae drain mainly into submandibular, submental, and deep cervical nodes (Fig. 8.275).

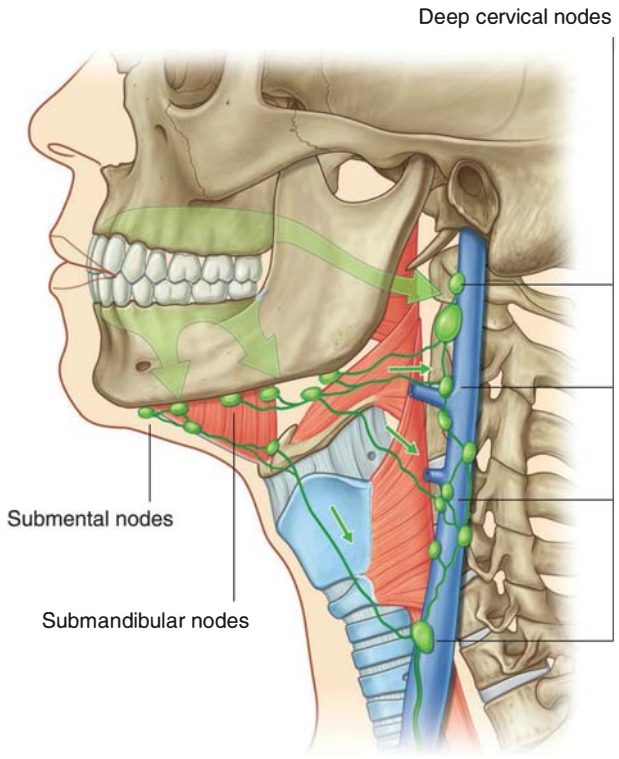


Fig. 8.275 Lymphatic drainage of the teeth and gums.

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Innervation

All nerves that innervate the teeth and gingivae are branches of the trigeminal nerve [V] (Figs. 8.276 and 8.277).

Inferior alveolar nerve

The lower teeth are all innervated by branches from the inferior alveolar nerve, which originates in the infratemporal fossa from the mandibular nerve [V_3] (Figs. 8.276 and 8.277). The inferior alveolar nerve and its accompanying vessels enter the mandibular foramen on the medial surface of the ramus of the mandible and travel anteriorly through the bone in the mandibular canal. Branches to the back teeth originate directly from the inferior alveolar nerve.

Adjacent to the first premolar tooth, the inferior alveolar nerve divides into incisive and mental branches:

- The **incisive branch** innervates the first premolar, the canine, and the incisor teeth, together with the associated vestibular (buccal) gingiva.
- The **mental nerve** exits the mandible through the mental foramen and innervates the chin and lower lip.

Anterior, middle, and posterior superior alveolar nerves

All upper teeth are innervated by the anterior, middle, and posterior superior alveolar nerves, which originate directly or indirectly from the maxillary nerve [V_2] (Figs. 8.276 and 8.277).

The posterior superior alveolar nerve originates directly from the maxillary nerve [V_2] in the pterygopalatine fossa, exits the pterygopalatine fossa through the pterygomaxillary fissure, and descends on the posterolateral surface of the maxilla. It enters the maxilla through a small foramen approximately midway between the pterygomaxillary fissure and the last molar tooth, and passes through the bone in the wall of the maxillary sinus. The posterior superior alveolar nerve then innervates the molar teeth through the superior alveolar plexus formed by the posterior, middle, and anterior alveolar nerves.

The middle and anterior superior alveolar nerves originate from the infra-orbital branch of the maxillary nerve [V_2] in the floor of the orbit:

- The middle superior alveolar nerve arises from the infra-orbital nerve in the infra-orbital groove, passes through the bone in the lateral wall of the maxillary sinus, and innervates the premolar teeth via the superior alveolar plexus.
- The anterior superior alveolar nerve originates from the infra-orbital nerve in the infra-orbital canal, passes through the maxilla in the anterior wall of the maxillary sinus, and via the superior alveolar plexus, supplies the canine and incisor teeth.

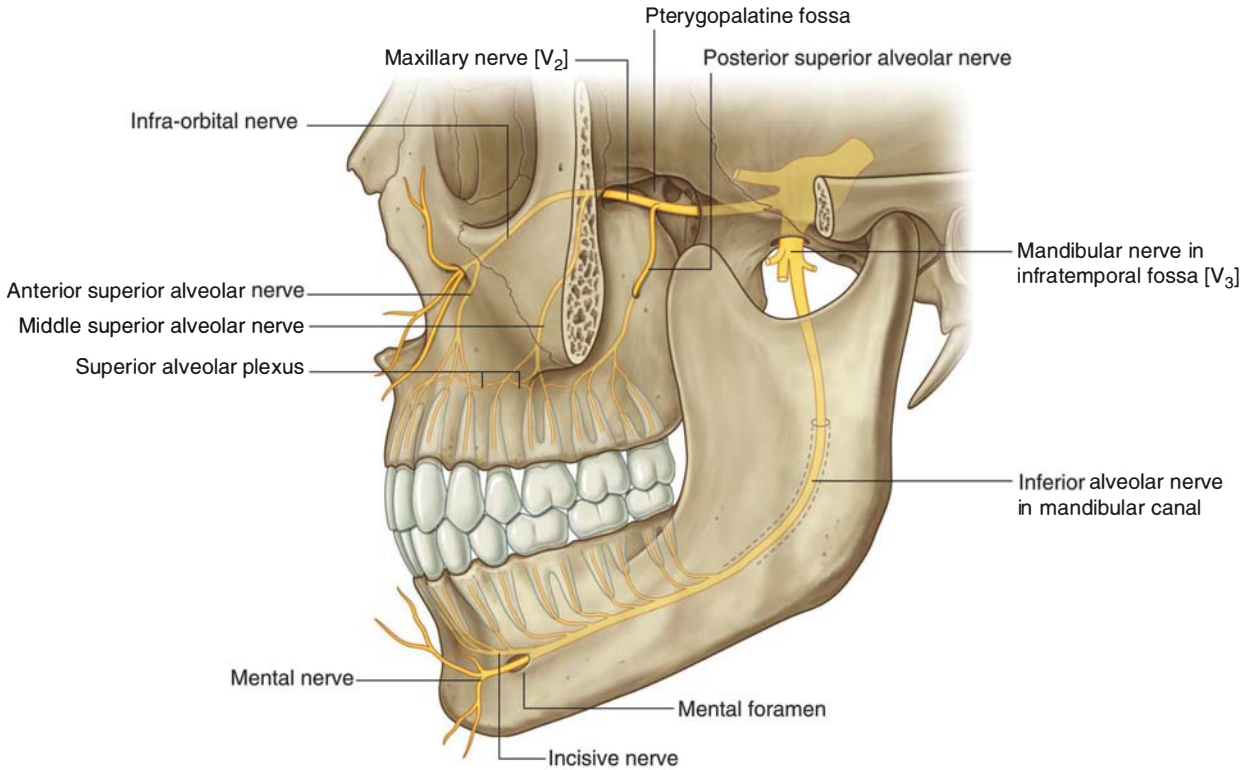


Fig. 8.276 Innervation of the teeth.

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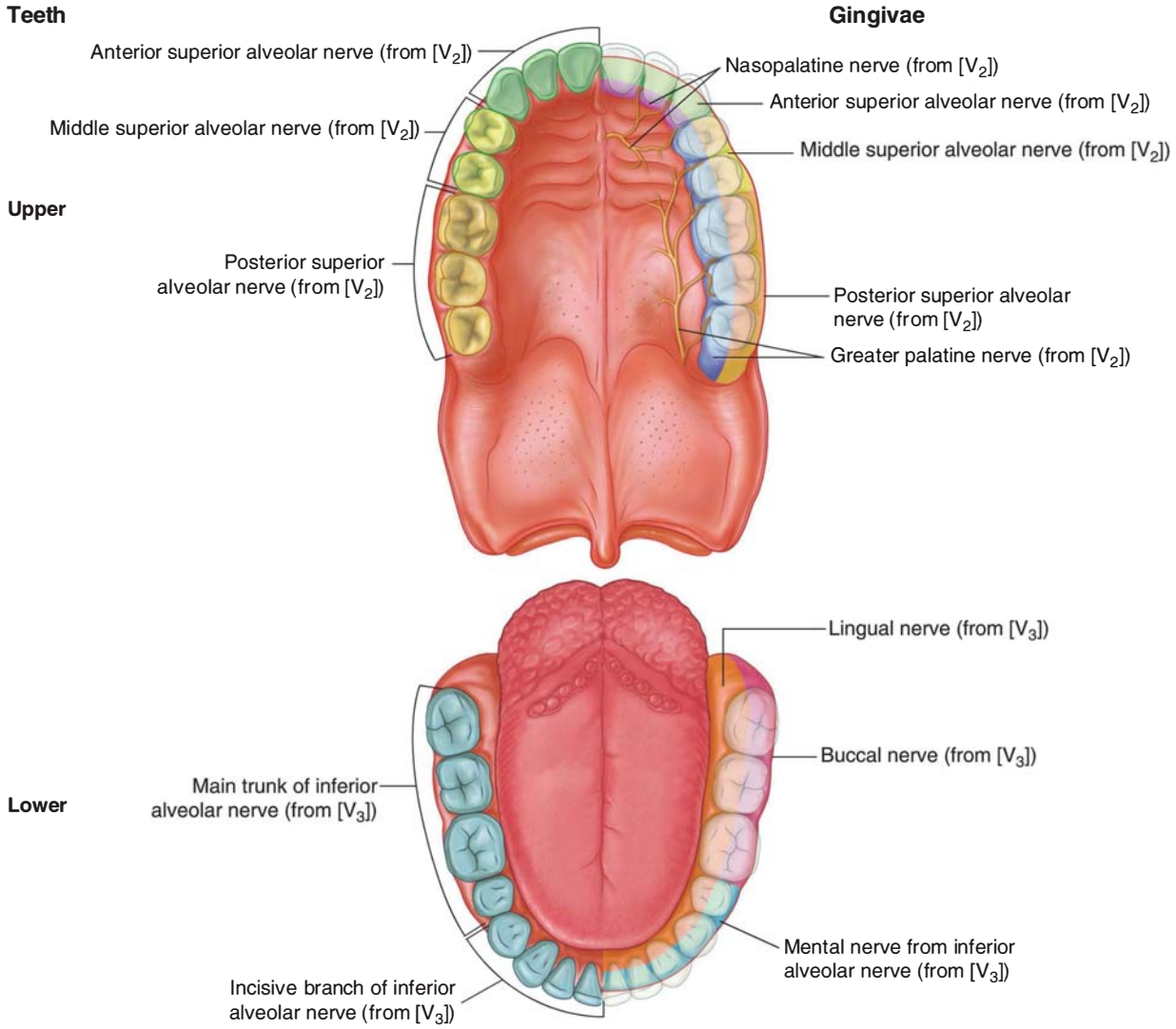


Fig. 8.277 Innervation of the teeth and gums.

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Innervation of gingivae

Like the teeth, the gingivae are innervated by nerves that ultimately originate from the trigeminal nerve [V] (Fig. 8.277):

- Gingiva associated with the upper teeth is innervated by branches derived from the maxillary nerve [V_2].
- Gingiva associated with the lower teeth is innervated by branches of the mandibular nerve [V_3].

The gingiva on the buccal side of the upper teeth is innervated by the anterior, middle, and superior alveolar nerves, which also innervate the adjacent teeth. Gingiva on the palatal (lingual) side of the same teeth is innervated by the nasopalatine and the greater palatine nerves:

- The nasopalatine nerve innervates gingiva associated with the incisor and canine teeth.
- The greater palatine nerve supplies gingiva associated with the remaining teeth.

The gingiva associated with the (buccal) side of the mandibular incisor, canine, and premolar teeth is innervated by the mental branch of the inferior alveolar nerve. Gingiva on the buccal side of the mandibular molar teeth is innervated by the buccal nerve, which originates in the infratemporal fossa from the mandibular nerve [V_3]. Gingiva adjacent to the lingual surface of all lower teeth is innervated by the lingual nerve.

Surface anatomy

Head and neck surface anatomy

Skeletal landmarks in the head and neck are used for locating major blood vessels, glands, and muscles, and for locating points of access to the airway.

Neurological examination of the cranial and upper cervical nerves is carried out by assessing function in the head and neck.

In addition, information about the general status of body health can often be obtained by evaluating surface features, the eye and the oral cavity, and the characteristics of speech.

Anatomical position of the head and major landmarks

The head is in the anatomical position when the inferior margins of the bony orbits and the superior margins of the external acoustic meatuses are in the same horizontal plane (Frankfort plane).

In addition to the external acoustic meatus and the bony margin of the orbit, other features that are palpable include the head of the mandible, zygomatic arch, zygomatic bone, mastoid process, and external occipital protuberance (Fig. 8.278).

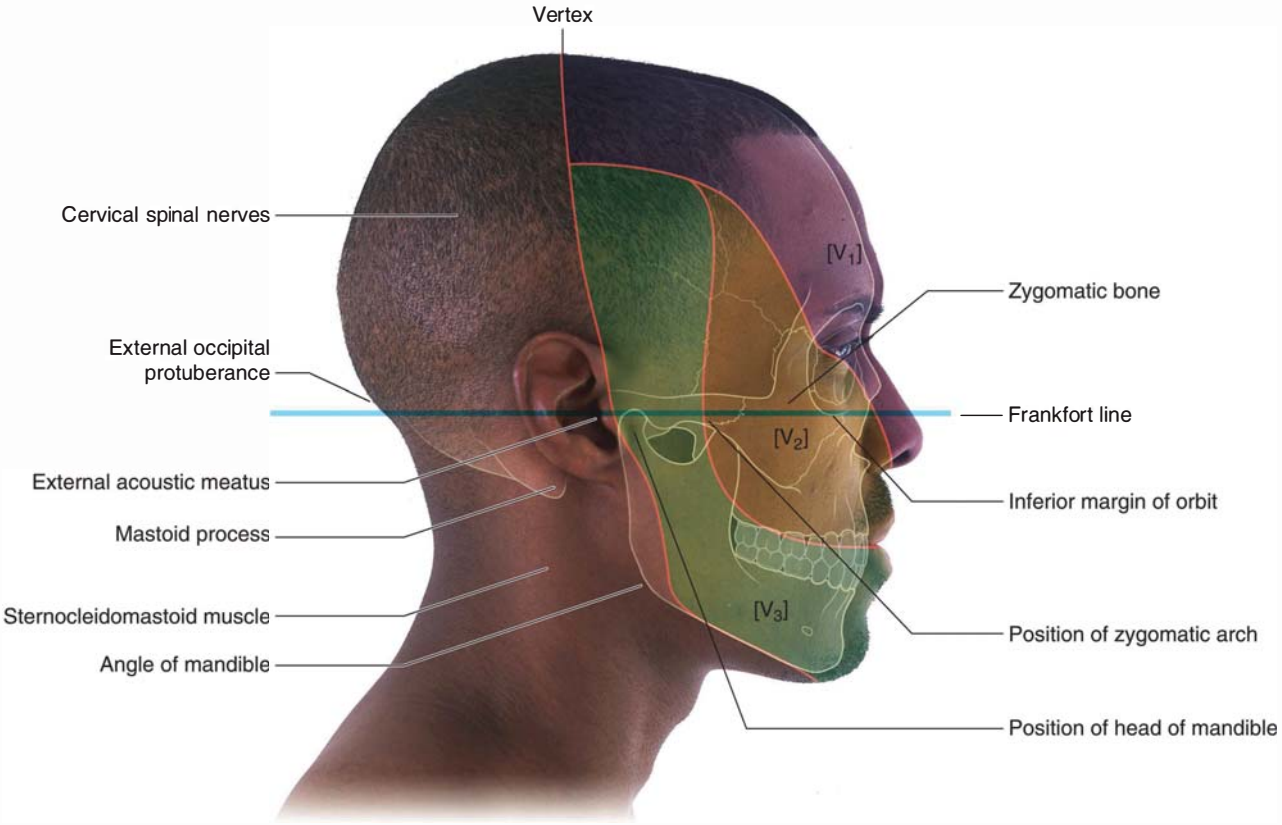


Fig. 8.278 Anatomical position of the head and major landmarks. Lateral head and neck of a man.

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The head of the mandible is anterior to the external ear and behind and inferior to the posterior end of the zygomatic arch. It is best found by opening and closing the jaw and palpating the head of the mandible as it moves forward onto the articular tubercle and then back into the mandibular fossa, respectively.

The zygomatic arch extends forward from the region of the temporomandibular joint to the zygomatic bone, which forms a bony prominence lateral to the inferior margin of the anterior opening of the orbit.

The mastoid process is a large bony protuberance that is easily palpable posterior to the inferior aspect of the external acoustic meatus. The superior end of the sternocleidomastoid muscle attaches to the mastoid process.

The external occipital protuberance is palpable in the midline posteriorly where the contour of the skull curves sharply forward. This landmark marks the point superficially where the back of the neck joins the head.

Another clinically useful feature of the head is the vertex. This is the highest point of the head in the anatomical position and marks the approximate point on the scalp where there is a transition from cervical to cranial innervation of the scalp. Anterior to the vertex, the scalp and face are innervated by the trigeminal nerve [V]. Posterior to the vertex, the scalp is innervated by branches from cervical spinal nerves.

Visualizing structures at the CIII/CIV and CVI vertebral levels

Two vertebral levels in the neck are associated with important anatomical features (Fig. 8.279).

The intervertebral disc between the CIII and CIV vertebrae is in the same horizontal plane as the bifurcation of the common carotid artery into the internal and external carotid arteries. This level is approximately at the upper margin of the thyroid cartilage.

Vertebral level CVI marks the transition from pharynx to esophagus and larynx to trachea. The CVI vertebral level therefore marks the superior ends of the esophagus and trachea and is approximately at the level of the inferior margin of the cricoid cartilage.

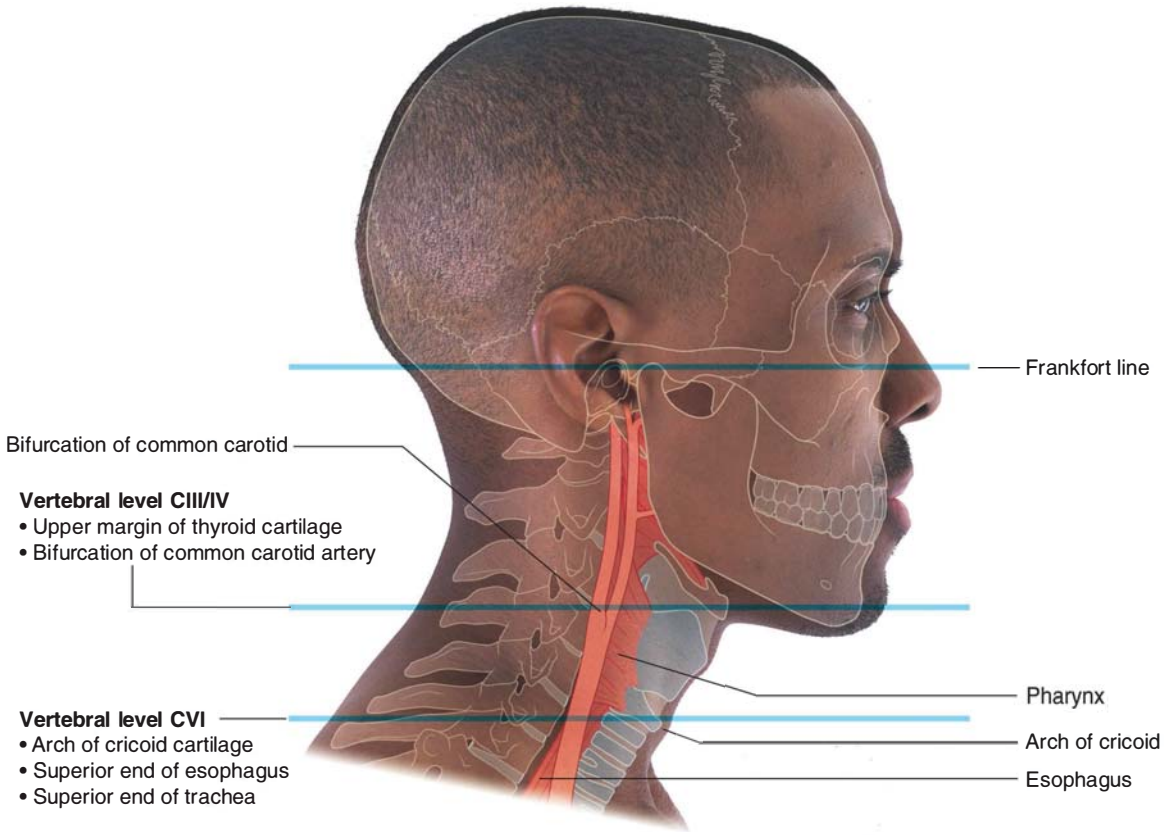


Fig. 8.279 Visualizing structures at the CIII/IV and CVI vertebral levels. Lateral head and neck of a man.

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How to outline the anterior and posterior triangles of the neck

The boundaries of the anterior and posterior triangles on each side of the neck are easily established using readily visible bony and muscular landmarks (Fig. 8.280).

The base of each anterior triangle is the inferior margin of the mandible, the anterior margin is the midline of the neck, and the posterior margin is the anterior border of the sternocleidomastoid muscle. The apex of each anterior triangle points inferiorly and is at the suprasternal notch.

The anterior triangles are associated with structures such as the airway and digestive tract, and nerves and vessels that pass between the thorax and head. They are also associated with the thyroid and parathyroid glands.

The base of each posterior triangle is the middle one-third of the clavicle. The medial margin is the posterior border of the sternocleidomastoid muscle, and the lateral margin is the anterior border of the trapezius muscle. The apex points superiorly and is immediately posteroinferior to the mastoid process.

The posterior triangles are associated with nerves and vessels that pass into and out of the upper limbs.

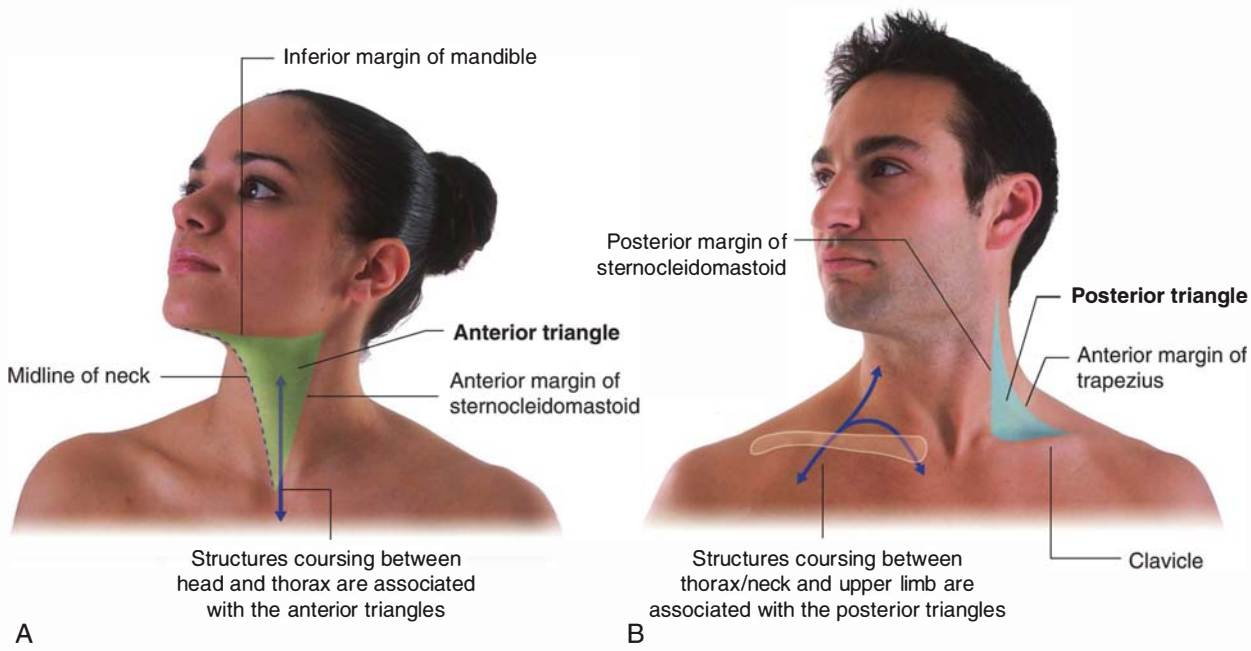


Fig. 8.280 How to outline the anterior and posterior triangles of the neck. **A.** In a woman, anterolateral view. The left anterior triangle is indicated. **B.** In a man, anterior view of the posterior triangle.

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How to locate the cricothyroid ligament

An important structure to locate in the neck is the cricothyroid ligament (cricovocal membrane, cricothyroid membrane) (Fig. 8.281) because artificial penetration of this membrane in emergency situations can provide access to the lower airway when the upper airway above the level of the vocal folds is blocked.

The ligament can be easily found using palpable features of the larynx as landmarks.

Using a finger to gently feel laryngeal structures in the midline, first find the thyroid notch in the superior margin of the thyroid cartilage and then move the finger inferiorly over the laryngeal prominence and down the anterior surface of the thyroid angle. As the finger crosses the inferior margin of the thyroid cartilage in the midline, a soft depression is felt before the finger slides onto the arch of the cricoid cartilage, which is hard.

The soft depression between the lower margin of the thyroid cartilage and the arch of the cricoid is the position of the cricothyroid ligament.

A tube passed through the cricothyroid ligament enters the airway just inferior to the position of the vocal folds of the larynx.

Structures that may occur in or cross the midline between the skin and the cricothyroid ligament include the pyramidal lobe of the thyroid gland and small vessels, respectively.

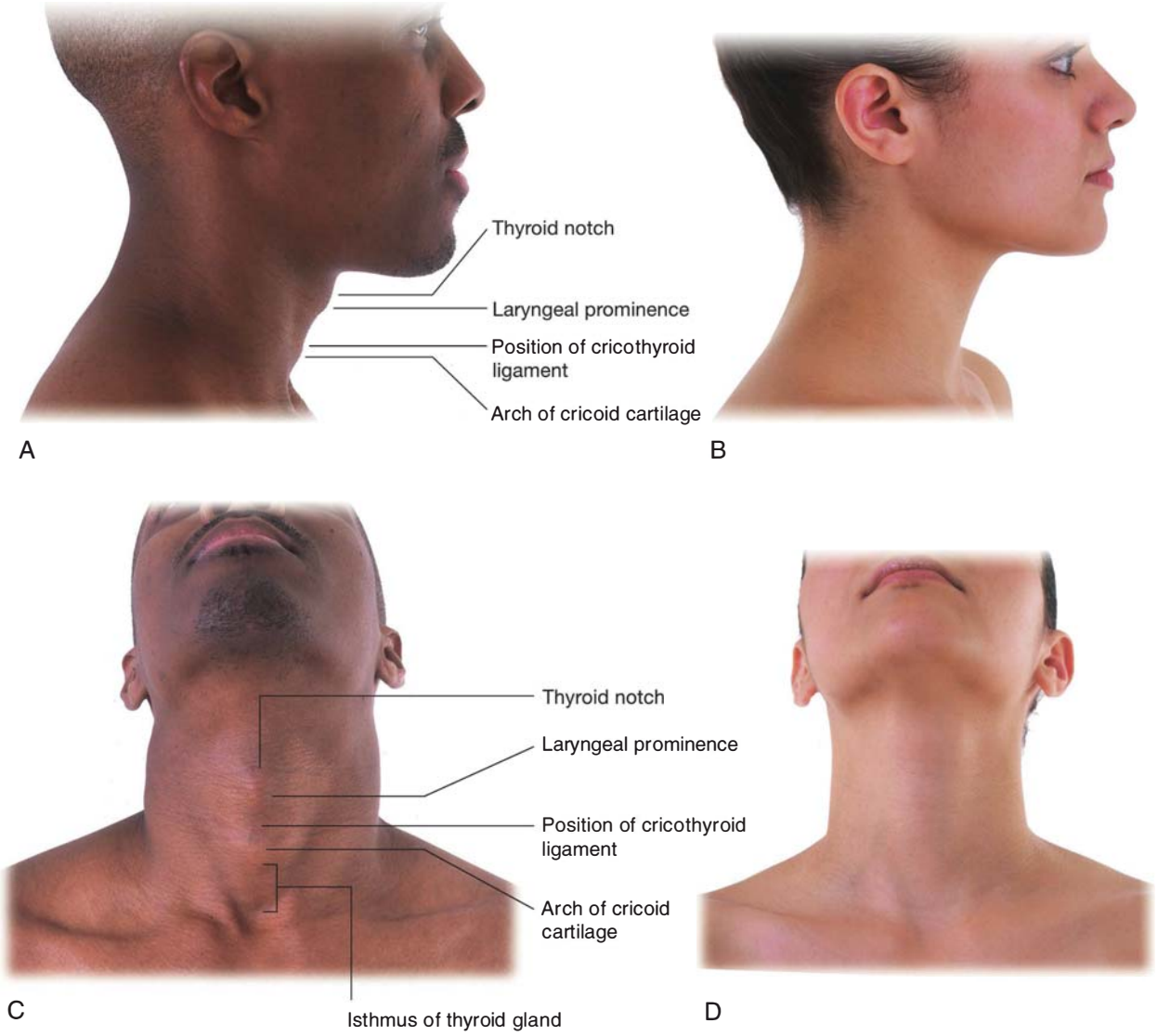


Fig. 8.281 How to locate the cricothyroid ligament. **A.** In a man, lateral view of head and neck. **B.** In a woman, lateral view of head and neck. **C.** In a man, anterior neck with the chin elevated. **D.** In a woman, anterior neck with the chin elevated.

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Inferior to the cricoid cartilage, the upper cartilage of the larynx can sometimes be palpated above the level of the isthmus of the thyroid gland that crosses the trachea anteriorly.

The landmarks used for finding the cricothyroid ligament are similar in men and women; however, because the laminae of the thyroid cartilage meet at a more acute angle in men, the structures are more prominent in men than in women.

How to find the thyroid gland

The left and right lobes of the thyroid gland are in the anterior triangles in the lower neck on either side of the airway and digestive tract inferior to the position of the oblique line of the thyroid cartilage (Fig. 8.282). In fact, the sternothyroid muscles, which attach superiorly to the oblique lines, lie anterior to the lobes of the thyroid gland and prevent the lobes from moving upward in the neck.

The lobes of the thyroid gland can be most easily palpated by finding the thyroid prominence and arch of the cricoid cartilage and then feeling posterolateral to the larynx.

The isthmus of the thyroid gland crosses anterior to the upper end of the trachea and can be easily palpated in the midline inferior to the arch of the cricoid.

The presence of the isthmus of the thyroid gland makes palpating the tracheal cartilages difficult in the neck. Also, the presence of the isthmus of the thyroid gland and the associated vessels found in and crossing the midline makes it difficult to artificially enter the airway anteriorly through the trachea. This procedure, a tracheostomy, is a surgical procedure.

Estimating the position of the middle meningeal artery

The middle meningeal artery (Fig. 8.283) is a branch of the maxillary artery in the infratemporal fossa. It enters the skull through the foramen spinosum and is within the dura mater lining the cranial cavity.

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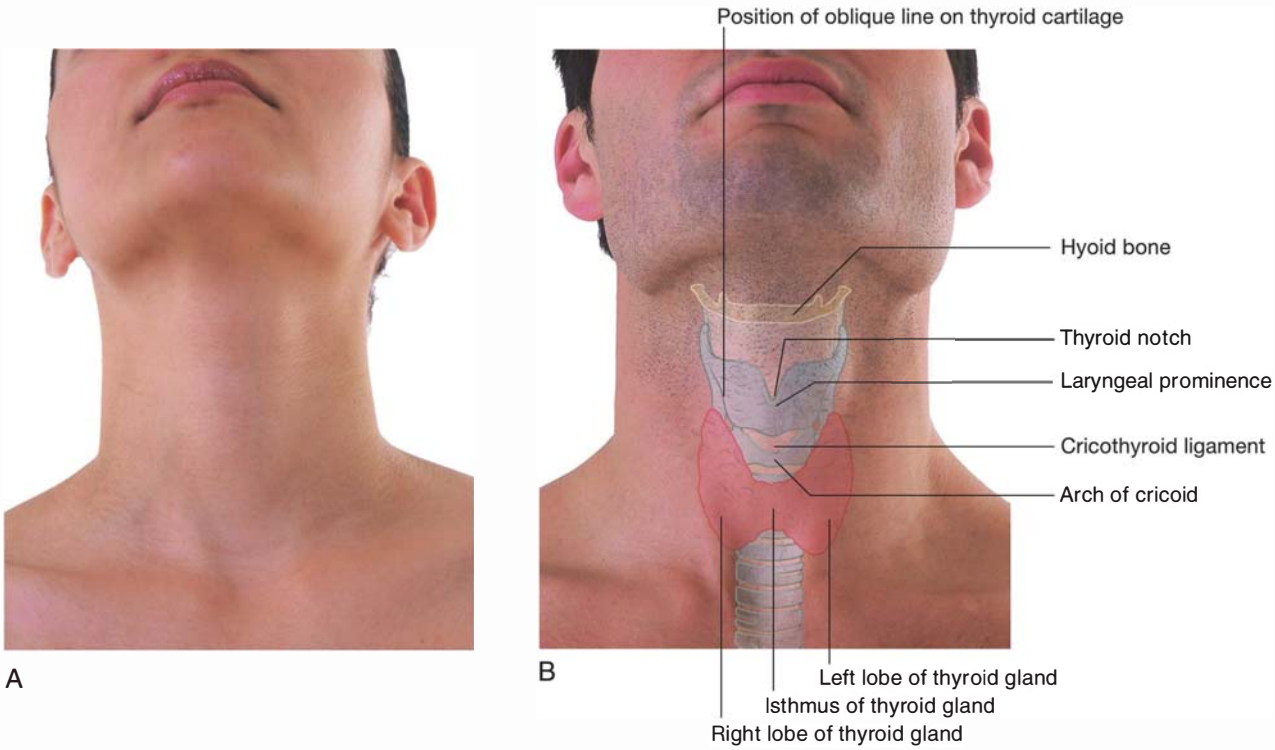


Fig. 8.282 How to find the thyroid gland. **A.** In a woman, anterior view of neck. **B.** In a man, anterior view of neck.

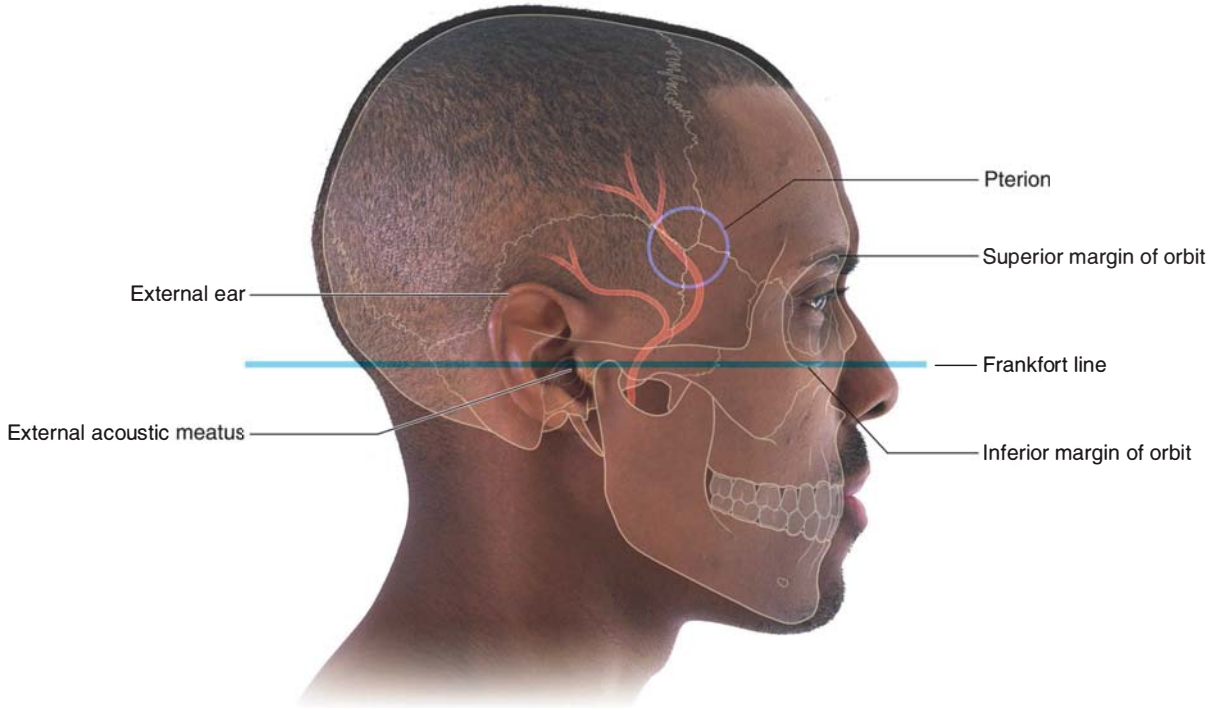


Fig. 8.283 Estimating the position of the middle meningeal artery. Lateral head and neck of a man.

In lateral blows to the head the middle meningeal artery can be ruptured, leading to extradural hemorrhage and eventual death if not treated.

The anterior branch of the middle meningeal artery is the part of the vessel most often torn. This branch is in the temple region of the head, approximately midway between the superior margin of the orbit and the upper part of the external ear in the pterion region. The pterion is a small circular area enclosing the region where the sphenoid, frontal, parietal, and temporal bones of the skull come together.

Lateral blows to the head can fracture the internal table of bone of the skull and tear the middle meningeal artery in the outer layer of dura mater that is fused to the cranium. Blood under pulsatile arterial pressure leaks out of the vessel and gradually separates the dura from the bone, forming a progressively larger extradural hematoma.

Major features of the face

The major features of the face are those related to the anterior openings of the orbit, the nasal cavities, and the oral cavity (Fig. 8.284).

The palpebral fissures are between the upper and lower eyelids and can be opened and closed. The oral fissure is the gap between the upper and lower lips and can also be opened and closed.

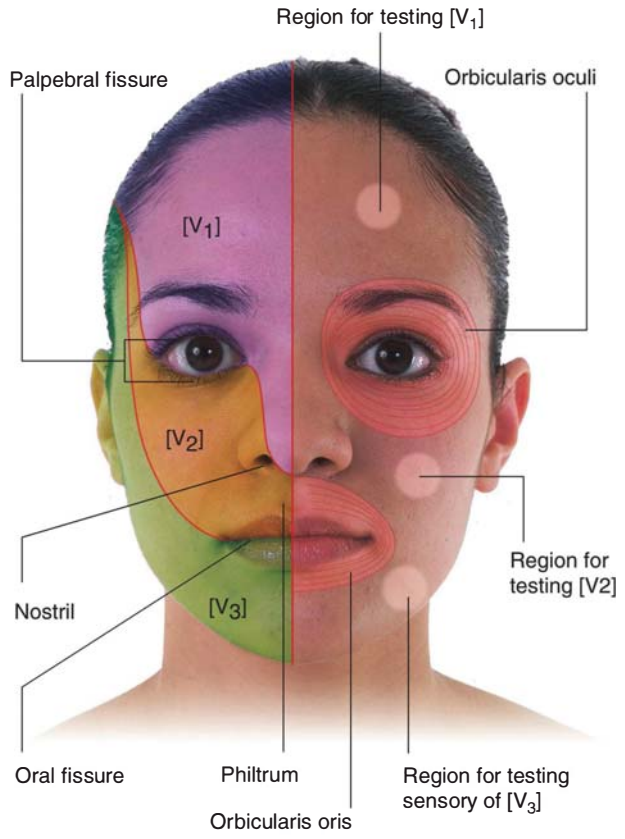


Fig. 8.284 Major features of the face. Anterior head and neck of a woman.

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The sphincter muscles of the oral and palpebral fissures are the orbicularis oris and orbicularis oculi muscles, respectively. These muscles are innervated by the facial nerve [VII].

The nares are the anterior apertures of the nasal cavities and are continuously open.

The vertical groove in the midline between the external nose and the upper lip is the philtrum.

Sensory innervation of the face is carried by the trigeminal nerve [V]. The three divisions of this nerve are represented on the face and can be tested by touching the forehead (the ophthalmic nerve [V_1]), the anterior cheek (the maxillary nerve [V_2]), and skin over the anterior body of the mandible (the mandibular nerve [V_3]).

The eye and lacrimal apparatus

Major features of the eye include the sclera, cornea, iris, and pupil (Fig. 8.285). The cornea is continuous with the sclera and is the clear circular region of the external covering of the eye through which the pupil and iris are visible. The sclera is not transparent and is normally white.

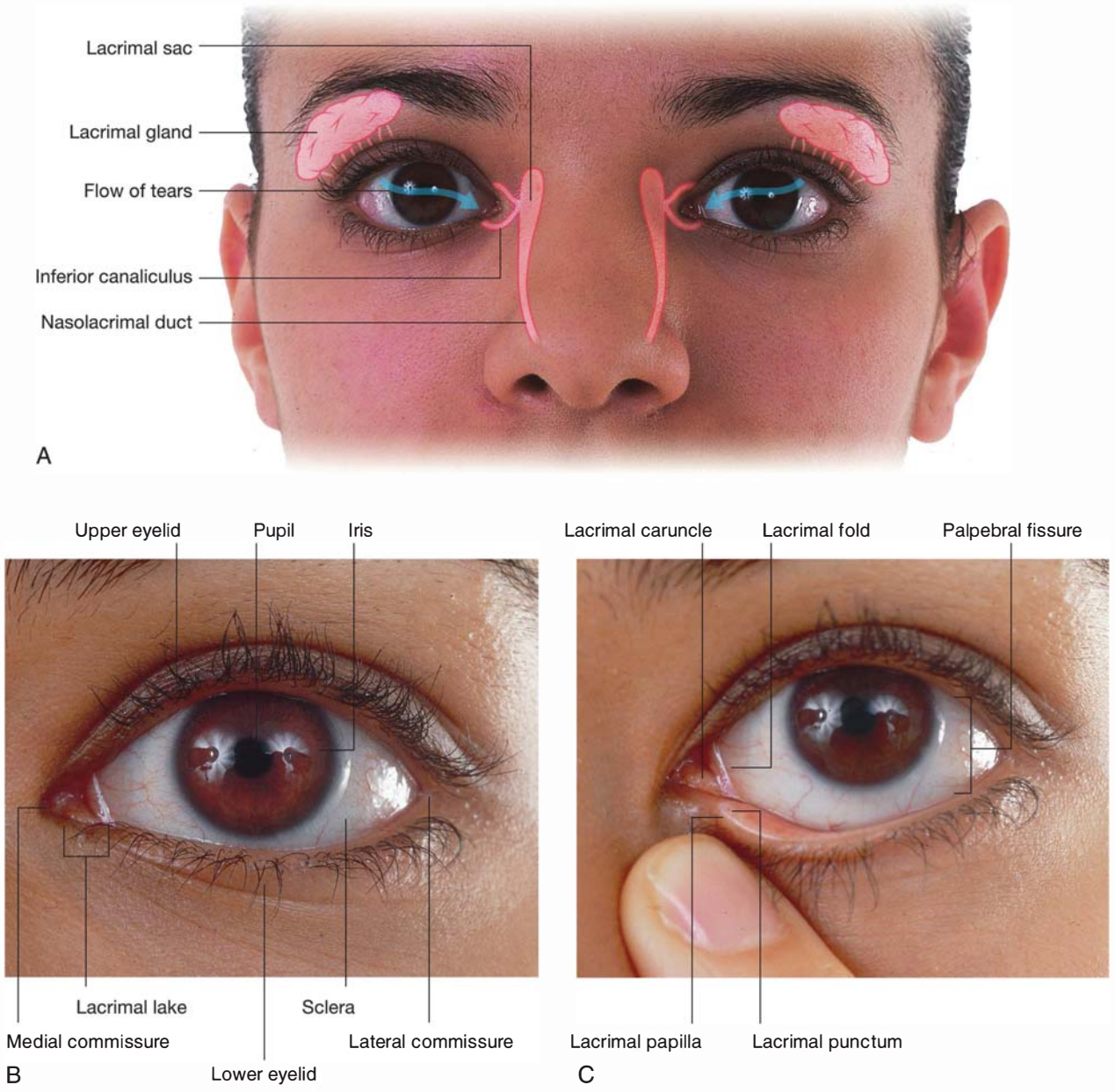


Fig. 8.285 Eye and lacrimal apparatus. **A.** Face of a woman. Lacrimal apparatus and the flow of tears are indicated. **B.** Left eye and surrounding structures. **C.** Left eye and surrounding structures with lower eyelid pulled down to reveal the lacrimal papilla and lacrimal punctum.

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The upper and lower eyelids of each eye enclose between them the palpebral fissure. The eyelids come together at the medial and lateral palpebral commissures on either side of each eye.

At the medial side of the palpebral fissure and lateral to the medial palpebral commissure is a small triangular soft tissue structure (the lacrimal lake).

The elevated mound of tissue on the medial side of the lacrimal lake is the lacrimal caruncle, and the lateral margin overlying the sclera is the lacrimal fold.

The lacrimal apparatus consists of the lacrimal gland and the system of ducts and channels that collects the tears and drain them into the nasal cavity. Tears hydrate and maintain the transparency of the cornea.

The lacrimal gland is associated with the upper eyelid and is in a small depression in the lateral roof of the orbit just posterior to the orbital margin. The multiple small ducts of the gland open into the upper margin of the conjunctival sac, which is the thin gap between the deep surface of the eyelid and the cornea.

Tears are swept medially over the eye by blinking and are collected in small openings (lacrimal puncta), one on each of the upper and lower eyelids near the lacrimal lake.

Each punctum is on a small raised mound of tissue (a lacrimal papilla), and is the opening of a small canal (lacrimal canaliculus) that connects with the lacrimal sac.

The lacrimal sac is in the lacrimal fossa on the medial side of the orbit. From the lacrimal sac, tears drain via the nasolacrimal duct into the nasal cavity.

External ear

The external ear (Fig. 8.286) consists of the auricle and the external acoustic meatus. The auricle is supported by cartilage and is covered by skin. The external auditory meatus is near the anterior margin of the auricle.

The auricle is characterized by a number of depressions, eminences, and folds. The folded outer margin of the auricle is the helix, which ends inferiorly as the lobule. A smaller fold (the antihelix) parallels the contour of the helix and is separated from it by a depression (the scaphoid fossa).

The tragus is a small eminence anteroinferior to the external acoustic meatus. Opposite the tragus and at the end of the antihelix is another eminence (the antitragus). The depression between the tragus and antitragus is the intertragic incisure.

The deepest depression (the concha) is bracketed by the antihelix and leads into the external acoustic meatus. Other depressions include the triangular fossa and the cymba conchae.

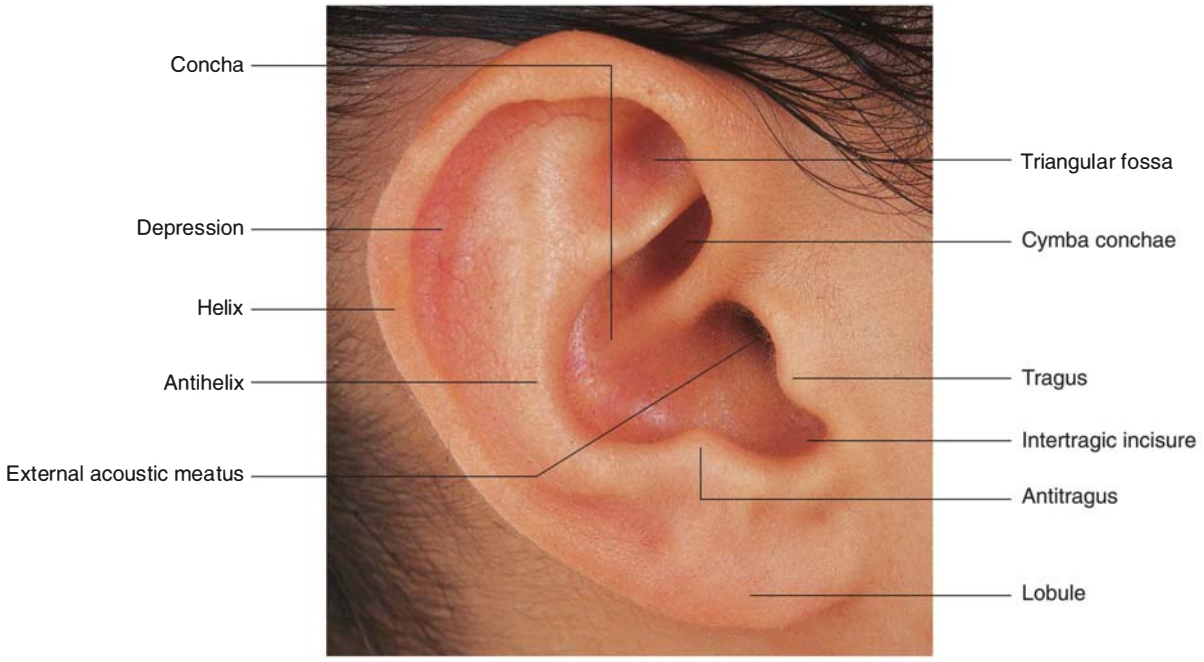


Fig. 8.286 External ear. Lateral view of the right ear of a woman.

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Pulse points

Arterial pulses can be felt at four locations in the head and neck (Fig. 8.287).

- Carotid pulse—the common or external carotid artery can be palpated in the anterior triangle of the neck. This is one of the strongest pulses in the body. The pulse can be obtained by palpating either the common carotid artery posterolateral to the larynx or the external carotid artery immediately lateral to the pharynx midway between the superior margin of the thyroid cartilage below and the greater horn of the hyoid bone above.
- Facial pulse—the facial artery can be palpated as it crosses the inferior border of the mandible immediately adjacent to the anterior margin of the masseter muscle.
- Temporal pulse—the superficial temporal artery can be palpated anterior to the ear and immediately postero-superior to the position of the temporomandibular joint.
- Temporal pulse—the anterior branch of the superficial temporal artery can be palpated posterior to the zygomatic process of the frontal bone as it passes lateral to the temporal fascia and into anterolateral regions of the scalp. In some individuals pulsations of the superficial temporal artery can be seen through the skin.

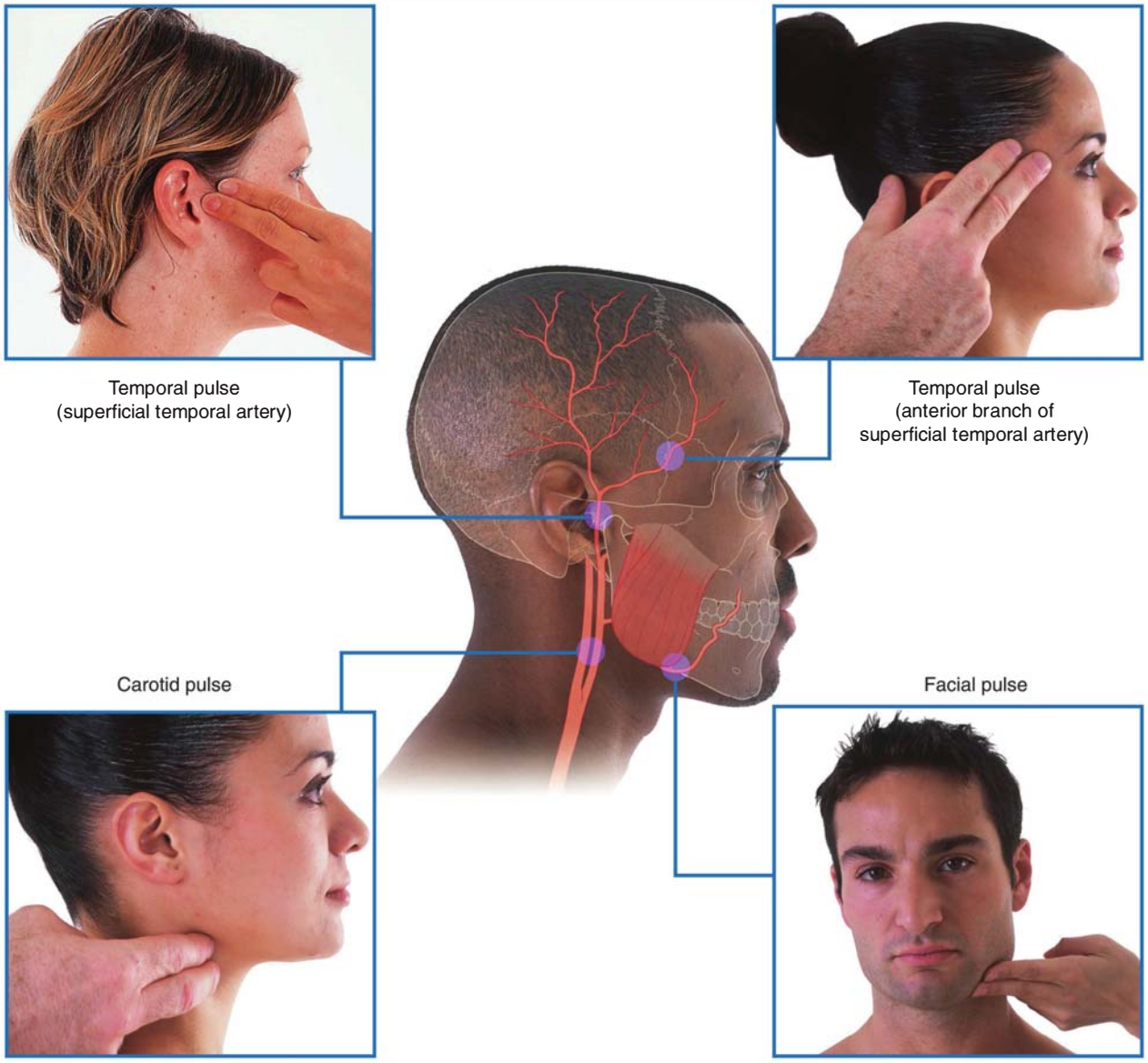


Fig. 8.287 Where to take arterial pulses in the head and neck.

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Clinical cases

Case 1

MULTINODULAR GOITER

A 50-year-old overweight woman came to the doctor complaining of hoarseness of voice and noisy breathing. She was also concerned at the increase in size of her neck. On examination she had a slow pulse rate (45 beats per minute). She also had an irregular knobby mass in the anterior aspect of the lower neck, which deviated the trachea to the right.

A clinical diagnosis of a multinodular goiter and hypothyroidism was made.

Enlargement of the thyroid gland is due to increased secretion of thyroid-stimulating hormone, which is usually secondary to diminished output of thyroid hormones. The thyroid undergoes periods of activity and regression, which can lead to the formation of nodules, some of which are solid and some of which are partially cystic (colloid cysts). This nodule formation is compounded by areas of fibrosis within the gland. Other causes of multinodular goiter include iodine deficiency and in certain circumstances, drugs that interfere with the metabolism and production of thyroxine. The typical symptom of a goiter is a painless swelling of the thyroid gland. It may be smooth or nodular, and occasionally it may extend into the superior mediastinum as a retrosternal goiter.

The trachea was deviated.

The enlargement of the thyroid gland due to a multinodular goiter may not be symmetrical. In this case there was significant asymmetrical enlargement of the left lobe of the thyroid deviating the trachea to the right.

The patient had a hoarse voice and noisy breathing.

If the thyroid gland enlargement is significant it can compress the trachea, narrowing it to such an extent that a "crowing sound" is heard during inspiration (stridor).

Other possible causes for hoarseness include paralysis of the vocal cord due to compression of the left recurrent laryngeal nerve from the goiter. Of concern is the possibility of malignant change within the goiter directly invading the recurrent laryngeal nerve. Fortunately, malignant change is rare within the thyroid gland.

When patients have a relatively low production of thyroxine such that the basal metabolic rate is reduced they become more susceptible to infection, including throat and upper respiratory tract infections.

On examination the thyroid gland moved during swallowing.

Characteristically, an enlarged thyroid gland is evident as a neck mass arising on one or both sides of the trachea. The enlarged thyroid gland moves on swallowing because it is attached to the larynx by the pretracheal fascia.

The patient was hypothyroid.

Hypothyroidism refers to the clinical and biochemical state in which the thyroid gland is underactive (hyperthyroidism refers to an overactive thyroid gland). Some patients have thyroid masses and no clinical or biochemical abnormalities—these patients are euthyroid.

The hormone thyroxine controls the basal metabolic rate; therefore, low levels of thyroxine affect the resting pulse rate and may produce other changes, including weight gain, and in some cases depression.

The patient was insistent upon surgery.

After discussion about the risks and complications, a subtotal thyroidectomy was performed. After the procedure the patient complained of tingling in her hands and feet and around her mouth, and carpopedal spasm. These symptoms are typical of tetany and are caused by low serum calcium levels.

The etiology of the low serum calcium level was trauma and bruising of the four parathyroid glands left in situ after the operation. Undoubtedly the trauma of removal of such a large thyroid gland produced a change within the parathyroid gland, which failed to function appropriately. The secretion of parathyroid hormone rapidly decreased over the next 24 hours, resulting in increased excitability of peripheral nerves, manifest by carpopedal spasm and orofacial tingling. Muscle spasms can also be elicited by tapping the facial nerve [VII] as it

(continues)

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Case 1 *(continued)*

emerges from the parotid gland to produce twitching of the facial muscles (Chvostek's sign).

The patient recovered from these symptoms due to a low calcium level over the next 24 hours.

At her return to the clinic the patient was placed on supplementary oral thyroxine, which is necessary after removal of the thyroid gland.

The patient also complained of a hoarse voice.

The etiology of her hoarse voice was damage to the recurrent laryngeal nerve.

The recurrent laryngeal nerve lies close to the thyroid gland. It may be damaged in difficult surgical procedures, and this may produce unilateral spasm of the ipsilateral vocal cord to produce a hoarse voice.

Since the thyroidectomy and institution of thyroxine treatment, the patient has lost weight and has no further complaints.

Case 2

PAROTID DUCT CALCULUS

A 25-year-old man complained of significant swelling in front of his right ear before and around mealtimes. This swelling was associated with considerable pain, which was provoked by the ingestion of lemon sweets. On examination he had tenderness around the right parotid region and a hard nodule was demonstrated in the buccal mucosa adjacent to the right upper molar teeth.

A diagnosis of parotid duct calculus was made.

The formation of stones in the salivary glands is not uncommon, but it is more likely in the submandibular gland than in the parotid gland because the saliva is more mucinous and the duct has a long upward course from the floor of the mouth. Nevertheless, stones do form in the parotid gland and the parotid ducts. Notably, most parotid duct calculi and submandibular duct calculi occur in mouths with excellent dental hygiene and mucosa.

An ultrasound scan was performed.

An initial ultrasound scan demonstrated a stone in the distal end of the right parotid duct with evidence of ductal dilatation (Fig. 8.288). Assessment of the gland also demonstrated dilated ducts within the gland and evidence of intraparotid lymphadenopathy.

The patient was treated with antibiotics.

A course of antibiotics was given to remove the bacteria that had produced the inflammation. On return to the doctor some days later the gland was normal in size and there was no evidence of inflammation or infection.

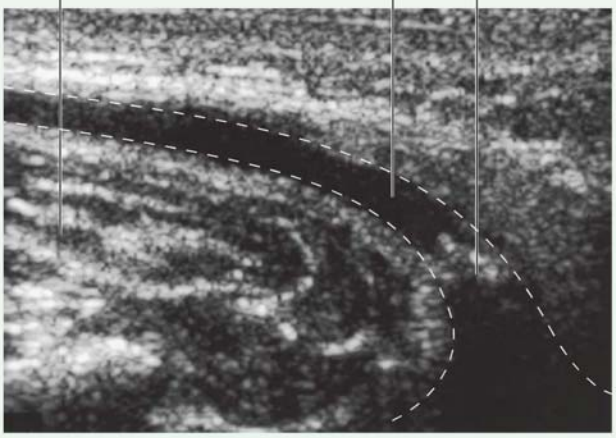


Fig. 8.288 Ultrasound scan (axial view) demonstrating a stone in a dilated parotid duct.

An operation was necessary.

The stone was at the distal end of the parotid duct and it would seem logical and straightforward to make a small incision at the sphincter in the buccal mucosa and deliver the stone, thus permitting the gland to drain normally. Unfortunately, in this patient's case the gland was significantly destroyed by the chronic obstruction and bacterial infection. Furthermore, smaller calculi were also demonstrated in the gland at ultrasound. On direct questioning it appeared that the patient had had numerous attacks over the previous 4–5 years and it was decided that the parotid gland should be removed surgically.

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Case 2 *(continued)*

The patient consented for removal of the parotid gland and a discussion of the possibility for loss of facial function and facial paralysis was had with the patient at this time.

Within the parotid gland the facial nerve [VII] divides into its five terminal branches. At operation the gland is displayed and extremely careful dissection is necessary to peel away the parotid gland from the branches of the facial nerve [VII]. This procedure was made more difficult by the chronic inflammatory change within the gland. After the procedure the patient made a good recovery, though there was some mild paralysis of the whole of the right side of the face. Importantly, taste to the anterior two-thirds of the tongue was preserved. The taste fibers to the anterior two-thirds of the tongue travel in the chorda tympani nerve, which is a branch of the facial nerve [VII]. This nerve leaves the facial nerve [VII] to join the lingual nerve proximal to the parotid gland; therefore, any damage to the facial nerve [VII] within the parotid gland does not affect special sensation (taste).

Over the following week the paralysis improved and was likely due to nerve bruising during the procedure. The patient remained asymptomatic.

Case 3

EXTRADURAL HEMATOMA

A 33-year-old man was playing cricket for his local Sunday team. As the new bowler pitched the ball short, it bounced higher than he anticipated and hit him on the side of his head. He immediately fell to the ground unconscious, but after about 30 seconds he was helped to his feet and felt otherwise well. It was noted he had some bruising around his temple. He decided not to continue playing and went to watch the match from the side. Over the next hour he became extremely sleepy and was eventually unrousable. He was rushed to hospital.

When he was admitted to hospital, the patient's breathing was shallow and irregular and it was necessary to intubate him. A skull radiograph demonstrated a fracture in the region of the pterion. No other abnormality was demonstrated other than minor soft tissue bruising over the left temporal fossa.

A CT scan was performed.

The CT scan demonstrated a lentiform area of high density within the left cranial fossa.

A diagnosis of extradural hemorrhage was made.

Fractures in the region of the pterion are extremely dangerous. A division of the middle meningeal artery passes deep to this structure and is subject to laceration and disruption, especially in conjunction with a skull injury in this region. In this case the middle meningeal artery was torn and started to bleed, producing a large extradural clot.

The patient's blood pressure began to increase.

Within the skull there is a fixed volume and clearly what goes in must come out (e.g., blood, cerebrospinal fluid). If there is a space-occupying lesion, such as an extradural hematoma, there is no space into which it can decompress. As the lesion expands, the brain becomes compressed and the intracranial pressure increases. This pressure compresses vessels, so lowering the cerebral perfusion pressure. To combat this the homeostatic mechanisms of the body increase the blood pressure to overcome the increase in intracerebral pressure. Unfortunately, the increase in intracranial pressure is compounded by the cerebral edema that occurs at and after the initial insult.

An urgent surgical procedure was performed.

Burr holes were placed around the region of the hematoma and it was evacuated. The small branch of the middle meningeal artery was ligated and the patient spent a few days in the intensive care unit. Fortunately the patient made an uneventful recovery.

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Case 4

STENOSIS OF THE INTERNAL CAROTID ARTERY

A 60-year-old woman was brought to the emergency department with acute right-sided weakness, predominantly in the upper limb, which lasted for 24 hours. She made an uneventful recovery, but was extremely concerned about the nature of her illness and went to see her local doctor.

A diagnosis of a transient ischemic attack (TIA) was made.

A TIA is a neurological deficit resolving within 24 hours. It is a type of stroke.

Neurological deficits may be permanent or transient. Most transient events resolve within 21 days; any failure of resolution beyond 21 days is an established stroke.

An investigation into the cause of the TIA was undertaken.

Eighty-five percent of all strokes result from cerebral infarction, of which most are due to embolization.

A duplex Doppler scan of the carotid vessels was performed.

The majority of emboli originate from plaques that develop at and around the carotid bifurcation. Emboli consist of platelet aggregates, cholesterol, and atheromatous debris. Emboli may also arise from the heart secondary to cardiac tumors or myocardial infarction.

The lesion in the brain was on the left side.

The motor cortex for the whole of the right side of the body is represented in the left motor strip of the brain, which sits on the precentral gyrus.

The duplex Doppler ultrasound scan demonstrated a significant narrowing (stenosis) of the left internal carotid artery with evidence of plaque formation and abnormal flow in this region. The narrowing was approximately 90%.

Treatment required an operation.

A carotid endarterectomy (removal of the stenosis and the atheromatous plaque) was planned. This procedure is indicated in the presence of an ulcerating plaque with stenosis. The procedure was carried out under general anesthetic and a curvilinear incision was placed in the left side of the neck. The common carotid, external carotid, and internal carotid arteries were displayed. All vessels were clamped and a shunt was placed from the common carotid artery into the internal carotid artery to maintain cerebral blood flow during the procedure. The internal carotid artery was opened and the plaque excised.

After the procedure the patient did extremely well and suffered no further cerebral events. However, a new medical student examined the patient the following day and demonstrated a number of interesting findings. These included altered skin sensation inferior to the left mandible, altered sensation on the left side of the soft palate, a paralyzed left vocal cord, inability to shrug the left shoulder, and a tongue that deviated to the left.

The etiology of these injuries was due to localized nerve trauma.

This constellation of neurological deficits can be accounted for by trauma to the nerves that are close to the carotid bifurcation. The changes in skin sensation can be accounted for by a neurapraxia due to damage to cervical nerves. The alteration in sensation in the soft palate is due to neurapraxia of the glossopharyngeal nerve [IX]. The paralyzed left cord results from neurapraxia of the recurrent laryngeal nerve, while the inability to shrug the shoulder is due to neurapraxia of the accessory nerve [XI]. Deviation of the tongue can be accounted for by damage to the hypoglossal nerve [XII].

Most of these changes are transient and are usually due to traction injuries during the surgical procedure.

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Case 5

POSTERIOR COMMUNICATING ARTERY ANEURYSM

A 33-year-old fit and well woman came to the emergency department complaining of double vision and pain behind her right eye. She had no other symptoms. On examination of the right eye the pupil was dilated. There was a mild ptosis. Testing of eye movement revealed that the eye turned down and out and the pupillary reflex was not present.

These findings revealed that the patient had an ipsilateral third nerve palsy (palsy of the oculomotor nerve [III]).

The oculomotor nerve [III] is the main motor nerve to the ocular and extra-ocular muscles. It arises from the midbrain and pierces the dura mater to run in the lateral wall of the cavernous sinus. The oculomotor nerve [III] leaves the cranial cavity and enters the orbit through the superior orbital fissure. Within this fissure it divides into its superior and inferior divisions.

The site of the nerve lesion needs to be assessed.

Third nerve palsy may involve the nucleus of the oculomotor nerve [III], which typically spares the pupil and is painless. The pupillary reflexes are supplied from the autonomic fibers of the Edinger–Westphal nucleus, which pass through the ciliary ganglion.

The lesion cannot be a primary oculomotor nerve [III] nuclear injury.

As both the pupillary reflexes and vision are affected, the lesion is likely to be along the course of the oculomotor nerve [III]. Medical conditions such as diabetes mellitus and vascular disease may produce an isolated oculomotor nerve [III] injury, but they are not associated with pain.

The lesion was caused by an aneurysm.

One of the commonest causes of a third nerve palsy is pressure on the nerve from a posterior communicating artery aneurysm, which lies parallel to the nerve on the anterior aspect of the brainstem. As the aneurysm abuts the outside of the oculomotor nerve [III], it involves the parasympathetic fibers, which lead to a predominance of the loss of pupillary function over general function.

The aneurysm was imaged with an angiogram.

The patient initially underwent CT and MRI scanning. Currently, the definitive test for assessment of aneurysms arising from the circle of Willis and its branches is a digital subtraction angiogram. The angiogram demonstrated the posterior communicating artery aneurysm. The patient underwent surgery and made an excellent recovery.

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Case 6

RECURRENT EPISTAXIS

A 10-year-old boy was brought to an ENT surgeon (ear, nose, and throat surgeon) with epistaxis (nose bleeding). The bleeding was associated with his nose picking habit. However, the bleeding was profuse and on two occasions required hospital admission and nasal packing.

On inspection an indurated area was noted.

The typical findings are an indurated area in the anterior inferior aspect of the nasal septum (Kiesselbach's area). This is a very vascular area that has a considerable number of veins, which are often traumatized during nose picking.

The patient underwent treatment.

Typical treatment is cauterization of these prominent veins in Kiesselbach's area, which is usually performed by a simple local analgesia and the application of silver nitrate.

Unfortunately, the boy was involved in a fight the next day and again developed severe epistaxis, which again was difficult to control.

Not only is there a rich venous plexus around Kiesselbach's area, but there is also a significant arterial supply, which is provided from the nasal septal branches of the posterior and anterior ethmoidal arteries and the branches of the greater palatine artery. These are supplemented from the septal branches of the superior labial artery.

In most cases treatment is conservative.

Conservative treatment usually involves packing the nasal cavity until bleeding has stopped and correcting any bleeding abnormality. In patients with bleeding refractory

(continues)

Case 6 *(continued)*

to medical treatment a series of maneuvers have been employed, including ligating the anterior and posterior ethmoidal arteries through a medial incision in the canthus orbit, or by ligating other major arteries supplying the nasal cavity. Unfortunately, many of these procedures fail because of the rich and diverse origin of blood supply to the nasal cavity.

Determination of the specific site of bleeding can be achieved radiologically.

By placing a catheter from the femoral artery through the aorta and into the carotid circulation the sphenopalatine artery can be easily cannulated from the maxillary branch of the external carotid artery. Bleeding can usually be demonstrated and the vessel can be embolized using small particles.

Fortunately in this young boy's case, bleeding stopped after further medical management and he remained asymptomatic.

Case 7

COMPLICATION OF ORBITAL FRACTURE

A 35-year-old man was involved in a fight and sustained a punch to the right orbit. He came to the emergency department with double vision.

The double vision was only in one plane.

Examination of the orbits revealed that when the patient was asked to look upward the right eye was unable to rotate superiorly when adducted. There was some limitation in general eye movement. Assessment of the lateral rectus muscle (abducent nerve [VI]), superior oblique muscle (trochlear nerve [IV]), and the rest of the eye muscles (oculomotor nerve [III]) was otherwise unremarkable.

The patient underwent a CT scan.

A CT scan of the facial bones demonstrated a fracture through the floor of the orbit (Fig. 8.289).

A careful review of this CT scan demonstrated that the inferior oblique muscle had been pulled inferiorly with the fragment of bone in the fracture. This produced a tethering effect, so when the patient was asked to gaze in the upward direction, the left eye was able to do so but the right eye was unable to because of the tethered inferior oblique muscle.

The patient underwent surgical exploration to elevate the small bony fragment and return the inferior oblique to its appropriate position. On follow-up the patient had no complications.

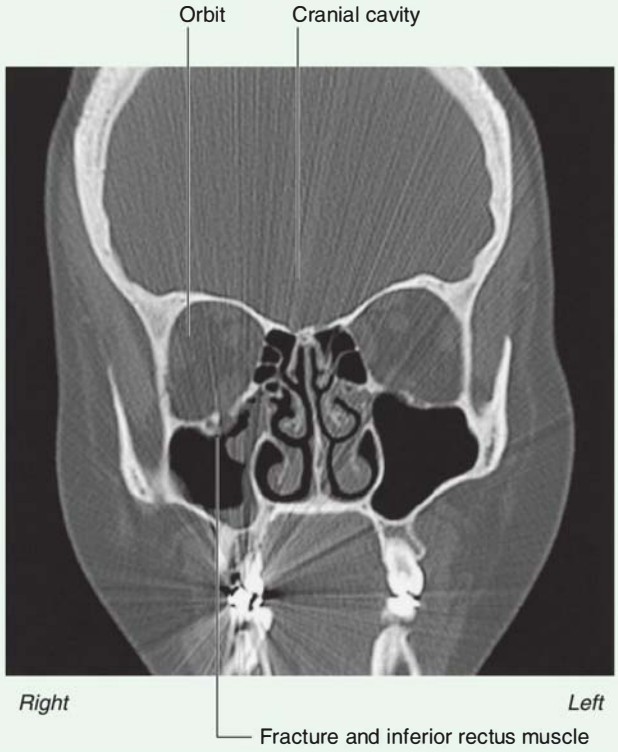


Fig. 8.289 Coronal CT scan demonstrating an orbital blowout fracture.

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Case 8

PITUITARY MACROADENOMA

A 30-year-old woman came to her doctor with a history of amenorrhea (absence of menses) and galactorrhea (the production of breast milk). She was not pregnant and appeared otherwise fit and well.

Serum prolactin was measured.

Prolactin is a hormone produced by the pituitary gland and necessary for the production of breast milk postpartum. This hormone was markedly elevated.

Further clinical tests demonstrated visual field defects.

The patient went to see an optometrist who performed a visual field assessment and demonstrated a reduction in the lateral aspects of the normal visual fields. This was bilateral and symmetrical—a bilateral temporal hemianopia.

The visual pathways have now determined the site of the lesion.

Visual information from the temporal fields is projected onto the medial aspect of the retina bilaterally. The visual information from the medial aspects of the retina is carried in fibers that cross the midline through the optic chiasm to the opposite side.

The lesion is in the area of the optic chiasm.

Any disruption of the optic chiasm produces the field defect of bitemporal hemianopia. Tumors of the optic chiasm are unusual, though gliomas do occur. More frequently, compression of the optic chiasm by tumors in the vicinity is the usual cause for bitemporal hemianopia.

A pituitary tumor was diagnosed.

The optic chiasm is anterior and extremely close to the pituitary gland. Given that the patient is producing excessive amounts of prolactin (a pituitary tumor) and there is loss of the function of the chiasm, the most likely clinical explanation is an exophytic pituitary tumor compressing the optic chiasm.

An MRI scan was performed and demonstrated a large tumor (macroadenoma) of the pituitary gland.

Drug treatment was commenced and the tumor shrank (Fig. 8.290). The endocrinological effects of the prolactin secretion also stopped.

Follow-up scans were performed.

Over the ensuing few years the tumor shrank. Unfortunately, the patient again began to secrete prolactin and surgery was performed.

A transsphenoidal approach was undertaken.

With meticulous accuracy a series of very fine instruments was passed through the nasal cavity into the sphenoid bone. The bone was drilled and via this approach the pituitary gland was removed.

Extreme care must be taken because on both sides of the pituitary gland is the cavernous sinus through which the internal carotid artery, oculomotor nerve [III], trochlear nerve [IV], trigeminal nerve [V], and abducent nerve [VI] pass.

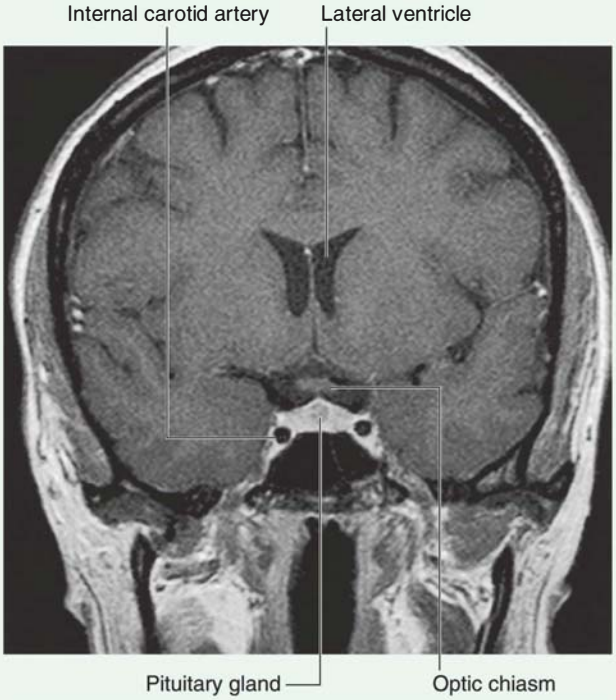


Fig. 8.290 Coronal MRI showing pituitary macroadenoma.

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